ZOO502 – MIDTERM MERGED PPT HANDOUTS (LECTURE 01 TO 132)

Animal Physiology & Behavior

Prepared By:

All About Biotech

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Note:

Students must follow the course announcement for midterm syllabus & any up-gradation in the course.

General Themes in Physiology

The Spectrum of Study of Physiology

Lecture no 1

The Spectrum of Study of Physiology

Physiology

- Study of functions of tissues, organs and organ systems of animals.

- To understand the mechanisms operating in animals.

The Spectrum of Study of Physiology

Physiology Obeys the Laws of Physics and Chemistry

- Ohm's Law
- Boyle's Law
- Ideal Gas Law
- Law of Gravity
- Kinetic & potential energy
- Inertia
- Momentum
- Velocity
- Drag

The Spectrum of Study of Physiology

Curiosity Underlies the Learning of Physiology

- How a Humming bird's heart beats 20 times per second during hovering flight?
- How can insects see in ultraviolet spectrum?
- How do kangaroo rats survive in deserts?

The Spectrum of Study of Physiology

From Animal Physiology to Human Physiology

- An insight into the physiological processes of humans.
- Human species has a linked evolutionary history.

Examples:

- Human heart beat
- Electrical nerve impulse

The Spectrum of Study of Physiology

From Animal Physiology to Human Physiology

- Animal physiology lays foundation of human physiology.
- Human physiology lays foundation of scientific medical practice.
- Developing treatment of human diseases.

The Spectrum of Study of Physiology

Animal Models for Human Diseases

- Diabetic mice
- Congenitally fat rats
- Zebra fish embryos with heart defects

- Provide insight into the physiological processes of disease

General Themes in Physiology

Central Themes in Physiology

Lecture no 2

Central Themes in Physiology

Major Goals of Study of Physiology

- Explore the basic physiological processes
- Their evolution by selective forces

They provide insight about:

- Patterns of physiological evolution
- Adaptive value of physiological processes

Central Themes in Physiology

Five Major Themes

1. Structure-function relationships
2. Adaptation, Acclimatization and Acclimation
3. Principle of homeostasis
4. Feedback control systems
5. Conformity and Regulation

General Themes in Physiology

Structure-Function Relationships

Lecture no 3

Structure-Function Relationships

Function is Based on Structure

- Structural design is matched to functional demands
- Such relationships arise through evolution and natural selection

Structure-Function Relationships

Example

A frog's feeding process involves:

- Skeletal muscles of limbs for jumping on prey
- Smooth muscles of stomach for grinding food
- Cardiac muscles of heart for distribution of food

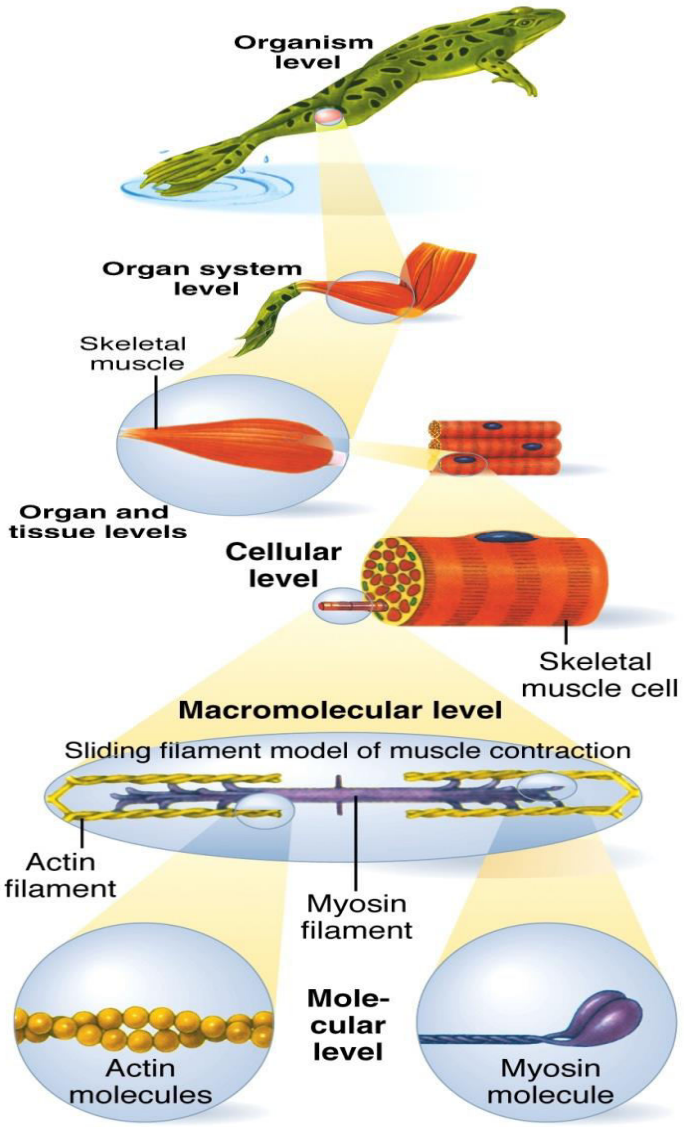
All these structures are adapted for their functions

Structure-Function Relationships

Structural Specialization for Functions

- Skeletal muscles—evolved and adapted for movement of bones
- Smooth muscles of digestive tract—specialized for grinding and mixing.
- Cardiac muscles—specialized to pump and circulate blood

Structure-Function Relationships



Eckert Animal Physiology (4th ed) by Randall

Structure-Function Relationships Exist At All Levels of Biological Organization

- System level—Muscles
- Cellular level—Muscle cells
- Macromolecular level—Sarcomeres
- Molecular level—Actin & Myosin

General Themes in Physiology

Adaptation, Acclimatization and Acclimation

Lecture no 4

Adaptation, Acclimatization and Acclimation

Adaptation

- A slowly occurring evolutionary process.
- Involves thousands of generations.
- Physiology of species becomes well matched to the environment.
- Ensures survival of the species.
- Generally not reversible.

Adaptation, Acclimatization and Acclimation

Acclimatization

- An adaptive change within an individual animal.
- Happens due to chronic exposure to new, naturally occurring environmental conditions.

Adaptation, Acclimatization and Acclimation

Acclimation

- Changes are induced experimentally in the laboratory

Both of these are:

- Acquired characters
- Restricted to one or few members of species
- Not inheritable
- Have no evolutionary significance
- Reversible

Adaptation, Acclimatization and Acclimation

Acclimatization

Example

- An animal migrates to a high altitude area
- Faces low oxygen partial pressure

Effects

- Lung ventilation rate increases initially
- Ventilation drops to normal in few days
- Animal has acclimatized to the high-altitude

Adaptation, Acclimatization and Acclimation

Acclimation

Example

- Simulating high-altitude conditions in lab.
- An animal physiologist places the animal in a hypobaric chamber.

Effects

- Breathing rate reacts in the same way.
- Animal becomes acclimated to the experimental conditions.

Adaptation, Acclimatization and Acclimation

Adaptation: Example

- Bar-headed goose
- Able to fly above the peaks of Mount Everest
- Species is naturally adapted to high altitude
- It is due to natural selection that operated for thousands of years



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General Themes in Physiology

Principle of Homeostasis

Lecture no 5

Principle of Homeostasis

Homeostasis

- Walter Cannon (1929)
- The tendency of an organism to maintain relative internal stability

Principle of Homeostasis

Environmental Fluctuations

- Most habitats are quite hostile to animal cells

Examples

- Freshwater is more dilute
- Seawater is more salty
- Environmental temperature may be too hot or too cold

Principle of Homeostasis

Need for Homeostatic Mechanisms

- Environmental fluctuations are disruptive to the functions of cells, tissues and organs
- So, physiological regulatory systems to maintain relatively stable conditions are a necessity.

Principle of Homeostasis

Homeostasis—Definition

- To keep internal fluctuations in a narrow range.
- The ability to protect internal environment from the harms of fluctuations in external environment.
- Homeostasis does not mean to keep a fixed internal environment.

Principle of Homeostasis

Homeostasis—Examples

- Homeostatic regulation of water
- Homeostatic regulation of temperature
- Regulation of pH
- Regulation of glucose concentration
- Regulation of osmotic pressure
- Regulation of oxygen level
- Regulation of ion conc.

Principle of Homeostasis

Mechanism of Homeostasis

- Variables
- Set point
- Living control systems VS physical control systems
- Three components: Receptors, Control centre, effectors

Principle of Homeostasis

Mechanism of Homeostasis—Example

- Temperature control system in air conditioners and water heating geysers VS Endothermic animals

General Themes in Physiology

Feedback Control Systems

Lecture no 6

Feedback Control Systems

Definition

- Mechanism of self-regulation of Biological processes.

Principle

- Output or product itself regulates the process

Significance

- Maintenance of homeostasis

Feedback Control Systems

Mechanism

Feedback controls regulate particular variables e.g.,

- Temperature
- Salinity
- pH

This regulation requires:

- Continuous sampling of controlled variables
- Respective corrective actions.

Feedback Control Systems

Types of Feedback Systems

Two Types:

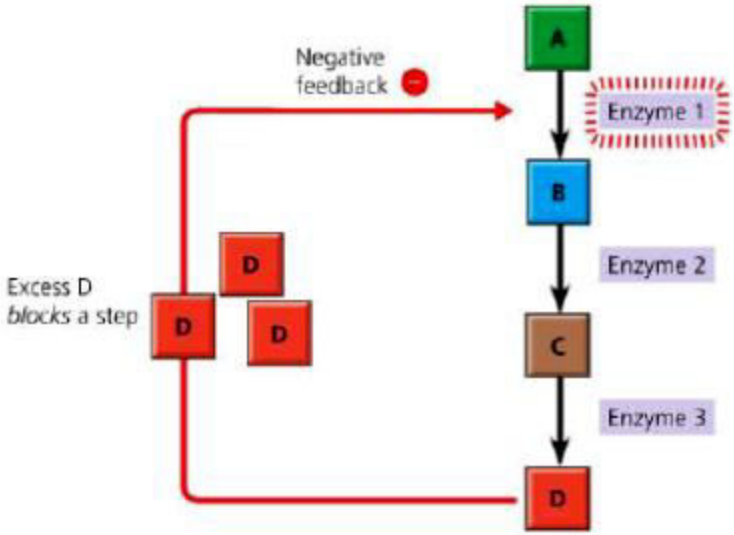
1. Negative feedback systems
2. Positive feedback systems

Feedback Control Systems

Negative Feedback Systems

- The most common form of regulation
- The end product works to stop or slow down the process

Feedback Control Systems



Negative Feedback Example~1

- Breakdown of sugar in the cells generates ATP.
- Excess ATP "feeds back" and inhibits an enzyme near the beginning of pathway.
- ATP production stops.

Biology Neil A. Campbell Jane B. Reece 8th ed 2008

Feedback Control Systems

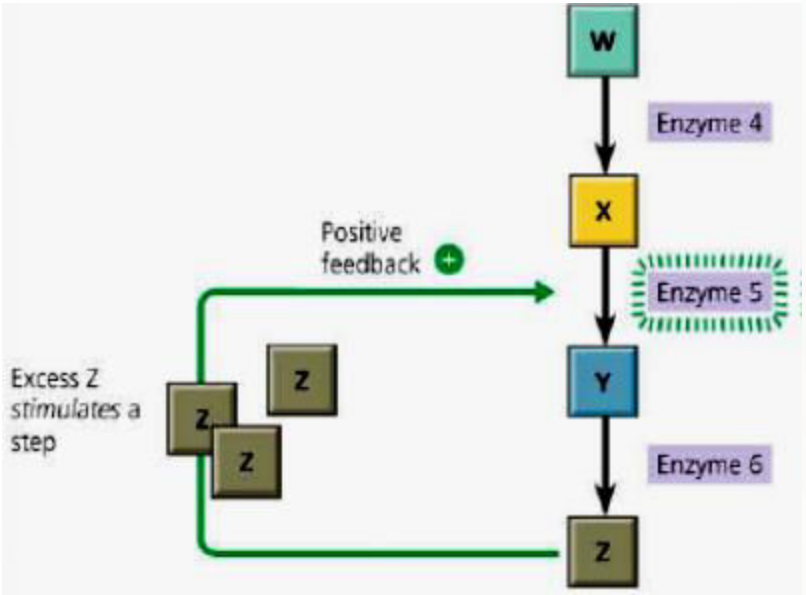
Negative Feedback Example~2

- Control of blood sugar (glucose) by insulin.
- Receptors sense a rise in blood sugar.
- Control center (pancreas) secretes insulin.
- Insulin lowers blood sugar levels.
- As blood sugar level reaches homeostasis, pancreas stops releasing insulin.

Feedback Control Systems

Positive Feedback Systems

- An end product speeds up the process of its production.



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Feedback Control Systems

Positive Feedback Examples

1. Clotting of blood in response to an injury
2. Release of Oxytocin during labor

General Themes in Physiology

Conformity and Regulation

Lecture no 7

Conformity and Regulation

Conformity and Regulation

Two categories of animal responses to changes in external environment:

- Conformity
- Regulation

Conformity and Regulation

Conformers

- Animals that conform to external changes
- Are unable to maintain homeostasis for internal conditions

Types

- Osmoconformers
- Oxyconformers
- Thermoconformers

Conformity and Regulation

Regulators

- Use internal control mechanisms to regulate internal conditions
- Maintain homeostasis

Types

- Osmoregulators
- Oxyregulators
- Thermoregulators

Membrane Physiology

Membrane Permeability

Lecture no 8

Membrane Permeability

Selective Permeability

- Structural features of plasma membrane (PM) confer it with selective permeability.
- PM regulates movement of substances across it.
- Fundamental to the functioning of living cells and maintenance of intracellular physiological conditions.

Membrane Permeability

Membrane Permeability For Various Substances

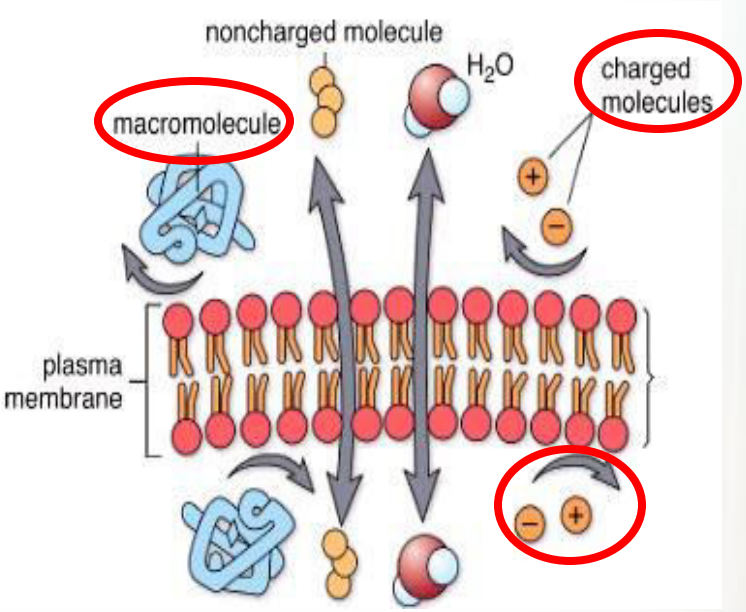
- Varies greatly with type and size of molecules.

Non-Polar, Lipid-Soluble Substances

- Cross passively by dissolving in the lipid bilayer of membrane.
- Rate of penetration depends on lipid solubility and size of molecules.

Membrane Permeability

2



Polar Substances & Ions

- Have difficulty in passing through because of hydrophobic interior of lipid bilayer.

Macromolecules

- Cannot cross plasma membrane due to size.

Membrane Permeability

Passive Permeability and Permeability Constant

- Permeability of the membrane is the rate at which a substance passively penetrates the membrane.
- If we assume that a continuous concentration gradient exists for a diffusible substance across the membrane, then:

$$\frac{dQ_s}{dt} = P(C_I - C_{II})$$

Here:

- dQ_s/dt = the amount of substanc... area of the membrane per unit ti...
- C_I and C_{II} are the respective concentrations of the substance on two sides of the membrane
- P is the permeability constant of the substance

Membrane Permeability

Permeability can be Altered

- Hormones and some other molecules can alter the permeability to certain substances.
- Hormones react with receptor sites and influence channel size or carrier mechanisms.

Membrane Permeability

Examples

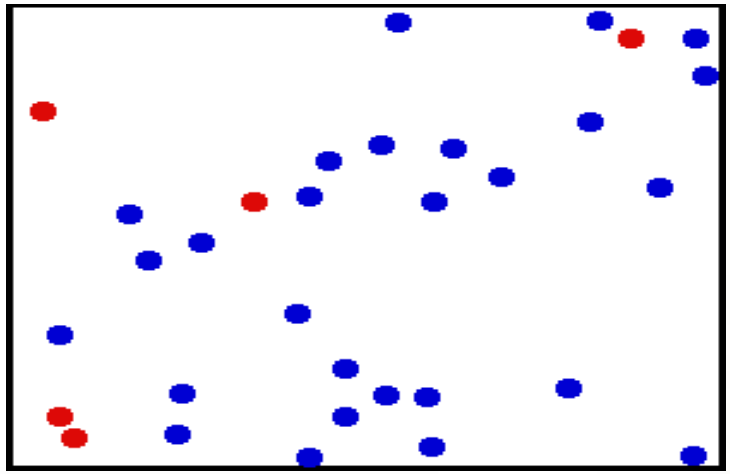
- Antidiuretic hormone can increase water permeability of renal collecting duct in mammals up to 10 times.
- Neurotransmitters induce large increase in permeability to ions by acting on integral membrane proteins in nerve and muscle cells.

Membrane Physiology

Diffusion and Membrane Flux

Lecture no 9

Diffusion and Membrane Flux



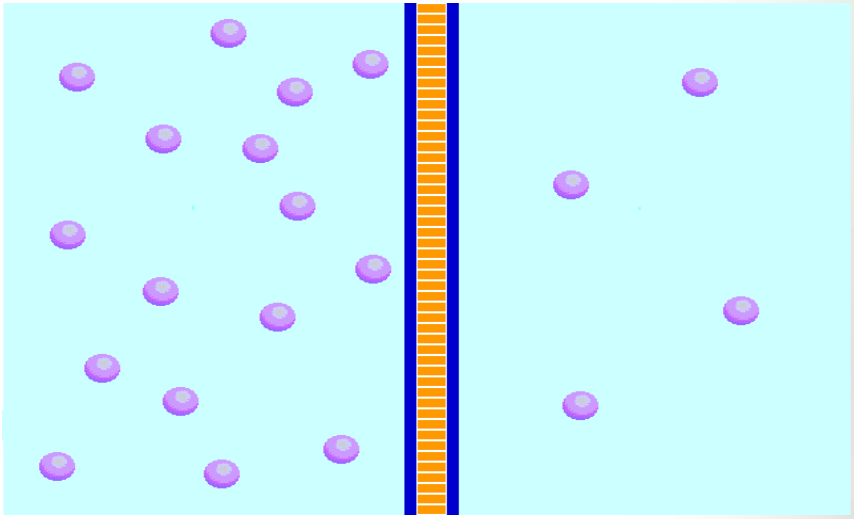
Molecular Motion

- Molecules have thermal energy due to which they are in continuous motion.
- This is called thermal motion.
- One result of thermal motion is the diffusion.

Diffusion and Membrane Flux

Diffusion—Definition

- "the random thermal motion of molecules resulting in their dispersion from regions of higher concentration to regions of lower concentration".



Diffusion and Membrane Flux

Diffusion—Applications

- Much of the traffic across cell membranes occurs by diffusion.
- When a substance is more concentrated on one side of a membrane, there is a tendency for it to diffuse across the membrane, down its concentration gradient.
- Example: Diffusion of Oxygen

Diffusion and Membrane Flux

Diffusion—Mechanism

- Diffusion is a passive transport and occurs without the expense of energy.
- Concentration gradient itself acts as potential energy and drives diffusion.
- Rate of diffusion varies for different molecules due to permeability differences.

Diffusion and Membrane Flux

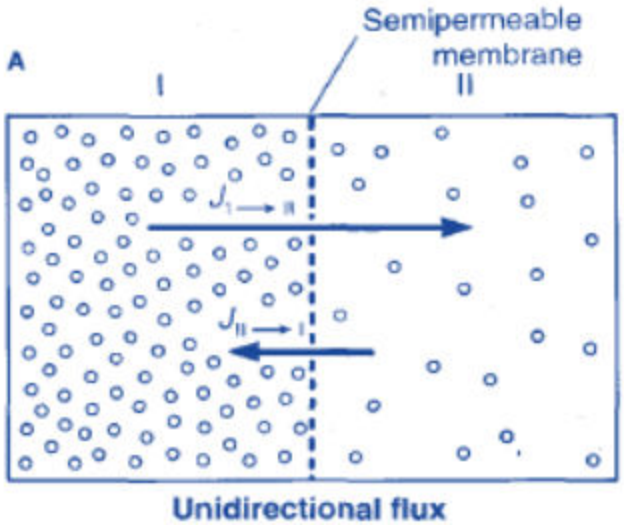
Fick Diffusion Equation

- The rate of diffusion of a solute is calculated by the Fick Diffusion Equation.

$$\frac{dQ_s}{dt} = D_s A \frac{dC_s}{dx}$$

- dQ_s/dt = rate of diffusion
- D_s is the diffusion coefficient of "s". It varies with the nature and molecular weight of substance and solvent.
- "A" is the cross sectional area through which "s" is diffusing.
- dC_s/dx is the concentration gradient of "s" i.e. the change in concentration (dC_s) with distance (dx). It is clearly very important, as it determines the rate at which "s" will diffuse.

Diffusion and Membrane Flux



Unidirectional flux

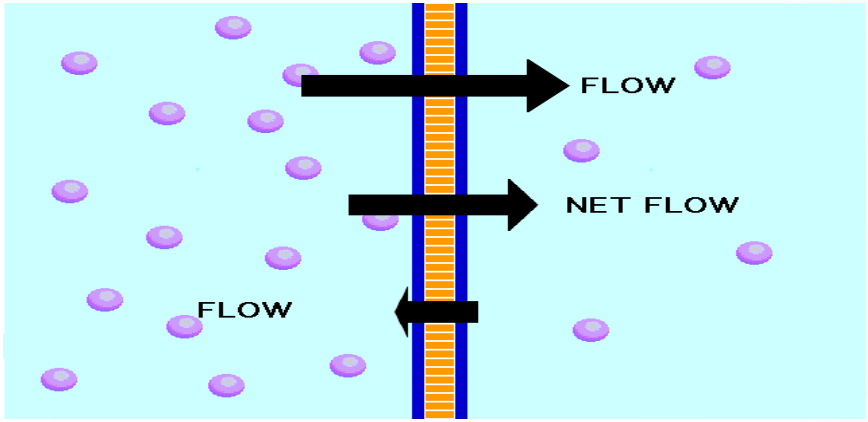
Membrane Flux

- "The amount of solute that passes through a unit area of membrane every second in one direction".

$$J = \frac{dQ_s}{dt}$$

- If a solute occurs on both sides of a membrane, it exhibits a flux in each direction.

Diffusion and Membrane Flux



Membrane Flux

- Flux in one direction is independent of the flux in opposite direction.
- If influx and efflux are equal, net flux is zero.
- If flux is greater in one direction, then net flux is the difference between the two fluxes.

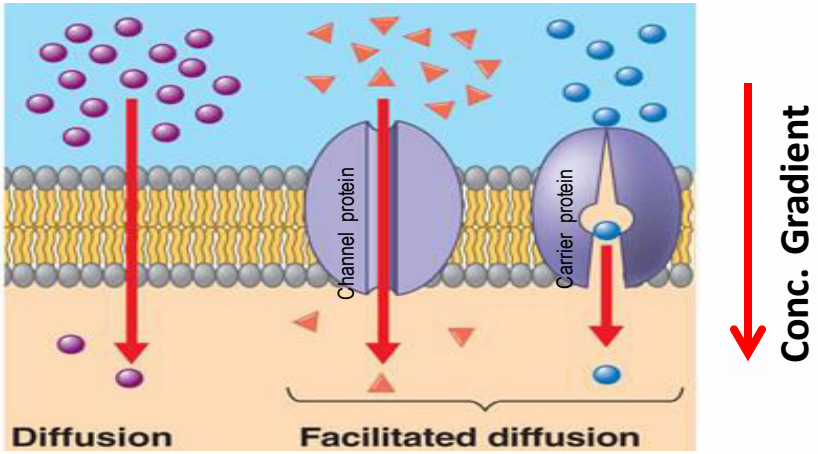
Membrane Physiology

The Facilitated Diffusion

Lecture no 10

The Facilitated Diffusion

1



Definition

- Passive diffusion through the membrane down the conc. gradient with the help of transport proteins.
- It is a passive process, not requiring energy in the form of ATP.

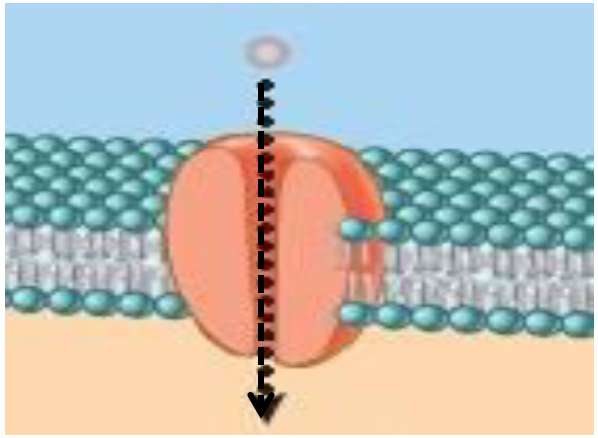
The Facilitated Diffusion

Transport Proteins

- Transport proteins are very specific.
- They transport some substances but not others.
- Two types of transport proteins are:
  - Channel proteins
  - Carrier proteins

The Facilitated Diffusion

1



Channel Proteins

- Extend across the cell membrane.
- Have water-filled pores: membrane channels.

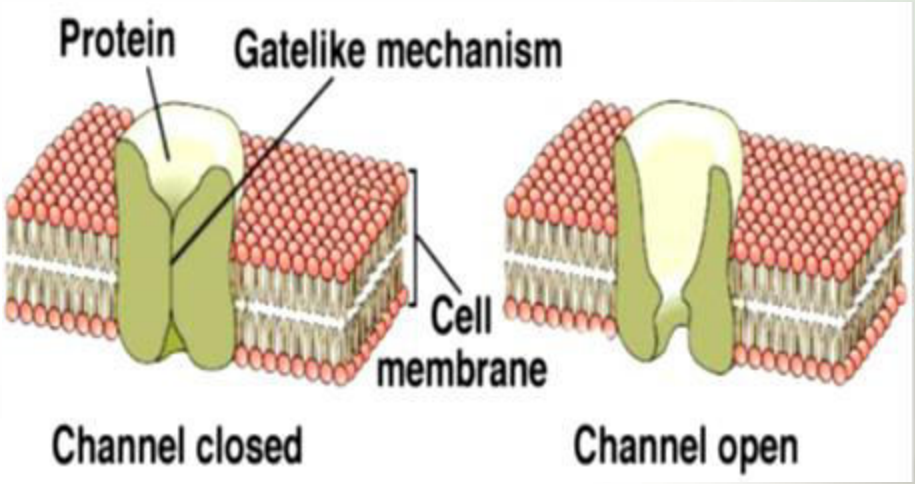
Membrane Channels

- Allow specific molecule or ion to pass through and cross the membrane.
- Diameter of channels is less than 1.0 nm.

The Facilitated Diffusion

Membrane Channels

- Can opening and close.
- Opening and closing is regulated.
- Molecules can cross the membrane when these pores are open.



The Facilitated Diffusion

Channel Proteins: Types

- Water channel proteins
- Ion channel proteins
- Specialized channel proteins

The Facilitated Diffusion

Water Channel Proteins

- Known as Aquaporins
- Facilitate diffusion of water and small ions (e.g. Na^+, K^+, Ca^{2+} and Cl^-).
- Have hydrophilic passageways that facilitate quick flow of water molecules and ions in dissolved form.

The Facilitated Diffusion

Ion Channel Proteins

- Allow specific ions.
- Gated channels
- Open or close in response to an electrical or chemical stimulus.
- Chemical stimulus is a substance other than that is to be transported. e.g. neurotransmitters for Na^+ gated ion channels.
- Electrical stimulus activates K^+ ion channels

The Facilitated Diffusion

Specialized Channel Proteins

- Many channels specialized for the facilitated diffusion of small, uncharged polar molecules e.g. CO_2, NO, and CO.

The Facilitated Diffusion

Carrier Proteins

- Movement of many polar molecules (e.g. sugars, amino acids, nucleotides and certain metabolites).
- They are very selective about which species of molecules they facilitate.

The Facilitated Diffusion

Carrier Proteins–Types

- Exist in many forms in all types of membranes.
- Various functional types:
- Uniporters
- Coupled transporters
- Symporters
- Antiporters

The Facilitated Diffusion

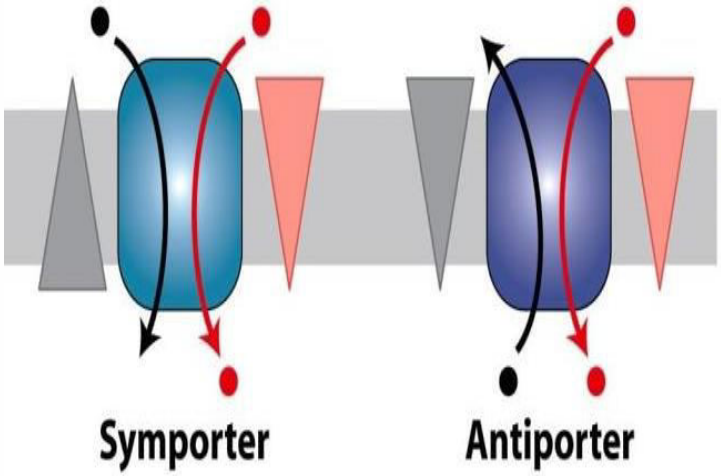
Carrier Proteins–Types

- Uniporters transport a single solute from one side of the membrane to the o



- Coupled transporters transfer one and simultaneously or sequentially transfer a second solute.

The Facilitated Diffusion



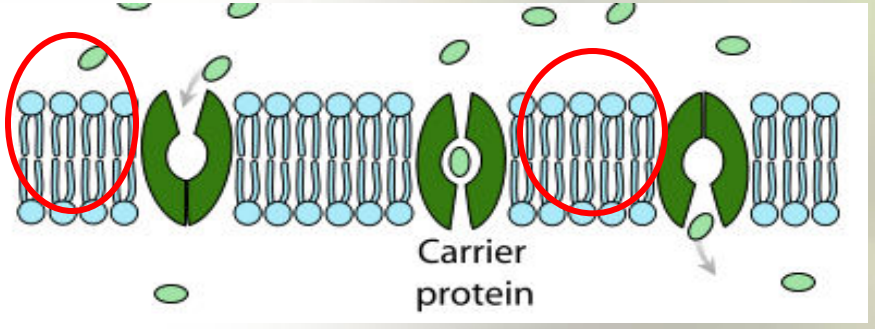
Carrier Proteins–Types

- Symporters are coupled transporters that transfer two solutes in the same direction.
- Antiporters are coupled transporters that transfer two solutes in opposite directions.

The Facilitated Diffusion

Carrier Proteins: Transport Mechanism

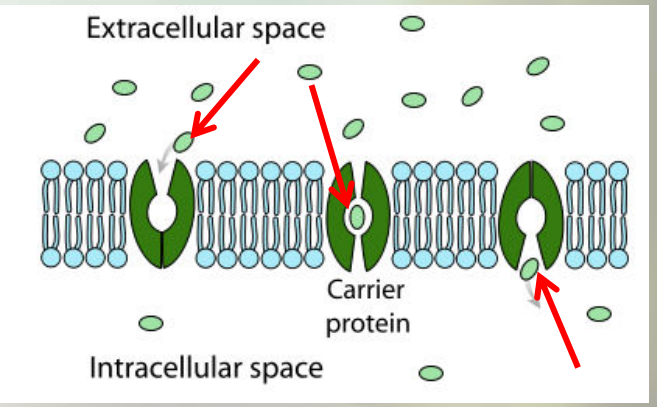
- A carrier protein has the capacity to alternate between two conformations.
- It has a solute binding site that can sequentially translocate from one side of the bilayer to the other.



The Facilitated Diffusion

Carrier Proteins: Transport Mechanism

- This is due to change in the shape of protein molecule, triggered by binding and releasing of transported molecule.



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Membrane Physiology

Donnan Equilibrium

Lecture no 11

Donnan Equilibrium

Discovery

- The phenomenon of unequal distribution of ions across the two sides of a differentially permeable membrane.
- Discovered by Frederick Donnan in 1911.
- Given the name Donnan equilibrium.

Donnan Equilibrium

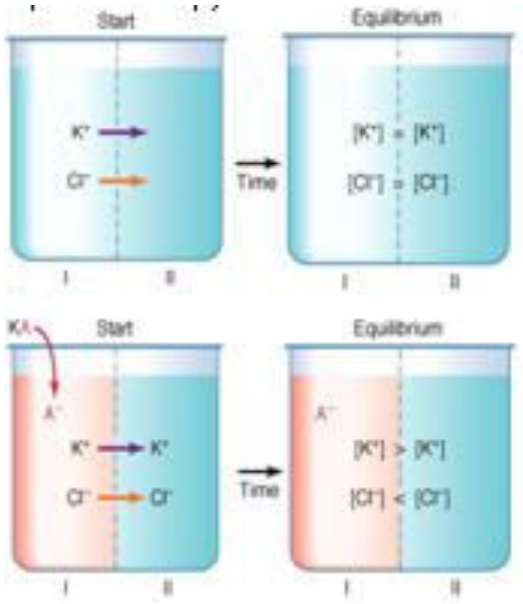
Definition

- "If diffusible solutes are separated by a membrane that is freely permeable to water and electrolytes but totally impermeable to one species of ion, the diffusible solutes become unequally distributed between the two compartments".

Donnan Equilibrium

Example—An Experiment

- Take pure water in two compartments, separated by a membrane. Add KCl to one of them. KCl will ionize.
- Ions diffuse through the membrane until the conc. of K^+ and Cl^- becomes equal on both sides of the membrane.
- Now add a salt "KA" in I. It will produce K^+ and A^-.
- K^+ is diffusible, but A^- is not. So A^- will remain confined to I.
- K^+ and Cl^- quickly redistribute to a new equilibrium.
- Presence of A^- will cause an unequal distribution of ions.
- K^+ will be more conc. in I having A- while Cl^- will be less conc. in it.



Donnan Equilibrium

Explanation

- Donnan equilibrium is characterized by a reciprocal distribution of the anions and cations.
- This happens because the gradients are not only chemical but are also electrical.
- There must be electro-neutrality within both compartments. i.e. positive charges must equal negative charges.

Donnan Equilibrium

Donnan Effect

- Donnan equilibrium implies that:
- if there is a nondiffusible solute in one of the sol. separated by semi-permeable membrane, the conc. of sol. on both sides do not equalize.
- Conc. of sol. with non-diffusible solute remains high even at equilibrium.
- This is Donnan effect.

Donnan Equilibrium

Donnan Effect & Living Cells

- Cells contain non-diffusible anionic colloids with proteins and organic phosphates.
- So there is a high conc. of non-diffusible anions inside the cell.
- This creates Donnan Eq.
- As an implication, there are more ions inside the cell than outside.

Donnan Equilibrium

Effects of Donnan Equilibrium

- The unequal distribution of solutes results in the movement of water into the cell, increasing its hydrostatic pressure.
- Unequal distribution of ions results in the development of electrical potential across the plasma membrane.

Membrane Physiology

Ion Distribution Across The Plasma Membrane

Lecture no 12

Ion Distribution Across The Plasma Membrane

Ion Distribution Across The Plasma Membrane

- Plasma membrane maintains different concentration of ions inside and outside the cell.
- This results in potential difference across the membrane.
- Major ions that are unequally distributed are: K^+, Na^+, Cl^- and Ca^{2+}.

Ion Distribution Across The Plasma Membrane

Distribution of K^+ Ions

- The conc. of K^+ is maintained 10-30 times more in the cytosol than extracellular fluid.
- This happens because cell membranes are more permeable to K^+.
- As a result K^+ are the most conc. inorganic ions in the cytosol.

Ion Distribution Across The Plasma Membrane

Distribution of Na^+ and Cl^- Ions

- Intracellular conc. of Na^+ & Cl^- is maintained about ten times lower than the extracellular conc.
- The lower conc. of Na^+ is due to lower membrane permeability for Na^+.
- Membrane permeability to Cl^- varies. In some cells it is higher while in others it is lower.

Ion Distribution Across The Plasma Membrane

Distribution of Ca^{2+} Ions

- Intracellular conc. of Ca^{2+} is maintained several orders of magnitude below the extracellular conc.
- This difference is due to outward active transport of Ca^{2+} and sequestering of Ca^{2+} ions into mitochondria and ER.
- Cytosolic conc. of Ca^{2+} remains below 10^{-6} M

Ion Distribution Across The Plasma Membrane

Donnan Equilibrium and Unequal Ion Distribution

- Differential permeability of the membrane to different ion species can be explained with Donnan equilibrium.

Ion Distribution Across The Plasma Membrane

Donnan Equilibrium and Unequal Ion Distribution

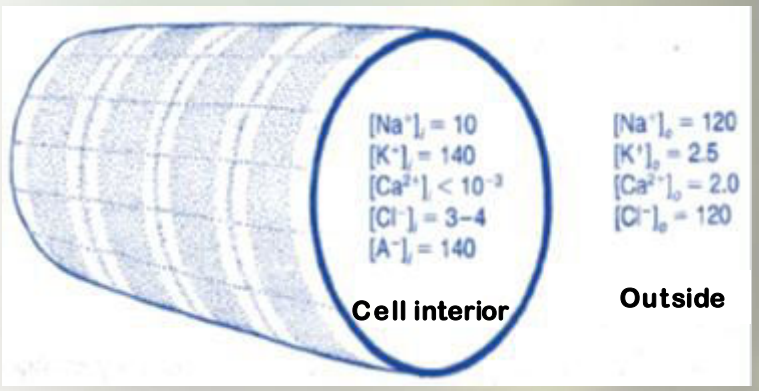
- Cytosol has nondiffusible peptides and proteins that have many carboxyl and other anionic groups.

- Non-diffusible anions produce Donnan effect i.e. unequal distribution of ions across the membrane.

Ion Distribution Across The Plasma Membrane

Donnan Equilibrium and Unequal Ion Distribution

- A multitude of anions inside the cell, cause unequal distribution of almost all ions across the membrane.



Membrane Physiology

Active Transport and Na^+/K^+ Pump

Lecture no 13

Active Transport and Na^+/K^+ Pump

Active Transport

- All living cells spend energy to maintain the transmembrane conc. of solutes far away from equilibrium.
- This involves uphill movement of solutes against conc. gradients, utilizing ATP energy.
- As it is an energy requiring process, it is called as active transport.

Active Transport and Na^+/K^+ Pump

Membrane Pumps

- Active transport of solutes is carried out by specialized carrier proteins.
- They can move solutes against their conc. gradients.
- Mechanisms through which carrier proteins actively transport substances are called membrane pumps.

Active Transport and Na^+/K^+ Pump

Na^+/K^+ Pump: Model of Active Transport

- The system that maintains steep conc. gradients for Na^+ and K^+ in the cell is known as Na^+/K^+ Pump.
- This pump demonstrates all the features of active transport.

Active Transport and Na^+/K^+ Pump

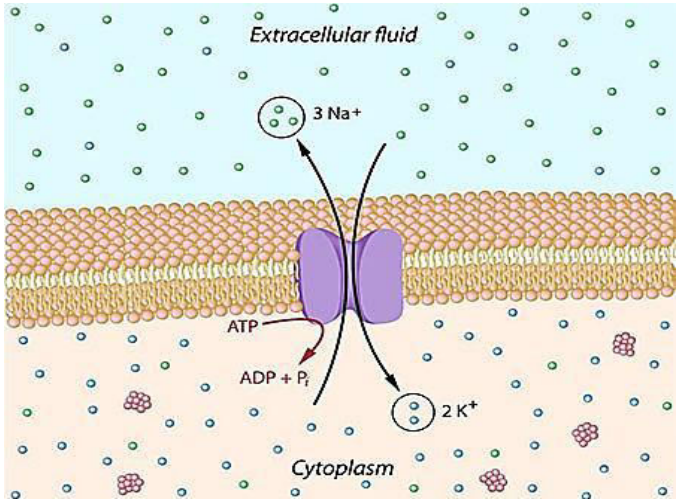
Role of Na^+/K^+ Pump

- Concentration of K^+ is 10~20 times higher inside the cell than outside, while the opposite happens for Na^+.
- These conc. differences are maintained by the Na^+/K^+ pump found in the plasma membranes of virtually all animal cells.

Active Transport and Na^+/K^+ Pump

Nature & Function of Na^+/K^+ Pump

- This pump is a carrier protein "Na^+/K^+ ATPase".
- This protein is an antiport that couples the transfer of Na^+ and K^+ in opposite directions.
- The transfer process involves obligatory exchange of three Na^+ ions from inside with two K^+ ions from outside the cell at the expense of one ATP molecule.
- This unequal stoichiometry of Na^+/K^+ pump makes it an electrogenic pump because of net transport of charge across the membrane, that contributes to membrane potential.



http://e21usa.com/wp-content/uploads/2015/03/sodium-potassium-pump-image.jpg

Active Transport and Na^+/K^+ Pump

Important Features of Active Transport

- Active transport takes place against substantial concentration gradients.
- Active transport systems exhibit a high degree of selectivity.
- Active transport requires ATP as the source of chemical energy.

Active Transport and Na^+/K^+ Pump

Important Features of Active Transport

- Metabolic poisons that stop the production of ATP bring active transport to a halt.
- Hydrolysis of ATP is carried out by specific "ATPases" present in the membrane.

Active Transport and Na^+/K^+ Pump

Important Features of Active Transport

- Na^+/K^+ activated ATPases are associated with Na^+/K^+ pump, while Calcium-activated ATPases are associated with calcium-pumping membranes.

Active Transport and Na^+/K^+ Pump

Important Features of Active Transport

- Certain membrane pumps e.g. Na^+/K^+ pump are antiports. They exchange one kind of molecule or ion from one side of the membrane for another kind of molecular or ion from the other side.

Active Transport and Na^+/K^+ Pump

Important Features of Active Transport

- Some pumps perform electrical work by producing a net flux of charge.
- e.g. Na^+/K^+ pump produces a net outward movement of one positive charge per cycle in the form of three Na^+ ions exchanged for only two K^+ ions.

Active Transport and Na^+/K^+ Pump

Important Features of Active Transport

- Ionic pumps that produce net charge movement are known as rheogenic. because they produce a transmembrane electric current.
- If the current produces a measurable effect on the voltage across the membrane, the pump is said to be electrogenic.

Active Transport and Na^+/K^+ Pump

Important Features of Active Transport

- The sodium~potassium pump is the major electrogenic pump of animal cells.
- Active transport follows Michaelis~Menten kinetics and exhibits competitive inhibition by analog molecules. These are characteristics of enzymatic reactions.

Membrane Physiology

Membrane Potential and Membrane Excitation

Lecture no 14

Membrane Potential and Membrane Excitation

Membrane Potential

- All electrical phenomena in the cells, particularly neurons depend on transmembrane potential difference, generally called as membrane potential, V_m.
- It is electrochemical in nature.
- Basis of generation of signals which are used in communication and muscles in contraction.

Membrane Potential and Membrane Excitation

Membrane Potential

- Membrane potential arises from two features found in all eukaryotic cells:

1. Maintenance of conc. difference of various ions inside and outside the cell through passive and active mechanisms.
2. Ion selective channels that are permeable to different ionic species.

Membrane Potential and Membrane Excitation

Membrane Potential

- Membrane potential is always taken as the intracellular potential relative to extracellular potential.
- It is expressed in millivolts (mV).
- Values vary with type and physiological state of the cell, ranging between -30 to -100 mV.

Membrane Potential and Membrane Excitation

Electric Field Across the Membrane

- The entire potential difference is localized to the regions adjacent to two surfaces of the membrane.
- This potential difference constitutes an electrical gradient that acts as an energy source to move ions across the membrane.

Membrane Potential and Membrane Excitation

Electric Field Across the Membrane

- Electric field (E) is measured in terms of voltage (V) and (d). So that:

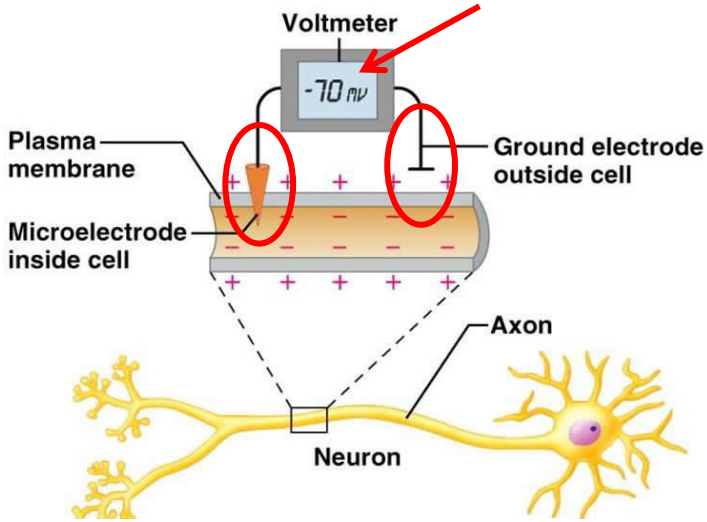
$$E = V/d$$

- Considering the membrane thickness of only 5 nm, this electric field is very large.

Membrane Potential and Membrane Excitation

Measurement of Membrane Potential: Voltage Clamp Method

- Electric current is generated due to a net flux of charged particles.
- Using the Voltage Clamp method, current can be directly detected by using two electrodes, one placed in the cytosol and other in the extracellular medium.
- Electrodes detect the voltage or potential difference.
- Classic instrument used was an oscilloscope.
- Now advanced digital equipment is used.



Membrane Potential and Membrane Excitation

Membrane Excitation

- A stimulus decreases potential difference across the membrane of an excitable cell e.g. a neuron, muscle or any sensory cell.
- This decrease in potential difference i.e. depolarization causes the opening of voltage-gated Na^+ channels.

Membrane Potential and Membrane Excitation

Membrane Excitation

- This results in an influx of sodium ions into the cell and an action potential is triggered.
- The opening of voltage-gated Na^+ channels in response to depolarization, and the resulting flow of Na^+ ions into the cell, is known as membrane excitation.

Membrane Physiology

Role of Ion Channels

Lecture no 15

Role of Ion Channels

Ion Channels

- Many proteins in plasma membrane form ion channels that function in the electrical responses of excitable cells. e.g.
  - Resting K^+ selective channels
  - Voltage-gated ion selective channels
  - Ligand-gated ion channels
  - Stimulus-activated ion channels

Role of Ion Channels

Resting K^+ Selective Channels

- Uniformly distributed over the entire membrane of excitable cells.
- Always remain open.
- Largely responsible for maintaining the V_{rest}.
- Responsible for passive change in V_m during AP in response to depolarization and hyperpolarization.

Role of Ion Channels

Voltage-Gated Ion Selective Channels

- Many types.
- Make the cell membranes excitable.
- Responsible for nearly all active electrical signals in living tissues.
- Mainly localized to particular areas of excitable cells (e.g. axonal membrane of neurons).

Role of Ion Channels

Voltage-Gated Ion Selective Channels

- Active changes in V_m in response to depolarization depend on the opening or closing of these channels.
- Control the flow of ionic currents generated due to electrochemical gradients
- Exhibit ion selectivity, allowing only one or a few species of ions to pass through them.

Role of Ion Channels

Voltage-Gated Ion Channels: Types

- Voltage-gated Na^+ channels
- Voltage-gated Ca^{2+} channels
- Voltage-gated K^+ channels
- Ca^{2+} activated K^+ channels.
- Named for the ionic species that normally moves through them.

Role of Ion Channels

Voltage-Gated Na^+ Channels

- Fast-acting channels.
- Activated by depolarization and produce rising phase of action potential.

Role of Ion Channels

Voltage-Gated Ca^{2+} Channels

- Activated by depolarization but more slowly than Na^{+} channels.
- Allow Ca^{2+} ions to enter the cell, where they act as second messenger.

Role of Ion Channels

Voltage-Gated K^+ Channels

- Known as “delayed rectifiers”.
- Allow potassium ions to flow out of the cell and rapidly repolarize the membrane to terminate an action potential.

Role of Ion Channels

Ca^{2+} Activated K^{+} Channels

- Activated by depolarization and elevated cytoplasmic concentration of Ca^{2+} ions.
- Remain open as long as cytoplasmic calcium ion concentration remains high.

Role of Ion Channels

Ligand-Gated Ion Channels

- Activated when specific ligand molecules bind to receptor proteins.
- Ligands are second messenger molecules and neurotransmitters.
- Ligand binding results in conformational change that causes opening of channel gate and thus ion flux across the membrane.

Role of Ion Channels

Stimulus-Activated Ion Channels

- Ion channels activated by specific stimulus energies.
- Found in sensory receptor cells.

Role of Ion Channels

Stimulus-Activated Ion Channels: Examples

- Specific ion channels in photoreceptors, stimulated by light.
- Ion channels in taste buds and olfactory neurons, activated by chemicals.
- Ion channels in mechanoreceptors, activated by mechanical strain.

Membrane Physiology

Resting Membrane Potential

Lecture no 16

Resting Membrane Potential

Resting Potential Values

- The non-excited or "resting" cell has a potential difference, V_{rest} across its membrane.
- Values of V_{rest} vary in different types of cells.
- Most neurons have an RMP of -70mV, when no impulse is being conducted.

Resting Membrane Potential

Ions Responsible for RMP

- Plasma membranes are permeable to more than one type of ionic species.
- All diffusible ions affect the potential across a membrane in proportion to the permeability.
- Nondiffusible ion species have no effect on MP.
- Major ions responsible for V_{rest} : K^+, Na^+ & Cl^-.

Resting Membrane Potential

Calculating the Effect of Ions on RMP

- The effect of single ion gradient on membrane potential is predicted by the Nernst equation.
- Effect of multiple ions on membrane potential is calculated with Goldsman's equation.
- Both equations apply to all excitable cells i.e. neurons and muscle cells.

Resting Membrane Potential

Nernst Equation

- According to Nernst, the equilibrium potential of a diffusible ion depends on:
  - Absolute temperature
  - Valence of the ion
  - Log of ratios of ion conc. on the two sides of membrane.

Resting Membrane Potential

Nernst Equation

$$E_x = \frac{RT}{zF} \ln \frac{[X]_I}{[X]_{II}}$$

- E_x = equilibrium potential for ion X.
- R = gas constant
- T = absolute temperature
- z = valence of ion X
- F = Faraday constant
- ln = Natural log
- $[X]_I$ and $[X]_{II}$ = conc. of ion X on two sides of membrane.

Resting Membrane Potential

Goldman's Equation

$$E_{ions} = \frac{RT}{F} \ln \frac{P_K[K^+]_o + P_{Na}[Na^+]_o + P_{Cl}[Cl^-]_i}{P_K[K^+]_i + P_{Na}[Na^+]_i + P_{Cl}[Cl^-]_o}$$

- E_{ions} = equilibrium potential of ions
- P_K, P_{Na}, and P_{Cl} = permeability constants for ion species.
- $[K^+]_i$ and $[K^+]_o$ = the concentrations outside and inside the cell.

Resting Membrane Potential

Role of Ion Channels

- The always open resting K^+ selective channels and voltage-gated Na^+ channels play important role in producing and maintaining the transmembrane potential difference.

Resting Membrane Potential

Role of K^+ Channels

- The K^+ selective channels remain open in the resting membrane.
- So, the membrane is highly permeable to K^+.
- Due to this RMP are more sensitive to changes in the $[K^+]_o$.

Resting Membrane Potential

Role of Na^+ Channels

- The Na^+ channels remain closed in the resting membrane.
- Due to this, the resting membrane is relatively nondiffusible to Na^+.
- So large changes in $[Na^+]_o$ have little effect on the resting potential.

Resting Membrane Potential

Role of Active Transport

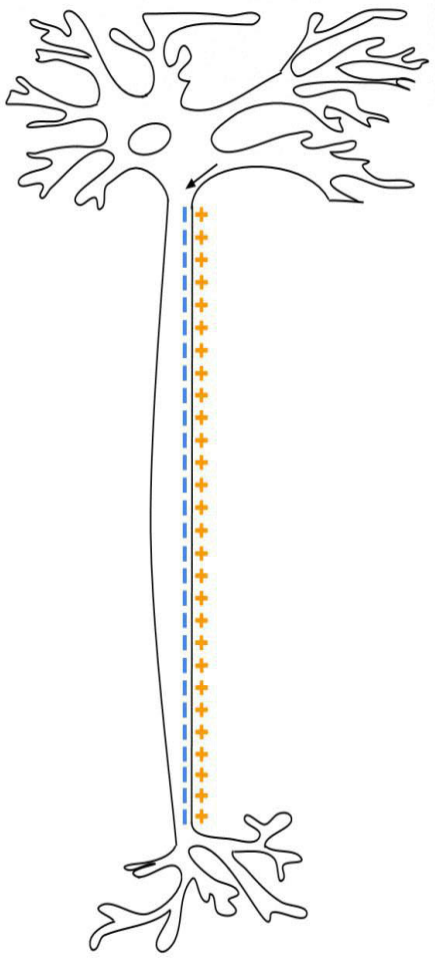
- Active transport moves ions across the cell membrane against their concentration gradients.
- So it results in asymmetrical distribution of ions.
- The asymmetrical distribution of ions is the basis of resting membrane potential.
- Example: Na^+/K^+ pump.

Membrane Physiology

Action Potentials

Lecture no 17

Action Potentials



https://upload.wikimedia.org/wikipedia/commons/9/95/Action_Potential.gif

Definition

- Action potentials are the type of signals that neurons use to send information over long distances.
- Actions potentials are large but brief changes in membrane potential that are propagated along axons without decrement.

Action Potentials

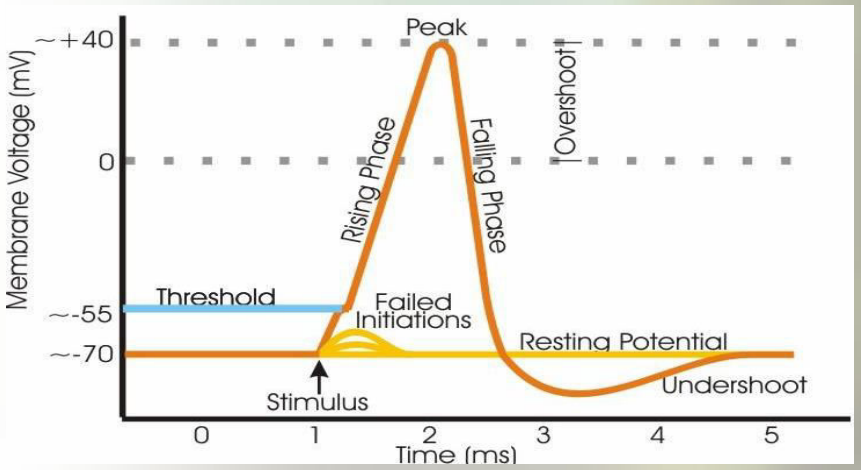
Importance of Action Potentials

- APs are responsible for every sensation, every memory, every thought — indeed every impulse.
- They control:
  - Effector responses
  - Activation of voltage-gated ion channels
  - Muscle contraction
  - Exocytosis

Action Potentials

Phases of AP

1. Stimulation
2. Rising phase (Depolarization)
3. Peak phase
4. Falling phase (Repolarization)
5. Undershoot (afterhyperpolarization)



http://slideplayer.com/4667714/15/images/6/Action+potential.jpg

Action Potentials

Stimulation

- When a stimulus is received at axon hillock, it causes some voltage-gated Na^+ channels to open in the neuronal membrane.
- Resultantly Na^+ start to diffuse in.
- This results in a local depolarization due to increase in the V_m.

Action Potentials

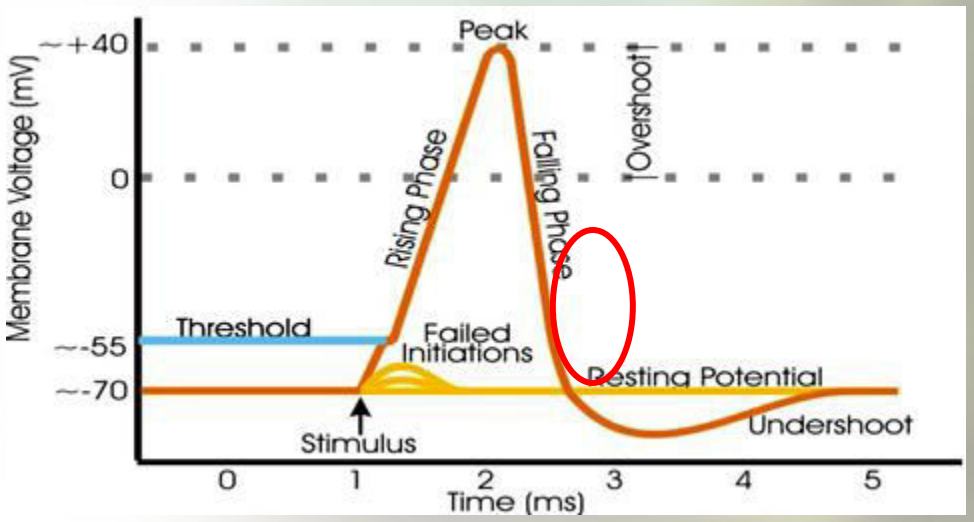
Stimulation

- A depolarization in the range of -55 to -30 mV is the threshold that triggers an AP.
- If the intensity of stimulus is less, it causes a subthreshold depolarization that fails to generate an AP.

Action Potentials

Rising Phase (Depolarization)

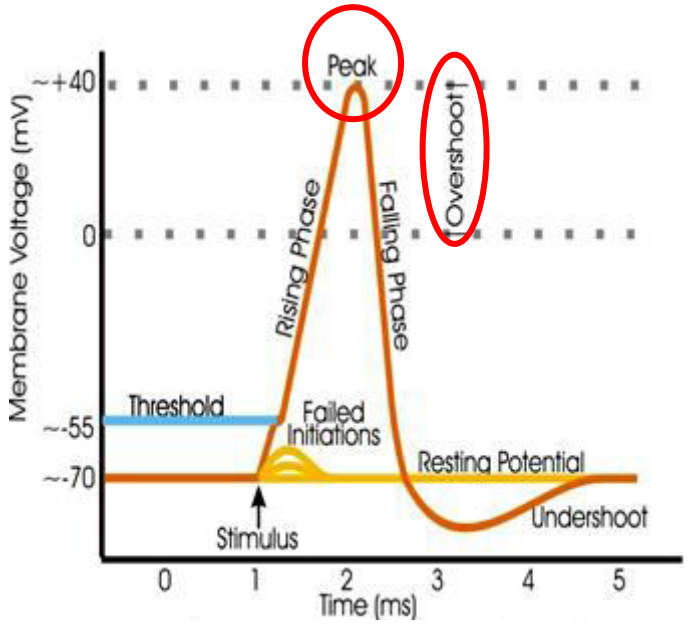
- Due to +ve feedback, more Na^+ channels open.
- Na^+ current dominates
- Membrane potential becomes +ve inside.
- Going towards sodium equilibrium pot. (E_{Na}).



Action Potentials

The Peak Phase

- As the Na^+ channels become maximally open, positive feedback slows down.
- Membrane potential reaches a maximum, close to the E_{Na} ~ +55mV.
- At this stage further depolarization stops.
- The very brief period for which V_m is inside-positive is called the overshoot.



Action Potentials

Falling phase (Repolarization)

- Voltage-gated Na^+ channels start to close.
- No further influx of Na^+.
- Voltage-gated potassium channels begin to open.
- K^+ outflow starts.
- Changes in Na^+ & K^+ permeability cause V_m to drop towards E_K ~58mV.
- Result: reversal of V_m to -ve inside: repolarization.

Action Potentials

Undershoot: Afterhyperpolarization

- Membrane repolarization initiates closing of Volt-gated K^+ channels.
- These channels do not close immediately in response to change in V_m.
- Closing is voltage and time dependent.
- So, they are called "delayed rectifiers".

Action Potentials

Undershoot: Afterhyperpolarization

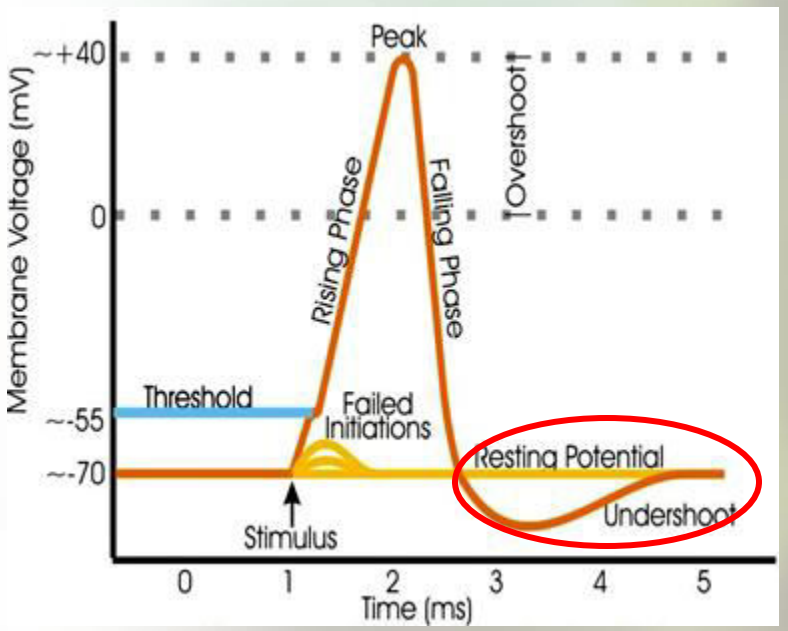
- Due to this delay, K^+ continues to flow out of the cell even after the membrane has fully repolarized.
- Thus, V_m dips below the normal V_{rest} for a brief period i.e. membrane becomes hyperpolarized.

Action Potentials

Undershoot: After hyperpolarization

- This brief hyperpolarization is termed as undershoot or after hyperpolarization.

- It persists until the K^+ permeability returns to its normal value.



Membrane Physiology

General Properties of Action Potentials (I)

Lecture no 18

General Properties of Action Potentials

General Properties

- Action potentials are also called as spikes and nerve impulses.
- They are generated by the plasma membranes of neurons, muscle cells, some receptor and secretory cells.
- Action potentials differ in different types of cells due to different channel properties.

General Properties of Action Potentials

General Properties

- The shape, magnitude, and time course of all APs produced by a particular cell type are essentially identical.
- Action potentials propagate along nerve fibers without any reduction in amplitude.
- The speed of propagation is constant

General Properties of Action Potentials

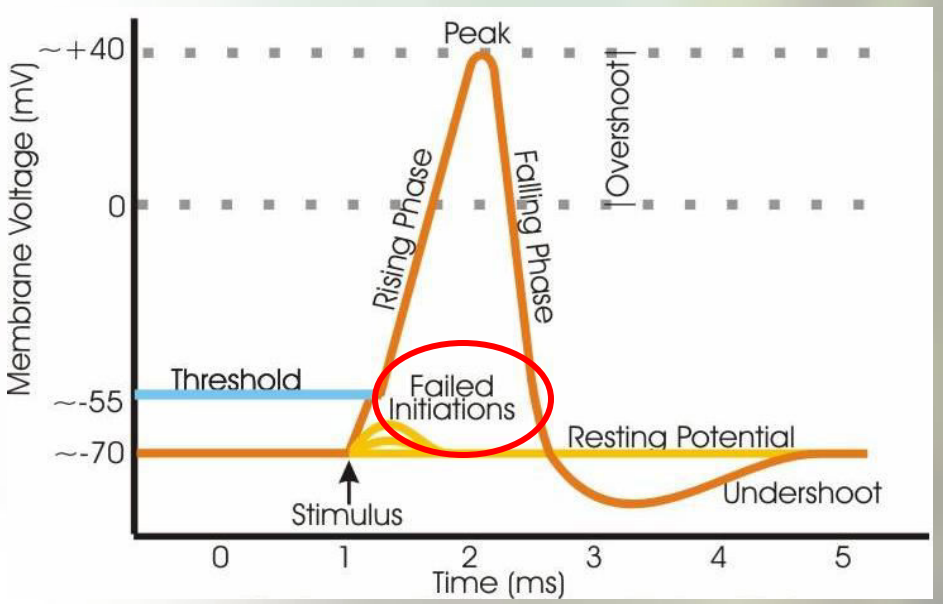
Threshold Potential

- The minimum strength of stimulus that can produce an action potential.
- Threshold potentials of most neurons fall between -30 mV and -50 mV.

General Properties of Action Potentials

Subthreshold Stimulus

- A stimulus that is too weak to produce an action potential.
- It results in small depolarization that causes failed, abortive & nonpropagated excitation called “a local response”.



General Properties of Action Potentials

APs are "All-or-None" Events

- An excitable membrane responds to a stimulus either with a full-fledged action potential that spreads along the nerve fiber, or it does not respond at all.
- This property is the all-or-none law.
- There is no "in-between" AP.

General Properties of Action Potentials

APs are "All-or-None" Events

- This principle works similar to a gun fire.
- Just as squeezing the trigger halfway (subthreshold) does not cause the gun to fire halfway or squeezing it harder (suprathreshold) does not produce a greater explosion.

General Properties of Action Potentials

APs are Regenerative

- Once the threshold potential is reached, AP becomes regenerative i.e. the event becomes self-perpetuating.
- Membrane potential continues to change without any further stimulus.

General Properties of Action Potentials

Time Duration of APs

- In mammalian neurons, APs typically last only a millisecond or so.
- In many invertebrate species, APs can last for 10-100 milliseconds.
- In vertebrate heart muscle cells, the duration of an AP can be as long as half a second (i.e. 500-milliseconds).

General Properties of Action Potentials

Number of Ions Involved in an AP

- Only a small number of ions moving across the membrane cause potential changes during different phases of APs.
- e.g. Na^+ that cross the membrane during depolarization in squid giant axon represent only 0.00003% of the total intracellular Na^+ ions.

Membrane Physiology

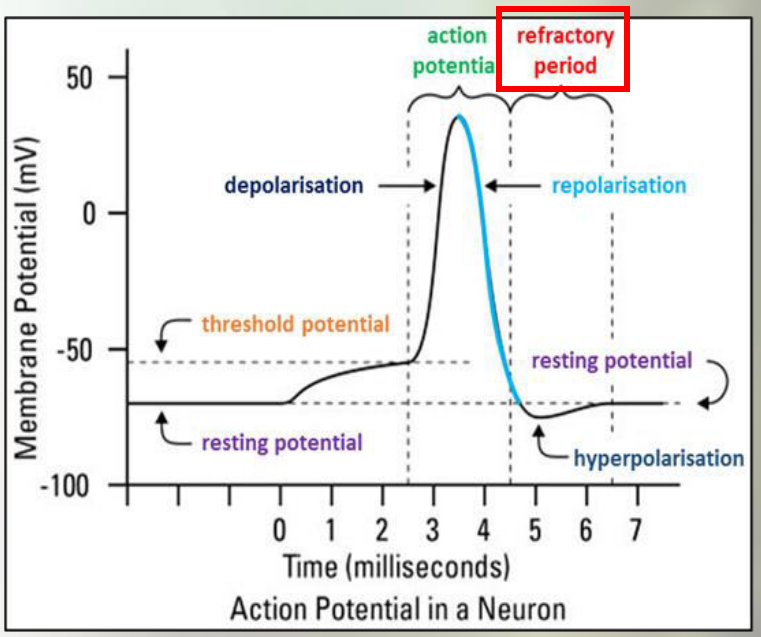
General Properties of Action Potentials (II)

Lecture no 19

General Properties of Action Potentials

Refractory Period

- Each AP is followed by a refractory period.
- "The time during which a subsequent AP is impossible or difficult to fire".
- Two types: Absolute and relative refractory period



General Properties of Action Potentials

Absolute Refractory Period

- "The period during which no new action potential can be fired".
- A membrane that has just fired an action potential cannot fire another one immediately.
- Because the ion channels have not returned to the deactivated state.

General Properties of Action Potentials

Relative Refractory Period

- "The period in which APs are difficult to evoke and require a higher threshold and stronger depolarization".
- When some but not all of the ion channels have recovered, axon can be stimulated to produce another AP, with a higher threshold.

General Properties of Action Potentials

Significance of Refractory Period

- Interval between two APs cannot be shorter than the Absolute RP.
- Only a certain number of APs can be produced in a nerve fibre.
- It decreases fatigue in a nerve fibre.
- It permits propagation of discrete impulses and prevents fusion or summation of impulses.

General Properties of Action Potentials

Accommodation

- A characteristic of excitable membranes.
- "If a neuron is stimulated by a series of sub-threshold depolarizations, a decrease in excitability occurs and the threshold potential increases".
- It is due to change in the sensitivity of membrane channels.

General Properties of Action Potentials

Adaptation

- "The reduction in the frequency of action potentials that is typically seen in a neuron during a sustained stimulus".

- Two types:
  - Phasic response
  - Tonic response

Action Potentials

Adaptation: Phasic response

- The type of adaptation in which neurons adapt rapidly and stop generating more APs after initial one or two APs.

Action Potentials

Adaptation: Tonic Response

- The type of adaptation in which neurons adapt slowly and fire repetitively to a subthreshold stimulus, but with gradually decreasing frequency.

Membrane Physiology

Ionic Basis of Action Potential

Lecture no 20

Ionic Basis of Action Potential

Sodium and Potassium Channels

- An action potential results from ion movements through voltage-gated sodium and potassium channels.
- Membrane depolarization opens both types of channels, but they respond independently and sequentially.

Ionic Basis of Action Potential

Unequal Distribution of Ions

- The production of an action potential depends on the unequal distribution of ionic species across the membrane.
- This generates an electrochemical gradient that provides a source of potential energy.

Ionic Basis of Action Potential

Gating of Ion Channels

- Neurons contain voltage-gated ion channels that open or close in response to stimuli.
- This gating of ion channels forms the basis of nearly all electrical signaling in the nervous system.

Ionic Basis of Action Potential

Ionic Currents Through Ion Channels

- Opening or closing of ion channels alters the membrane's permeability to particular ions.
- This allows flow of ionic currents through the channels, driven by electrochemical gradients.
- These ionic currents alter the membrane potential.

Ionic Basis of Action Potential

Role of Na^+ and K^+ Channels

- Na^+ and K^+ channels are involved in producing action potentials.
- These two channel types are quite different from the Na^+/K^+ pump and other passive channels involved with RMP.

Ionic Basis of Action Potential

Role of Na^+ and K^+ Channels

- Both channels have different properties from one another.
- But their activities are interdependent that are responsible for essentially all features of the action potential.

Membrane Physiology

Ion channels—
Localization and
Characterization

Lecture no 21

Ion channels—Localization & Characterization

Hodgkin and Huxley Predicted Ion Channels

- A. L. Hodgkin and A. F. Huxley carried out experiments on giant axons of squids in 1940s and 1950s.
- They found that g_{Na} and g_K change during an action potential.
- They predicted presence of membrane channels in neurons.

Ion channels—Localization & Characterization

Molecular Nature of Ion Channels

- After Hodgkin and Huxley, technical advances contributed to understanding the nature of membrane channels.
- Techniques for measuring ionic currents were developed (e.g patch-clamping).

Ion channels—Localization & Characterization

Molecular Nature of Ion Channels

- Techniques of protein chemistry and molecular biology helped to identify channel proteins.
- As a result the molecular, protein nature of ion channels was established.

Ion channels—Localization & Characterization

Role of Neurotoxins in Localization of Ion Channels

- Naturally occurring neurotoxins can bind to specific channels.
- They have been extensively used in localization and characterization of voltage-dependent channels.

Ion channels—Localization & Characterization

Role of Neurotoxins in Localization of Ion Channels

- An important toxin is tetrodotoxin (TTX) obtained from the Japanese puffer fish.
- TTX selectively blocks fast-acting, voltage-gated Na^+ channels.
- Radioactively labeled TTX is used to estimate density of Na^+ channels.

Ion channels—Localization & Characterization

Use of Antibodies in Localization of Ion Channels

- Now-a-days, antibodies have been developed that bind with the channel proteins.
- These proteins can be viewed directly.

Membrane Physiology

Voltage-Gated Sodium (Na^+) Channels

Lecture no 22

Voltage-Gated Sodium (Na^+) Channels

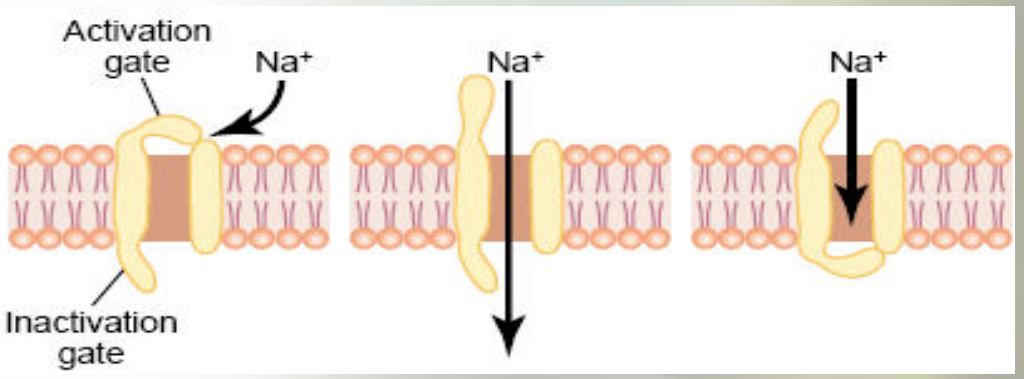
Nature

- Highly sensitive to voltage changes, so called fast-acting Sodium channels.
- Activated by a stimulus generated depolarization.
- Their opening results in inward flow of Na+ ions
- Produce rising phase of an AP.

Voltage-Gated Sodium (Na^+) Channels

Channel Gates

- Consist of channel proteins that are inserted in the lipid bilayer of membrane.
- The channel has two gates: one on the outside called activation gate, and other on the inside called inactivation gate.



Voltage-Gated Sodium (Na^+) Channels

Density of Na^+ Channels in Axons

- Na^+ channels are not very densely packed in the axonal membranes.
- About 500 Na^+ channels per μm^2, occupying only 1% of total surface area.
- Still, each channel can pass up to 10^7 Na^+ ions per second providing enough I_{Na}.

Voltage-Gated Sodium (Na^+) Channels

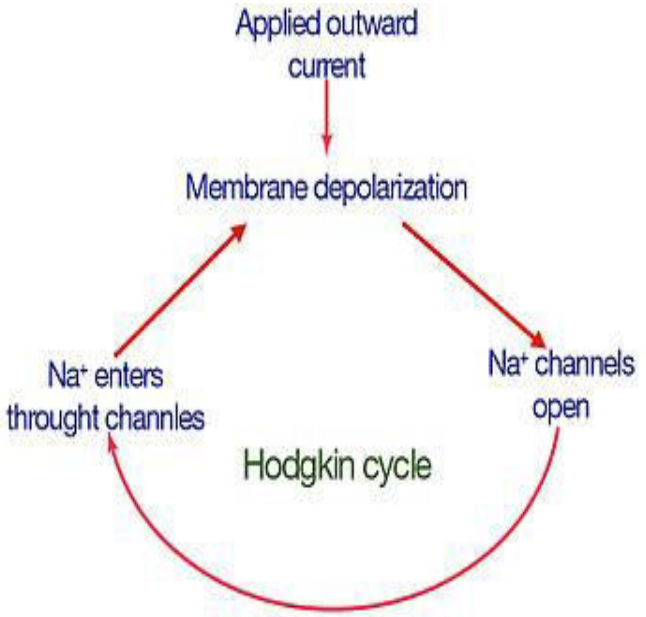
Opening of Na^+ Channels Cause Increase in Na^+ Conductance

- Changes in V_m affect the number of Na^+ channels open at any instant.
- More the number of open channels, more is the Na^+ conductance (g_{Na}).
- So, changes in g_{Na} occur as a function of V_m and time.

Voltage-Gated Sodium (Na^+) Channels

Hodgkin Cycle

- A stimulus causes local depolarization of the membrane.
- Na^+ channels open in response to initial depolarization.
- This increases Na^+ conductance that allows Na^+ to enter the cell, which cause further depolarization of membrane and decrease in V_m.
- This relationship between Na^+ conductance (g_{Na}) and V_m represents a type of positive feedback system and is termed as the Hodgkin cycle.
- It results in triggering of an action potential.



Voltage-Gated Sodium (Na^+) Channels

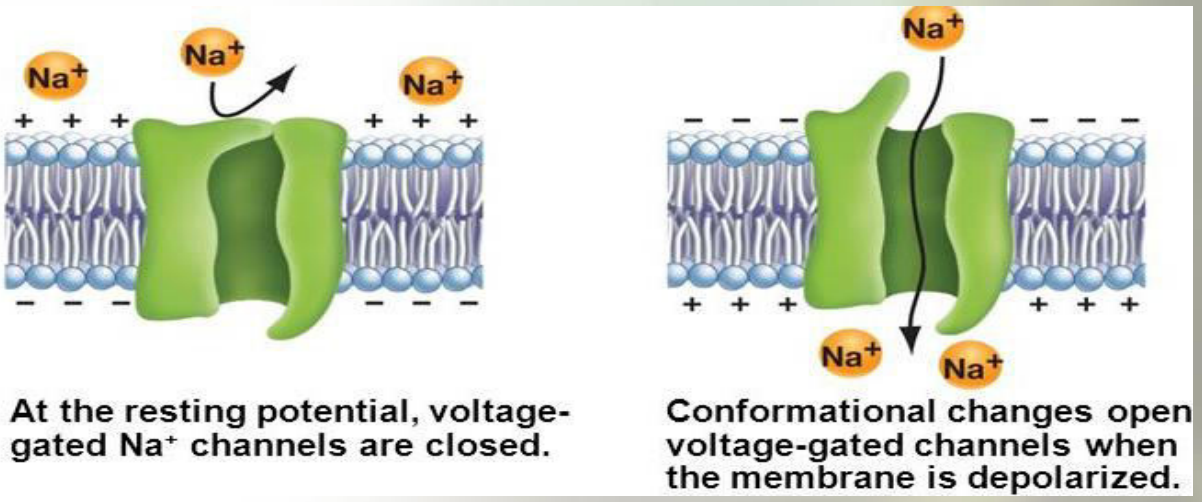
Na^+ Channels: Mechanism of Opening

- Depolarization causes changes in V_m.
- Changes in V_m regulate the opening of channels.
- As channel proteins bear a net charge, so, a change in V_m produces an emf on the charge.
- The emf causes the charge to move in space.

Voltage-Gated Sodium (Na^+) Channels

Na^+ Channels: Mechanism of Opening

- Charge movement causes conformational change in the protein molecule
- This results in opening of channel and increase in g_{Na} through the channels.



http://slideplayer.com/slide/3994975/13/images/4/Voltage-gated+channels.jpg

Voltage-Gated Sodium (Na^+) Channels

The Gating Current (I_g)

- The movement of charge results in a small gating current (I_g).
- The gating current produced by Na^+ channels is I_{Na} and is associated with the opening and closing of voltage-gated Na^+ channels.

Voltage-Gated Sodium (Na^+) Channels

Selectivity of Na^+ Channels

- The selection of ions by channels depends on:
  - size
  - charge

  of the permeating ions.
- The cation-selective Na^+ and K^+ channels have -ve charges at their outer ends.
- So they attract cations and repel anions.

Voltage-Gated Sodium (Na^+) Channels

Selectivity of Na^+ Channels

- A channel's selectivity is indicated by its relative permeability for various ion species.
- The permeability of Na^+ channel for Na^+ ion is 1.00, while its permeability for Li^+ is 0.93 and for K^+ ions it is only 0.09.

Membrane Physiology

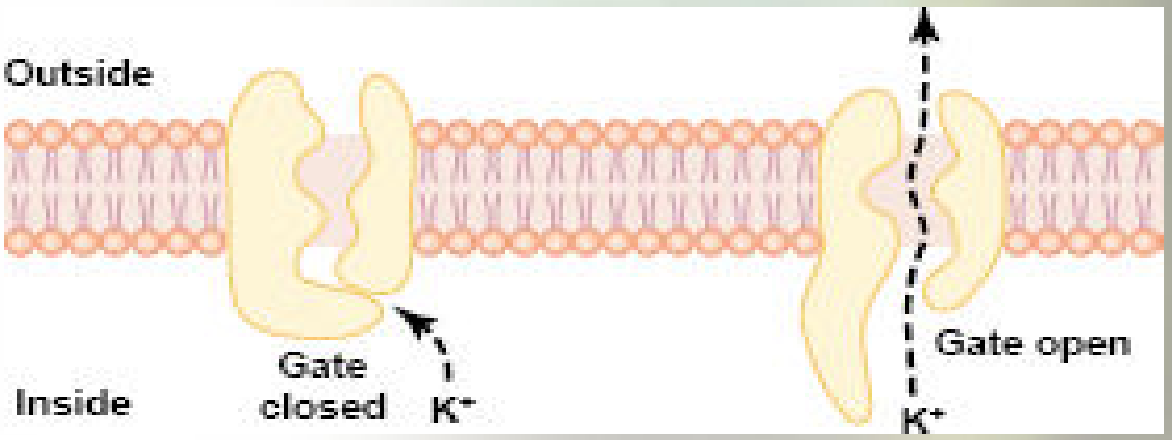
Voltage-Gated Potassium (K^+) Channels

Lecture no 23

Voltage-Gated Potassium (K^+) Channels

Features

- Slightly smaller than the sodium channels.
- Size: 0.3 by 0.3 nm as compared to Na^+ channel size of 0.3 by 0.5 nm.
- Channel gates are on the intracellular ends.
- Opening of gates results in the outflow of K^+ ions.



Textbook of Medical Physiology (11thed) by Guyton

Voltage-Gated Potassium (K^+) Channels

Opening of K^+ Channels

- These channels remain closed in a resting neuron.
- They open during the action potential when inside of the membrane becomes positively charged due to depolarization.

Voltage-Gated Potassium (K^+) Channels

Delayed Rectifiers

- Voltage-gated K^+ channels respond more slowly to voltage changes as compared to the Na^+ channels. So they are known as “delayed rectifiers”.

Voltage-Gated Potassium (K^+) Channels

Effect on Membrane Potential

- Opening of K^+ channels results in increased conductance for potassium ions (g_K)
- gK begins to increase when AP is near its peak, and remains high in the falling phase.
- Outward flow of K^+ ions brings V_m close to E_K i.e. ~90 mV

Voltage-Gated Potassium (K^+) Channels

Hyperpolarization

- As the V_m approaches E_K, membrane polarization increases.
- This increase in the magnitude of membrane potential is called hyperpolarization.
- Hyperpolarization results in decrease in g_K and termination of an action potential.

Voltage-Gated Potassium (K^+) Channels

Role in Action Potential

- Actual role of voltage-gated K+ channels is not to generate APs.
- They are involved in the acceleration of membrane repolarization.
- Activation of more K^+ channels shortens the duration of APs.
- It helps the neurons to generate APs at a higher frequency.

Voltage-Gated Potassium (K^+) Channels

Number of K^+ Channels in Neurons

- The neurons involved in generating APs of longer duration have lower number of these channels.
- Some myelinated mammalian neurons entirely lack these channels.

Membrane Physiology

Voltage-Gated Calcium (Ca^{2+}) Channels

Lecture no 24

Voltage-Gated Calcium (Ca^{2+}) Channels

Features

- Voltage-gated Ca^{2+} channels occur in virtually all cell types.
- Their molecular structure is strikingly similar to voltage-gated Na^{+} channels.
- They have a selective permeability to Ca^{2+} ions.
- Activation of these channels allows Ca^{2+} to rush into the cell.

Voltage-Gated Calcium (Ca^{2+}) Channels

Slow Activation of Ca^{2+} Channels

- Ca^{2+} channels remain closed at V_{rest}.
- They are activated with depolarization but 10 to 20 times more slowly than Na^{+} channels.
- They are called slow channels, in contrast to the fast Na^{+} channels.

Voltage-Gated Calcium (Ca^{2+}) Channels

Role of Ca^{2+} Channel Activation

- In neurons and skeletal muscles, they carry a part of the inward regenerative depolarizing current along with Na^{+} current.
- In these cells, the I_{Ca} is not strong enough to produce an all-or-none AP without help from I_{Na}.

Voltage-Gated Calcium (Ca^{2+}) Channels

Ca^{2+} Channels in Smooth Muscles

- In some smooth muscles, the fast Na^{+} channels are hardly present
- So the APs are caused entirely by activation of slow calcium channels.
- In such cells, the membrane has greater number of these voltage-gated calcium channels.

Voltage-Gated Calcium (Ca^{2+}) Channels

Action Potentials due to Ca^{2+} Channels

- The APs generated due to calcium flow occur in the same self-regenerative way as for the Na^{+} channels.
- However, Ca^{2+} channels open more slowly and also remain open much longer.
- This results in prolonged APs in these muscle.

Voltage-Gated Calcium (Ca^{2+}) Channels

Role of Ca^{2+} Ions

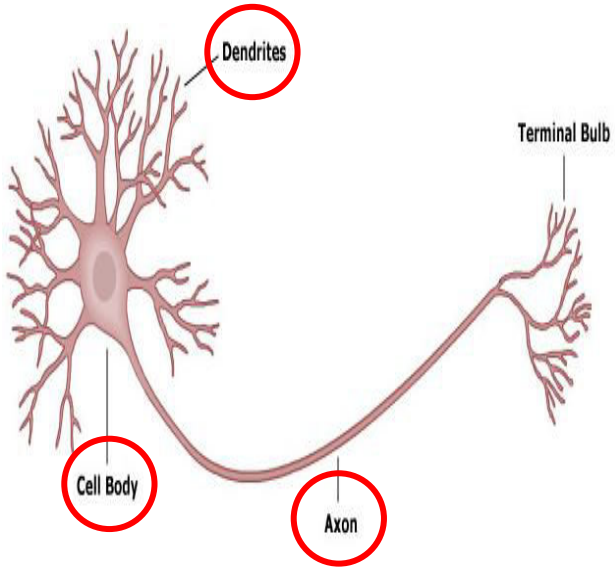
- Ca^{2+} ions that enter the cell through Ca^{2+} channels have two major roles:
- Propagating electrical signals.
- Acting as intracellular messengers that trigger subsequent intracellular events e.g. release of neuro-transmitters and contraction of muscles.

Nerve Physiology

Structural Features of Neuron

Lecture no 25

Structural Features of Neuron



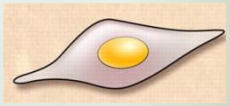
Three Basic Components of Neuron

- Structural and functional units of nervous system.
- Vary considerably in shape and size.
- Three basic components:
  1. A cell body or soma
  2. Dendrites
  3. Axon

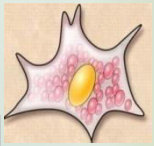
Structural Features of Neuron

Cell Body or Soma

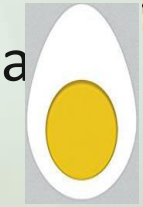
- Vary in size and shape

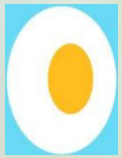


- Fusiform

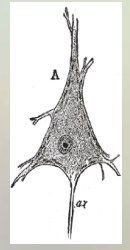


- Stella





- Oval
- Rounded



- Pyramidal
- Size range 5µm-135µm

Structural Features of Neuron

Functions of the Cell Body

- Main nutritional part of the nerve cell.
- Responsible for synthesis of materials necessary for growth and metabolic maintenance of the cell.

Structural Features of Neuron

Structural Components of Soma

Cell body contains:

- Nucleus
- All cell organelles
- Nissl's bodies
- Neurofibrils

Nissl's bodies

- Consist of a group of ribosomes and rough endoplasmic reticulum
- Are associated with protein synthesis

Structural Features of Neuron

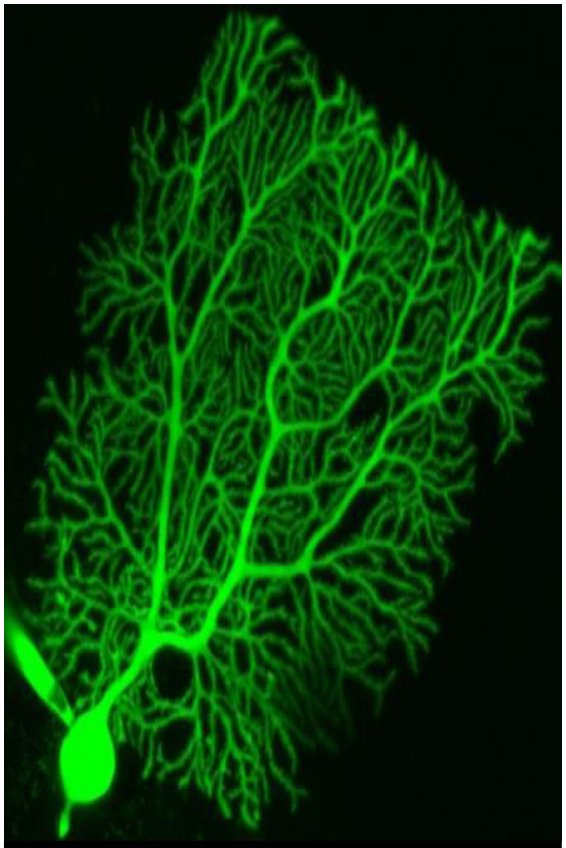
Neurofibrils

- Delicate threads running from cell body into the axon and dendrites.

Neurofibrils—Functions

- Give support and shape to the neuron.
- Transport substances from cell body to the distal cell processes.

Structural Features of Neuron



A Purkinje cell

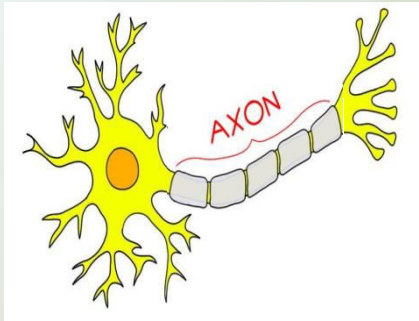
http://www.riken.jp/~/media/riken/research/rikenresearch/figures/hi_4309.jpg

Dendrites

- Short, thin, branched, cytoplasmic processes.
- Extend from cell body.
- Receive signals from other neurons.
- Conduct impulses towards the cell body.
- Greek “Dendron”--Tree
- Neurons with extensive dendritic tree receive many inputs.
- Purkinje cells of brain

Structural Features of Neuron

Axon

- Long and thick process with constant diameter.
- 

1

- Lengths vary from few mm to more than a meter.

Structural Features of Neuron

Axon—Functions

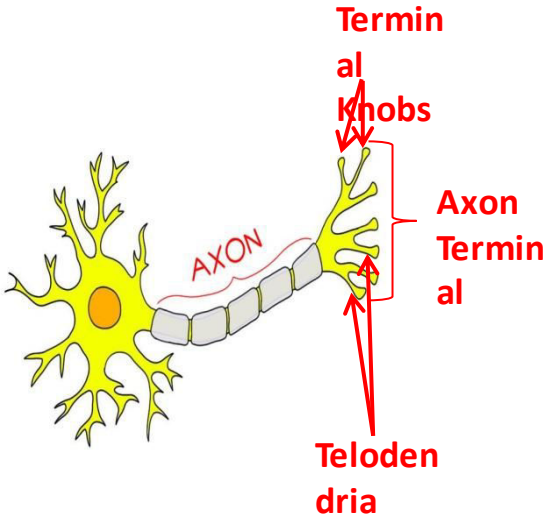
- Specialized to conduct signals away from the cell body.
- Carry information for long distances with high fidelity and without loss.

Structural Features of Neuron

Axon—Structural Components

- Axons contain a jelly-like semi-fluid substance Axoplasm.
- A plasma membrane called as Axolemma.
- Have mitochondria and ER but lack Nissl's granules.

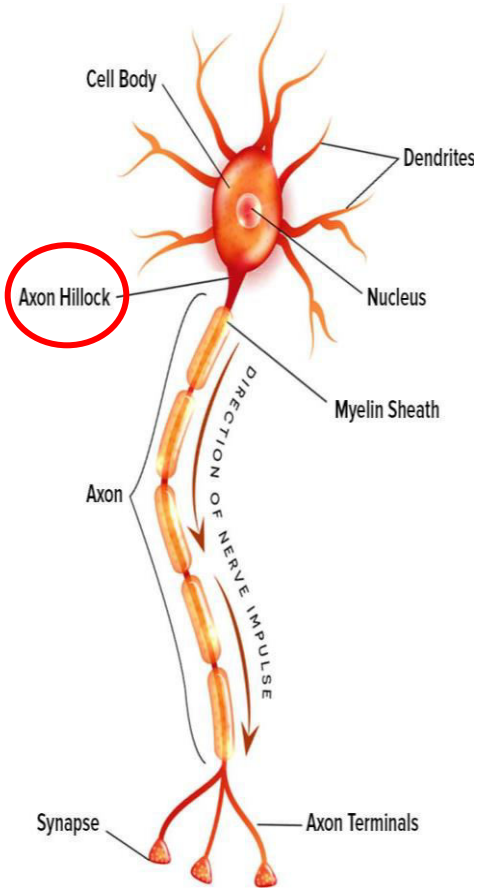
Structural Features of Neuron



Axon Terminals

- Branched terminal ends of axons.
- At each terminal are small extensions telodendria with enlarged ends terminal knobs.
- Terminal knobs produce neurotransmitters.
- Axon terminals allow transmission of signals to other neurons, glands or muscle fibers.

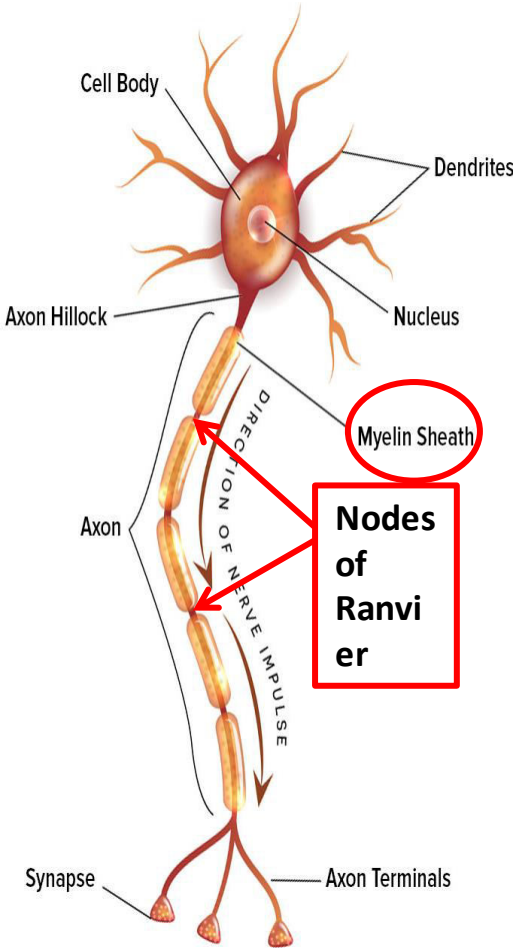
Structural Features of Neuron



Axon Hillock

- At the junction of cell body and axon is a cone-shaped region called axon hillock.
- Here signals are generated that travel down the axon.

Structural Features of Neuron



Myelin Sheath

- A fatty myelin sheath cover the axons.
- It is secreted by Schwann cells.
- Myelin sheath is not continuous.
- There are non-myelinated points called Nodes of Ranvier.

Structural Features of Neuron

Saltatory Impulse

- Impulses jump over node to node.
- Such impulses are known as saltatory impulses.
- This pattern in insulated axons increases the speed of nerve impulse.

END

Nerve Physiology

Transmission of Signals in the Nervous System

Lecture no 26

Transmission of Signals in the Nervous System

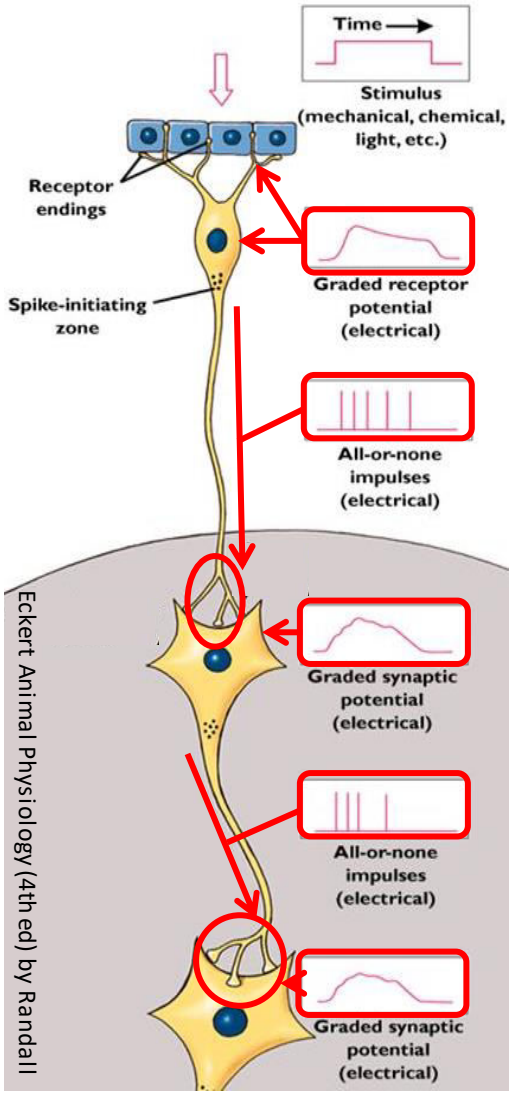
Methods of Signal Transmission

- Neuronal signals are transmitted in two forms:
  1. Graded, electro-tonically conducted potentials
  2. Action potentials
- Both alternate as signal is passed along one neuron and transmitted on to another neuron.

Transmission of Signals in the Nervous System

Coding of Signals During Transmission

- A signal is coded alternately in graded potentials and in all-or-none APs.
- Graded potentials are produced for short distance conduction at sensory and postsynaptic membranes.
- APs are generated for long-distance conduction.
- This also involves inter-conversion of elect. & chem. methods at synapses.



Transmission of Signals in the Nervous System

Signal Transmission: Example

- Transmission of a signal generated by a stimulus that is received by a sensory neuron present in a receptor organ.

Transmission of Signals in the Nervous System

Receptor Potential

- Stimulus is received at receptor endings of a sensory neuron.
- It causes a change in V_m (i.e. depolarization) in proportion to its strength.
- This potential change at receptor site is known as “receptor potential”.
- It is “graded” as it varies in a continuous fashion.

Transmission of Signals in the Nervous System

Receptor Potential

- Time course and amplitude of a receptor potential are closely related to the time course and intensity of the stimulus.
- So, receptor potential is an electrical neuronal analog of the stimulus.

Transmission of Signals in the Nervous System

Spreading the Signals: Passive Electrotonic Transmission

- The signal spreads away from the receptor site passively through electrotonic transmission
- It decays over a relatively short distance.

Transmission of Signals in the Nervous System

Decay in Passive Electro-tonic Transmission

- The decay happens b/c the receptor part of the neuronal membrane lacks voltage-gated ion channels.
- So, APs cannot be produced and signals cannot be propagated regeneratively in the receptor endings of sensory neuron.

Transmission of Signals in the Nervous System

Distant Transmission: Regenerative APs:

- For distant transmission, sensory signals are transformed into APs at the spike initiating zone (axon hillock) that contains voltage gated ion channels.

Transmission of Signals in the Nervous System

Synaptic Transmission:

- As the signal reaches axon terminals, it is transformed from electrically encoded signal to chemical signal (neurotransmitter molecules).
- This chemical signal is transmitted across the synapses to next neuron.

Transmission of Signals in the Nervous System

Graded Postsynaptic Potentials:

- Neurotransmitters cause change in the V_m of postsynaptic neuron.
- The change in V_m happens as the chemical signal is reconverted into an electrical signal.
- V_m generated in the postsynaptic neuron is the postsynaptic potential (psp).

Transmission of Signals in the Nervous System

Graded Postsynaptic Potentials:

- psp is a graded signal, reflecting the properties of original stimulus.
- This graded psp brings the spike-initiating zone of postsynaptic neuron to threshold, triggering an AP.

Transmission of Signals in the Nervous System

Transmission of Signals in the Nervous System

- Following this course of transmission, signals travel from receptors through PNS and reach the CNS.

END

Nerve Physiology

Effort by: AARISH ALI

Transmission of Signals in a Single Neuron

Lecture no 27

Transmission of Signals in a Single Neuron

Signal Reception and Integration

- A nerve cell receives input signals through:
  - Dendrites
  - Membrane of soma
- Soma integrates messages from all inputs.
- It determines if the signal should be trans-mitted to next neuron passively or actively through AP.

Transmission of Signals in a Single Neuron

Spread of Information Through a Neuron

- Information received by a neuron is in the form of a stimulus-generated local depolarization.
- It spreads through:
- Passive electrotonic conduction in nonspiking local circuit neurons.
- Active regenerative APs in neurons with voltage gated ion channels.

Transmission of Signals in a Single Neuron

Non-spiking Local-circuit Neurons

- Small neurons that lack voltage-gated ion channels.
- Have only the resting K^+ ion channels.
- Incapable of producing APs.
- Conductance depends on passive electrical properties: capacitance and resistance (cable properties).

Transmission of Signals in a Single Neuron

Cable Properties

- Passive electrical properties make axons comparable to electric wires.

Role of Cable Properties

- Affect the speed and distance of transmission of electrical signals through the axon.

Transmission of Signals in a Single Neuron

Implication of Cable Properties

- Current flowing along an axon decays with distance.
- The decay happens because of:

1. Resistance of cytoplasm and cell membrane to flow of electrical signals.
2. Absence of insulation around the axonal membrane that allows leakage of current (K^+).

Transmission of Signals in a Single Neuron

Passive Electrotonic Conduction of Signals

- Local-circuit neurons are only few millimeters in length.
- So, graded signals can be transmitted through passive, electrotonic conduction to the axon terminals without the aid of APs.

Transmission of Signals in a Single Neuron

Passive Electrotonic Conduction of Signals

- Amplitude of signals is attenuated as they spread through the cell.
- But signals are still large enough to modulate the release of neurotrans-mitter at the terminals.

Transmission of Signals in a Single Neuron

Neurons Conducting APs

- Neurons with functional voltage-gated ion channels have active electrical properties.
- They carry electrical signals without decrement by producing regenerative APs.
- Such neurons have longer axons and transmit signals to long distances.

END

Nerve Physiology

Propagation of Action Potentials

Lecture no 28

Propagation of Action Potentials

Propagation of AP is Necessary

- An AP contains the information of the stimulus in electrical form.
- To carry this information to the central parts of nervous system is the basic requirement of communication system.
- So the propagation of AP along the axon must happen.

Propagation of Action Potentials

Regenerative Property of APs

- An AP occurs over a region of only few mm.
- To propagate, events of AP must regenerate.
- AP has the property of regenerating itself as it travels down from cell body to the synaptic terminals.
- Due to this property, APs act as a mode of long-distance transmission.

Propagation of Action Potentials

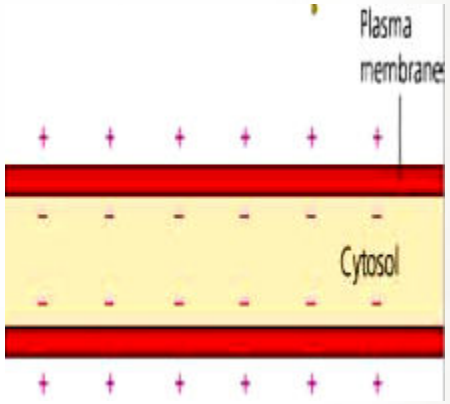
Mechanism of Propagation

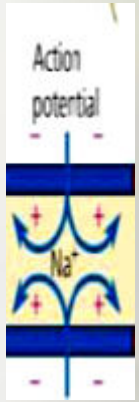
- The AP elicited at any one point on an excitable membrane, spreads excitation to the adjacent portions of the membrane.
- This results in its propagation along the membrane.

Propagation of Action Potentials

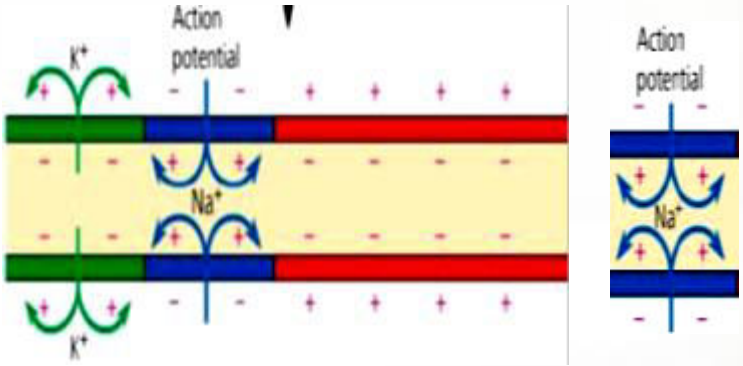
Mechanism of Propagation

- At the site where an AP is initiated, an inflow of Na^+ current starts.
- This initiates the rising phase of an AP.





Propagation of Action Potentials



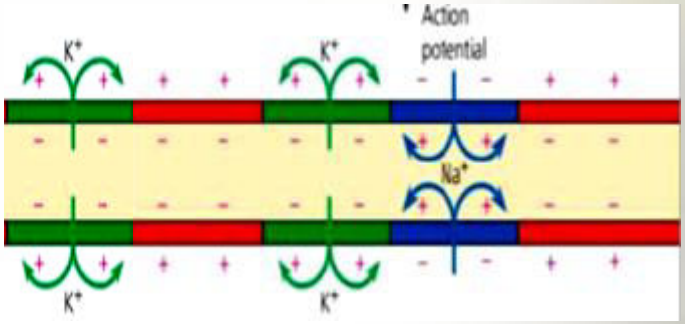
Mechanism of Propagation

- The Na^+ current in one region, affects about 1-3 mm of the adjacent area.
- This causes excitation and depolarization in this region too, initiating AP.

Propagation of Action Potentials

Mechanism of Propagation

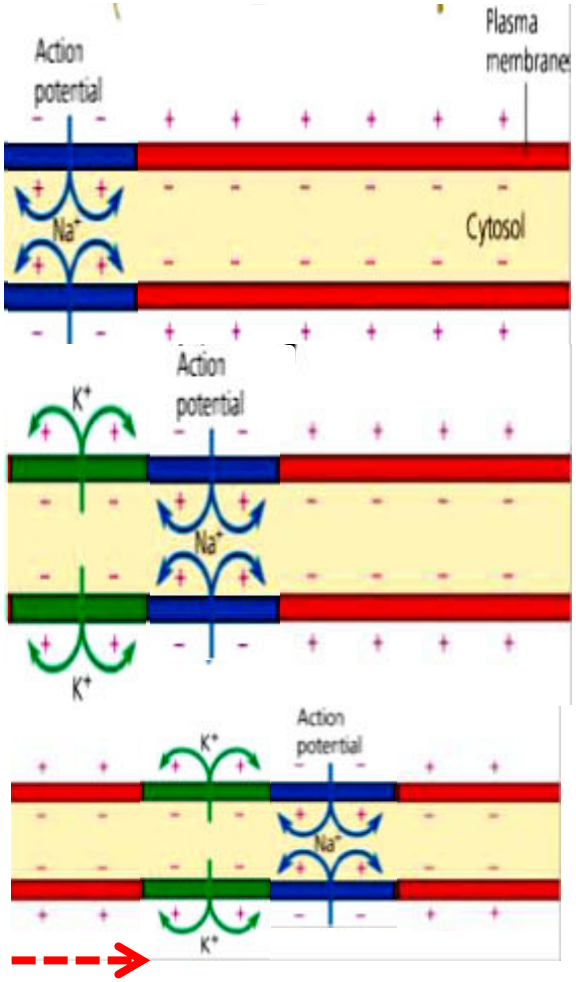
- Immediately behind the traveling zone of depolarization is a zone of repolarization due to outflow of K^+.



Propagation of Action Potentials

Mechanism of Propagation

- Depolarization-repolarization events spread in the further regions of the membrane.
- In this way, local currents of ions cause AP to be propagated along the length of the axon.



Propagation of Action Potentials

Propagation is Unidirectional

- The Na^+ channels become inactivated in the repolarized zone.
- So, the inward current that depolarizes the axon membrane ahead of the AP cannot produce another AP behind it.

Propagation of Action Potentials

Propagation is Unidirectional

- This prevents APs from traveling back toward the cell body.
- Thus, an AP that starts at the axon hillock moves in only one direction—towards the synaptic terminals.

Propagation of Action Potentials

END

Propagation Without Decrement

- At each position along the axon, the process is identical.
- That is, the shape and magnitude of the AP remain constant.
- It ensures propagation of initial excitation without decrement.

Nerve Physiology

Speed of Propagation

Lecture no 29

Speed of Propagation

Speed of Propagation of APs vs Electric Current

- The AP is an electro-chemical current, produced due to the flow of ions through the pores of channel proteins.
- This kind of current travels much more slowly than electricity that is due to the flow of electrons.

Speed of Propagation

Speed of Propagation in Invertebrate Axons

- Narrow axons: few cm/s
- Giant axons of arthropods and molluscs: 30 m/sec

Speed of Propagation

Speed of Propagation in Vertebrate Axons

- Vertebrate axons have narrow diameters but have myelin sheath around their axons.
- They conduct impulses at higher speed.
- APs can travel as fast as 120 m/s in large axons.
- In small diameter, non-myelinated fibers, speed may be as low as 0.25 m/s.

Speed of Propagation

Length Constant Affects Speed of Propagation

- The speed varies with length constant b/c the resistance to electrical current flow is inversely proportional to the cross-sectional area.
- The length constant increases with axon diameter. So wider axons conduct APs more rapidly than narrow ones.

Speed of Propagation

Length Constant: Evolutionary Trends

- Conduction velocity has increased with evolutionary increase in the length constant of axons.
- In invertebrates length constant has been increased by an increase in axonal diameter, that reduces the internal longitudinal resistance.

Speed of Propagation

Length Constant: Evolutionary Trends

- In vertebrates length constant is increased by:

1. Axon myelination
2. Forming nerves having tens of thousands of axons in a single nerve.

END

Nerve Physiology

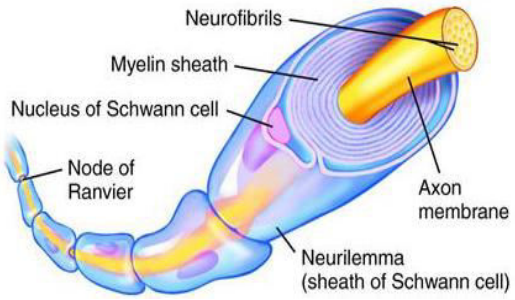
Axon Myelination and Saltatory Conduction

Lecture no 30

Axon Myelination and Saltatory Conduction

Myelin Sheath

- A thick, multi-layered membranous structure that surrounds many vertebrate axons.
- Mainly composed of a fatty substance sphingomyelin.



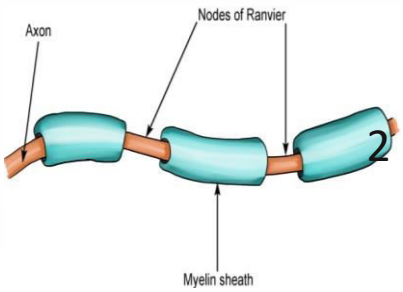
https://beyondthedish.files.wordpress.com/2015/12/myelin_sheath.jpg

Axon Myelination and Saltatory Conduction

Deposition of Myelin Sheath

- Myelin sheath is deposited around the axon during the course of development.
- Two types of specialized glial cells deposit myelin sheath :
  - Oligodendrocytes in CNS
  - Schwann cells in PNS

Axon Myelination and Saltatory Conduction



internode

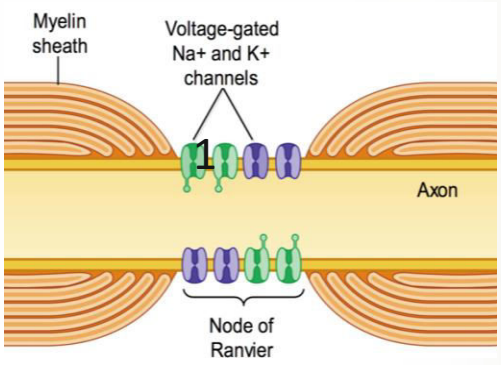
Nodes of Ranvier

- Myelin sheath is not continuous.
- Gaps are present along its length every 1-3 mm.
- Gaps have un-insulated area Nodes of Ranvier (2-10 µm)
- Myelinated areas b/w nodes: internodes

Axon Myelination and Saltatory Conduction

Nodes of Ranvier

- Voltage-gated Na^+ channels are restricted to nodes of Ranvier.
- Extracellular fluid is in contact with membrane.
- Ion exchange occurs through these nodes.



https://d1yboe6750e2cu.cloudfront.net/i/a31fa732f1bb6a28f3c5ec2834d8cff1b35c230e

Axon Myelination and Saltatory Conduction

Functions of Myelin Sheath

- Electrical insulation
- Increase the speed of conduction
- Space efficiency
- Saltatory conduction

Axon Myelination and Saltatory Conduction

Electrical Insulation

- The sheath mainly contains lipid substance sphingomyelin that is a poor conductor of electrical current.
- It provides electrical insulation for the axon.
- It decreases ion flow about 5000-fold.
- So, it acts as an analogue to the plastic insulation around electrical wires.

Axon Myelination and Saltatory Conduction

Increase the Speed of Conduction

- Insulation with myelin enhances the efficiency of longitudinal spread of current.
- So, it results in rapid transmission of nerve impulse.

Axon Myelination and Saltatory Conduction

Increase the Speed of Conduction

- It renders thin axons higher conduction speed than very thick non-myelinated axons.
- e.g. 20 µm myelinated axon has faster speed than 40x thicker squid giant axon.

Axon Myelination and Saltatory Conduction

Space Efficiency

- Thousands of myelinated axons can be packed into space occupied by just one giant axon.

Axon Myelination and Saltatory Conduction

Saltatory Conduction

- Ions cannot flow through internodes that have thick myelin sheath, so APs are not generated in these regions.
- Ions flow only through the nodes of Ranvier.

Axon Myelination and Saltatory Conduction

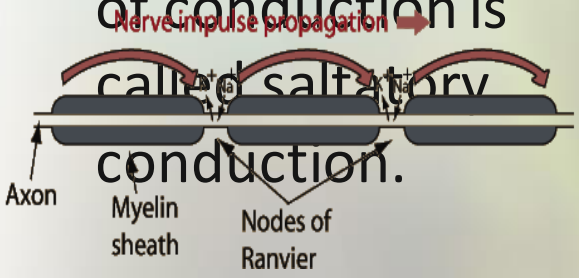
Saltatory Conduction

- Inward Na^+ current produced at a node during the rising phase of AP travels all the way to the next node.
- Here it depolarizes the membrane and regenerates the AP.

Axon Myelination and Saltatory Conduction

Saltatory Conduction

- AP appears to jump along the axon from node to node.
- This mechanism of conduction is called saltatory conduction.



http://hyperphysics.phy-astr.gsu.edu/hbase/Biology/imgbio/myelin2.gif

Axon Myelination and Saltatory Conduction

END

Significance of Saltatory Conduction

- Increases the velocity of nerve transmission 5 to 50 fold.
- Conserves energy by reducing the loss of ions up to 100 times.

Nerve Physiology

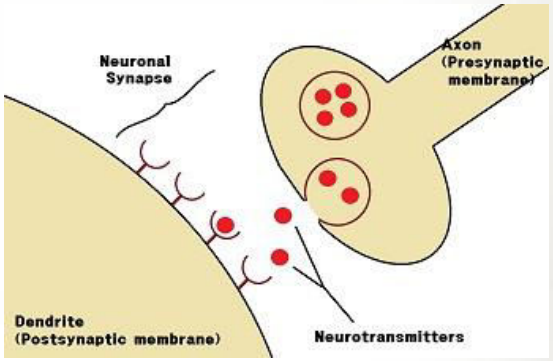
Synapses and Their Types: Electrical Synapses

Lecture no 31

Synapses and Their Types: Electrical Synapses

Synapses

- Junctions between axon terminals of one neuron and the dendrites of another neuron.
- transmission of signals takes place from one neuron to another through these junctions.



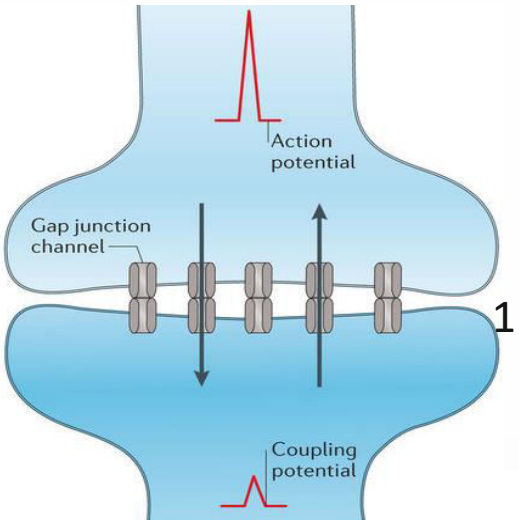
http://study.com/cimages/multimages/16/Neuronal_Synapse.jpg

Synapses and Their Types: Electrical Synapses

Types of Synapses

- Electrical synapses
- Chemical synapses

Synapses and Their Types: Electrical Synapses



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http://www.nature.com/nrn/journal/v15/n4/images/nrn3708-f1.jpg

Electrical Synapses

- At ES, the PMs of pre- and postsynaptic cells are in close apposition.
- They are electrically coupled by channel proteins: gap junctions.

Synapses and Their Types: Electrical Synapses

Mechanism of Signal Transmission

- Electrical current can flow directly from one cell into the other through gap junctions purely by electrical means, without involving any chemical transmitter.

Synapses and Their Types: Electrical Synapses

Mechanism of Signal Transmission

- It is just like signal transmission along a single axon, involving passive spread of local circuit current that depolarizes and excites the next neuron.
- So, the transfer of information through these synapses is much rapid.

Synapses and Their Types: Electrical Synapses

Limited Efficacy of Electrical Synapses

- Although electrical signal in pre-synaptic cell rapidly transmits a signal in the postsynaptic cell—this signal is somewhat attenuated.
- So, a single presynaptic AP is unable to provide enough current to elicit an AP in postsynaptic cell, reducing its efficacy.

Due to this attenuation

Synapses and Their Types: Electrical Synapses

Advantage of Electrical Synapses

- Particularly effective in the synchronization of electrical activity within a group of cells.
- They can transmit information across a series of cell-cell junctions.
- e.g. in the myocardium of heart, in which signals are passed between muscle cells.

Synapses and Their Types: Electrical Synapses

Occurrence of electrical synapses

- Electrical synapses are relatively rare and most signaling between neurons takes place through chemical synapses.

Synapses and Their Types: Electrical Synapses

Occurrence of electrical synapses

Electrical transmission has been discovered in:

- Many cells of central nervous system
- Retina of the eye
- Smooth muscle fibers
- Cardiac muscle fibers
- Most receptor cells
- Some axons of PNS

END

Nerve Physiology

Synapses and Their Types: Chemical Synapses

Lecture no 32

Synapses and Their Types: Chemical Synapses

The chemical synapses

- Involved in chemical synaptic transmission through neuro-transmitters.
- More common mode of synaptic transmission.

Types

- Fast chemical synapses
- Slow chemical synapses

Synapses and Their Types: Chemical Synapses

Fast Chemical Synapses

- Found in the CNS
- At neuromuscular junctions.
- Produce immediate but short-lived response.

Not Faster than Electrical Synapses

- Although named "fast", considerably slower than transmission across electrical synapses.

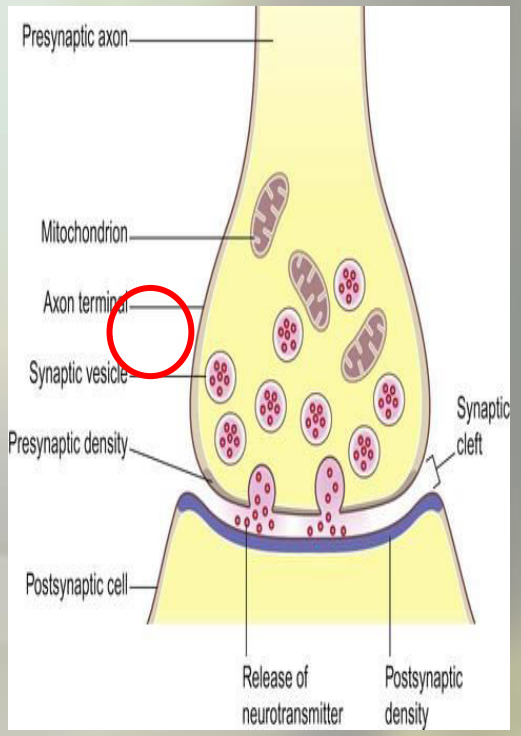
Synapses and Their Types: Chemical Synapses

Neurotransmitters at Fast Chemical Synapses

- Neurotransmitters involved in fast chemical synapses are typically small molecules (e.g. Acetylcholine).
- They are stored in small, clear synaptic vesicles in the axon terminals.

Release of Neurotransmitters

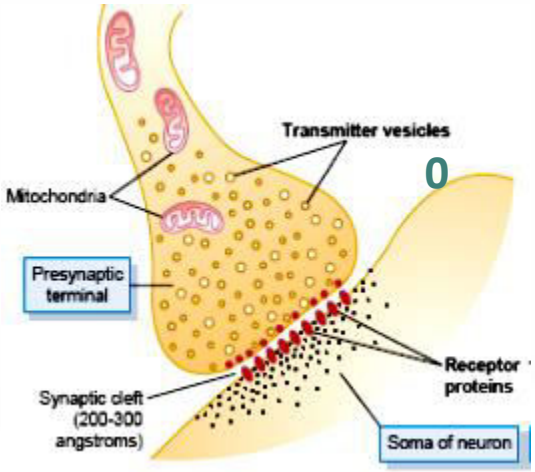
- APs in presynaptic neuron cause release of neurotrans-mitter molecules at axon terminals.
- Release occurs by exocytosis into the synaptic cleft through specialized sites on the membrane.



Synapses and Their Types: Chemical Synapses

Synaptic Cleft

- A narrow, fluid-filled space, about 20 nm wide.
- It separates the membranes of pre- and postsynaptic neurons.

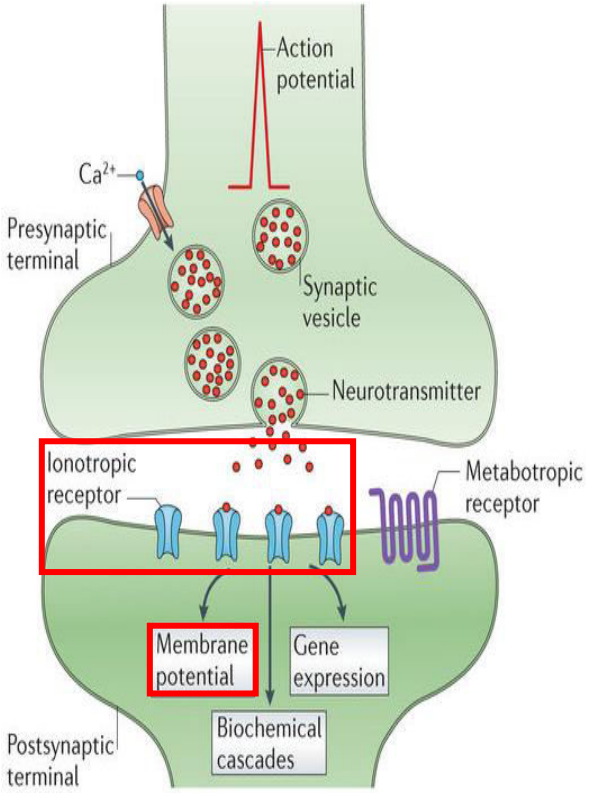


Textbook of Medical Physiology (11th ed) by Guyton

Synapses and Their Types: Chemical Synapses

Mode of Action of Neurotransmitters

- Neurotransmitter binds to specific protein receptors—ligand-gated ion channels in the postsynaptic membrane.
- Result: channels open & allow ionic current to flow into the postsynaptic cell and change its V_m.
- An AP is initiated.



Synapses and Their Types: Chemical Synapses

Slow Chemical Synapses

- Onset of postsynaptic response is slower (hundreds of milliseconds).
- Last much longer (from seconds to hours)

Synapses and Their Types: Chemical Synapses

Neurotransmitters at Slow Chemical Synapses

- Large molecules
- Synthesized from amino acids

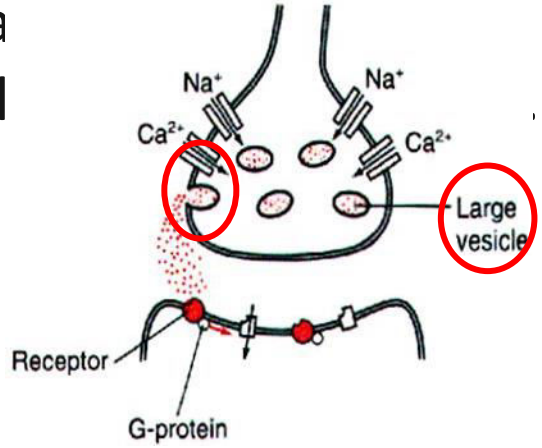
Types

- Biogenic amines: derived from a single amino acid.
- Neuropeptides: Consist of several amino acid residues.

Synapses and Their Types: Chemical Synapses

Packaging and Release of Neurotransmitters

- Vesicles in the slow system are larger.
- Synthesized in the cell body and transported to the nerve terminals.
- Release occurs at sites that are located away from the sites of release of fast neurotra
  morphol



Eckert Animal Physiology (4th ed) by Randall

Synapses and Their Types: Chemical Synapses

Mode of Action of Neurotransmitters

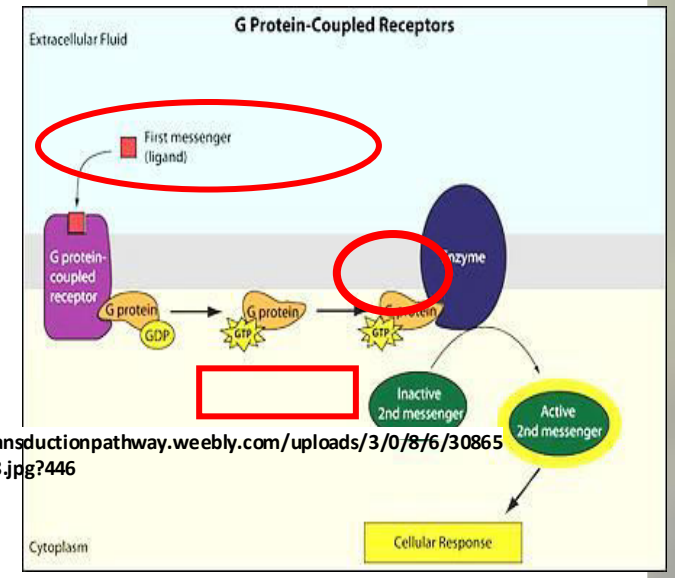
- Slow response transmitters don’t act through ligand-gated channels.
- They act through G protein-linked receptors

Synapses and Their Types: Chemical Synapses

Mode of Action of Neurotransmitters

- A neurotransmitter binds to its receptor, forming a neurotransmitter-receptor complex that activates a G protein.
- G protein activates a signal transduction pathway through a second messenger e.g. cAMP.

- In response, second messenger elicits a cellular response that modifies functions of channels and many in-tracellular processes.



http://signaltransductionpathway.weebly.com/uploads/3/0/8/6/30865779/612632978.jpg?446

Synapses and Their Types: Chemical Synapses

Some Neurons are Involved in Both Types of Transmission

- Physiological and anatomical evidence suggest existence of dual role neurons and transmitters.
- So a single neurotrans-mitter may act both through ligand-gated channels and G-protein-coupled receptors.

END

Nerve Physiology

Mechanism of Release of Neurotransmitters

Lecture no 33

Mechanism of Release of Neurotransmitters

Mechanisms of Release

- The mechanisms of release of neurotrans-mitter into the synaptic cleft are common for both fast and slow synaptic transmission.
- Two basic patterns of release:
  - Release with Action Potential
  - Nonspiking release

Mechanism of Release of Neurotransmitters

Release with AP

- Most neurons release neurotransmitters when an action potential reaches their axon terminals.

Mechanism of Release

- As an AP arrives at axon terminals, it activates voltage-gated Ca^{2+} channels, allowing Ca^{2+} to enter the terminal.

Mechanism of Release of Neurotransmitters

Mechanism of Release

- Ca^{2+} ions bind with protein molecules on the inner surface of the membrane at special “release sites”.
- Binding causes release sites to open through the membrane.
- This allows the vesicles to release their transmitter into the synaptic cleft.

Mechanism of Release of Neurotransmitters

Mechanism of Release

- An AP and influx of Ca^{2+} into the terminal is essential for transmitter release
- When AP ends and influx of Ca^{2+} drops, the release of neurotransmitter is stopped.

Mechanism of Release of Neurotransmitters

Quantal Release of Neurotransmitter

- Neurotransmitters are generally released in tiny packets called quanta.
- Each quantum may consist of about 2000 to 10,000 molecules of transmitter molecules.

Mechanism of Release of Neurotransmitters

Depolarization-Release Coupling

- Probability of quantal release increases if the presynaptic membrane is depolarized due to AP.
- Amount of transmitter released varies directly with depolarization of presynaptic terminal: more depolarization—more transmitter.

Mechanism of Release of Neurotransmitters

Nonspiking Release

- Some neurons release neurotransmitter from their terminals even in the absence of APs.
- In these neurons info. transfer occurs by elect-rotonically conducted graded potentials.
- Amount of transmitter released depends on V_m And depolarization.

END

Nerve Physiology

Excitatory and Inhibitory Postsynaptic Potentials

Lecture no 34

Excitatory and Inhibitory Postsynaptic Potentials

Excitatory PSPs

- A synaptic potential that makes the postsynaptic neuron more likely to fire an action potential.
- It results from the flow of positively charged Na^+ or Ca^{2+} ions into the postsynaptic cell.
- It happens when ligand-gated ion channels open due to the binding of neurotransmitters in fast chemical transmission.

Excitatory and Inhibitory Postsynaptic Potentials

Excitatory Postsynaptic Current

- The flow of ions that causes an EPSP is known as EPSC.
- Current through a single ion channel is too small to generate a potential difference in the post synaptic cell.
- Currents through many channels are summed up (summation) to produce an EPSP.

Excitatory and Inhibitory Postsynaptic Potentials

EPSP

- Larger EPSPs result in greater membrane depolarization increasing the likelihood of post-synaptic cell to reach threshold to fire an AP.
- Neurotransmitter most often associated with EPSP in CNS is Glutamate
- ACh is most common excitatory transmitter at neuromuscular junctions

Excitatory and Inhibitory Postsynaptic Potentials

Inhibitory Postsynaptic Potentials

- A synaptic potential that makes a postsynaptic neuron less likely to generate an AP.
- IPSPs result from the inflow of -ive ions or outflow of +ive ions.
- Inhibitory synaptic currents are carried by channels that are permeable to K^+ or Cl^-.

Excitatory and Inhibitory Postsynaptic Potentials

Inhibitory Transmitters

- IPSPs can occur at all chemical synapses that release inhibitory neurotransmitters e.g. GABA and Glycine.
- They bind to receptors that induce a change in the permeability of post-synaptic membrane to particular ions, causing either Cl^- inflow or K^+ outflow.

Excitatory and Inhibitory Postsynaptic Potentials

Mechanism of Inhibition

- Ionic currents cause the postsynaptic V_m to become more -ive than RMP i.e. hyperpolarized.
- For an AP, depolarization of postsynaptic membrane is needed that requires V_m to become more +ive than RMP.
- Hyperpolarization makes AP less likely to occur in postsynaptic neuron.

Excitatory and Inhibitory Postsynaptic Potentials

Channel properties vs Neurotransmitters

- There is nothing inherently excitatory or inhibitory about a particular transmitter
- Properties of channels opened by transmitter and the type of ions that flow, determine how a transmitter affects the postsynaptic cell.

Excitatory and Inhibitory Postsynaptic Potentials

END

Channel properties vs Neurotransmitters Example: Acetylcholine

- ACh is excitatory at the neuromuscular junctions
- It opens channels that allow Na^+ to flow in and K^+ to flow out.
- ACh is inhibitory at the parasympathetic neurons in heart.
- It affects K^+ selective channels & prolongs hyperpolarization.

Nerve Physiology

Neurotransmitters: Diversity and Classification

Lecture no 35

Neurotransmitters: Diversity and Classification

Diversity

- By the mid-1960s, only three neurotransmitters had been identified.
- Acetylcholine (ACh)
- Norepinephrine
- γ-aminobutyric acid (GABA).

Neurotransmitters: Diversity and Classification

Diversity

- Today, more than 100 neurotransmitters have been identified.
- They vary in:
  - Size
  - Molecular weight
  - Chemical structures

Neurotransmitters: Diversity and Classification

Classification Based on Chemical Structure

Two groups:

- Small, low molecular weight neurotrans-mitters
- Large, high molecular weight neurotrans-mitters derived from amino acids

Neurotransmitters: Diversity and Classification

Small, Low M. Weight Neurotransmitters

- Acetylcholine (ACh)
- Amino acids
  GABA, Glycine,
  Glutamate, Aspartate
- Biogenic amines
  Norepinephrine,
  Epinephrine, Dopamine,
  Serotonin & Histamine.
- Gases
  NO & CO

Neurotransmitters: Diversity and Classification

Large, High M. Weight Neurotransmitters

- Neuropeptides
- Larger molecules Constructed of amino acids.
- More than 40 neuropeptide transmitters identified in mammalian CNS

Neurotransmitters: Diversity and Classification

Neuropeptides

Include

- Many hypothalamic and pituitary peptide hormones
- Substance-P
- Endorphins
- Enkephalins
- Many amino acid derivatives

Neurotransmitters: Diversity and Classification

Classification Based on Mode of Action

- Fast, Direct Neurotransmitters
- Slow, Indirect Neurotransmitters

Neurotransmitters: Diversity and Classification

Fast, Direct Neurotransmitters

- They act directly on ion channel proteins.
- They change the conductance of postsynaptic membrane for various ions.
- Include: acetylcholine, and amino acids (glutamate, aspartate, glycine and γ-Aminobutyric acid)

Neurotransmitters: Diversity and Classification

Slow, Indirect Neurotransmitters

- They work through an indirect biochemical pathway that involves G proteins.
- They change the state of a second messenger that results in changes in conductance of ion channel proteins.

Neurotransmitters: Diversity and Classification

Slow, Indirect Neurotransmitters

Include:

- Biogenic amines
- Neuropeptides

- Many of these also act as neuromodulators.
- Neuromodulators affect neighboring neurons and modify their behavior at once.

END

Nerve Physiology

Fast, Direct Neurotransmitters

LECTURE NO 36

Fast, Direct Neurotransmitters

Fast, Direct Neurotransmitters

Types:

1. Fast excitatory synaptic transmitters
2. Fast inhibitory transmitters

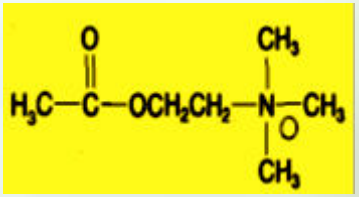
Fast, Direct Neurotransmitters

Fast Excitatory Synaptic Transmitters

- Act by opening ion channels in postsynaptic cell membrane.
- Include:
  - Acetylcholine (ACh)
  - Glutamate
  - Aspartate

Fast, Direct Neurotransmitters

Acetylcholine



- Most familiar fast acting neurotransmitter.
- In most instances, has an excitatory effect.
- Acts as inhibitory trans-mitter in some instances.
- e.g. at peripheral para-sympathetic nerve endings where it inhibits heart by vagus nerves.

Fast, Direct Neurotransmitters

Cholinergic Neurons

- Neurons that release Acetylcholine are cholinergic.
- These include many neurons of vertebrates and invertebrates.

Fast, Direct Neurotransmitters

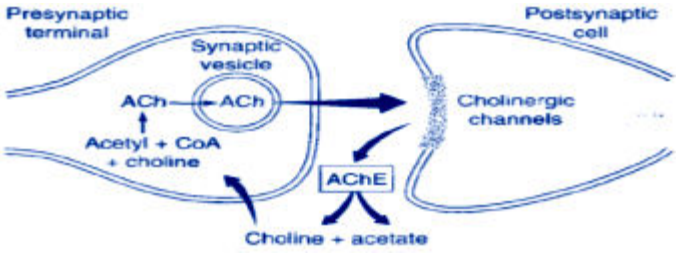
Acetylcholine: Mode of Action

- When ACh is released into the synaptic cleft, it binds to the ligand-gated ACh-specific receptors in the postsynaptic membrane.
- Binding causes Na^+ and K^+ ion channels to open briefly and produce an excitatory PSP.

Fast, Direct Neurotransmitters

Role of Enzyme Acetylcholinesterase (AChE)

- AChE is abundantly present in the synaptic cleft.
- It causes termination of transmission at cholinergic synapses by hydrolyzing ACh into choline and acetate.
- Choline is actively reabsorbed by the presynaptic membrane and recycled by condensation with acetyl CoA to form new molecules of ACh.



Fast, Direct Neurotransmitters

Inhibition of Acetylcholinesterase

- Many insecticides and nerve gases block AChE producing dangerous health effects.
- When AChE is blocked, acetylcholine piles up in the synaptic cleft causing continuous stimulation.
- It results in disruption of neuro-muscular systems.
- Death can follow.

Fast, Direct Neurotransmitters

Acetylcholine Agonists

- Molecules that mimic the action of a neuro-transmitter are called agonists.
- Acetylcholine agonists are:
  - Carbachol
  - Nicotine
  - Muscarine
- They can activate cholinergic synapses.

Fast, Direct Neurotransmitters

ACh Antagonists

- Molecules that block the action of a neuro-transmitter are called antagonists.
- They have structural features in common with a transmitter.
- D-tubocurarine, active agent in South American blow-dart poison curare blocks transmission at cholinergic synapses.

Fast, Direct Neurotransmitters

Glutamate (Glutamic Acid)

- Most common excitatory neurotransmitter in the vertebrate brain.
- In insects and crustaceans it is released at fast excitatory neuromuscular junctions.

Fast, Direct Neurotransmitters

Aspartate (Aspartic Acid)

- Excitatory transmitter, primarily localized to the ventral spinal cord.
- Produced in the mitochondria and transported to cytoplasm, and packaged into synaptic vesicles.
- Forms excitatory/ inhibitory pair with glycine in spinal cord.

Fast, Direct Neurotransmitters

Fast Inhibitory Transmitters

- Glycine
- γ-Aminobutyric acid (GABA)

Glycine

- Secreted mainly at inhibitory synapses in the spinal cord.
- Always acts as an inhibitory transmitter.

Fast, Direct Neurotransmitters

γ -Aminobutyric acid

- Plays important role as an inhibitory transmitter in the vertebrate CNS.
- Forms an excitatory/ inhibitory pair with glutamate in the brain.
- Also released at the inhibitory motor synapses in crustaceans and annelids
- Produces inhibitory PSP by increasing permeability to Cl^-.

END

Nerve Physiology

Slow, Indirect Neurotransmitters

LECTURE NO 37

Slow, Indirect Neurotransmitters

Classes

- Biogenic amines
- Neuropeptides

Slow, Indirect Neurotransmitters

Biogenic Amines

- Act through second messengers
- Produce slow synaptic transmission

Include:

- Catecholamines (Norepinephrine, Epinephrine and Dopamine)
- Serotonin (an indolamine)
- Histamine (an imidazole)

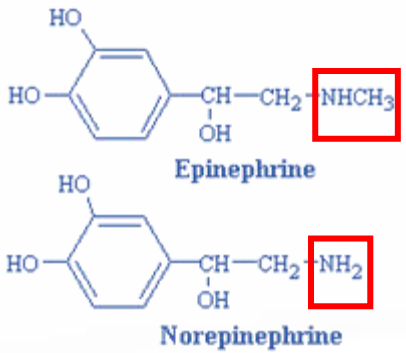
Slow, Indirect Neurotransmitters

Norepinephrine and Epinephrine

- Adrenergic neurons
- Norepinephrine is the primary excitatory transmitter in post-ganglionic cells of sympathetic system.
- Epinephrine is excitatory at some synapses and inhibitory at others depending on post-synaptic membrane.

Neurons that use

Slow, Indirect Neurotransmitters



Norepinephrine and Epinephrine

- Both are also released by chromaffin cells of the vertebrate adrenal medulla.
- Both are structurally very similar and have similar pharmacological actions.

Slow, Indirect Neurotransmitters

Synthesis & inactivation of Norepinephrine

- It is synthesized from phenylalanine.
- two ways of inactivation:

1. Taken up by cytoplasm of presynaptic neuron, and repackaged into synaptic vesicles or inactivated by monoamine oxidase.
2. Also deactivated within the synaptic cleft by methylation.

Slow, Indirect Neurotransmitters

Biogenic Amine Analogues

- Several psychoactive drugs have molecular structures similar to the biogenic amines and act at synapses that use these transmitters.

Examples

1. Mescaline that induces hallucinations.
- interferes with norepine-phrine at synapses in CNS

Slow, Indirect Neurotransmitters

Biogenic Amine Analogues: Examples

2. Amphetamine.
- potent CNS stimulant
- Mimics norepinephrine
- Interacts with adrenergic neurotransmission

3. Cocaine
- interferes with the inactivation of norepinephrine

Slow, Indirect Neurotransmitters

Neuropeptides

- Include more than 40 peptide molecules.
- Synthesized are released in vertebrate CNS.

Slow, Indirect Neurotransmitters

Neuropeptides

- Some are also produced outside CNS in neural and non-neural tissues e.g.
  - intestinal endocrine cells
  - autonomic neurons
  - sensory neurons
- Many of these are also found in invertebrate nervous systems.

Slow, Indirect Neurotransmitters

Neuropeptide: Examples

- Hypothalamic and pituitary peptide hormones
- Gastrointestinal hormones (glucagon, gastrin, cholecystokinin)
- Substance-P
- Endorphins
- Enkephalins

Slow, Indirect Neurotransmitters

Neuropeptide Release

- A single neuropeptide species may be released in three ways:
- As a neurotransmitter
- As a neurosecretory substance
- As a hormone

Slow, Indirect Neurotransmitters

Neuropeptide Release

- Peptides acting as neurotransmitters are released into the synaptic cleft.
- Neurosecretory peptides are liberated into the circulation and carried by the blood to their target neurons.
- Peptide hormones are released from endocrine tissue and target non-neuronal tissues.

Slow, Indirect Neurotransmitters

Effectiveness of Neuropeptides

- Neuropeptides are more potent transmitters than small neurotransmitters.

This is due to:

1. They can bind to receptors at much lower conc. (10^{-9} M vs 10^{-5} M).

Slow, Indirect Neurotransmitters

Effectiveness of Neuropeptides

2. They act through second messenger pathways that provide amplification. So, a small amount can produce a large effect.

3. Their actions are slowly terminated. So remain available to receptors for long time.

END

Nerve Physiology

Neuropeptides: Endorphins and Enkephalins

LECTURE NO 38

Neuropeptides: Endorphins and Enkephalins

Pharmacological Actions

- Naturally occurring signaling molecules produced within the CNS.
- Act as analgesics i.e. reduce perception of pain
- Induce euphoria during stress.
- Decrease urine output
- Depress respiration

Neuropeptides: Endorphins and Enkephalins

Endorphin Receptors (Opioid Receptors)

- Surface membranes of many CNS neurons contain receptors that bind endorphins and enkephalins.
- Narcotic opiates (opium, morphine and heroin) have similar structures and mimic endorphins.

Neuropeptides: Endorphins and Enkephalins

Endorphin Receptors (Opioid Receptors)

- These opiates also bind to endorphin receptors in the brain.
- They produce similar effects i.e. reduce pain and cause euphoria by changing emotional state
- So these receptors are also called opioid receptors.

Neuropeptides: Endorphins and Enkephalins

Endogenous Opioids

- The exogenous opiates i.e. opium and its derivatives produce sense of pleasure.
- The levels of endorphin and enkephalin also rise in brain naturally in response to pleasurable activities.
- Due to these similarities they are called endogenous opioids.

Neuropeptides: Endorphins and Enkephalins

Basis of Analgesic Action

- Endorphins and Enkephalins can block the release of neuro-transmitters from nerve endings that are involved in the perception of pain.

Neuropeptides: Endorphins and Enkephalins

Basis of Opioid Addiction

- When opioid molecules bind to the receptors, they elicit an intense feelings of pleasure—the basis of narcotic use of opiates.
- However, repeated doses of exogenous opiates provoke compensatory changes in neuronal metabolism.

Neuropeptides: Endorphins and Enkephalins

Basis of Opioid Addiction

- So the removal of opiates shifts the nervous system into a state that is perceived as extreme discomfort until the opiate is re-administered.
- This metabolically induced dependence is termed addiction.

Neuropeptides: Endorphins and Enkephalins

Naloxone and opioid receptors

- Naloxone is a drug which acts as a competitive blocker of the opioid receptors.
- It antagonizes narcotic effect of opiates.
- It binds tightly to the opiate receptors without activating it.
- So it blocks opiates to act on their target cells.

Neuropeptides: Endorphins and Enkephalins

Naloxone and opioid receptors

- The antagonistic properties of naloxone have proved to be a useful tool in studies of opioid receptors and the responses mediated by these receptors

END

Nerve Physiology

Receptors in Fast, Direct Neurotransmission

LECTURE NO 39

Receptors in Fast, Direct Neurotransmission

Postsynaptic Cell Receptors

- Membranes of the postsynaptic cells have receptor proteins to which neurotransmitters bind and elicit a response directly.
- Properties of these receptors and their specificity determine the type of responses.

Receptors in Fast, Direct Neurotransmission

Postsynaptic Cell Receptors

- The receptors of fast, direct neurotransmitters are ligand-gated ion channels
- On binding of its ligand, the synaptic channel opens, allowing a minute ionic current to pass through the channel.
- Sum of such currents generates PSP.

Receptors in Fast, Direct Neurotransmission

Example: Acetylcholine Receptors

- AChRs are channel proteins at vertebrate neuromuscular junctions

- Types of AChRs:
  1. Nicotinic AChRs

  Activated by nicotine

  2. Muscarinic AChRs

  Activated by muscarine (a toadstool poison)

Mas-ca-rinic

Receptors in Fast, Direct Neurotransmission

Nicotinic AChRs

- Were the first ligand-gated ion channels to be purified chemically and studied electrically.
- In addition to acetylcholine, they are also activated by nicotine but not with muscarine.

Receptors in Fast, Direct Neurotransmission

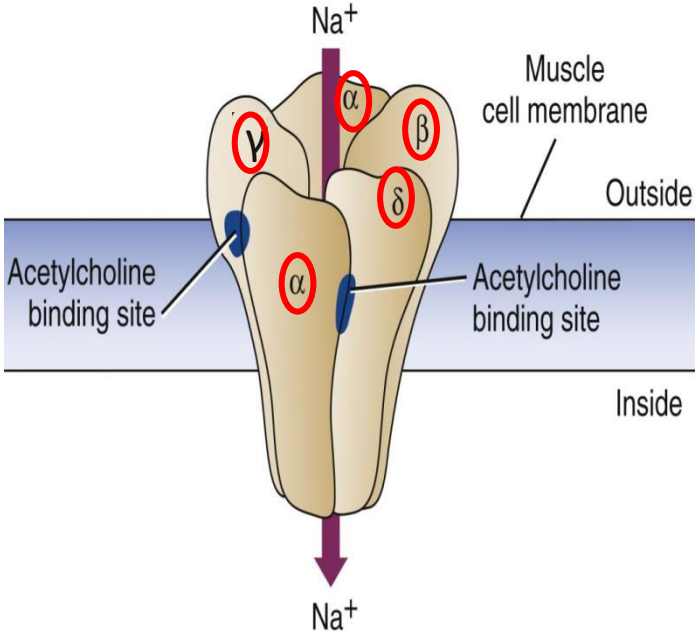
Nicotinic AChRs

Nicotinic receptors are found at:

- Synapses in autonomic ganglia of sympathetic and parasympathetic nervous systems.
- Non-autonomic nerve endings of neuro-muscular junctions.

Receptors in Fast, Direct Neurotransmission

Structure of nAChRs

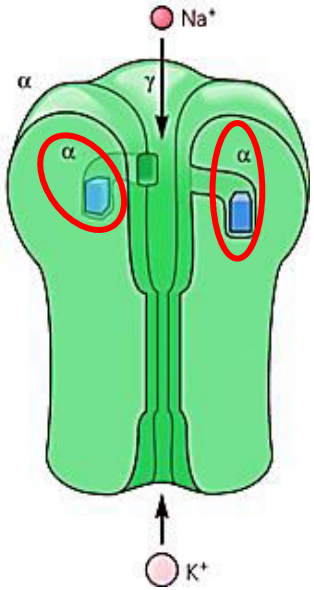


https://basicmedicalkey.com/wp-content/uploads/2017/01/B9781416066279000068_gr1.jpg

- Each nicotinic AChR consists of five subunits that associate and form a channel at the center.
- There are two identical α subunits.
- Other three subunits are β, γ, and σ.

Receptors in Fast, Direct Neurotransmission

Structure of nAChRs



- Receptor sites are on each of the two α-subunits.
- When ligand molecules i.e. ACh, carbachol or nicotine bind, the channel becomes activated and opens allowing Na^+ and K^+ to flow.

Receptors in Fast, Direct Neurotransmission

Muscarinic AChRs

- Activated by muscarine.
- They are quite different from nAChRs as they are not ion channel proteins.
- They belong to the family of G protein-coupled receptors.
- So ACh acting through these channels follows the slow, indirect mechanism.

Receptors in Fast, Direct Neurotransmission

Other Ligand-Gated Channels

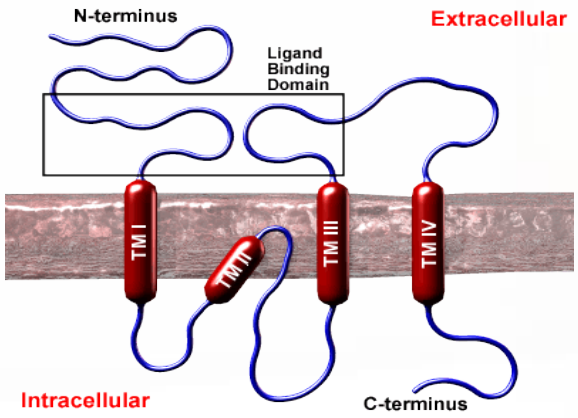
- For the fast, direct transmitters other than ACh, many other ligand-gated channels are involved; e.g. receptors for glycine and GABA .
- Each type of receptor is expressed in a unique and characteristic pattern within the mammalian brain.

Receptors in Fast, Direct Neurotransmission

Homology of Channels

- The receptor channel proteins for ACh, Glycine & GABA form a family having structural homology that indicates their evolutionary relatedness. e.g.
  - Pentameric structure
  - Composed of two to four types of subunits
  - One type of subunit binding the ligand.

Receptors in Fast, Direct Neurotransmission



http://www.bristol.ac.uk/media-library/sites/synaptic/migrated/images/iglur-structure.gif

Glutamate Receptors

- Glutamate receptors form a separate family.
- Having only a slight resemblance to AChRs.
- This family includes three types of fast-acting glutamate receptors.

Receptors in Fast, Direct Neurotransmission

Glutamate Receptors

- There is intense interest in this receptor family b/c glutamate is the most common excitatory neurotransmitter in mammalian CNS.
- Glutamate receptors participate in synaptic modifications that underlie learning and memory.

END

Nerve Physiology

Receptors in Slow, Indirect Neurotransmission

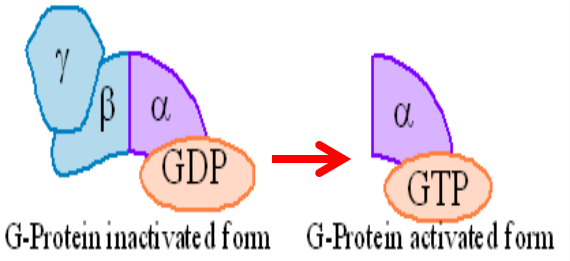
LECTURE NO 40

Receptors in Slow, Indirect Neurotransmission

G Protein-Linked Receptors

- Neurotransmitters that produce slow post-synaptic response bind to the receptors that are linked to a G protein.
- These receptors act by activating the G protein that uses GTP as energy source and regulates the activity of associated effector proteins.

Receptors in Slow, Indirect Neurotransmission



1

G Proteins

- The G proteins family consists of about 20 different proteins that are composed of three subunits: α, β and γ.
- The α subunit of inactive molecule binds to a GDP.
- When active, GDP is converted to GTP.

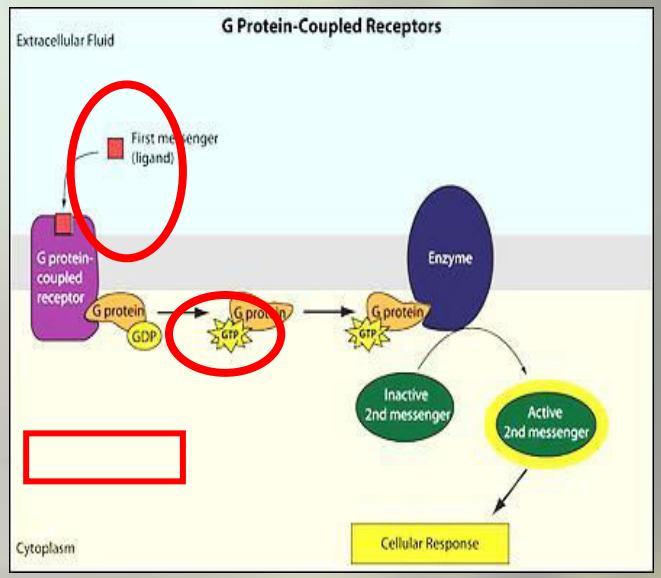
Receptors in Slow, Indirect Neurotransmission

Receptor Molecules

- Receptor molecules span the membrane.
- Neurotransmitter binds on the extracellular face.
- Cytoplasmic face activates the linked G-protein.
- Activated G protein regulates the activity of effector

proteins that control the active conc. of intracellular second messengers.

- 2nd messengers are responsible for cellular response.



http://signaltransductionpathway.weebly.com/uploads/3/0/8/6/30865779/612632978.jpg?446

Receptors in Slow, Indirect Neurotransmission

ACh as Slow, Indirect Neurotransmitter

- ACh acts as slow N.T on the atrial cells of heart.
- It acts on the muscarinic receptors and activates a G-protein.
- The G protein activates, opening K^+ channels.
- As a result, hyperpolarization of atrial cells is prolonged—decreasing cardiac activity & regulating cardiac cycle.

END

Nerve Physiology

Neuromodulation

LECTURE NO 41

Neuromodulation

Definition

- A physiological process by which neurotrans-mitters released by a small group of neurons diffuse through large areas of the CNS and interact and modulate the effect of many neurons simultaneously.
- This is in contrast to the normal synaptic transmission.

one presynaptic neuron directly influences a single postsynaptic neuron.

Neuromodulation

Neuromodulators

- Neurotransmitters involved in altering the cellular properties and efficacy of synaptic transmission of multiple postsynaptic neurons.
- They act through G-protein linked receptors that use voltage-gated ion channels for ion movements.
- Generation of response is slow, but lasts longer.

like slow synaptic transmission

seconds to minutes

Neuromodulation

Neuromodulators

Major neuromodulators in the CNS include:

- Dopamine
- Serotonin
- Acetylcholine
- Histamine
- Norepinephrine
- Endorphins
- Enkephalins
- Substance-P

Neuromodulation

Basis of Neuromodulatory Action

- A neuromodulator is released as a neurotransmitter.
- But it is not reabsorbed by pre-synaptic neuron or broken down into its metabolites.
- it stays for a significant interval of time in CSF.
- influencing or modulating activity of several neurons.

Neuromodulation

Mechanism of Action of Neuromodulators

- They do not produce EPSPs or IPSPs.
- They cause long-term changes that slightly modify, depress or enhance the action of neurotransmitter at the synapse.
- They may act as autocrine or paracrine agents.

Neuromodulation

Mechanism of Action of Neuromodulators

- As autocrine agent, they bind to the cell that produced them (presynaptic cell) and affect the amount of neurotransmitter released.

Neuromodulation

Mechanism of Action of Neuromodulators

- As paracrine agent, they bind to the receptors on one or more post-synaptic cells to cause the release of neurotransmitter.

Neuromodulation

Effects of Neuromodulation

- Development of complex behavioral patterns e.g. happiness, exploration, revenge, reward, greed.
- Processes of thinking, cognition, planning, learning and memory.
- Behavioral problems: mood swings, sleep disturbances, feelings of stress, anxiety, anger and depression.

Neuromodulation

Pharmacological Applications

- Vast pharmacological applications for the treatment of complex and challenging nervous diseases. e.g.
- ADHD
- Narcolepsy
- Epilepsy
- Depression
- Dementia
- Alzheimer's disease
- Parkinson's disease

END

Nerve Physiology

Neural Integration

LECTURE NO 42

Neural Integration

Definition

- Neurons in the brain receive thousands of synaptic inputs from other neurons. Before the generation of an output.
- These neurons add up these inputs through a process that is called neuronal integration.

Neural Integration

Neural Integration

Neural integration occurs through two processes:

- Temporal summation
- Spatial summation

Neural Integration

Temporal Summation

- When two EPSPs occur at a single synapse in such rapid succession that the postsynaptic neuron's membrane potential has not returned to the resting potential before the arrival of a second EPSP, the EPSPs are added together.
- This is known as temporal summation.

Explanation

- When a presynaptic terminal fires, released transmitter produces an EPSP in the postsynaptic neuron.
- This opens membrane channels for just a millisecond.
- Changed PSP lasts up to 15 milliseconds.
- During rapid firing rate, channels open again adding to the PSP that increases to a greater level.

Neural Integration

Spatial Summation

- EPSPs produced simultaneously by different synapses on the same postsynaptic neuron can also add together.
- This effect is known as spatial summation.

Neural Integration

Spatial Summation: Explanation

- Effect of excitation produced by a single presynaptic terminal almost never excites the postsynaptic neuron.
- It is b/c the transmitter released by a single terminal can cause an EPSP of about 0.5 to 1 millivolt.
- An EPSP of 10 to 20 millivolts is normally required to reach threshold for excitation.
- To provide this amount of EPSP, many presynaptic terminals are stimulated at the same time.
- Their effects can summate even if they are spread spatially on the soma.
- Summated EPSP can cause excitation.
- This effect of summing widely spaced multiple PSPs is called spatial summation.

Neural Integration

Significance of Summation

- Due to summation, the postsynaptic potential becomes much higher.
- This ensures that an output AP is produced that can generate a response.

END

Receptor Physiology

Sensory Stimuli, Sensory Organs and Receptor Cells

LECTURE NO 43

Sensory Stimuli, Sensory Organs & Receptor Cells

or
collect

3-components
of a sensory
system

Sensory Stimuli, Sensory Organs & Receptor Cells

- Stimulus is an external or internal factor that can provoke response through nervous system.
- Sensory organs receive the info. of the stimulus.
- Sensory organs have receptor cells which are stimulated by stimulus and generate a sensory signal and transmit this signal to the N.S.

Sensory Stimuli, Sensory Organs & Receptor Cells

Sensory Stimuli

- Detectable changes in the internal or external environment.
- Cause a neuro-physiological response.
- Sensory inputs
- Gathered constantly from the environment.
- Keep the animal aware of its external or internal environment.

Sensory Stimuli, Sensory Organs & Receptor Cells

Threshold Level of Stimulus

- Level of stimulus must exceed the threshold to be detected.
- Subthreshold stimuli are not detected by the receptors.
- If the intensity of a stimulus reaches threshold, information is transmitted to the CNS.

Sensory Stimuli, Sensory Organs & Receptor Cells

Nature of Stimuli

- All stimuli represent some form of energy.

Energy may be:

- Mechanical

  light, sound, vibration, gravitation, pressure

- Chemical

  odor, taste, allergens

Sensory Stimuli, Sensory Organs & Receptor Cells

Types of Stimuli

Two types:

- External stimuli

  Stimuli of odor, touch, light, sound and gravitation

- Internal stimuli

  Stimuli of pain, homeostatic imbalances and blood pressure

Sensory Stimuli, Sensory Organs & Receptor Cells

from external environment

Sensory Organs

- Specialized structures where sensory receptors are concentrated.
- Specialized for receiving a particular type of stimulus.
- Gather sensory info. more accurately than isolated receptor cells and can transmit it to the nervous system.

Sensory Stimuli, Sensory Organs & Receptor Cells

Sensory Organs

- Positioned at many locations both on the surface and inside the body.
- Most sensory organs are concentrated at anterior end of the animal.
- Major sensory organs of human body are: eyes, ears, nose, tongue and skin.

Sensory Stimuli, Sensory Organs & Receptor Cells

Receptor cells

- Cells specialized to respond to particular kinds of stimuli.
- Receive information from outside the body as well as from inside the body.
- Send this information to the nervous system.

Sensory Stimuli, Sensory Organs & Receptor Cells

Features of Receptor cells

1. Have Receptor Molecule

- They have a chemical group or molecule (such as a protein) on the cell surface or in the cell interior that has an affinity for a specific chemical group, molecule, or virus.
- Such molecule or group is known as receptor.

Sensory Stimuli, Sensory Organs & Receptor Cells

Features of Receptor cells

2. Have Capacity of Stimulus Transduction

- A physical or chemical stimulus received by the receptor cell is converted (transduced) into electrical signals that can travel through the nervous system.

Sensory Stimuli, Sensory Organs & Receptor Cells

Features of Receptor cells

3. Linked to the Nervous System

- Receptor cells are innervated with nerve endings or are closely associated with neurons.
- So they are able to transmit the signal to the nervous system directly.

Sensory Stimuli, Sensory Organs & Receptor Cells

END

Location of Receptor cells

- Some receptor cells are concentrated in sensory organs .
  e.g. olfactory, visual, gustatory and sound receptors are in nose, eyes, tongue and ear.
- Many receptor cells are scattered in the skin and also in the deeper parts of the body.

Receptor Physiology

Sensations and Quality of Stimulus

LECTURE NO 44

Sensations and Quality of Stimulus

Sensations

- Subjective phenomena closely associated with the stimulus.
- When signals are transmitted to brain, it interprets or perceives these signals.
- Subjective description of this neuronal perception is termed as sensation.

Sensations and Quality of Stimulus

Sensations

- Interpretation of sensations is based on experience and learning i.e. previous exposure and its interpretation stored in the memory.

Feelings are Sensations

- Pain
- Color
- Taste (e.g. sweet)
- Noise
- Melody
- Bad or good odor

Sensations and Quality of Stimulus

Quality of Stimulus

- Features that characterize a stimulus and distinguish it from another.
- Human sensations depend on the quality of stimulus.

Sensations and Quality of Stimulus

Quality of Stimulus Examples

- Mechanical stimulation producing sensation of touch is different from the stimulus of light that produces a visual response.
- Stimuli of one type also differ in some features.
  e.g.,
  Light can be red or blue
  Sound can be high or low

Sensations and Quality of Stimulus

Sensations are Subjective

- Human perception of sensations for a stimulus is subjective. i.e. the described qualities are not really inherent in the stimuli themselves.
- e.g. “sweetness” of sugar or red color of light are just perceptions that are not inherent in the stimuli themselves.

Sensations and Quality of Stimulus

END

Sensations Depend on Neuronal Processing

Subjective sensations depend entirely on:

- Properties of receptor cells that send different types of signals for different types of stimuli.
- Neuronal processing of the stimulus that produces recognizable sensations.

Receptor Physiology

Sensory Modalities and Receptor Types

LECTURE NO 45

Sensory Modalities and Receptor Types

Sensory Modalities

- Sensory modalities are the types of sensory information that we can distinguish.
- Also known as “senses”.

Sensory Modalities and Receptor Types

Human Sensory Modalities

- Include five major senses
- Perceived through five sense organs having specific receptors

1. Sense of vision perceived through eyes having photoreceptors.
2. Sense of hearing perceived through ears having mechano-receptors.

Sensory Modalities and Receptor Types

Human Sensory Modalities

3. Sense of touch perceived through skin having Meissner's corpuscles.
4. Sense of taste perceived through tongue having gustatory receptors.
5. Sense of smell perceived through nasal cavity having olfactory receptors.

Sensory Modalities and Receptor Types

Human Sensory Modalities

- Sensory modalities also include many interoceptive (internal) receptors.
- Interoceptive receptors constitute internal sensory systems.
- They respond to signals from within the body.

Sensory Modalities and Receptor Types

Interoceptive Receptors

These include:

- Receptors of vestibular system that monitor orientation of the body.
- Present in semicircular canals in inner ear.
- Thermoreceptors that keep track of thermal state of the body.
- They are found scattered on the skin.

Sensory Modalities and Receptor Types

Interoceptive Receptors

- Chemoreceptors that keep track of chemical state of the body.
- Have various types distributed in specific parts of the body.
- Proprioceptors that monitor position of the muscles and joints.
- They are located in the movable joints.

Sensory Modalities and Receptor Types

Interoceptive Receptors

- Nociceptors perceive the sensation of pain.
- They are distributed evenly in the skin and deeper body parts.
- Pacinian corpuscles receive pressure stimulus.
- They are situated deep in the body.
- In the limbs, they receive vibrations.

Sensory Modalities and Receptor Types

Interoceptive Receptors: Importance

- They communicate information to the brain by pathways that often are not brought into consciousness.
- Play crucial role in providing information to the brain about the state of body and its position in space.

Sensory Modalities and Receptor Types

Interoceptive Receptors: Importance

- Imagine: how complicated walking would be if we had to pay conscious attention to the position of every muscle and joint ?

Sensory Modalities and Receptor Types

Sensory Modalities in Nonhuman Animals

- Many animals possess other sensory modalities that are unavailable to human beings.

Examples

- Pit organs
- Electroreceptors
- Magnoreceptors

Sensory Modalities and Receptor Types

Pit Organs

- Found in some snakes.
- Can detect heat energy emitted from mammalian bodies.
- Snakes use this sense to identify their prey due to temperature difference.

Sensory Modalities and Receptor Types

Electroreceptors

- Can detect very low frequency electric signals.
- Found in some electric fishes.
- They use this sense to communicate with one another in murky waters.

Sensory Modalities and Receptor Types

Magnoreceptors

- Detect earth's magnetic field.
- Present in many migratory birds.
- Used as a navigational guide.

END

Receptor Physiology

Properties of Receptor Cells

LECTURE NO 46

Properties of Receptor Cells

Properties of Sensory Receptor Cells

- Selectivity
- Transduction
- Sensitivity
- Transmission

Properties of Receptor Cells

Selectivity of Receptor Cells

- Receptor cells are highly selective for a specific kind of stimulus energy.
- They are selective because their membranes can receive and respond to only certain type of energy.

Properties of Receptor Cells

Receptor Selectivity Example-1

- Light may strike any part of the body
- But only the eyes contain specialized photoreceptor cells.
- These cells can receive and respond to the stimulus of light and transduce photons into neuronal energy.

Properties of Receptor Cells

Receptor Selectivity Example-2

- Mechanoreceptors are selective to stimuli that cause slight distortion in their membrane.
- It happens due to the presence of certain proteins in their membrane which are activated through physical stimulation.

Properties of Receptor Cells

Capacity of Transduction

- Receptor cells have capacity to transduce sensory input.
- A physical or chemical stimulus is transduced into the electrical nerve impulse.
- Transduction involves a conformational change in receptor molecules.
- The receptor molecules are typically proteins.

Properties of Receptor Cells

Sensitivity of Receptors

- Receptors are extremely sensitive to their stimuli.
- They can receive very weak stimuli and still transduce these signals into nerve impulses that contain much larger amounts of energy.
- It is because they can amplify the received signal.

Properties of Receptor Cells

Sensitivity of Receptors

- Receptor cells contain intracellular machinery for amplification of weak stimuli.
- Activation of receptor molecules initiates a cascade of chemical reactions in the cell.
- This results in amplification of the signal by many orders of magnitude.

Properties of Receptor Cells

Transmission of Signals

- After receiving and processing the signal in the cell, transmission of signals is the final step in all receptor cells.
- It involves opening of ion channels that cause flow of ionic current through the cell membrane.
- Flowing current transmits the signals to the cells of nervous system.

END

Receptor Physiology

Sensory Transduction

LECTURE NO 47

Sensory Transduction

Definition

- The physical or chemical energy of stimulus is converted into electrical signals i.e nerve impulse by sensory receptor cells.
- This conversion is called sensory transduction.

Sensory Transduction

Sensory Transduction Systems

- All sensory transduction systems contain related molecules and operate through similar cellular mechanisms.
- They perform three basic functions:
  - Detection of stimulus
  - Amplification of stimulus
  - Encoding of stimulus

Sensory Transduction

Detection of Stimulus

- The initial event in all sensory transduction systems.

Threshold of Detection

- The smallest amount of stimulus energy that produces a response in a receptor 50% of the time.

Sensory Transduction

Detection Sensitivity of Receptors

- Sensory receptors are highly sensitive.
- They are capable of detecting inputs near the theoretical lowest limits of the stimulus energy.

Sensory Transduction

Detection Sensitivity of Receptors—Examples

- Photoreceptors can detect and generate response to a single photon of light.
- Mechanoreceptor hair cells can detect displace-ments equal to the diameter of a hydrogen atom.
- Odor receptors can detect even a few molecules of odorant.

Sensory Transduction

Amplification of Stimulus

- Carried out if received stimulus energy is low.
- Occurs within the receptor cell.
- Mediated by intracellular mechanisms that involve a cascade of chemical reactions.
- Signal is amplified by many orders of magnitude.

Sensory Transduction

Encoding & Transmission of Stimulus

- Receptor cells encode sensory information into an electrical neuronal signal.
- This signal is transmitted to the brain via nervous system.
- Neuronal signal may be transmitted through:
  - Action potentials
  - Electrotonic conduction

Sensory Transduction

Role of Sensory Organs

- A single receptor can encode information about the intensity of a stimulus.
- But it cannot directly report the quality of the stimulus.
- e.g. a single photo-receptor cannot report whether a stimulating light is red or blue.

Sensory Transduction

Role of Sensory Organs

- Sensory organs contain a variety of receptor cells that respond differentially to stimuli with different qualities.
- For example, certain photoreceptors respond maximally to red light, while others respond to blue light.

Sensory Transduction

Role of Sensory Organs

- When receptor cells are grouped into organs, significantly more information about the stimulus can be conveyed.
- e.g. its absolute intensity, its spatial distribution, and other qualities such as wavelength of light or frequency of a sound.

END

Receptor Physiology

Range Fractionation

LECTURE NO 49

Range Fractionation

Definition

- In a sense organ, sensory receptors are arranged in an order of increasing sensitivity to different range of intensities of the stimulus.
- This hierarchical arrangement of receptors is known as range fractionation.

Range Fractionation

Range Fractionation and Stimulus Intensities

- Each individual receptor, in a sense organ, covers only a fraction of the total dynamic range of the stimulus intensities.
- Receptors work together in a hierarchical way to provide discrimination of stimulus intensities.

Range Fractionation

Due to
R. Fr

Recruitment Phenomenon

- An important implication of range fractionation.
- Most sensitive receptors are activated at stimulus intensities that are just above the threshold.
- Above that intensity, the most sensitive receptors become saturated.

Range Fractionation

Recruitment Phenomenon

- When stimulus energy is increased a little, the less sensitive receptors in the population join in and respond.
- With still greater stimulus intensities, another, formerly inactive lower-sensitivity population of receptors joins in.

Range Fractionation

Recruitment Phenomenon

- So, as the stimulus intensity is increased, less and less sensitive receptors become active, until the least sensitive sensory fibers are recruited.
- At such a stage, system becomes saturated and no further increase in intensity is detectable.

Range Fractionation

Recruitment Phenomenon

- At this saturation stage, all receptors respond maximally.
- This phenomenon of activation of receptors in a graded fashion is called recruitment.

Range Fractionation

Importance of Range Fractionation

- It results in increasing the range of a multi-neuronal sensory system than the range of any individual receptor.
- It increases the overall precision of sense organ.
- It enables the CNS to discriminate stimulus intensities.

Range Fractionation

Range Fractionation: Example

- High-sensitivity photoreceptors of eye are rod cells.
- The cone cells are low sensitivity.
- Rod photoreceptors are sensitive to low intensity dimmer light.
- Cone cells respond to bright light i.e. high intensity.

END

Receptor Physiology

Chemoreceptors: Taste and Smell Receptors

LECTURE NO 50

Chemoreceptors: Taste and Smell Receptors

Chemoreceptors

- Receptor cells specialized to acquire information about the chemical environment.
- Two categories
  - Gustatory (taste) receptors
  - Olfactory (smell) receptors
- Both types operate quite differently from one another.

Chemoreceptors: Taste and Smell Receptors

Gustatory (Taste) Receptors

- Respond to dissolved molecules that come in direct contact with the receptive structure.

Chemoreceptors: Taste and Smell Receptors

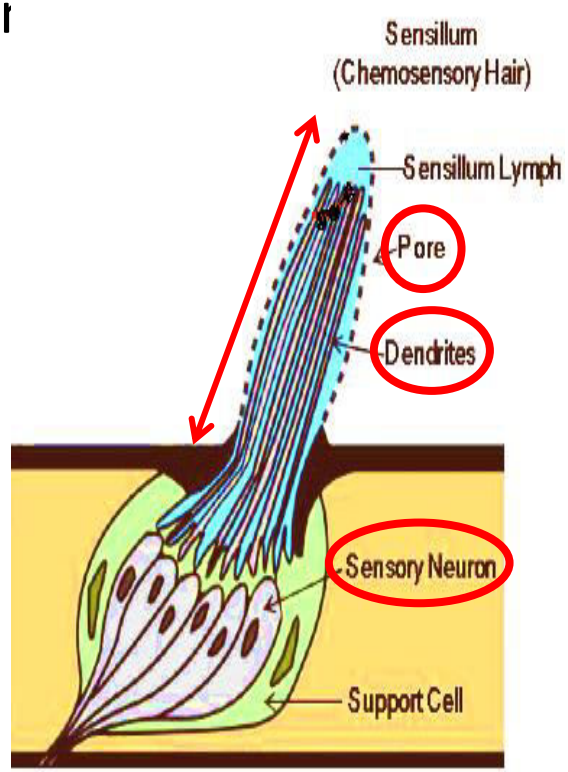
Taste Receptors in Insects

- Organs of taste in insects are sensory sensilla.
- Located on the feet and mouthparts.
- A sensillum contains several receptor cells.
- Each receptor cell is sensitive to a different chemical stimulus e.g., water, cations, anions, or carbohydrates.

Chemoreceptors: Taste and Smell Receptors

Taste Receptors in Insects

- The receptor cells of sensilla have hair-like appearance due to lor
- Dendrites are sent to the cuticle.
- The cuticle around sensilla has minute pores.
- Pores allow stimulant molecules to contact the dendrites.
- Stimulus is converted into electrical signal by the sensory cells' soma.



http://www.naro.affrc.go.jp/archive/nias/eng/research/h26/links/2015e-7_01.jpg

Chemoreceptors: Taste and Smell Receptors

Taste Receptors in Fishes

- Fishes have taste receptors on different locations.
- e.g. some fishes have modified pectoral fins with taste receptors at the tips of fin rays.
- They are used to locate food in the muddy bottom.

Chemoreceptors: Taste and Smell Receptors

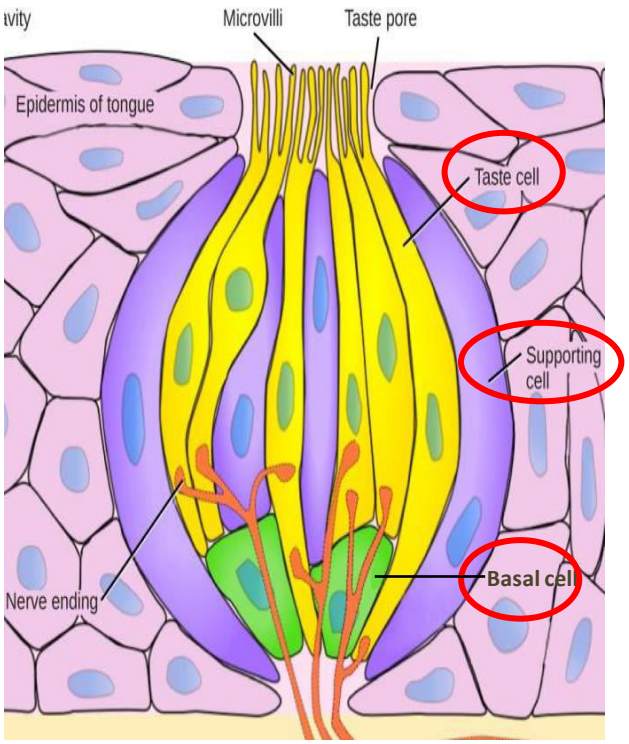
Taste Receptors in Terrestrial Vertebrates

- Located at the anterior region of digestive tract i.e.
  - On the tongue
  - Epiglottis
  - Back of the mouth
  - Pharynx
  - Upper esophagus
- The gustatory organs of vertebrates are called taste buds.

Chemoreceptors: Taste and Smell Receptors

Taste buds

- A taste bud is composed of about 50 types of modified epithelial cells, including:
  - Supporting cells (sustentacular cells)
  - Basal cells
  - Taste receptor cells
- basal cells are progenitor cells that give rise to new taste receptors.
  - They regularly generate new sensory taste receptor cells which have an active life of only 10 days.

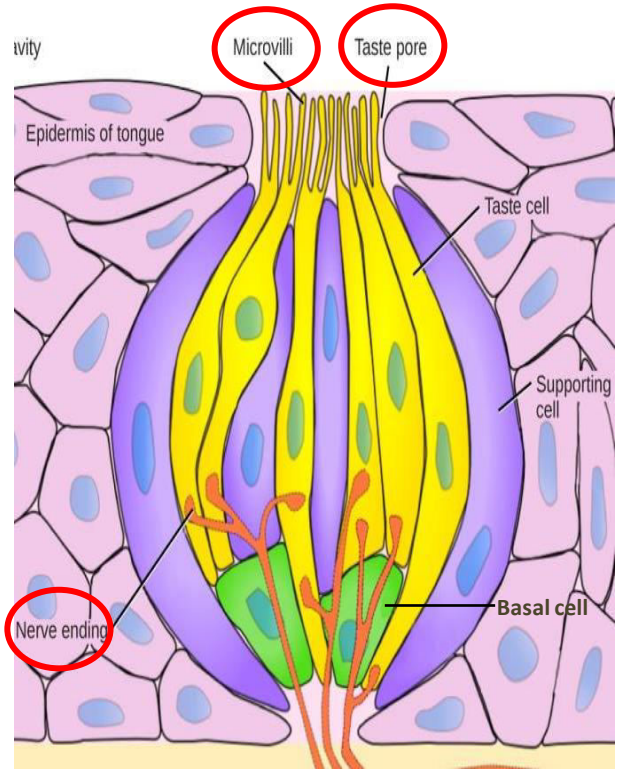


http://philschatz.com/biology-book/resources/Figure_36_03_04.jpg

Chemoreceptors: Taste and Smell Receptors

Taste Receptor Cells

- Outer tips of taste cells are arranged around a minute taste pore.
- At the tip of taste cell, several microvilli (taste hairs) protrude into the taste pore.
- Microvilli approach the cavity of the mouth.
- They provide receptor surface for taste.



http://philschatz.com/biology-book/resources/Figure_36_03_04.jpg

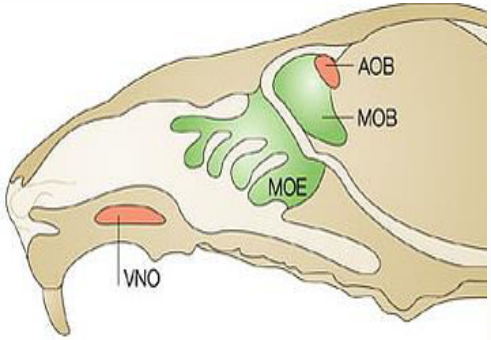
- A branching terminal network of taste nerve fibers surrounds the base of taste cells.

Chemoreceptors: Taste and Smell Receptors

Olfactory (Smell) Receptors

- Respond to airborne molecules that stimulate the receptor from distance.
- They detect odorants and pheromones.
- In insects, olfactory sensilla are present on their antennae.
- Vertebrate olfactory receptors are present in the nasal cavity.

Chemoreceptors: Taste and Smell Receptors



2

END

Olfactory (Smell) Receptors

Olfactory system has two distinct organs:

- Main Olfactory Epithelium (MOE) to detect odorants
- Vomeronasal Organ (VNO) to detect pheromones

Receptor Physiology

Mechanism of Taste Reception

LECTURE NO 51

Mechanism of Taste Reception

Sense of Taste

- Grouped into five primary sensations:
  - Salty
  - Sour
  - Sweet
  - Bitter
  - Umami
- All perceived tastes are due to combinations of these fundamental sensations.

Mechanism of Taste Reception

Taste Receptors

- A taste cell expresses only one type of taste receptor.
- Humans have more than 30 different types of receptors for bitter taste
- Only one type of receptor each for sweet and umami tastes.

Mechanism of Taste Reception

Taste Receptors

- Receptors of sweet, umami, and bitter tastes are G protein-coupled receptors.
- Receptors for salty and sour tastes are ion channel proteins.

Mechanism of Taste Reception

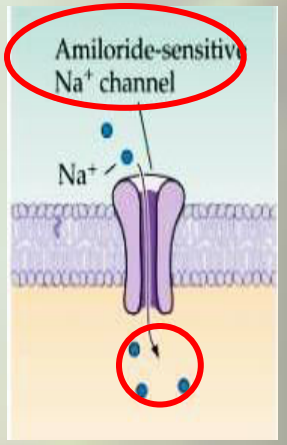
Salty Taste Reception

- Salty stimuli (e.g. NaCl) readily dissociate into Na^+ and Cl^- ions.
- Na^+ ions enter specific taste receptors through special Na^+ channels in these cells.
- These channels are different from voltage-gated Na^+ channels.

Mechanism of Taste Reception

Salty Taste Reception

- These Na^+ channels are blocked by the drug amiloride.

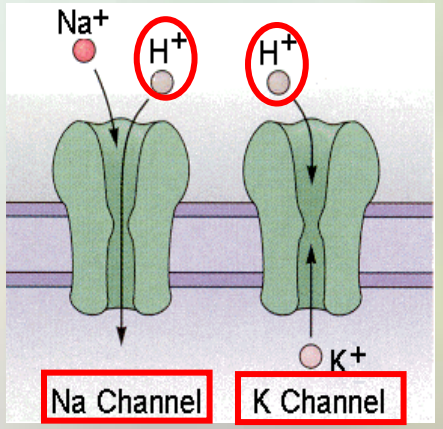


- Na^+ entering the channels directly depolarize receptor cell membrane.

Mechanism of Taste Reception

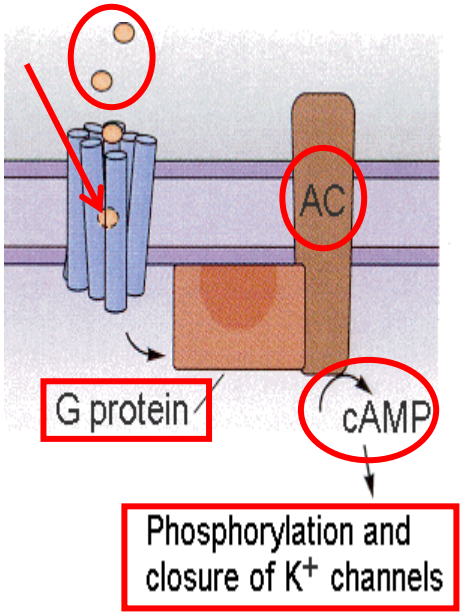
Sour Taste Reception

- Sour stimuli have excess H^+ ions.
- They act either through opening Na^+ channels or by blocking K^+ channels.



- In both cases, membrane is depolarized.

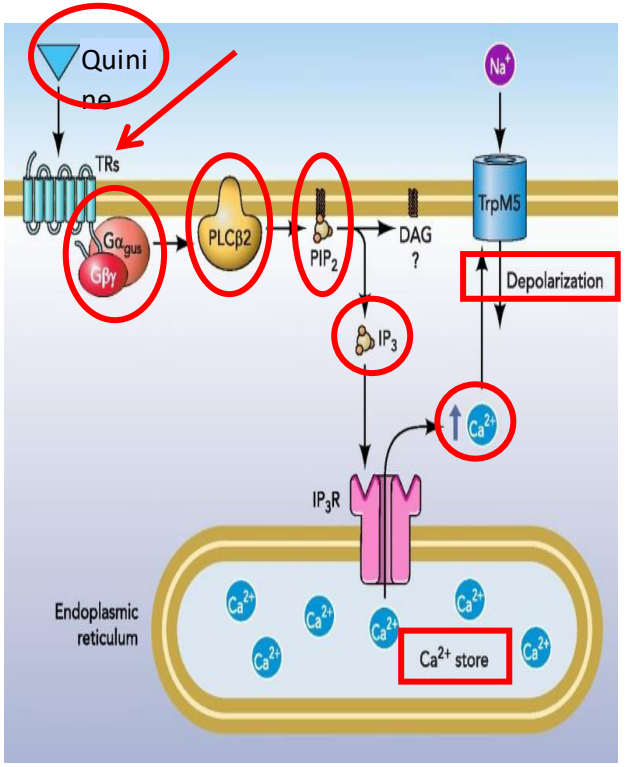
Mechanism of Taste Reception



Sweet Taste Reception

- When sweet compounds and Alanine (Ala) bind to receptors a G protein is activated.
- G protein activates adenylate cyclase that forms cAMP.
- Increased conc. of cAMP closes K^+ channels in the receptor membrane.
- Closing of K^+ channels depolarizes the receptor.

Mechanism of Taste Reception



http://physiologyonline.physiology.org/content/nips/28/1/51/F1.large.jpg

Bitter Taste Reception

- A bitter substance (e.g. quinine) binds to the receptor and activates a G protein.
- G protein is coupled with phospholipase C.
- PLC converts PIP_2 to $InsP_3$.
- $InsP_3$ causes release of Ca^{2+} from intracellular stores that cause the cell to depolarize.

Mechanism of Taste Reception

Umami Taste Reception

- The receptors for umami (savory or delicious) taste were discovered in the year 2000.
- Taste is produced by glutamate and MSG, found in meat and cheese.

Mechanism of Taste Reception

Umami Taste Reception

- Receptor for MSG is a G-protein coupled receptor, closely linked to sweet taste receptors.
- The G protein induces a cellular cascade involving cAMP and release of Ca^{2+} ions which cause depolarization.

Mechanism of Taste Reception

Release of Neurotransmitters

- In all cases of taste reception, depolarization of receptor cell generate APs and release of neurotransmitters.
- The neurotransmitters propagate the signal in nervous system.

Mechanism of Taste Reception

Transmission of Taste Signals

- All taste receptor are neurons & generate APs.
- They have no axons, so they cannot themselves carry information to CNS.
- They synapse with neurons of seventh, ninth and tenth cranial nerves i.e. facial , glossopharyngeal and vagus nerves.

Mechanism of Taste Reception

Labeled Line Coding

- Each receptor subtype for taste sensations is connected to a particular set of axons in the nerve.
- In such an arrangement, information about one taste e.g. "sweetness" would be carried by some specific subset of axons.
- Such a pattern is called labeled line coding.

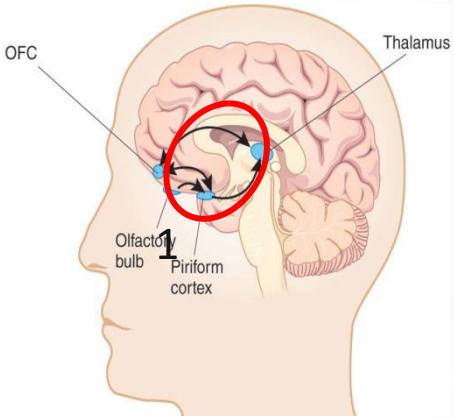
END

Receptor Physiology

Mechanism of Olfactory Reception

LECTURE NO 52

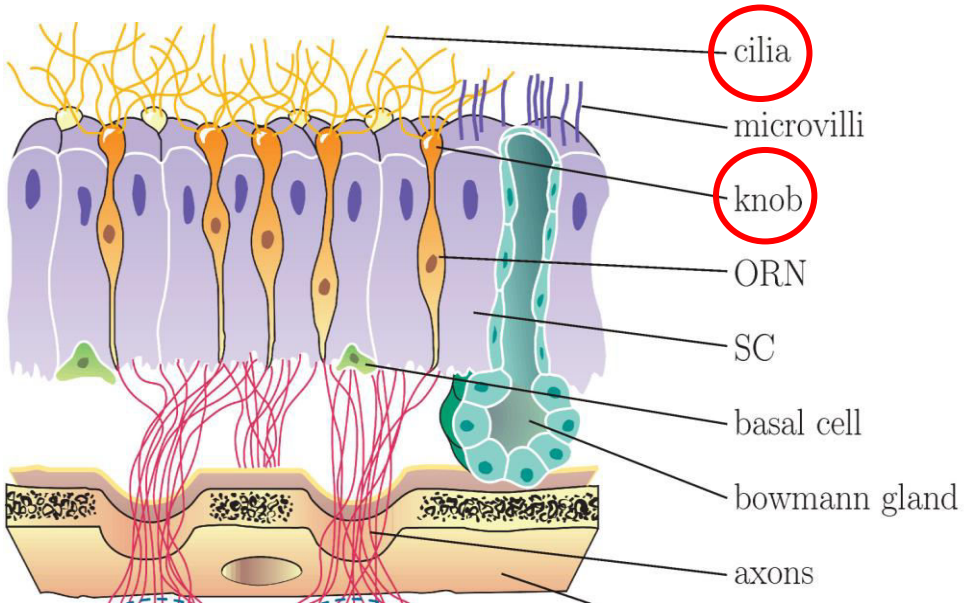
Mechanism of Olfactory Reception



Olfactory Receptors

- Olfactory receptors are neurons with long axons.
- They are located inside the nasal cavity.
- They send impulses directly to the olfactory bulb of the brain.

Mechanism of Olfactory Reception



https://chroniclesofxenopuslaevis.files.wordpress.com/2013/09/olfactory1.png

Olfactory Receptors

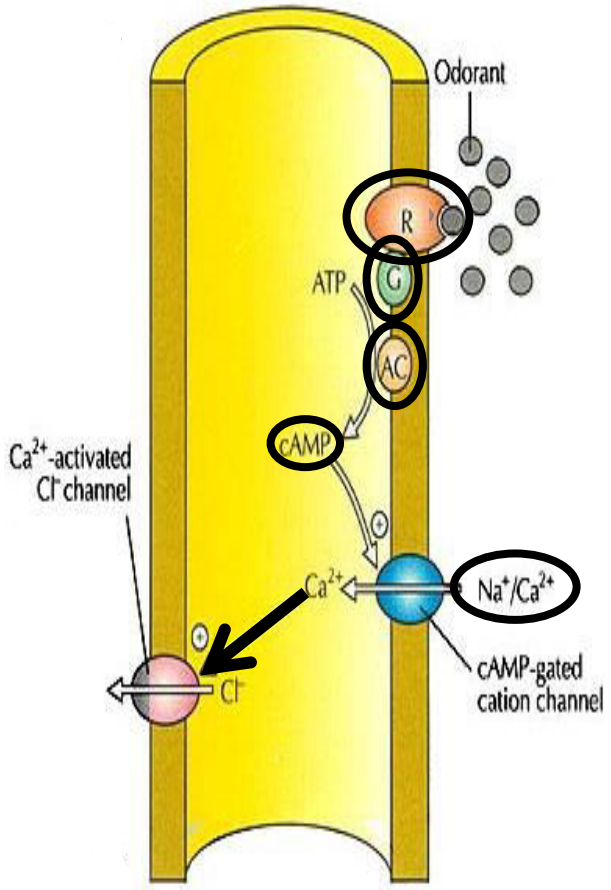
- A receptor neuron has a long thin dendrite that terminates in a small knob at the surface.
- The knob has 4 to 25 olfactory cilia (200 µm long), covered by proteinaceous mucus.

Mechanism of Olfactory Reception

Olfactory Transduction

- Odorant molecules are absorbed into the mucous layer and delivered to the cilia.
- They bind to a receptor protein in the cilia.

Mechanism of Olfactory Reception



http://www.cell.com/cms/attachment/483533/3374429/gr1.jpg

Olfactory Transduction

- The receptor protein is coupled to a G-protein.
- G-protein activates adenylyl cyclase that converts ATP into cAMP.
- cAMP opens channels that are permeable to both Na^+ and Ca^{2+}.
- Ca^{2+} inflow triggers opening of Cl^- channels.
- Both factors result in depolarization.

Mechanism of Olfactory Reception

Olfactory Transduction

- The action potentials are transmitted to the CNS through the olfactory nerve.
- This mechanism of transduction ensures amplification of excitatory effect of even the weakest odorant, thereby increasing the sensitivity of the olfactory receptors.

Mechanism of Olfactory Reception

Basis of Differentiating Smells

- The receptor protein in the olfactory cilia belongs to a very large family of proteins.
- These proteins are expressed only in olfactory epithelial cells.
- These proteins have small variation in their structure.

Mechanism of Olfactory Reception

Basis of Differentiating Smells

- These variations give rise to a large number of subtypes.
- Each subtype is associated with a different odorant.
- This forms the basis of ability to distinguish a wide variety of smells.

END

Receptor Physiology

Mechanoreception

LECTURE NO 54

Mechanoreception

Mechanoreception

- Sensory detection of physical stimuli that have mechanical energy.
- e.g. stretch, touch, pressure, sound and gravity (equilibrium).
- Mechanical stimuli cause physical changes (deformation, bending displacement, stretching) in receptive structures of mechanoreceptors.

Mechanoreception

Mechanoreceptors

- Consist of ion channels linked to external cell structures (sensory hairs) or internal cell structures (cytoskeleton).
- They can be extremely sensitive, responding to mechanical displacements of as little as 0.1 nm.

Mechanoreception

Mechanism of Mechanoreception

- Bending or stretching of external structure generates tension that alters permeability of the ion channels.
- Change in ion permeability alters the membrane potential.
- It results in depolarization or hyperpolarization.

Mechanoreception

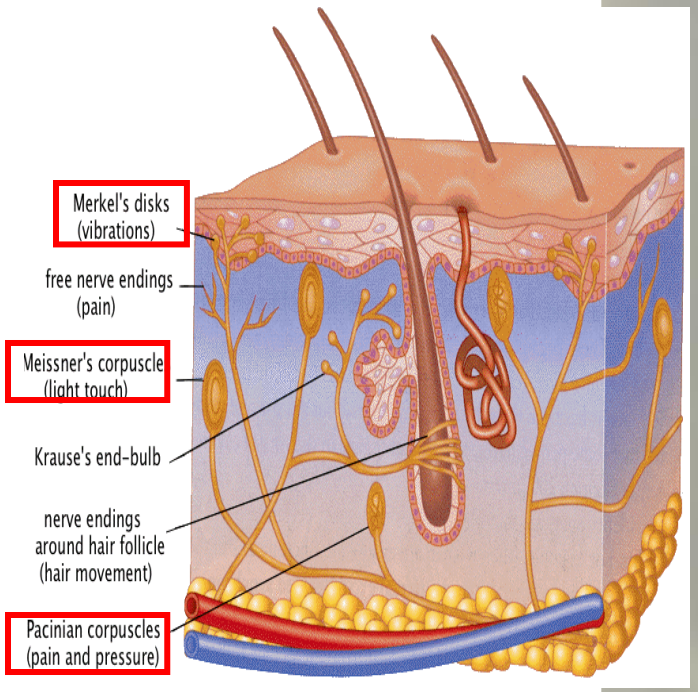
Mechanoreceptor Structure

- Simplest mechano-receptors consist of undifferentiated nerve endings.
- They are found in the connective tissue of skin.
- Complex mechano-receptors have accessory structures that transfer mechanical energy to the receptive membrane.

Mechanoreception

Mechanoreceptors for Touch, Vibration and Pressure

- Receptors for these senses are embedded in the skin.
- Gentle touch receptors situated at the base of hairs are hair end organs. They respond to displacement of hairs.
- Touch receptors in ridges of fingertips are Meissner's Corpuscles. They have encapsulated nerve endings.
- To receive pressure stimulus, Pacinian corpuscles are situated deep in skin.
- Merkel's Disks are associated with the reception of vibration.



Mechanoreception

Stretch Mechanoreceptors

- Found in muscles of arthropods and vertebrates.
- Have various types.
- Most common are proprioceptors that detect muscle movements.
- Consist of mechanically sensitive sensory nerve endings associated with specialized muscle fibers.

Mechanoreception

Stretch Mechanoreceptors

- When muscle is stretched, dendrites receive this stimulus.
- Action potentials are triggered in the sensory neuron and transmitted to the spinal cord.

Mechanoreception

Sound and Equilibrium Receptors

- Receptors found in vertebrate middle and inner ear.
- For both senses, particles or moving fluid cause deflection of cell surface structures.
- As a result mechano-receptor cells produce receptor potentials.

END

Receptor Physiology

Hair Cells

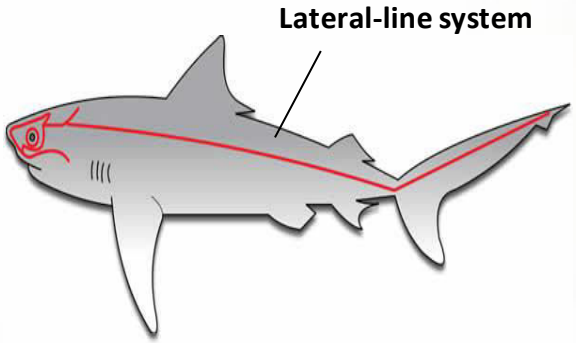
LECTURE NO 54

Hair Cells

Hair Cells

- Hair cells are ciliary cells.
- Found in several sensory organs of vertebrates.
- They are highly sensitive mechanoreceptors.
- Responsible for transducing mechanical stimuli into electrical signals.

Hair Cells



Systems Based on Hair Cells: Examples

- Lateral-line system of fishes and amphibians, involved in detection of motion in the surrounding water.
- This system is based on hair cells.

Hair Cells

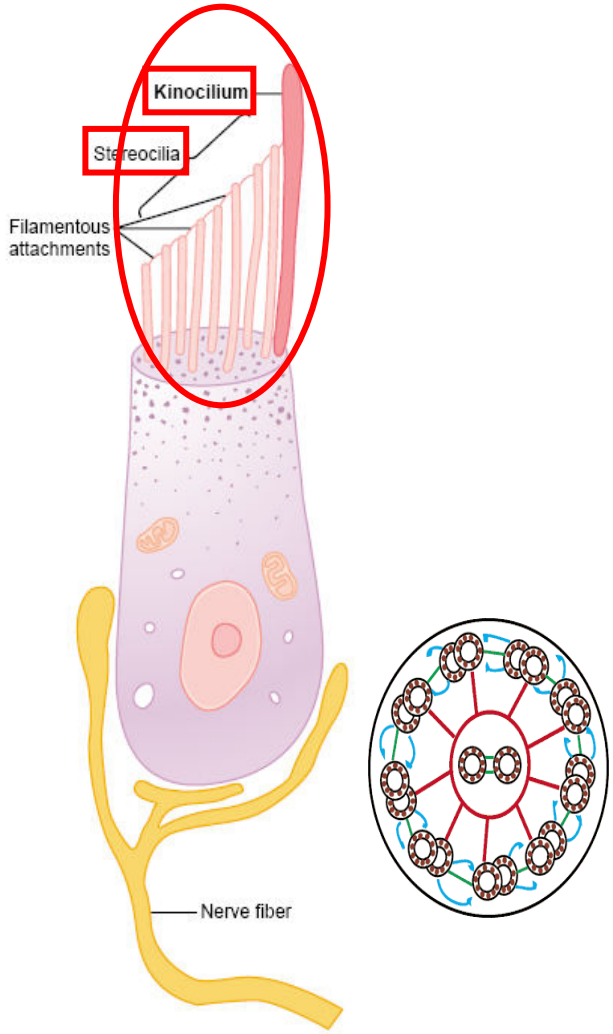
Systems Based on Hair Cells: Examples

- The organ of hearing and the organs of equilibrium are also based on hair cells.



Inner ear hair cells

Hair Cells



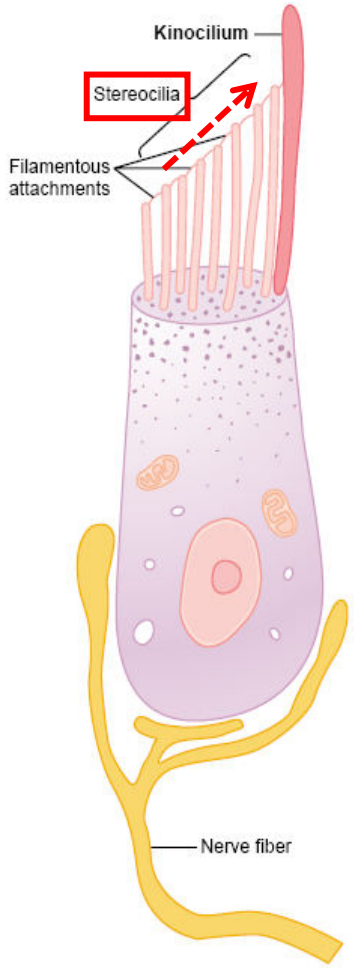
A Hair cell

Textbook of Medical Physiology (11th ed) by Guyton

Hair Cell Cilia

- Many cilia project from the apical end of each cell (reason of naming).
- Cilia are of two types:
  - Kinocilium
  - Stereocilia
- A hair cell has a single kinocilium.
- Kinocilium has a "9 + 2" arrangement of internal microtubules (similar to other motile cilia).

Hair Cells



A Hair cell

https://neupsykey.com/wp-content/uploads/2017/03/f0216-03.jpg

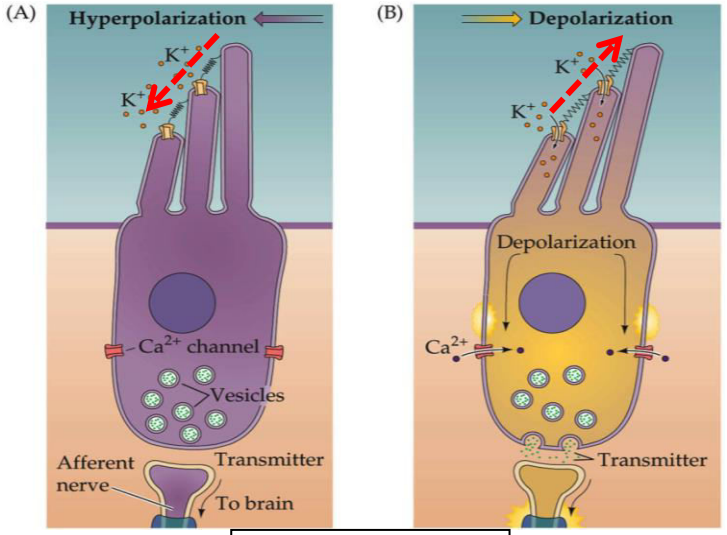
Hair Cell Cilia

- Each hair cell has 20-300 nonmotile stereocilia.
- Stereocilia are structurally and developmentally distinct from the kinocilium.
- They are not formed of microtubules.
- They are formed of actin filaments.
- The stereocilia are arranged in order of increasing length from one side of the cell to the other.

Hair Cells

Working Mechanism of Hair Cell

- The stimulus (pressure or force) moves bundles of stereocilia that produces an electrical signal.
- When cilia bend toward the tallest cilium, the cell depolarizes.
- When they bend in the opposite direction, the cell hyperpolarizes.



Neuroscience, 3rd ed (2004) by Purves & Dale

Hair Cells

Working Mechanism of Hair Cells

- Hair cells do not produce APs.
- They release neurotransmitters in a graded fashion.
- They form chemical synapses with afferent neurons which carry information into the CNS.

END

Receptor Physiology

Organs of Equilibrium

LECTURE NO 55

Organs of Equilibrium

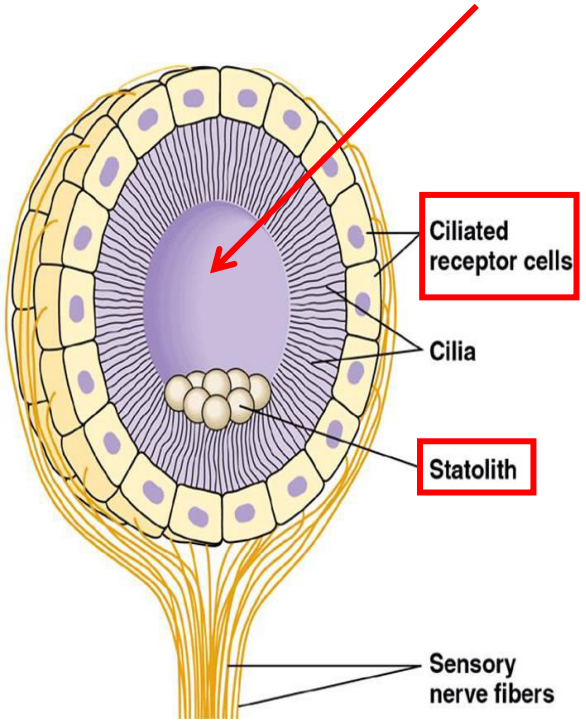
Organs of Equilibrium

- Detect an animal's position with respect to:
  - Gravity
  - Acceleration

Organs of Equilibrium

Statocyst: Invertebrate Organ of Equilibrium

- The simplest organ of equilibrium.
- Consists of a fluid filled cavity that:
  - is lined with ciliated mechanoreceptor cells
  - has a solid particle statolith inside it.
- Forms of this type are found in most invertebrate groups except insects.



http://slideplayer.com/8936534/27/images/26/Figure+49.21+The+statocyst+of+an+invertebrate.jpg

Organs of Equilibrium

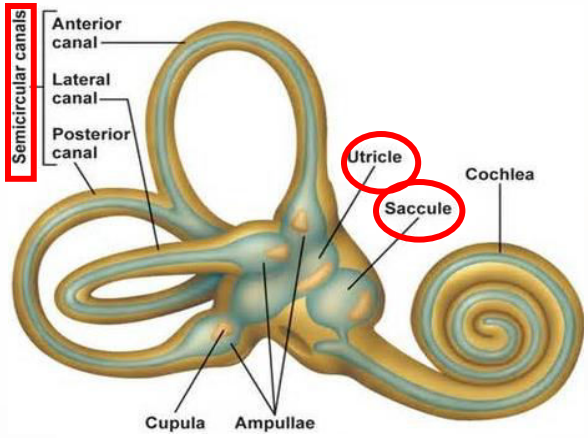
Working Mechanism of Statocyst

- When the position of animal changes, statolith strikes on the sensory mechanoreceptor cells of the statocyst that are stimulated.
- The receptor cells generate signals that travel to the CNS and set up reflex movements of the appendages.

Organs of Equilibrium

Vertebrate Organ of Equilibrium

- Called as vestibular apparatus, located in the inner ear. It has:
  - Saccule
  - Utricle
  - Semicircular canals

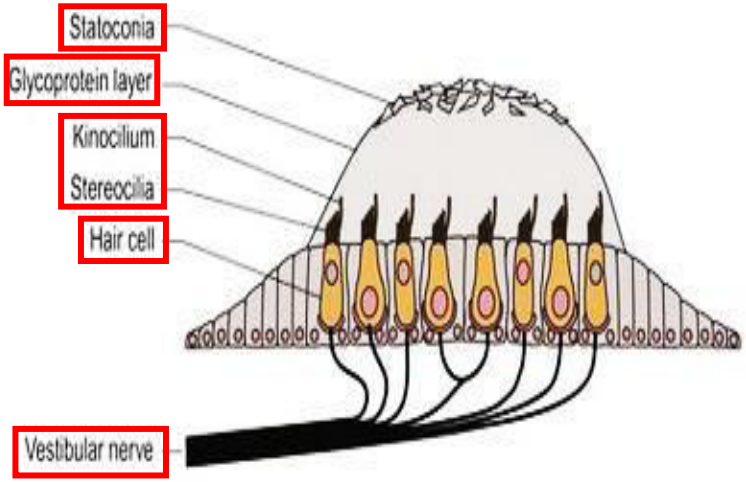


http://fitlifefusion.com/wp-content/uploads/2015/05/vestibular.jpeg

Organs of Equilibrium

Saccule and Utricle

- Their inner surfaces have a small sensory area “macula” covered by a gelatinous layer in which many $CaCO_3$ crystals otoliths (statoconia) are embedded.
- Each macula has thousands of hair cells whose cilia project into the gelatinous layer.
- Movement of otoliths due to gravity causes cilia to bend and generate signals that are transmitted through the vestibular nerve to the CNS.

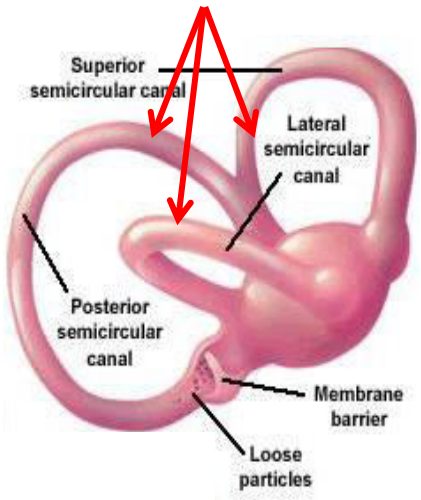


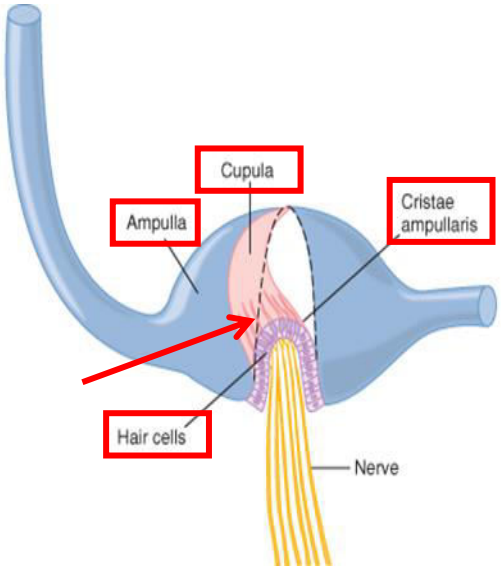
Organs of Equilibrium

Role of Saccule and Utricle

- Saccule and utricle maintain static equilibrium of the head.
- The macula of utricle lies in the horizontal plane on the inferior surface.
- It determines orientation of head in upright pose.
- Macula of saccule lies in the vertical plane.
- It determines head orientation lying pose.

Organs of Equilibrium





Textbook of Medical Physiology (11th ed) by Guyton

Semicircular Canals

- Three semicircular canals lay orthogonally in three mutually perpendicular planes.
- The canals are filled with a fluid "endolymph".
- Each canal has an enlargement at one end called the ampulla.
- Each ampulla has a small crest covered with loose gelatinous tissue mass "cupula".
- Hair cells are located on the ampullary crest whose cilia project into the cupula.

Organs of Equilibrium

Role of Semicircular Canals

- Hair cells in the semi-circular canals detect acceleration of the head.
- Orthogonal arrangement allows them to detect any movement of head in three dimensions.
- Rotation of head results in flow of endolymph that strikes the cupula resulting in displacement of cilia of hair cells.

Organs of Equilibrium

Role of Semicircular Canals

- Hair cells in the semi-circular canals detect acceleration of the head.
- Orthogonal arrangement allows them to detect any movement of head in three dimensions.
- Rotation of head results in flow of endolymph that strikes the cupula resulting in displacement of cilia of hair cells.

Organs of Equilibrium

END

Role of Semicircular Canals

- Displacement of cilia results in excitation of hair cells.
- Excitation causes membrane potential to change and generate an action potential.

Receptor Physiology

The Mammalian Ear

LECTURE NO 56

The Mammalian Ear

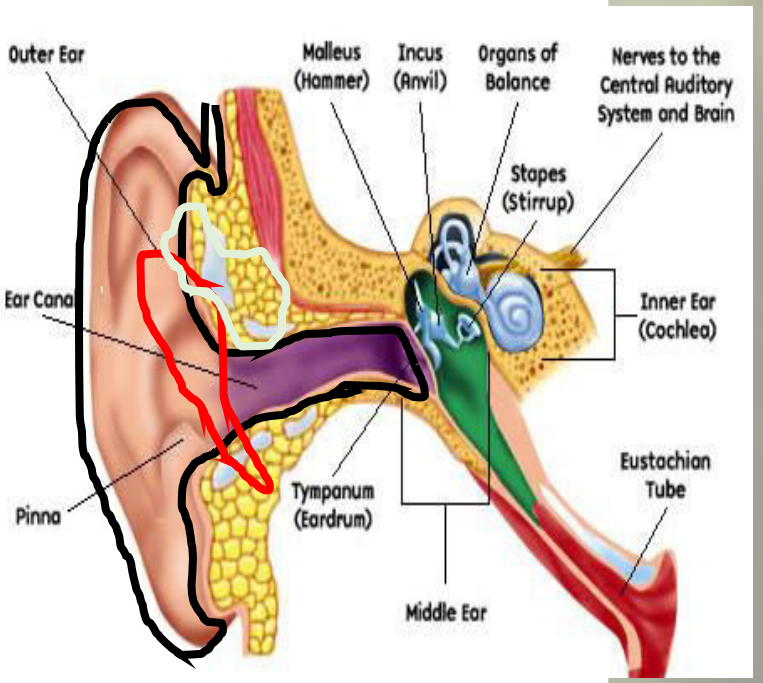
Ear

- Ear is the organ of hearing that detects sound within a particular frequency range.
- The human ear can detect sound frequencies lying between 20 to 20000 hertz.

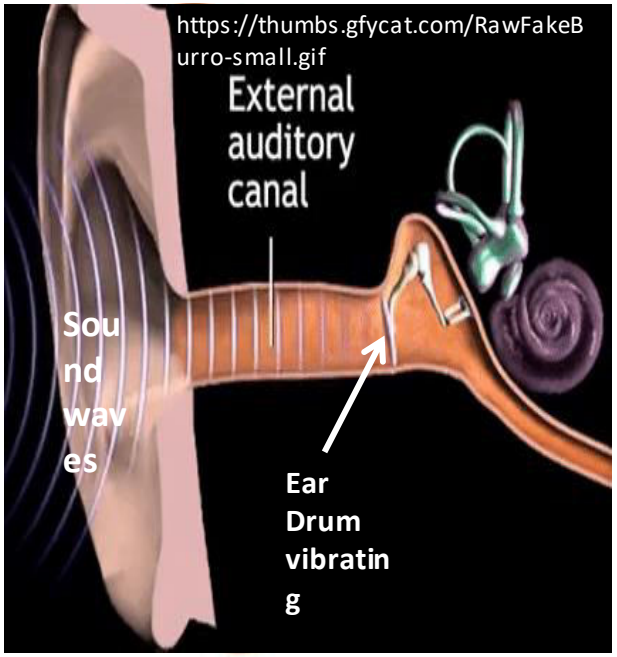
The Mammalian Ear

Functional Anatomy of Mammalian Ear

- Mammalian ear has three major divisions:
  - External ear
  - Middle ear
  - Inner ear
- The three parts are designed to conduct sound waves through three principally different media i.e., air, bone and fluid.
- The conducted waves are converted into electrical signals through a series of complex steps.



The Mammalian Ear



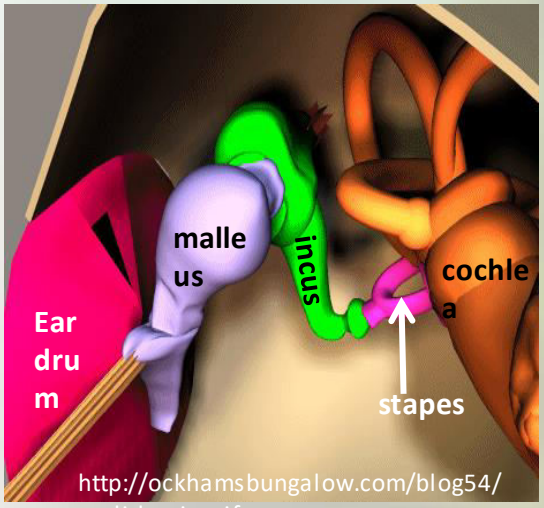
External Ear

- External ear includes:
  - The pinna (auricle)
  - External auditory meatus
  - Tympanic membrane (eardrum)
- These structures act as a funnel to collect sound waves from environment
- Sound is amplified and concentrated onto the eardrum that vibrates.

The Mammalian Ear

Middle Ear

- Contains three auditory ossicles malleus, incus, stapes in a series.
- Malleus attached to eardrum and stapes to oval window of cochlea.



http://ockhamsbungalow.com/blog54/earlid-anim.gif

The Mammalian Ear

Functions of Middle Ear

1. Receive sound waves as vibrations from eardrum and transmit them onto oval window of cochlea.
2. Amplify sound waves (more than 20 fold).

The Mammalian Ear

Inner Ear

- It consists of a bony labyrinth which is filled with perilymph and has an oval and a round window.
- Oval window receives sound vibrations through stapes.
- In the bony labyrinth lies the membranous labyrinth that is filled with endolymph.

The Mammalian Ear

END

Membranous Labyrinth

- It consists of three parts:
  - Three semicircular canals
  - Vestibule (sacculus and utricles)
  - Cochlea
- Cochlea is the sensory structure concerned with hearing.
- It transduces the mechanical energy of sound vibrations into electrical nerve impulse.

Receptor Physiology

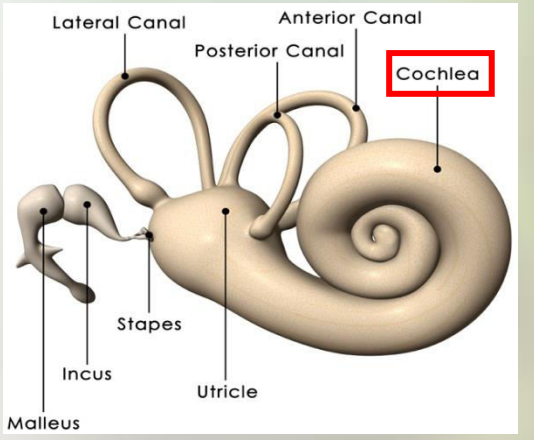
Cochlea

LECTURE NO 57

Cochlea

Cochlea

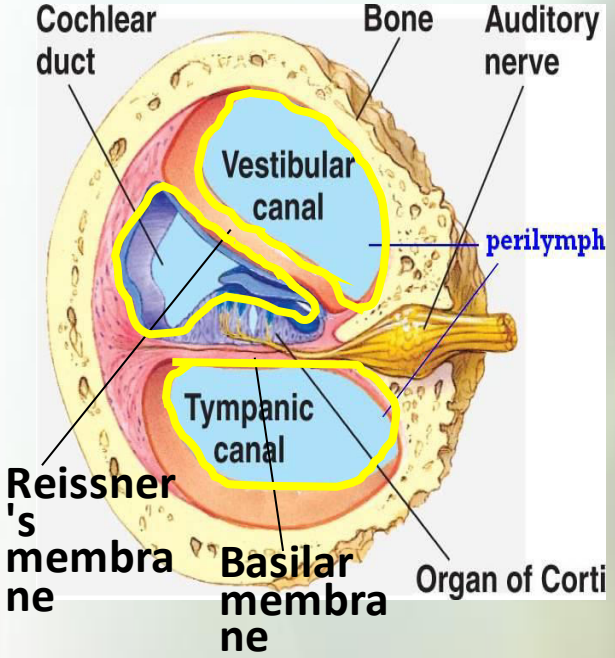
- Coiled sensory structure in the inner ear that is specialized for hearing.
- Takes vibrations of sound from middle ear bones and transforms them into nerve impulse.



Cochlea

Structure of Cochlea

- Cochlea has three fluid-filled canals (ducts) along its length:
- Upper, scala vestibuli (vestibular duct) filled with perilymph.
- Middle, scala media (cochlear duct), filled with endolymph.
- Lower, scala tympani (tympanic duct), filled with perilymph
- Partition between vestibular and cochlear ducts is the Reissner's membrane while partition between cochlear and tympanic ducts is the basilar membrane.



Cochlea

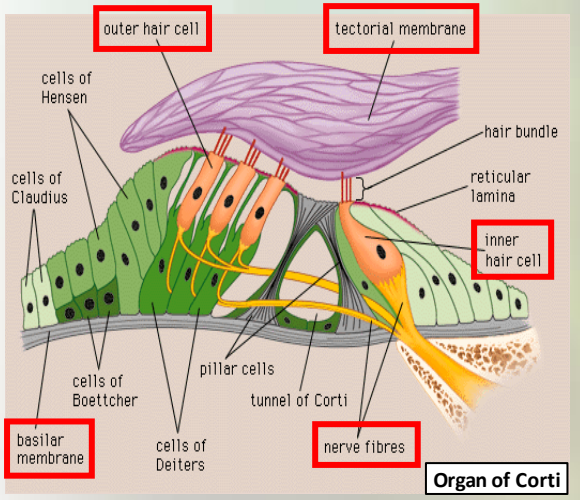
Organ of Corti

- Cochlear duct contains the sensory organ specialized for hearing: “the Organ of Corti”
- It contains hair cells and transduces auditory stimuli into sensory signals.
- It enables human ear to detect and distinguish different sound frequencies.

Cochlea

Organ of Corti

- It contains:
  - Hair cells
  - Basilar membrane
  - Auditory nerve endings
  - Tectorial membrane



https://media1.britannica.com/eb-media/00/14300-004-5FF07709.jpg

Cochlea

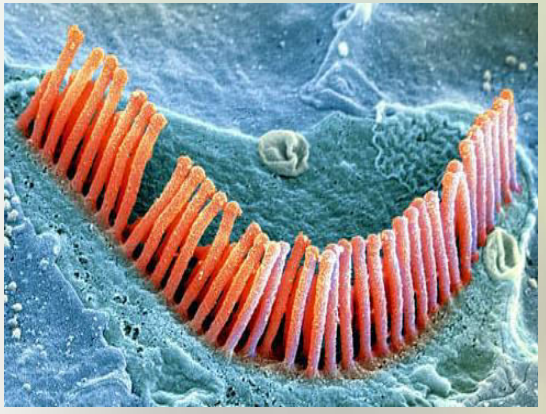
Hair Cells in Organ of Corti

- Two types:
  - Inner hair cells (IHC)
  - Outer hair cells (OHC)
- Both types differ in their functional anatomy.

Cochlea

Inner hair cells (IHC)

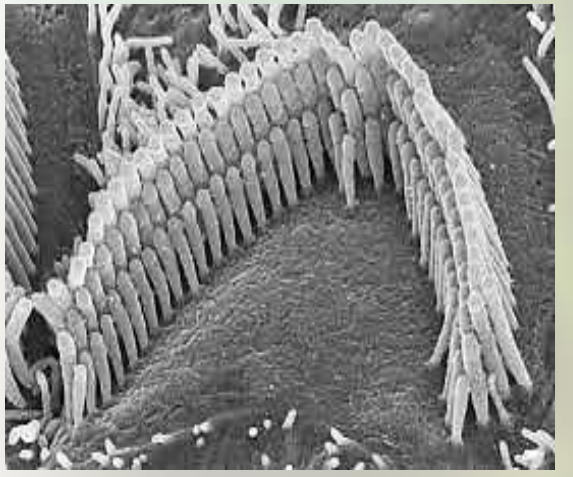
- There are about 3,500 IHCs arranged in a straight line or wide ‘U.
- They are true sensory cells which send impulses via auditory nerve.



Cochlea

Outer hair cells (OHC)

- About 12,000-25,000 OHCs in mammals.
- They are arranged in three or four rows forming a characteristic ‘W’ shape.



Cochlea

Outer hair cells (OHC)

- Outer hair cells have both sensory and motor elements.
- They contribute to hearing sensitivity and frequency selectivity.

Cochlea

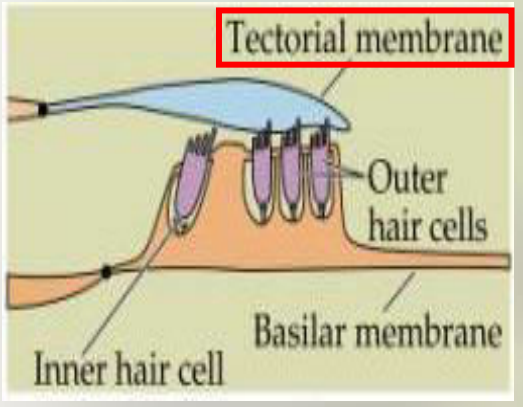
Basilar Membrane

- Basilar membrane forms the floor of cochlear duct and bears the organ of Corti.
- It is involved in the detection of sound according to its frequency range.

Cochlea

Tectorial Membrane

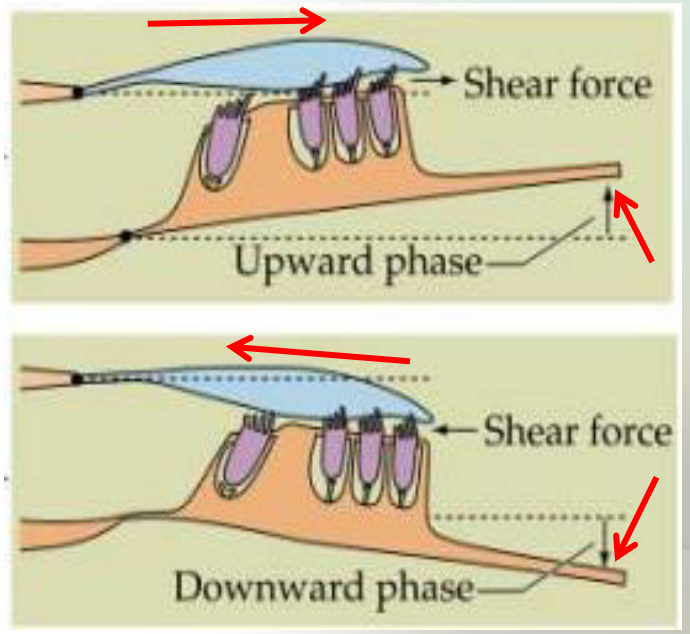
- It is a fibrous sheet lying on the apical surface of organ of Corti
- It is coated with a fine gelatinous mucus layer in which stereocilia of hair cells are embedded.



Cochlea

Tectorial Membrane

- When basilar membrane is displaced, tectorial membrane moves across the tops of the hair cells.
- It exerts a shearing force (perpendicular to the axis of the cilia) that bends the stereocilia of hair cells.



Cochlea

Tectorial Membrane

- Bending of hair cells generates nerve impulse that is transmitted to the sensory axons of the auditory nerve.

END

Receptor Physiology

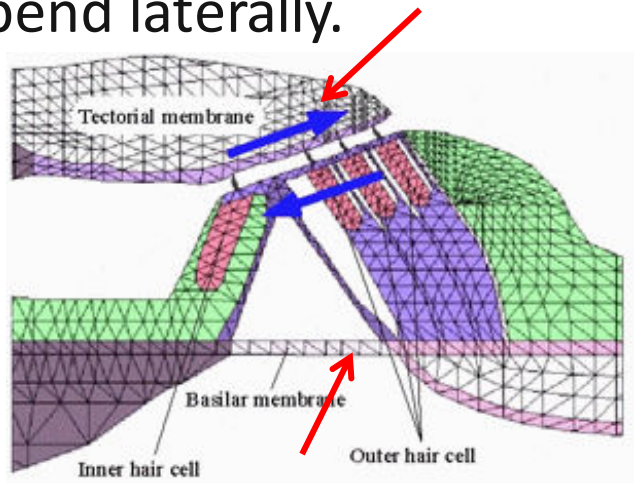
Sound Transduction by Cochlear Hair Cells

LECTURE NO 58

Sound Transduction by Cochlear Hair Cells

Reception of Sound at Cochlea

- Sound vibrations are received at the oval window through stapes.
- They cause displacement of basilar membrane.
- Tectorial membrane slides across the tips of the hair cells in the cochlear duct.
- Sliding of TM exerts a shearing force at the tips of stereocilia that bend laterally.



http://web.tbgu.ac.jp/ait/wada/wadalab/image/corti-e.gif

Sound Transduction by Cochlear Hair Cells

Mechanoelectrical Transduction

- Lateral bending of stereocilia causes the transduction events to happen.
- Mechanical deflection of stereocilia causes conformational changes in transduction ion channels in their tips.
- It causes these channels to open.

Sound Transduction by Cochlear Hair Cells

Mechanoelectrical Transduction

- A deflection of only 0.1-1.0 nm is the threshold of cochlear hair cells.
- When ion channels open, K^+ ions enter the cell from endolymph in cochlear duct.
- Inward K^+ current depolarizes (excites) the hair cells, producing hair cell receptor potential.

Sound Transduction by Cochlear Hair Cells

Release of Neurotransmitter

- Hair cell excitation also causes opening of voltage gated Ca^{2+} channels.
- Ca^{2+} influx causes transmitter (glutamate) release from basal end onto the auditory nerve endings.
- They send an electrical signal along the cochlear nerve.

Sound Transduction by Cochlear Hair Cells

Repolarization of Hair Cells

- K^+ entry via the transduction channels results in:
  - Opening of voltage gated Ca^{2+} channels
  - K^+ channels located in the membrane of soma of hair cell at the basal end.
- Opening of these K^+ channels causes K^+ efflux resulting in repolarization

Sound Transduction by Cochlear Hair Cells

Unusual Depolarization With K^+ Ions

- Cochlear hair cells use K^+ both for depolarization and repolarizaton.
- This is an unusual adaptation.
- This is b/c the basal and apical surfaces of hair cells are exposed to different extracellular ionic environments.

Sound Transduction by Cochlear Hair Cells

Unusual Depolarization With K^+ Ions

- Apical end with stereo-cilia protrudes into the cochlear duct filled with endolymph.
- Endolymph is K^+-rich, Na^+-poor fluid.
- Basal end of hair cell is exposed to perilymph in typmpanic duct.
- Perilymph, like other extracellular fluids is K^+-poor, Na^+-rich fluid.

Sound Transduction by Cochlear Hair Cells

Unusual Depolarization With K^+ Ions

- Difference in perilymph-endolymph composition results in endocochlear potential.
- This potential is 80 mV more positive in cochlear duct with endlymph than tympanic duct with perilymph.

Sound Transduction by Cochlear Hair Cells

Unusual Depolarization With K^+ Ions

- Hair cells have a V_{rest} of -60 mV.
- So the inside of hair cell is about 45 mV more negative than the perilymph (on its basal end).
- It is 140 mV more negative than the endolymph at its stereociliary end.

Sound Transduction by Cochlear Hair Cells

Unusual Depolarization With K^+ Ions

- This large electrical gradient across the stereocilia drives K^+ through open transduction channels into the hair cell, even though these cells already have a high internal K^+ concentration

END

Receptor Physiology

Frequency Analysis by Cochlea

LECTURE NO 59

Frequency Analysis by Cochlea

frequency
-wise

Tonotopic Sensitivity of Cochlea

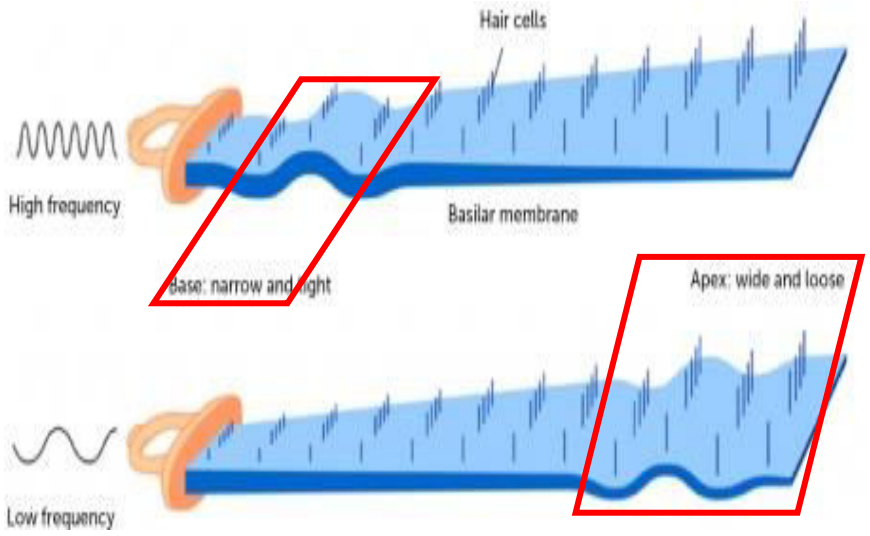
- Cochlea is tonotopically tuned organ.
- Tonotopic sensitivity is due to two reasons:

1. Variations in width and stiffness of basilar membrane.
2. Placement of hair cells on the basilar membrane.

Frequency Analysis by Cochlea

Variations in Width and Stiffness of Basilar membrane

- Basilar membrane is narrow and stiff at the base and wider and more flexible at the apex of cochlea.
- So, at different frequencies, its different portions vibrate stimulating particular hair cells and sensory neurons.



Frequency Analysis by Cochlea

Placement of Hair Cells

- Hair cells tuned to particular frequencies are placed at particular positions in narrow bands.
- The tuning is b/c of mechanical and channel properties.
- The resonance frequency of a cell is determined by the length of its stereocilia.
- Cells with long hairs are most sensitive to low frequency sounds, while cells with short hairs are tuned to high-frequency sounds.



Frequency Analysis by Cochlea

Frequency Analysis

- Sound vibrations encounter the basal membrane through its stiff base attached to the stapes.
- The base vibrates immediately in response to pressure changes.
- Vibrations travel along the membrane from the base toward its apex causing displacement of the membrane.

Frequency Analysis by Cochlea

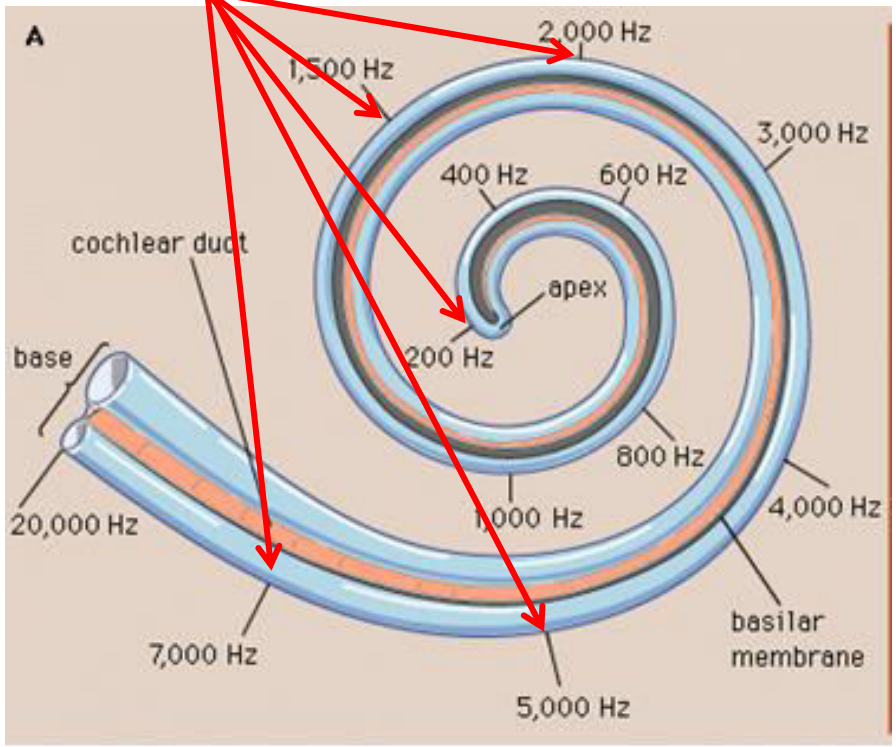
Frequency Analysis

- The region of maximal displacement varies with sound frequency.
- Membrane near the base under goes largest deformation and resonates optimally with high frequency tones.
- The distant regions of the membrane (near the apex) vibrate maximally in response to low frequency sounds.

Frequency Analysis by Cochlea

Frequency Analysis

- The frequencies of incoming sound waves are sorted out along the basilar membrane.
- Each frequency has its characteristic place.



Frequency Analysis by Cochlea

Frequency Analysis

- Very low frequencies (< 200 Hz) are compressed on to a relatively limited section at the apical end of the membrane.

END

Receptor Physiology

Electroreception

LECTURE NO 60

Electroreception

Electroreception

- All organisms produce weak electric fields due to the activity of nerves and muscles.
- Electroreception is the ability to detect electric fields generated by the animal itself or by other animals in the aquatic environment.

Electroreception

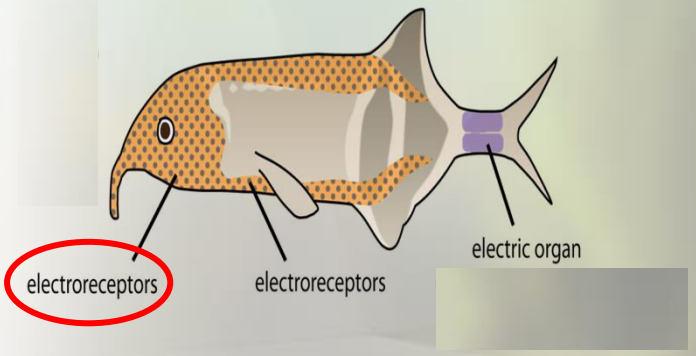
Electroreception

- Electroreception is an important sense in aquatic environments.
- It has been found in many fishes, some amphibians and a monotreme mammal duckbilled platypus.

Electroreception

Electroreceptors

- Electroreceptor cells are spread in the head and trunk regions of fishes and amphibians.
- They are linked to the lateral line system.



Electroreception

Electroreceptors

- The electroreceptors of platypus are present on its bill.
- They detect electric fields generated by the muscles of crustaceans, frogs, small fish, and other prey.

Effort by: AARISH ALI



Electroreception

Mechanism of Electroreception

- The current in water enters sensory pores in the epidermis of skin.
- an electroreceptor cell is present at the base of each pore.
- Each electroreceptor is actually a modified hair cell that has lost its cilia.

Electroreception

Mechanism of Electroreception

- The electroreceptor has synaptic connection with axons of the eighth cranial nerve that innervates the lateral-line system.
- The information sent from electroreceptors is processed in the cerebellum.

Electroreception

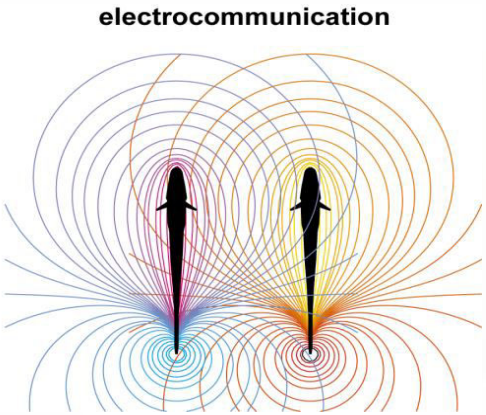
Uses of Electroreception

- Ability to detect electric fields helps a fish to find mates, capture prey, avoid predators and orient towards or away from certain objects.
- This is especially a valuable sense in deep, turbulent, or murky waters where vision is of little use.

Electroreception

Uses of Electroreception: Electrocommunication

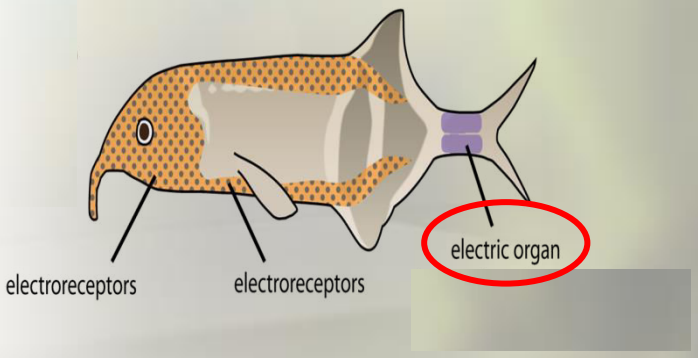
- Some fishes use this sense for electrocommu-nication i.e. detecting electric fields produced by the members of own species to communicate.



Electroreception

Uses of Electroreception: Electrolocation

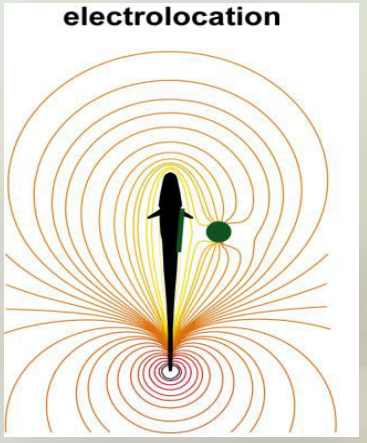
- Some fishes also use this sense for electrolocation.
- They possess specialized electric organs at one end of body that generate weak electrical fields.



Electroreception

Uses of Electroreception: Electrolocation

- The electrical pulses produced re-enter fish through epithelial pores.
- This generates an electric field around the fish body.



Electroreception

Uses of Electroreception: Electrolocation

- The electroreceptors detect any intruding object or animal when electric field is distorted.
- This helps these fishes to find prey and avoid predators within their range.

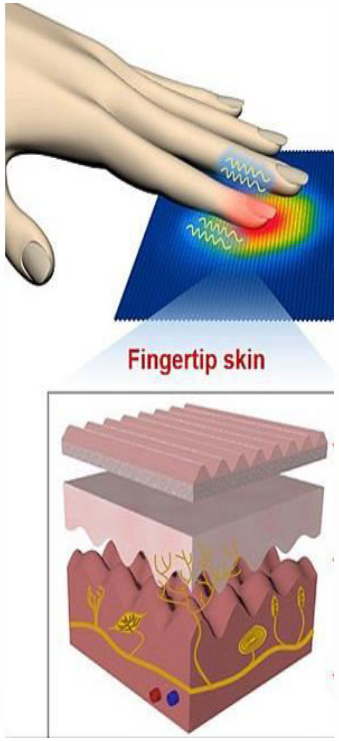
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Receptor Physiology

Thermoreception

LECTURE NO 61

Thermoreception



Thermoreception

- The sense by which an organism perceives hot or cold temperatures.

Thermoreception

Thermoreceptors

- Thermoreceptors are located in:
  - Skin
  - Tongue upper surface
  - Anterior hypothalamus

Thermoreception

Thermoreceptors

- Skin thermoreceptors detect changes in environmental temperature
- Thermoreceptors in anterior hypothalamus detect changes in body core temperature.

Thermoreception

Thermoreceptors

- Thermoreceptors are phasic-type receptors i.e.
  - They respond very rapidly to minute changes in temperature
  - But adapt and quit firing quickly, if the stimulus persists.

Thermoreception

Thermoreceptor Types

- There are two kinds of thermoreceptors in external skin and upper surface of tongue:
  - Cold receptors
  - Warmth receptors
- Both types of receptors are quite sensitive and enable humans to detect a change in skin temp. of as little as 0.01°C.

Thermoreception

Cold Receptors

- They are 3.5 times more common in skin than heat receptors.
- They consist of free nerve endings of neurons that have thin myelinated Aδ fibers.
- These fibers have faster conduction velocity (19m/s).
- They increase their firing rate when skin is cooled below body temperature

Thermoreception

Warmth Receptors

- They consist of free nerve endings of neurons that have unmyelinated C fibers.
- These fibers have low conduction speed (0.8 m/s).
- They increase their firing rate in response to temperatures above body temperature.

Thermoreception

TRP Proteins in Thermoreceptors

- The membranes of thermoreceptors have ion channel proteins belonging to the family of transient receptor potential (TRP) proteins.
- TRP family has many subfamilies of ion channels, each involved in different types of receptors.

Thermoreception

TRP Proteins in Thermoreceptors

- The TRP's involved in thermoreception are TRPA, TRPM and TRPV.
- TRPA and TRPM are involved in transduction of temperature in cold receptors.
- TRPV are involved in warmth reception.

Thermoreception

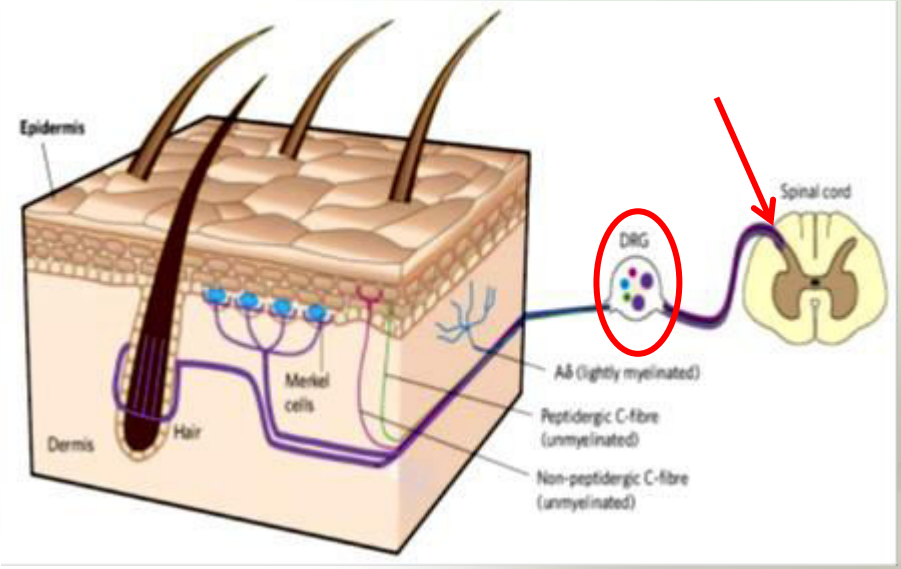
TRP Proteins in Thermoreceptors

- TRP ion channels allow an influx of many cations specially the Ca^{2+}ions.
- Increase in ion conc. depolarizes the membrane, and causes action potentials to be fired.

Thermoreception

Neural Pathway of Temperature Sensation

- Both warm and cool stimuli transduce information along the same neural pathway.
- Cell bodies of neurons of cutaneous thermoreceptors reside in the dorsal root ganglion (DRG) and the trigeminal ganglion on the dorsal horn of spinal cord.



Thermoreception

Neural Pathway of Temperature Sensation

- The neurons of dorsal horn of spinal cord communicate via synapses to the thalamus and then to the hypothalamus.
- The hypothalamus elicits proper thermoregulatory responses.

END

Receptor Physiology

Photoreception: Basics

LECTURE NO 62

Photoreception: Basics

Photoreception

- Transducing photons of light into electrical signals.
- These signals can be interpreted by the nervous system.

Photoreception: Basics

Photoreceptors

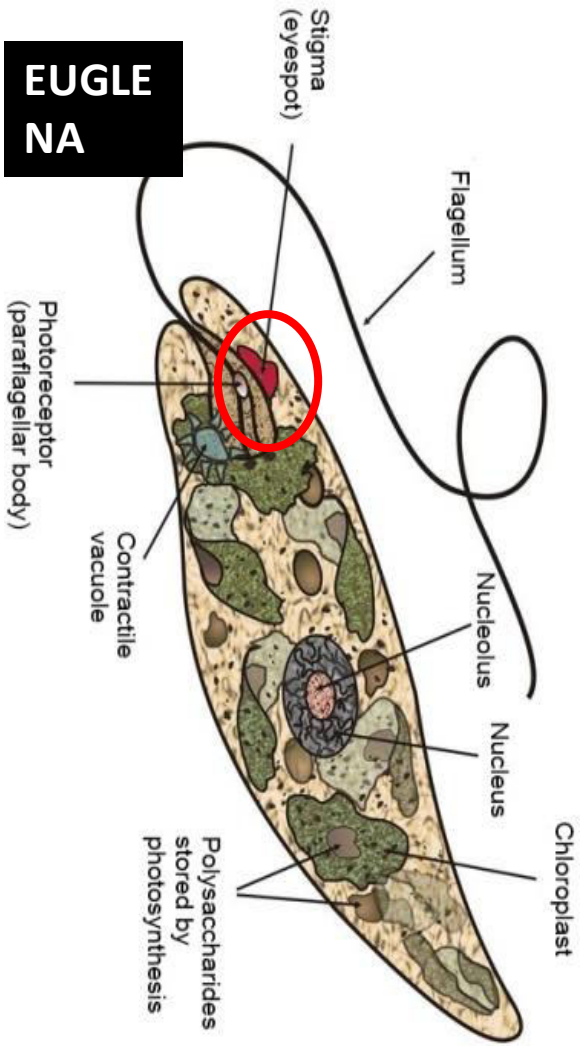
- Photoreceptors possess light-sensitive carotenoid pigments retinal and 3-dehydroretinal.
- Carotenoids are associated with opsin proteins to form rhodopsins.
- Rhodopsins absorb photons of light energy and produce a generator potential.

Photoreception: Basics

Photoreceptive Structures

- Complexity and arrangement of photoreceptors vary within animal kingdom.

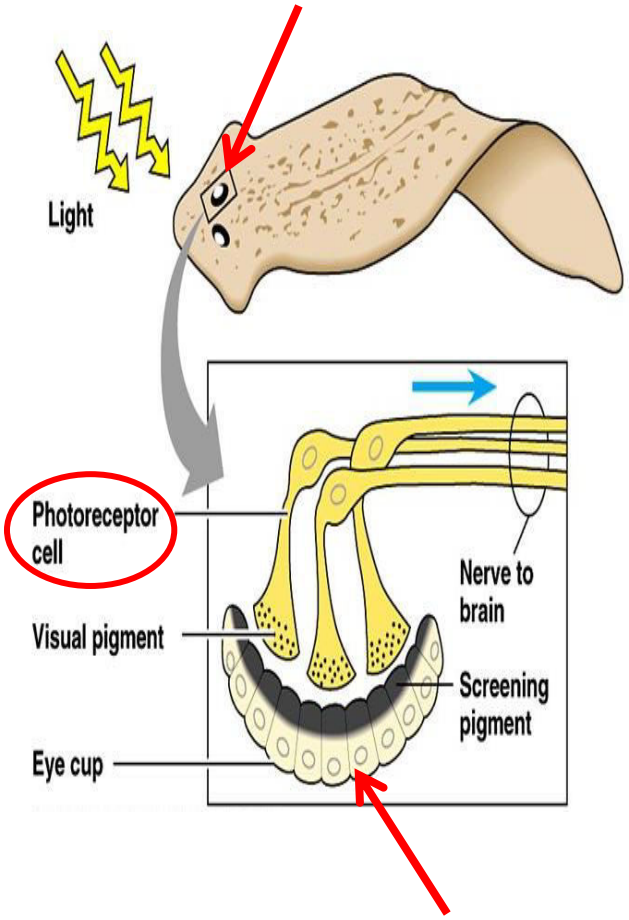
Photoreception: Basics



Eyespot or Stigma

- Simplest photoreceptive structure.
- Found in some protozoa e.g. euglena.
- It is a bright red colored organelle and has carotenoid pigments.
- It gives a sense of light and dark.
- Helps in phototaxis.

Photoreception: Basics



Eyecups or Ocelli

- Multicellular photo-receptive structures that consist of a cuplike depression containing photoreceptor cells.
- Found in cnidarians and flatworms e.g. Planaria.
- Cannot form image.
- Provide the animal a sense of direction only .

Photoreception: Basics

Eyes

- Photoreceptive organs in higher invertebrates and all vertebrates are image forming eyes.
- Give an animal more precise information about the surroundings.
- Eyes of vertebrates have cornea and lens which focus light onto the sensory retina and form sharp images.

Photoreception: Basics

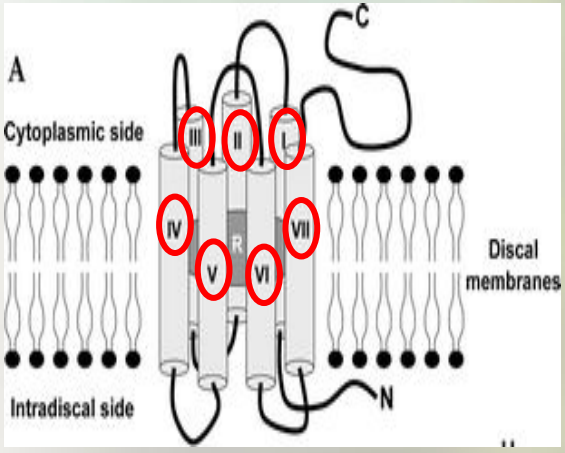
Pigments in Visual Transduction

- Visual transduction is based on a very highly conserved set of protein molecules in animal kingdom.
- These proteins are the opsins.
- They are found in the cell membranes of all photoreceptor cells.

Photoreception: Basics

Opsins

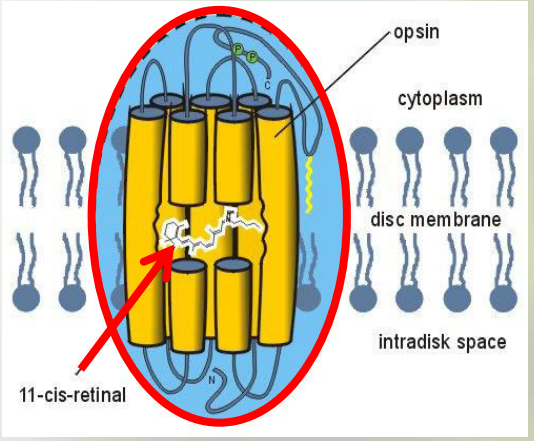
- An opsin molecule has seven transmembrane domains.
- They provide an optical pathway to capture photons within the photoreceptors.



Photoreception: Basics

Rhodopsin

- Opsins are coupled to light-absorbing photopigment Retinal.
- This form a functional pigment molecule rhodopsin.



Photoreception: Basics

Photopigments

- Photopigments are the molecules which are structurally altered by the absorption of photons.
- They activate a cascade of associated molecules, that results in opening of ion channels and generates graded receptor potential.

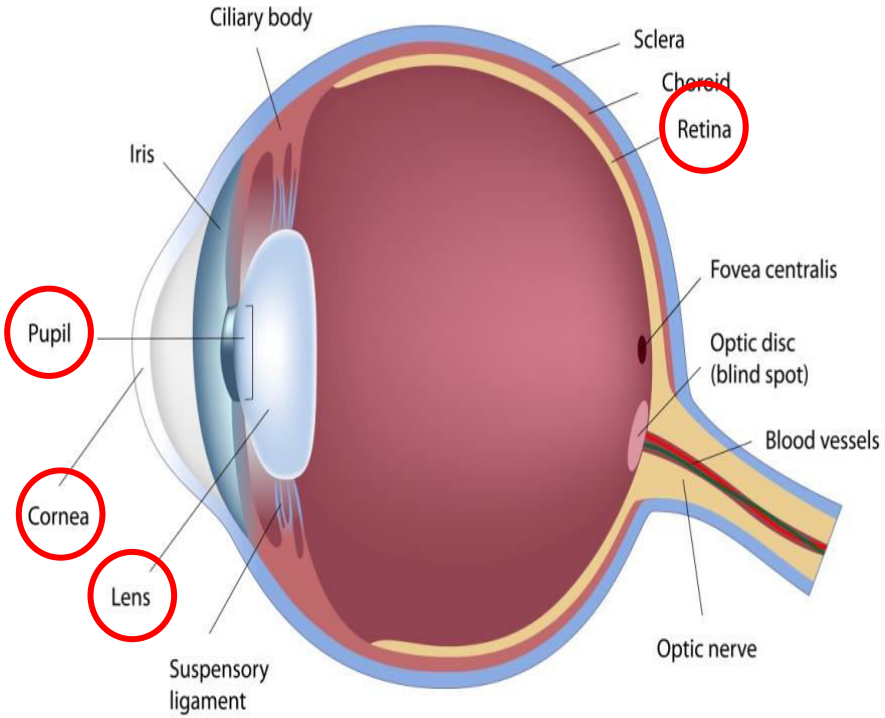
End

Receptor Physiology

The Vertebrate Eye: Structural Physiology

LECTURE NO 63

The Vertebrate Eye: Structural Physiology



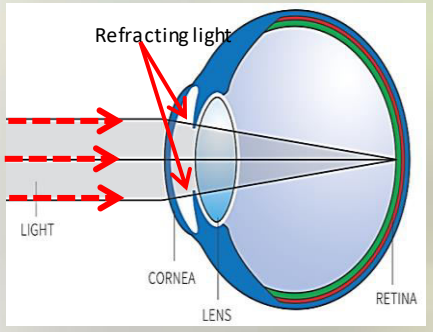
Structural Features of the Eye

- The parts of the eye involved in focusing and image formation are:
- Cornea, a biconvex lens, pupil and retina.

The Vertebrate Eye: Structural Physiology

Role of Cornea

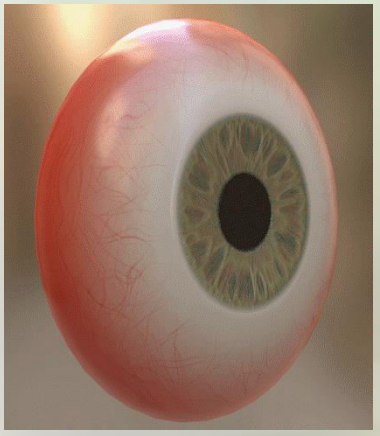
- Cornea is the clear outer surface of the eye.
- Light enters the eye through cornea.
- Its function is to bent and focus light rays onto the lens inside.
- 85% refraction occur here



The Vertebrate Eye: Structural Physiology

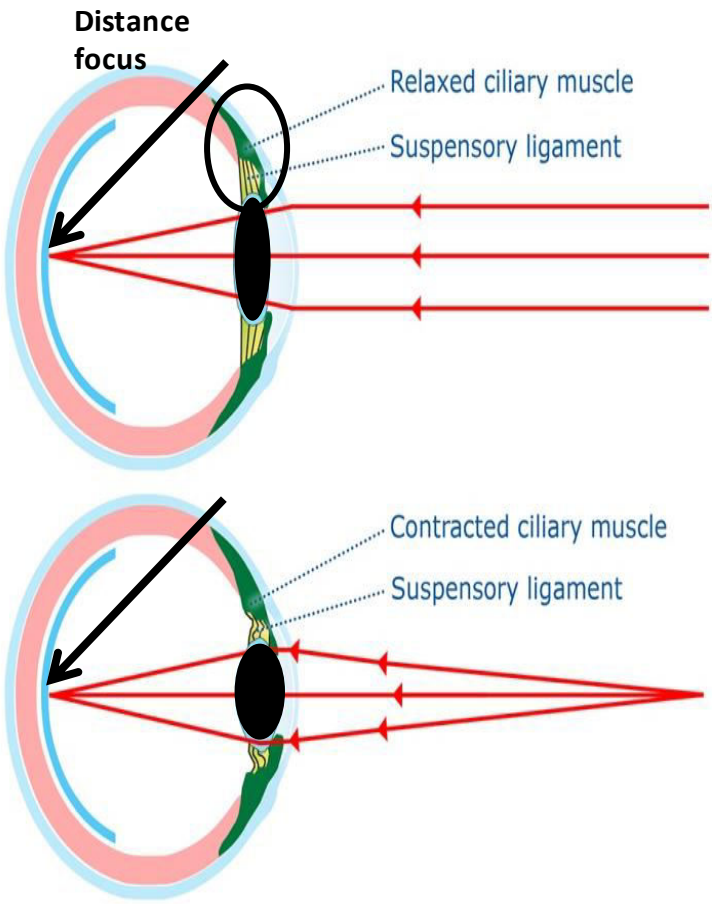
Role of Pupil

- Entry of light in the eye is controlled by increase or decrease in the diameter of pupil.
- Circular smooth muscle fibers in the iris control the diameter of pupil.



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The Vertebrate Eye: Structural Physiology



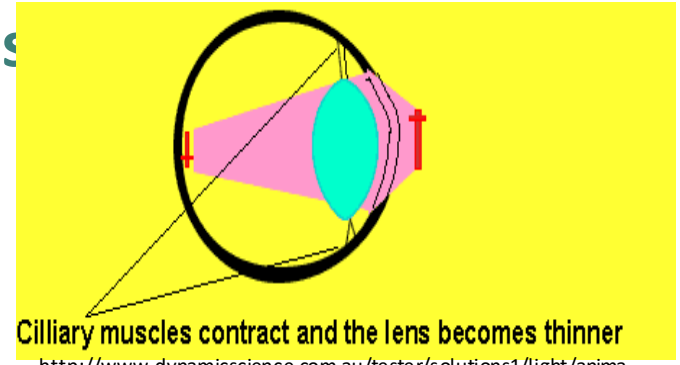
https://www.biocelearn.org.nz/system/images/images/000/000/053/original/Eye-focus-final-3000X2000.jpg?1457566795

Role of Lens

- Lens focuses the light rays on retina by bending and refracting.
- Image is focused by changing the curvature and thickness of the lens due to the suspensory ligaments (fibers of Zonula) and ciliary muscles.

The Vertebrate Eye: Structural Physiology

Role of Fibers of Zonula and Ciliary Muscles



http://www.dynamicscience.com.au/tester/solutions1/light/animatedeyeadjustment.gif

- Fibers of zonula held the lens in place.
- They change the shape of lens by exerting an outward tension on perimeter.
- Attached with fibers of zonula are the ciliary muscles
- They contract to pull fibers of zonula, which flatten the lens. This focuses distant objects on retina.
- When they relax, lens becomes rounded and near objects are focused.

The Vertebrate Eye: Structural Physiology

Role of Retina

- Retina is the sensory layer of eye with photoreceptor cells, rods and cones.
- Rods and cones transduce photon energy of light into the nerve impulse.
- From retina, the impulses are carried by optic nerve to the brain.

End

Receptor Physiology

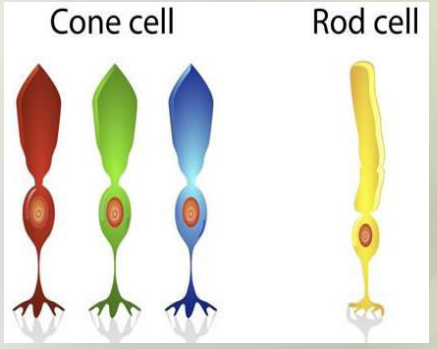
Visual Receptor Cells of Vertebrates

LECTURE NO 64

Visual Receptor Cells of Vertebrates

Photoreceptor Cells

- The photoreceptor cells capture the energy of light and transduce it into neuronal signals.
- They are located in the retina of vertebrate eye.
- They fall into two classes, rods and cones.



Visual Receptor Cells of Vertebrates

Rods

- Rods are more sensitive to light.
- Enable the animal to see in dark (night vision).
- They cannot distinguish colors, so provide achromatic vision (black and white) only.

Visual Receptor Cells of Vertebrates

Cones

- Cones are less sensitive to light.
- They and function best in bright light and provide high resolution with color vision.
- They contribute very little to night vision.

Visual Receptor Cells of Vertebrates

Types of Cones

- There are three types of cones.
- Each having a different sensitivity across the visible spectrum in the blue, green and red ranges.

Visual Receptor Cells of Vertebrates

Rods and Cones in Animal Retinas

- Relative numbers of rod and cones in the retina varies among animals.
- It correlates with the extent to which an animal is active at night.
- Most fishes, amphibians, reptiles and birds have strong color vision.

Visual Receptor Cells of Vertebrates

Rods and Cones in Animal Retinas

- Humans and primates also have well developed color vision.
- Human retina contains 125 million rods & 6 million cones.

Visual Receptor Cells of Vertebrates

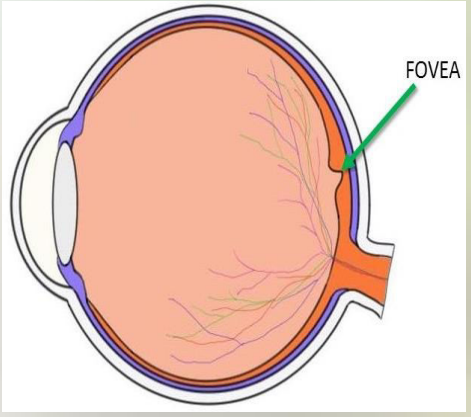
Rods and Cones in Animal Retinas

- Most nocturnal mammals have reduced capacity to see colors.
- Nocturnal mammals have high proportion of rods in retina that gives them keen night vision.

Visual Receptor Cells of Vertebrates

Fovea Centralis

- It is the center of visual field in many mammals.
- It is about 1 mm^2 central part of retina with very high density of cones but no rods.



Visual Receptor Cells of Vertebrates

Fovea Centralis

- It provides very detailed information about the visual field, i.e. high visual acuity.
- The ratio of rods to cones increases with distance from the fovea, with the peripheral regions having only rods.

Visual Receptor Cells of Vertebrates

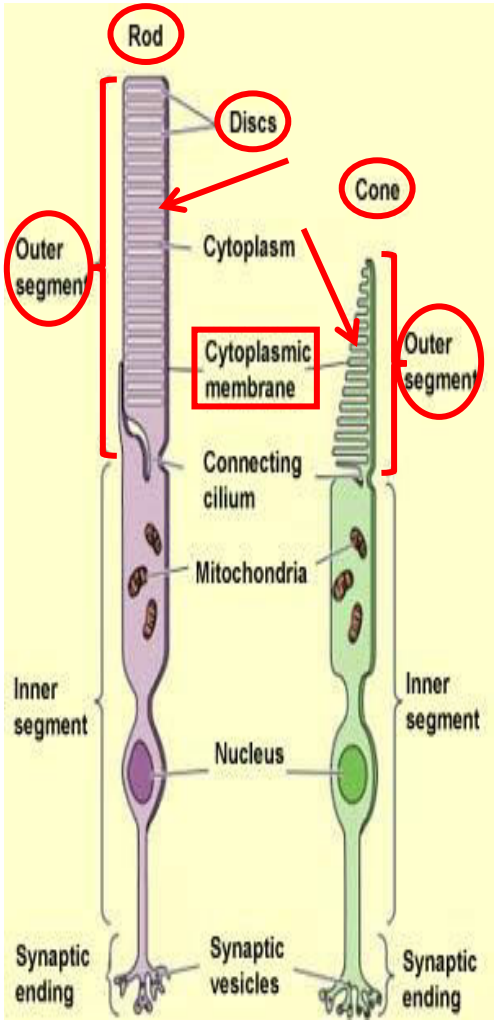
Structure of Rods and Cones

- General cell structure of rods and cones is basically similar.
- Rods are narrower and longer than the cones.
- Major functional segments of rods and cone are:
  - The outer segment
  - The inner segment
  - Synaptic body

Visual Receptor Cells of Vertebrates

The Outer Segment

- Light-sensitive photochemical (rhodopsin or photpsins) is found in the outer segment.
- It has large numbers of discs (upto 1000) formed by infolding of cell membrane.
- Lumen of lamellae of cones is open to the exterior.
- In rods, lamellae form stacked flattened disks.
- Photopigments are incorpora-ted as transmembrane proteins in membranes of

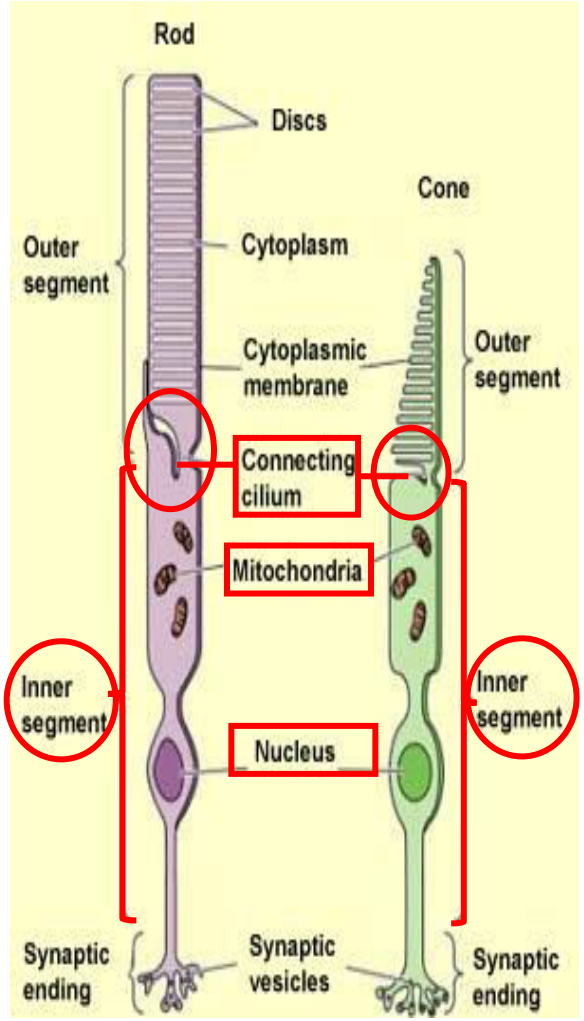


discs where primary steps of transduction occur.

Visual Receptor Cells of Vertebrates

The Inner Segment

- The vertebrate receptor cells contain a rudimentary cilium that connects the outer segment to the inner segment.
- The inner segment contains the cytoplasm with organelles and nucleus.



Visual Receptor Cells of Vertebrates

The Synaptic Body

- At the end of inner segment, there is a synaptic body or synaptic terminal
- It connects the rod or cone with neuronal cells.

End

Effort By

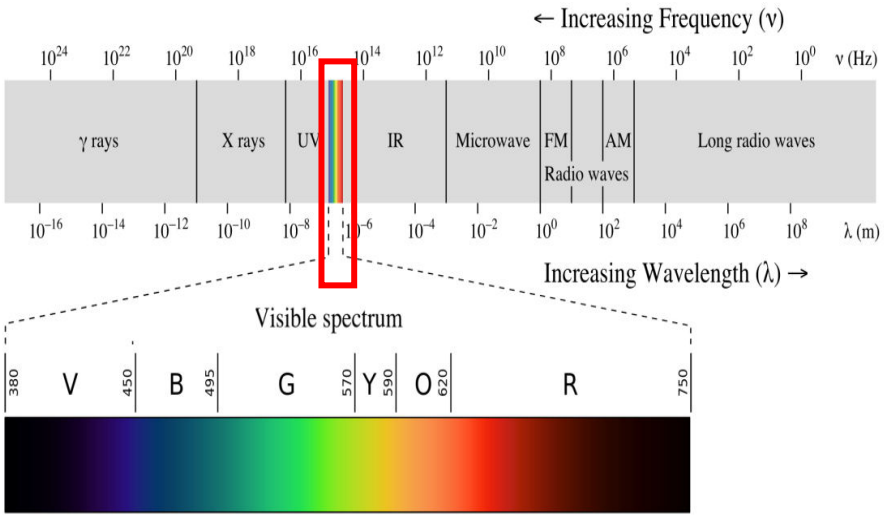
AARISH ALI

Receptor Physiology

Visual Pigments and Their Photochemistry-I

LECTURE NO 65

Visual Pigments and Their Photochemistry-I



Visible Light

- Visual receptor cells can detect only a part of the spectrum of electromagnetic radiation.
- Various wavelengths within the spectrum of visible light are perceived as different colors.
- Visible spectrum for human eyes lies between the wavelengths of 400-740 nm.

Visual Pigments and Their Photochemistry-I

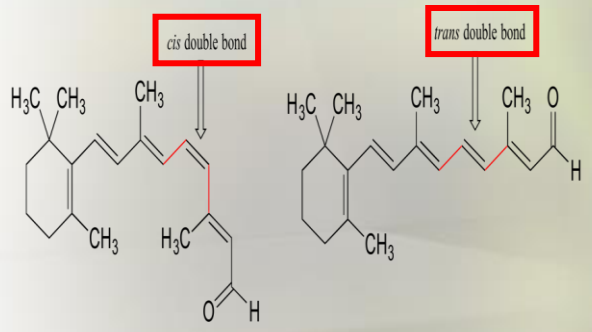
Visual Pigments

- Vision of an animal depends on the presence of visual pigments which can absorb photons of light and transduce them into chemical energy.
- Their light absorbance is due to the presence of alternating single and double bonds in a carbon chain or ring.

Visual Pigments and Their Photochemistry-I

Visual Pigments

- In a photoreceptor cell, a quantum of radiation absorbed by a photo-pigment, results in a cis-trans configuration change in a double bond.
- This starts a csacade of reactions resulting in transduction of light.



Visual Pigments and Their Photochemistry-I

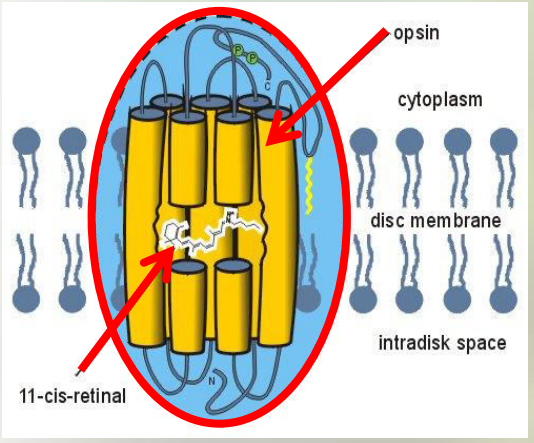
Visual Pigments

- The visual pigment in the rods is rhodopsin.
- The pigments in cones are color pigments Photopsins.
- All the pigments have only slightly different compositions.

Visual Pigments and Their Photochemistry-I

Chemistry of Rhodopsin

- A rhodopsin molecule consists of two major components:
  - Opsin protein
  - A light-absorbing molecule (retinal or 3-dehydroretinal)



Visual Pigments and Their Photochemistry-I

Chemistry of Rhodopsin

- Rhodopsin also includes:
  - an oligosaccharide chain of six-sugars
  - A variable number (30 or more) of phospholipid molecules.
- Both pigments retinal and 3-dehydroretinal are aldehydes of carotenoid vitamin A_1 and A_2.

Visual Pigments and Their Photochemistry-I

Chemistry of Rhodopsin

- The opsin along with phospholipids and oligo-saccharide chain, is bound to the photo-receptor membrane as an integral part.
- Carotenoid molecules (Retinal & 3-dehydro-retinal) attach or detach in the absence or presence of light.

End

Receptor Physiology

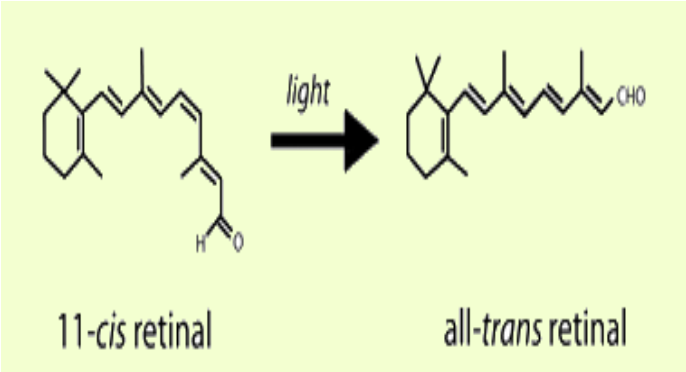
Visual Pigments and Their Photochemistry-II

LECTURE NO 66

Visual Pigments and Their Photochemistry-II

Isomerization of Rohodopsin

- The retinal molecule assumes two sterically distinct states in the retina.
- In the absence of light, retinal is in 11-cis configuration that binds to opsin covalently by a Schiff's base bond ($R_2C=NR$).
- In the presence of light, 11-cis retinal isomerizes into all-trans configuration.
- This cis-trans isomerization is light's only direct effect on visual pigment.

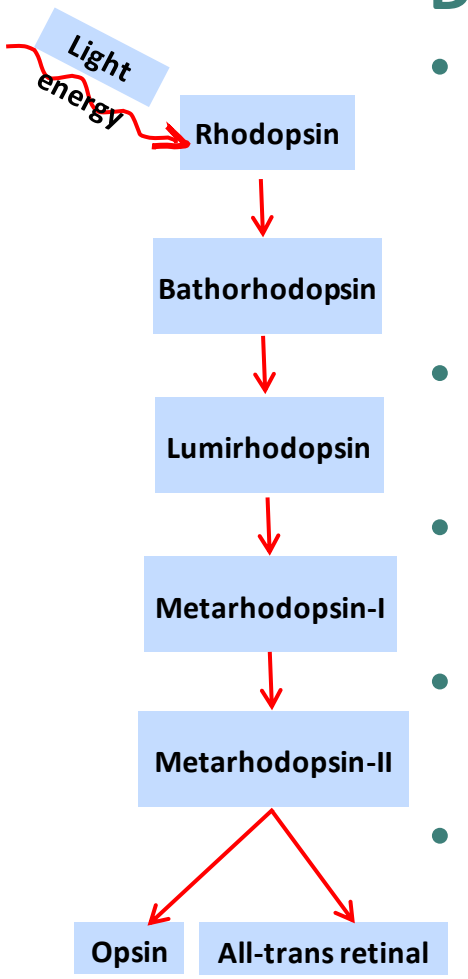


Visual Pigments and Their Photochemistry-II

Activation of Rhodopsin

- The cis to trans change in configuration destabilizes rhodopsin molecule and it starts to decompose.
- The decomposing state is the activated state of rhodopsin.
- It results in a series of biochemical reactions in the membrane resulting in transduction of light into electrical signal.
- Purple to yellow bleaching

Visual Pigments and Their Photochemistry-II



Decomposition of Rhodopsin

- As the light hits rhodopsin, it immediately produces bathorhodo-psin –a partially split combination of all-trans retinal and opsin.
- BR is extremely unstable and decays in nanoseconds to lumirhodopsin
- It decays in microseconds to metarhodopsin I.
- In about a millisecond it changes to metarhodopsin II.
- MR-II splits into opsin and all-trans retinal much more slowly (in seconds).

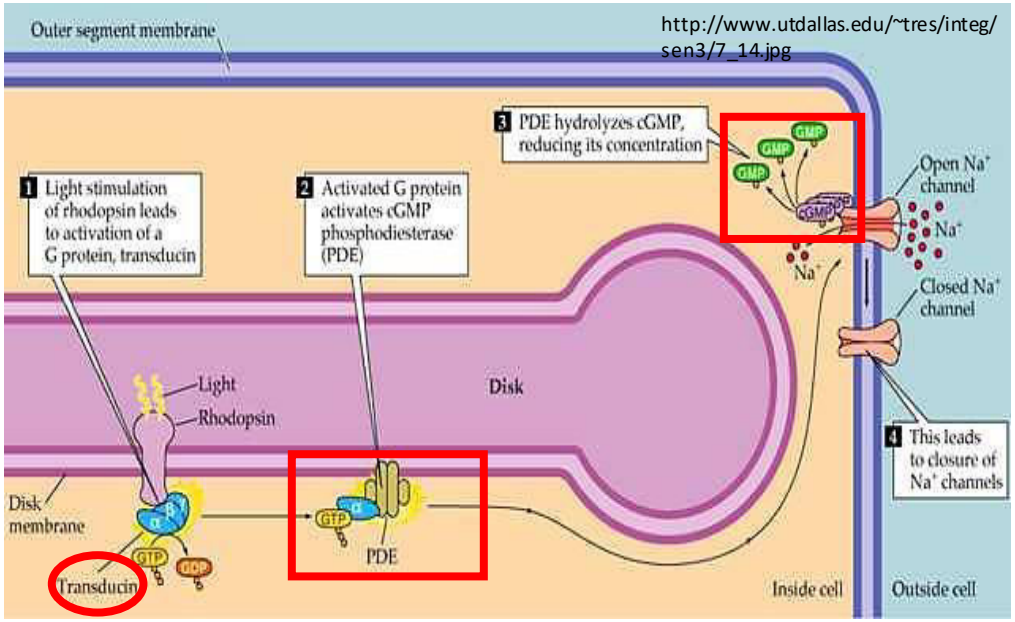
Visual Pigments and Their Photochemistry-II

Role of Metarhodopsin II in Visual Transduction

- It is the metarhodopsin II that is called activated rhodopsin.
- It excites changes in the rods which generate graded receptor potential and transmit the visual impulse into the nervous system

Visual Pigments and Their Photochemistry-II

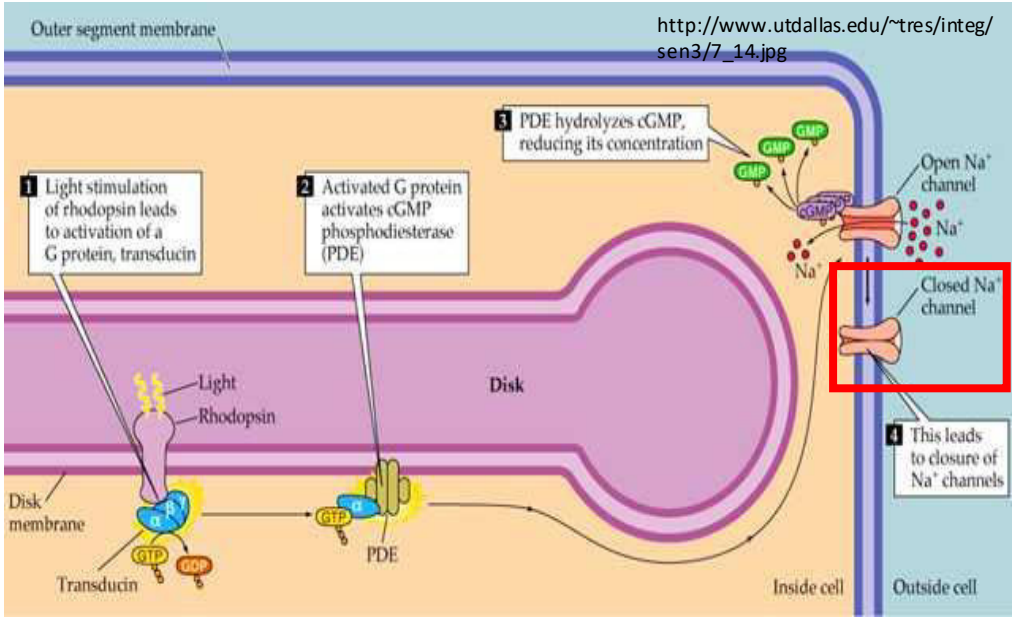
Visual Transduction



- Metarhodopsin II activates a G protein transducin, associated with the membrane.
- Activated transducin activates the enzyme phosphodiesterase, which hydrolyzes cGMP to 5'-GMP.

Visual Pigments and Their Photochemistry-II

Visual Transduction



- Drop in cGMP level results in drop in conductance of cations Na^+, Ca^{2+} and Mg^{2+} through permeating channels in membrane.
- As a result K^+ current dominates and causes the cell to hyperpolarize and generate graded receptor potential.

Visual Pigments and Their Photochemistry-II

Hyperpolarization in Photoreceptor Cells

- Production of graded receptor potential due to hyperpolarization is characteristic of photoreceptor cells.
- This behavior is different from all other sensory receptors in which receptor potential is generated due to depolarization.

End

Receptor Physiology

The Color Vision

LECTURE NO 67

The Color Vision

Types of Cone Cells

- Perception of color is based on three types of cone cells with one of the three visual pigments: blue, green or red.
- Each type of cone cell synthesizes only one of these types , making it selectively sensitive to that particular color.

The Color Vision

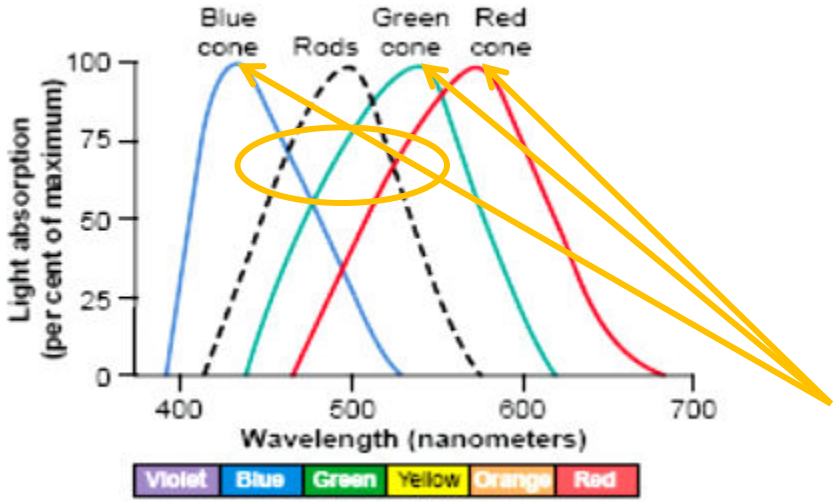
Photopsins

- The three visual color pigments are called photopsins.
- They are formed by binding of retinal to three types of opsins.
- These opsins have slight differences which let each photopsin to absorb light optimally at a distinct wavelength.

The Color Vision

Perception of Intermediate Hues

- Absorption spectra of blue, green and red pigments in the three types of cones show peak absorbance at wavelengths of 445, 535, and 570 nm.
- Ranges of these spectra overlap, enabling brain to perceive intermediate hues on simultaneous stimulation of two or more classes of cones.



The Color Vision

Genetic Basis of Color Blindness

- The three types of opsins in color pigments are encoded by three genes.
- Gene encoding the opsin in blue-cone pigment is located on an autosomal chromosome.
- The genes for red and green-cone pigments are closely linked on the X chromosome.

The Color Vision

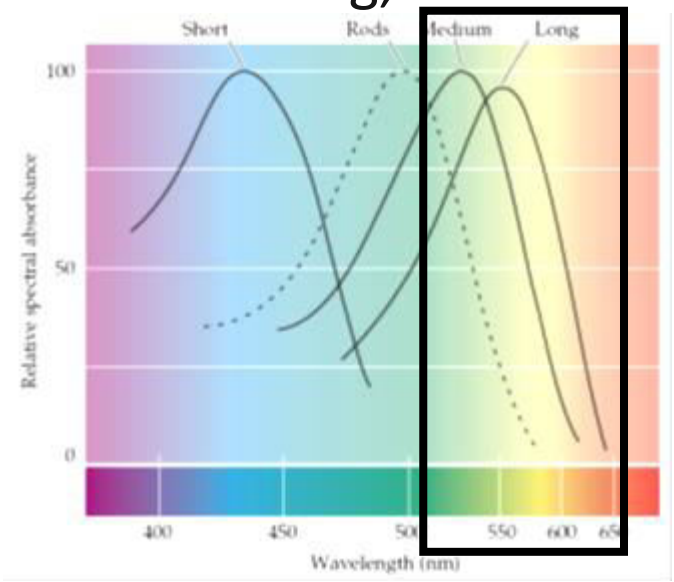
Genetic Basis of Color Blindness

- Color blindness is caused due to a mutation in one of the cone opsin genes that results in the absence of one type of pigmented cones.
- A person missing a single type of color receptive cones is unable to distinguish some colors.

The Color Vision

Red-Green Color Blindness

- The green, yellow, orange, and red colors lie between wavelengths of 525 and 675 nm.
- They are normally distinguished from one another by red and green cones.
- If either of these two cones is missing, the person cannot distingu these four colors.
- The person is especially unable to distinguish red from green and is said to have red-green color blindness.



The Color Vision

Protanope and Deuteranope

- A person with loss of red cones is called a protanope.
- In protanopia, overall visual spectrum is shortened at the long wavelength end because of a lack of red cones.

The Color Vision

Protanope and Deuteranope

- A color-blind person who lacks green cones is called a deuteranope.
- In deuteranopia, overall visual spectrum lies within normal range because red cones are available to detect the long wavelength red color.

End

Endocrine Physiology

Glands, Secretions and Mechanisms

LECTURE NO 68

Glands, Secretions and Secretory Mechanisms

Glands

- A gland is a cell or group of cells that secretes a particular chemical substance for use in the body or for discharge into the surroundings.
- Every animal has a large number of glands, which differ in structure and type of secretion.

Glands, Secretions and Secretory Mechanisms

Secretions

- Secretions are chemical substances, synthesized by glandular cells.
- They are released from the gland in response to an appropriate stimulus.
- The nature and amount of the secretion vary greatly among different glands.

Glands, Secretions and Secretory Mechanisms

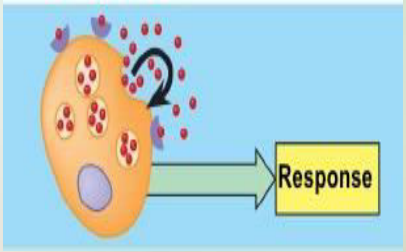
Types of Glandular Secretions

- Based on distance at which they effect their target:
  - Autocrine secretions
  - Paracrine secretions
  - Endocrine secretions
  - Exocrine secretions

Glands, Secretions and Secretory Mechanisms

Autocrine Secretion

- Refers to a secreted substance that affects the secreting cell itself.



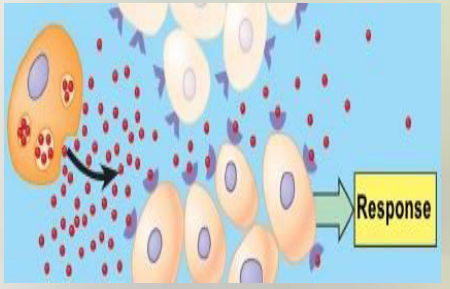
Glands, Secretions and Secretory Mechanisms

Paracrine Secretion

- Refers to a substance that has an effect on neighboring cells.

Example

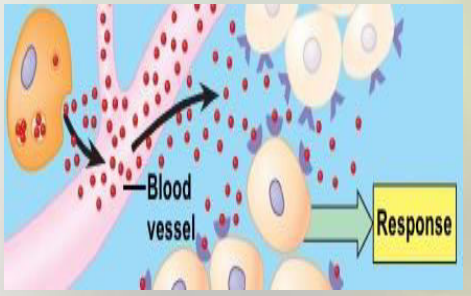
- histamine released by mast cells affects locally to induce vasodilation in inflammatory responses.



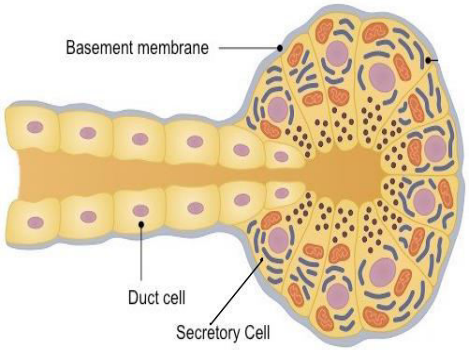
Glands, Secretions and Secretory Mechanisms

Endocrine Secretions

- Refer to the substances that are released into the bloodstream and act on distant target tissues.
- Endocrine secretions are generally known as hormones.



Glands, Secretions and Secretory Mechanisms



Exocrine Secretions

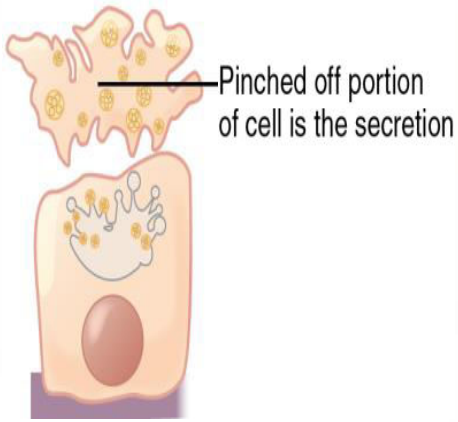
- Refer to a substance that is released via a duct through the gland to the external or internal epithelial surfaces.
- e.g. digestive secretions, milk, tears, perspiration, fluid containing sperms.

Glands, Secretions and Secretory Mechanisms

Secretory Mechanisms

- The substances to be exported out of the cell are stored in the form of secretory vesicles.
- Secretory vesicles are released as.
  - Apocrine secretion
  - Merocrine secretion
  - Holocrine secretion

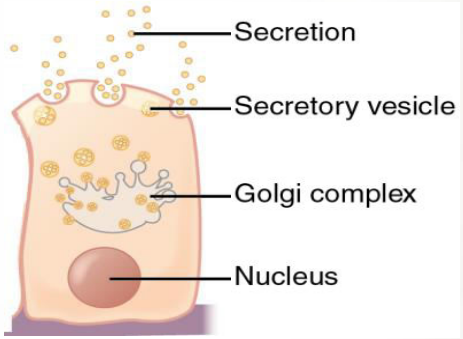
Glands, Secretions and Secretory Mechanisms



Apocrine Secretion

- Apical portion of the cell, containing the secretory vesicles is sloughed off and enters the lumen.
- Along with secretion, it loses some cytoplasm.
- e.g. mammary glands.

Glands, Secretions and Secretory Mechanisms



Merocrine Secretion

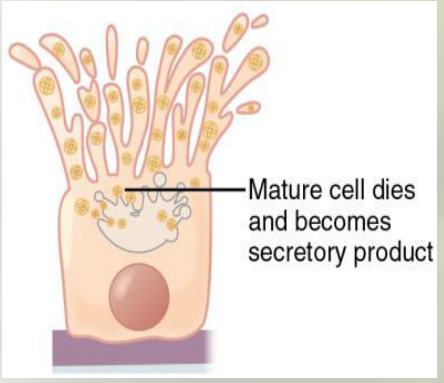
- The secretory vesicles release their material by exocytosis, without losing any part of the cell
- It is the most common method of secretion.
- e.g. sweat glands and many digestive glands.

Glands, Secretions and Secretory Mechanisms

Holocrine Secretion

- To release its contents, entire cell ruptures and breaks up.
- e.g. sebaceous glands

END



Endocrine Physiology

Types of Glands

LECTURE NO 69

Types of Glands

Types of Glands

- Glands are broadly classified into two types:
  - Endocrine glands
  - Exocrine glands

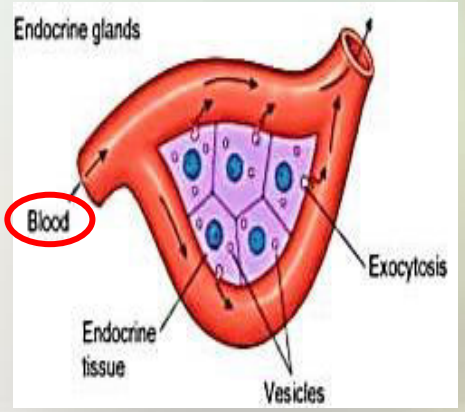
Endocrine Glands

- Endocrine glands are the ductless glands that secrete their products directly into the blood.
- Their secretions are called hormones.

Types of Glands

Endocrine Glands

- Various endocrine tissues are structurally and chemically diverse.
- However all endocrine tissues are richly vascularized.

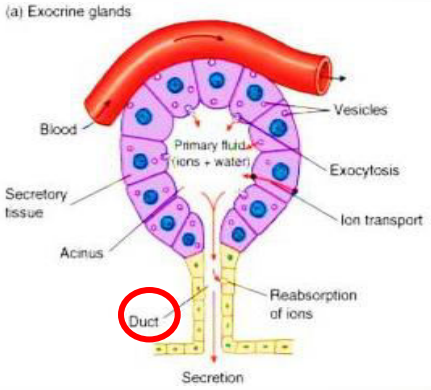


Types of Glands

Endocrine Glands

- Some endocrine glands contain more than one type of secretory cells, each producing a different hormone.
- The endocrine secretions play role in chemical coordination of the body.
- They also modulate short and long-term physiological processes.
- e.g. Pituitary, thyroid and adrenal glands.

Types of Glands



Exocrine Glands

- Exocrine glands produce fluid secretions that are delivered through ducts onto the epithelial surfaces of the body.

- The fluid secretions may be proteins (enzymes) or mucous or both.

Types of Glands

Examples of Exocrine Glands

- Salivary glands produce saliva that is delivered to the oral cavity through submandibular and parotid ducts.
- Pancreas produces enzyme-containing pancreatic juice that is delivered to the small intestine through pancreatic duct.

Types of Glands

Examples of Exocrine Glands

- Lacrimal glands produce tears that is delivered through lacrimal duct on the surface of eye.
- Mammary glands produce milk that is delivered through lactiferous ducts to the nipples.

END

Endocrine Physiology

Vertebrate Endocrine Overview

LECTURE NO 70

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands

- Vertebrates have a large number of endocrine glands and tissues that produce hormones.
- Some hormones are produced by only one particular gland located at specific site.

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands

- Some hormone-like substances e.g. prostaglandins and leukotrienes, are produced by all or nearly all tissues.
- Some hormones are produced by many selected tissues.

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands and Tissues		
Endocrine Gland	Hormone	Major Physiological Role
Hypothalamus (stimulatory hormones)	Corticotropin releasing hormone (CRH)	Stimulates ACTH release
	GH releasing hormone	Stimulates growth hormone release
	Gonadotropin releasing hormone (GnRH)	Stimulates release of FSH and LH
	TSH releasing hormone (TRH)	Stimulates TSH release and prolactin secretion

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands and Tissues		
Endocrine Gland	Hormone	Major Physiological Role
Hypothalamus (Inhibitory hormones)	MSH inhibiting hormone (MIH)	Inhibits melatonin stimulating hormone's secretion
	Prolactin inhibiting hormone (PIH)	Inhibits prolactin release
	Somatostatin	Inhibits release of growth hormone

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands and Tissues		
Endocrine Gland	Hormone	Major Physiological Role
Pituitary (Anterior)	Growth hormone (GH)	Promotes growth of body tissues Protein synthesis
	Prolactin (PRL)	Promotes milk production
	Thyroid stimulating hormone (TSH)	Stimulates thyroid hormone release
	Adrenocorticotropic hormone (ACTH)	Stimulates release of glucocorticoids by adrenal cortex

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands and Tissues		
Endocrine Gland	Hormone	Major Physiological Role
Pituitary (Anterior)	Luteinizing hormone (LH)	Stimulates androgens and progesterone production, induces ovulation
	Melanocyte-stimulating hormone (MSH)	Stimulates melanin pigment production in the skin.
Pituitary (Posterior)	Antidiuretic hormone (ADH)	Stimulates water reabsorption by kidneys Increases blood pressure by vasoconstriction
	Oxytocin	Stimulates milk ejection, uterine contractions during child birth

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands and Tissues		
Endocrine Gland	Hormone	Major Physiological Role
Thyroid (Follicular cells)	Thyroxine Triiodothyronine	Maintain metabolic rate and oxygen consumption; Growth and development
Thyroid (Parafollicular cells)	Calcitonin	Reduces blood Ca^{2+} levels
Parathyroid	Parathormone (PTH)	Increases blood Ca^{2+} levels; decreases blood phosphate level

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands and Tissues		
Endocrine Gland	Hormone	Major Physiological Role
Adrenal (Cortex)	Aldosterone	Increases blood Na^+ levels; increase K^+ secretion
	Cortisol, corticosterone cortisone	Role in carbohydrate metabolism increase blood glucose levels anti-inflammatory effects
Adrenal (Medulla)	Epinephrine Norepinephrine	Stimulate fight-or-flight response; increase blood glucose levels; increase metabolic activities

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands and Tissues		
Endocrine Gland	Hormone	Major Physiological Role
Pancreas (Islets of Langerhans)	Insulin	Reduces blood glucose levels; stimulates protein, glycogen and fat synthesis
	Glucagon	increases blood glucose levels; enhances gluconeogenesis and glycogenolysis
Pineal gland	Melatonin	Regulates some biological rhythms and protects CNS from free radicals Inhibits gonadal development (Antigonadotropic action)

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands and Tissues		
Endocrine Gland	Hormone	Major Physiological Role
Testes (Leydig cells)	Testosterone	Development of male secondary sexual characters
Testes (Sertoli cells)	Inhibin	Decreases pituitary FSH secretion
Ovaries	Estradiol (estrogen)	promotes uterine lining growth; female secondary sexual characteristics
	Progesterone	promote and maintain uterine lining growth

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands and Tissues		
Endocrine Gland	Hormone	Major Physiological Role
Placenta	Chorionic gonadotropin	Increases progesterone synthesis by corpus luteum
	Placental lactogen	Stimulates fetal growth and development. Increases mammary gland development in mother
Heart (Atrium)	Atrial natriuretic factor	Increases salt and water excretion by kidney to control blood volume and pressure

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands and Tissues		
Endocrine Gland	Hormone	Major Physiological Role
Gastro intestinal tract	Cholecystokinin	Stimulates secretions of pancreas. Stimulates gall bladder contraction
	Chymodenin	Stimulates secretion of chymotrypsinogen
	Gastrin	Stimulates HCl secretion in stomach
	Secretin	Stimulates secretion of bicarbonate
	Substance P	Acts as enteric neurotransmitter
	Motilin	Increases motility of intestinal villi

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands and Tissues		
Endocrine Gland	Hormone	Major Physiological Role
Most or all tissues	Leukotrienes	Control nucleotide formation
	Prostacyclins	Stimulate cAMP formation
	Prostaglandins	Stimulate cAMP formation
	Thromboxanes	Stimulates cGMP formation

Vertebrate Endocrine System: An Overview

Vertebrate Endocrine Glands and Tissues		
Endocrine Gland	Hormone	Major Physiological Role
Selected tissues	Endorphins	Produce pain killer effect
	Epidermal growth factor	Promotes epidermal cell proliferation
	Fibroblast growth factor	Promotes fibroblast proliferation
	Nerve growth factor	Promotes development of dendrites and axons
	Somatomedins	Promote cellular growth and proliferation

Vertebrate Endocrine System: An Overview

Identification of Glands

- Some glands are distinct and easily identifiable.
- However a large number of endocrine tissues are not distinct and are embedded in organs with quite different, nonendocrine functions.

END

Endocrine Physiology

Hormones and Their Properties

LECTURE NO 71

Hormones and Their Properties

Hormones

- Hormones are signaling molecules produced by endocrine glands.
- They are transported by blood to distant target organs
- They regulate they physiology and behavior of an animal.

Hormones and Their Properties

Properties of Hormones

- They act only on specific target cells or tissues.
- Hormonal action depend on the type of receptor.
- A hormone comes into contact with all tissues in the body during transport through blood, however, only the cells containing specific receptors are affected by it.

Hormones and Their Properties

Properties of Hormones

- The action of a hormone depends on the nature of enzyme cascade linked to the receptor.
- It also depends on the effector molecules expressed in a tissue.
- For these reasons, a hormone can act on two or more different types of tissues and generate different types of responses.

Hormones and Their Properties

Properties of Hormones

- Hormones are active at very low concentration.
- The amount of hormone produced by an endocrine gland is small.
- It is further diluted in the blood and interstitial fluid.
- The available conc. for the target cell lies between 10^{-8} to 10^{-12} M and the hormone is still effective.

Hormones and Their Properties

END

Properties of Hormones

- Hormonal effects are amplified, as the binding of a hormone molecule to its receptor leads to a cascade of enzymatic steps that amplify the effect.
- So just a few hormone molecules can influence thousands or millions of molecular reactions within a cell.

Endocrine Physiology

Chemical Types and Functions of

LECTURE NO 72

Chemical Types and Functions of Hormones

Chemical Types of Hormones

- Based on their structure and pathway for synthesis, hormones are divided into four groups:
  - Peptide and protein hormones
  - Amine hormones
  - Steroid hormones
  - Prostaglandins

Chemical Types and Functions of Hormones

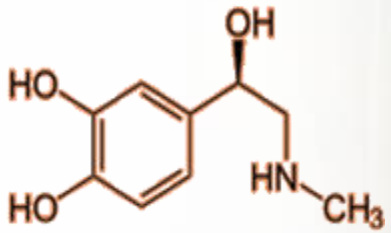
Peptide and Protein Hormones

- Composed of amino acids.
- They include very small peptides as well as large protein molecules.
- e.g. thyrotropin-releasing hormone, TRH, has only 3 amino acids and GH and prolactin with almost 200 amino acids.

Chemical Types and Functions of Hormones

Amine Hormones

- They are small molecules synthesized from a single amino acid tyrosine.
- They include:
  - Epinephrine
  - Norepinephrine
  - Thyroxine
  - Triiodothyronine

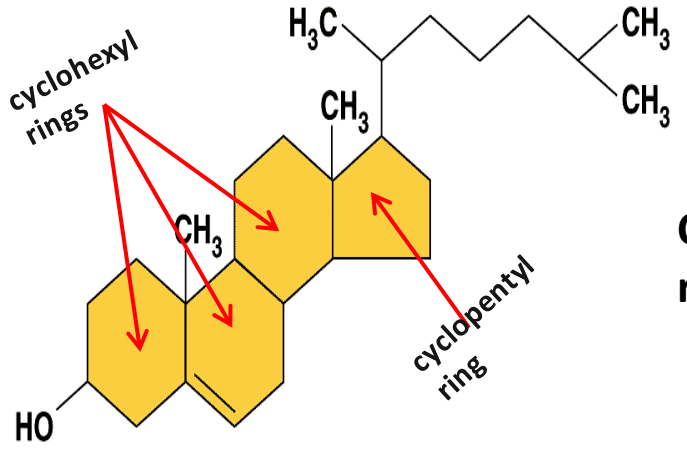


Epinephrine

Chemical Types and Functions of Hormones

Steroid Hormones

- Steroid hormones are synthesized from cholesterol.
- They consist of four fused carbon rings including three cyclohexyl rings and one cyclopentyl ring combined into a single structure.



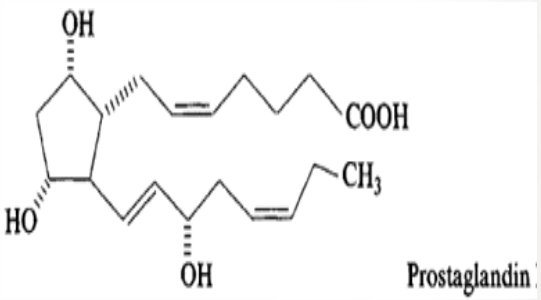
Choleste rol

- Examples: hormones of adrenal cortex (cortisol and aldosterone) hormones of ovaries (estrogen and progesterone) and hormone of testes (testosterone).

Chemical Types and Functions of Hormones

Prostaglandins

- Prostaglandins are cyclic unsaturated hydroxy fatty acids.
- They are synthesized in cellular membranes from 20-carbon fatty acid chains.



Prostaglandin

Chemical Types and Functions of Hormones

General Functions of Hormones

- Hormones coordinate long-term functions of animal tissues and organs. e.g.,
  - growth and maintenance
  - sexual activity
  - reproductive cycles
  - modification of behavior.

Chemical Types and Functions of Hormones

General Functions of Hormones

- They perform many regulatory functions. e.g.
  - osmoregulation
  - Regulation of blood sugar levels
  - control of metabolic rate

Chemical Types and Functions of Hormones

General Functions of Hormones

- Many rhythmic activities of animals are due to hormones. e.g.
  - Molting
  - Sleep-wake up cycles
  - Hunger
  - Seasonal Reproductive activations
  - Migration in birds

Chemical Types and Functions of Hormones

General Functions of Hormones

- Maintain homeostasis
- Coordinate body's responses to stress
- Mediate responses to many environmental stimuli.

Chemical Types and Functions of Hormones

END

General Functions of Hormones

- Slower, more-sustained activity of hormonal system complements the rapid-acting activity of the nervous system.
- This coordination between the two systems results in overall integration of metabolic and physiologic functions of the body.

Endocrine Physiology

Synthesis of

LECTURE NO 73

Synthesis of Hormones

Synthesis of Peptide and Protein Hormones

- Protein and peptide hormones are synthesized on the rough endoplasmic reticulum of endocrine cells.
- Mechanism of protein synthesis is followed.

Synthesis of Hormones

Synthesis of Peptide and Protein Hormones

- In the first step, they are synthesized as preprohormones which are larger proteins that are not biologically active.
- Preprohormones are cleaved to form smaller prohormones inside ER.

Synthesis of Hormones

Synthesis of Peptide and Protein Hormones

- Prohormones are then transferred to Golgi apparatus and packaged into secretory vesicles.
- In the vesicles, enzyme action cleaves them to produce smaller, biologically active hormones.

Synthesis of Hormones

Synthesis of Peptide and Protein Hormones

- The vesicles are stored within the cytoplasm or remain bound to the cell membrane until their secretion is needed.

Release of Peptide Hormones

- Release of peptide and protein hormones occurs by exocytosis.

Synthesis of Hormones

Synthesis of Steroid Hormones

- Steroid hormones are not stored in the glandular cells normally.
- They are synthesized from cholesterol instantly when a stimulus is received.
- For this purpose, cell maintains large stores of cholesterol which can be rapidly mobilized.

Synthesis of Hormones

Release of Steroid Hormones

- Steroid hormones are not stored in vesicles.
- Once they are synthesized they simply diffuse across the cell membrane and enter the blood.
- Simple diffusion out of the cell is easy because steroids are highly lipid soluble.

Synthesis of Hormones

Synthesis of Amine Hormones

- Amine hormones (thyroid and adrenal medullary hormones) are synthesized by the action of enzymes on amino acid tyrosine.
- Newly synthesized thyroid hormones are incorporated into macromolecules of the protein thyroglobulin.

Synthesis of Hormones

Synthesis of Amine Hormones

- Thyroglobulin is stored within the thyroid gland in large follicles.
- Catecholamines (epinephrine and norepinephrine) are formed in the adrenal medulla and stored in vesicles until secreted.

Synthesis of Hormones

Release of Amine Hormones

- Before secretion, the amines are split from thyroglobulin, and the freed hormones are released into the blood stream.
- After entering the blood, most of the thyroid hormones combine with plasma proteins which slowly releases them to the target tissues.

Synthesis of Hormones

Release of Amine Hormones

- Catecholamines are released from adrenal medullary cells by exocytosis.
- Once the catecholamines enter the blood, they can exist in the plasma in free form or in conjugation with other substances.

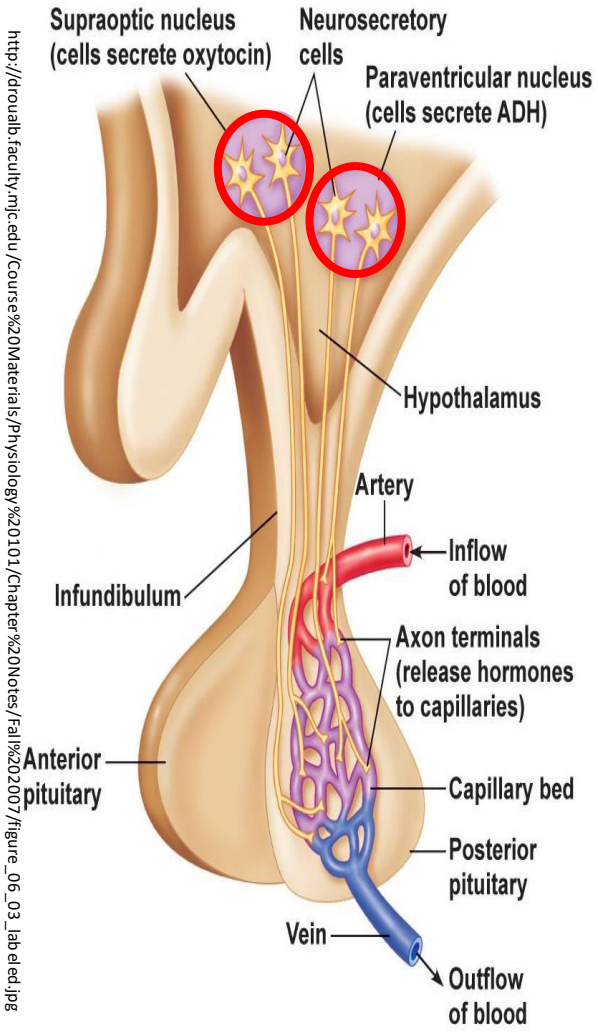
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Endocrine Physiology

Neuro-Endocrine Role of Hypothalamus

LECTURE NO 74

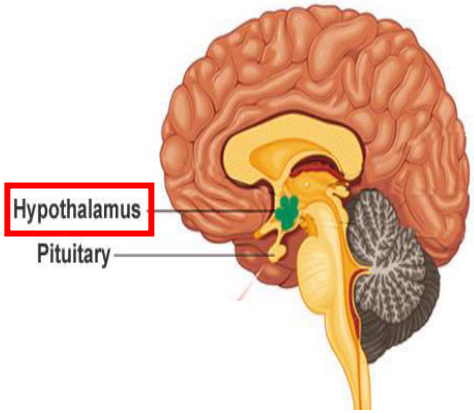
Neuro-Endocrine Role of Hypothalamus



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Hypothalamus as an Endocrine Tissue

- Part of brain having dual role as neuronal as well as endocrine tissue.
- Has specialized neuro-secretory cells which produce hormones.



Neuro-Endocrine Role of Hypothalamus

Hormones of Hypothalamus

- Two categories of hypothalamic hormones:

1. Hypophysiotropic hormones
2. Neurohypophyseal hormones

Neuro-Endocrine Role of Hypothalamus

Hypophysiotropic Hormones

- Seven hypothalamic hormones which act on anterior pituitary gland (adenohypophysis).
- They regulate secretion of adenohypophyseal hormones.
- They are of two types:

1. Hypothalamic releasing hormones (RHs)
2. Hypothalamic inhibiting hormones (RIHs)

Neuro-Endocrine Role of Hypothalamus

Hypothalamic Releasing Hormones (RHs)

- Stimulate the secretory activity of anterior pituitary. They include:

1. Corticotropin releasing hormone (CRH): stimulates ACTH release
2. GHRH: stimulates release of growth hormone
3. GnRH: stimulates the release of FSH and LH
4. TRH: stimulates TSH and prolactin secretion

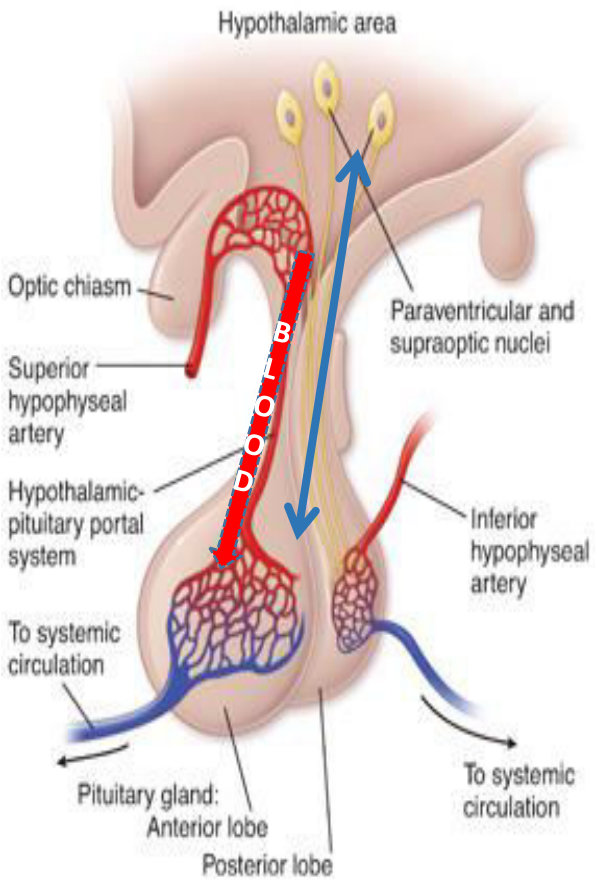
Neuro-Endocrine Role of Hypothalamus

Hypothalamic Inhibiting Hormones (RIHs)

- Inhibit the secretory activity of adenohypo-physis. They include:

1. MSH inhibiting hormone (MIH): inhibits the secretion of melanocyte stimulating hormone
2. Prolactin inhibiting hormone (PIH)
3. Somatostatin: inhibits the release of GH

Neuro-Endocrine Role of Hypothalamus

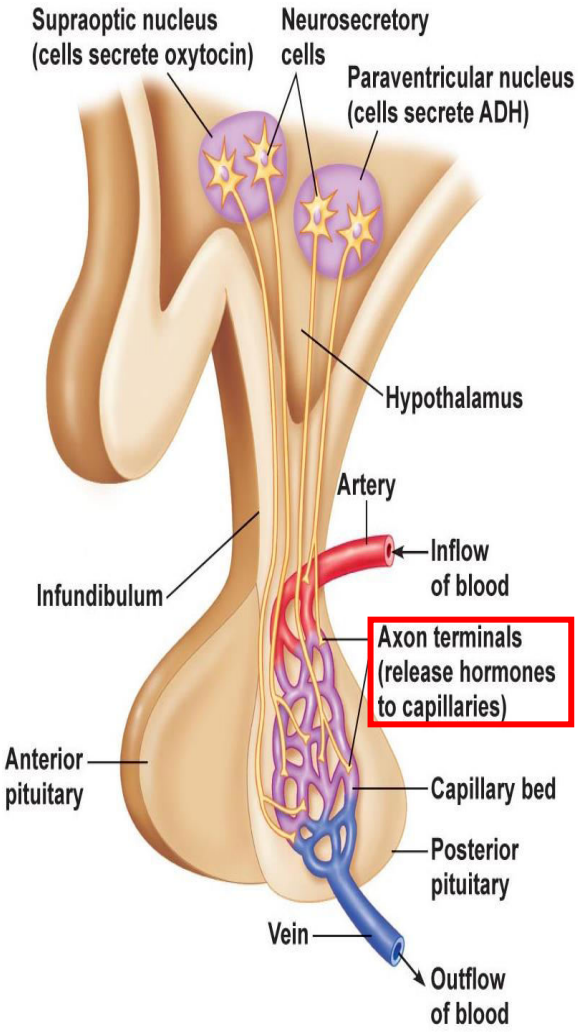


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Effective Conc. of Hypothalamic Hormones

- Hypothalamic hormones produce effect on the anterior pituitary in very low conc.
- It is b/c of close proximity and direct portal connection of hypothalamus and the anterior pituitary gland.

Neuro-Endocrine Role of Hypothalamus



http://droualb.faculty.mjc.edu/Course%20Materials/Physiology%20101/Chapter%20Notes/Fall%202007/figure_06_03_labeled.jpg

Neurohypophyseal Hormones

- These hormones include:
  - Oxytocin
  - Antidiuretic hormone (Vasopressin)
- These hormones are released directly in the posterior pituitary b/c the axon terminals of neurosecretory cells lie in the neurohypophysis.

Neuro-Endocrine Role of Hypothalamus

Neurohypophyseal Hormones

- The posterior lobe of pituitary stores these hormones and releases them as per requirement to act directly on target tissues.

END

Endocrine Physiology

Pituitary Gland and Its Hormones

LECTURE NO 75

Pituitary Gland and Its Hormones

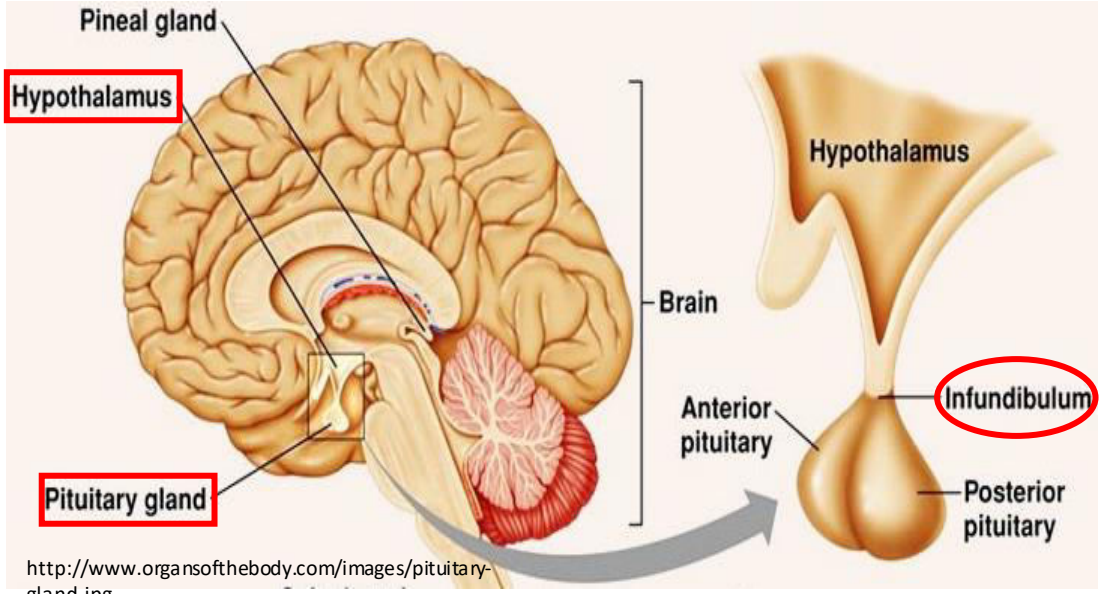
Pituitary Gland

- Also known as hypophysis.
- Often called as “master gland” secreting nine hormones.
- It hormones affect all tissues of the body, including secretions of many other endocrine glands.

Pituitary Gland and Its Hormones

Pituitary Gland

- A small gland, about 1 centimeter in diameter and 0.5 to 1 gram in weight.
- It lies below the hypothalamus as a small protrusion.
- It is connected to the hypothalamus by a pituitary stalk or infundibulum.

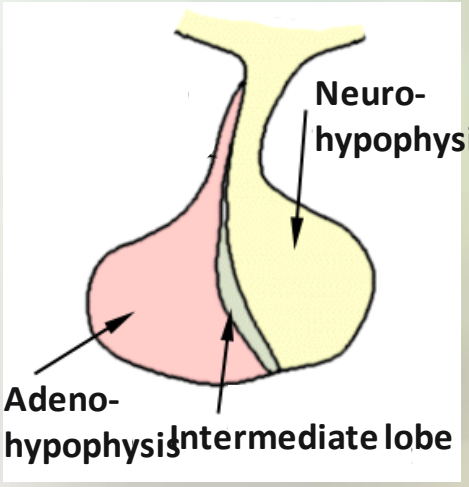


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Pituitary Gland and Its Hormones

Lobes of Pituitary Gland

- Anterior lobe (adenohypophysis)
- Intermediate lobe (*pars intermedia*)
- Posterior lobe (neurohypophysis)



Pituitary Gland and Its Hormones

Anterior Lobe (Adenohypophysis)

- The anterior pituitary is fleshy, glandular and highly vascularized in all animals.
- It contains five types of glandular cells that synthesize and secrete six hormones.
- Their secretions are controlled by regulatory hypothalamic hormones.

Pituitary Gland and Its Hormones

Glandular Cells and Hormones of Adenohypophysis

1. Somatotropes—produce growth hormone (GH)
2. Corticotropes—produce adrenocorticotropic hormone (ACTH)
3. Thyrotropes—produce thyroid-stimulating hormone (TSH)

Pituitary Gland and Its Hormones

Glandular Cells and Hormones of Adenohypophysis

4. Gonadotropes—produce gonadotropic hormones (luteinizing hormone, LH and follicle stimulating hormone, FSH)
5. Lactotropes—produce prolactin (PRL)

Pituitary Gland and Its Hormones

Adenohypophyseal Tropic Hormones

ACTH, TSH, LH and FSH are primarily tropic in their actions.

- They act on other endocrine tissues (thyroid, gonads, adrenal cortex).
- They regulate the secretory activity of these target glands.

Pituitary Gland and Its Hormones

Adenohypophyseal Gonadotropin Hormones

- LH and FSH are referred to as gonadotropins.
- They act on the gonads and stimulate secretion of gonadal hormones.

Pituitary Gland and Its Hormones

Adenohypophyseal Non-Tropic Hormones

- Growth hormone and prolactin are non-tropic hormones.
- Such hormones act directly on target tissues and produce effect.
- They do not act on other endocrine glands and do not cause the release of other hormones.

Pituitary Gland and Its Hormones

Intermediate lobe (*Pars Intermedia*)

- This lobe is avascular and almost absent in humans
- It is much developed in rodents (mice and rats) where it produces MSH.
- In humans MSH is secreted by *pars intermedia*, a part of adenohypophysis.
- Secretion of MSH is under the regulation of hypothalamic MIH.

Pituitary Gland and Its Hormones

Posterior Lobe (Neurohypophysis)

- The posterior pituitary is non-fleshy and non-glandular.
- It has neural composition and is considered as an extension of the hypothalamus.

Pituitary Gland and Its Hormones

Hormones of Neurohypophysis

- Posterior lobe releases two peptide hormones:
  - Antidiuretic Hormone (ADH) also known as Vasopressin
  - Oxytocin
- These hormones are synthesized by the cell bodies of neurons in the hypothalamus.

Pituitary Gland and Its Hormones

END

Hormones of Neurohypophysis

- ADH and oxytocin are released by axon terminals in the posterior pituitary.
- Posterior pituitary lobe stores these hormones and secretes them as per requirement.

Endocrine Physiology

Adenohypophysis: Tropic Hormones

LECTURE NO 76

Adenohypophysis: Tropic Hormones

Tropic Hormones

- Tropic hormones act on other endocrine tissues and regulate their secretory activity.

Adenohypophysis: Tropic Hormones

Tropic Hormones of Anterior Pituitary

- Adenohypophysis secretes four tropic hormones:

1. Adrenocorticotropic hormone (ACTH)
2. Thyroid stimulating hormone (TSH)
3. Luteinizing Hormone (LH)
4. Follicle stimulating hormone (FSH)

Adenohypophysis: Tropic Hormones

Adrenocorticotropic Hormone (ACTH)

- ACTH is a peptide hormone comprising single chain of 39 amino acids.
- Its principal function is to regulate secretion of corticosteroid hormones by the cortex of adrenal gland.

Adenohypophysis: Tropic Hormones

Adrenocorticotropic Hormone (ACTH)

- It also stimulates secretion of androgens by adrenal cortex.
- It plays role to maintain the size of endocrine tissues (zona fasciculata and zona reticularis) of adrenal cortex.

Adenohypophysis: Tropic Hormones

Adrenocorticotropic Hormone (ACTH)

- It also stimulates lipoprotein uptake into cortical cells for the biosynthesis of cholesterol and steroid hormones.
- In many organisms, ACTH also plays role in circadian rhythms.

Adenohypophysis: Tropic Hormones

Thyroid Stimulating Hormone (Thyrotropin)

- Thyrotropin or TSH is a glycoprotein.
- Its production is stimulated by thyrotropin-releasing hormone (TRH) from the hypothalamus.
- TSH is secreted throughout life but its levels are high during rapid growth and development.

Adenohypophysis: Tropic Hormones

Thyroid Stimulating Hormone (Thyrotropin)

- It acts on follicular cells of thyroid gland and helps to maintain their size.
- It controls the rate of secretion of thyroid hormones thyroxine (T_4) and triiodothyronine (T_3)
- These hormones, in turn, control the rate of metabolic reactions in the body.

Adenohypophysis: Tropic Hormones

Gonadotropins (LH and FSH)

- Both these hormones are glycoproteins.
- They are released under the influence of Gonadotropin-Releasing Hormone (GnRH) from hypothalamus.
- They work together in the reproductive systems of both males and females.

Adenohypophysis: Tropic Hormones

Gonadotropins (LH and FSH)

- They act on the gonads and control the growth of ovaries and testes.
- They also stimulate the secretion of gonadal hormones.
- FSH stimulates the development of ovarian follicles and regulates spermatogenesis in the testes.

Adenohypophysis: Tropic Hormones

Gonadotropins (LH and FSH)

- LH stimulates production of estrogen and progesterone by ovary.
- It causes ovulation and formation of corpus luteum.
- In males, LH is also called as ICSH (interstitial cell stimulating hormone).
- It stimulates testosterone production by the Leydig cells of testes.

END

Endocrine Physiology

Adenohypophysis: Non-Tropic Hormones

LECTURE NO 77

Adenohypophysis: Non-Tropic Hormones

Non-Tropic Hormones of Adenohypophysis

- Adenohypophysis secretes three hormones that act directly on their target tissues. These are:

1. Melanocyte stimulating hormone (MSH)
2. Prolactin (PRL)
3. Growth hormone (somatotropin)

Adenohypophysis: Non-Tropic Hormones

Melanocyte Stimulating Hormone (MSH)

- MSH is a peptide hormone.
- In lower vertebrates, large amounts of MSH are released from highly developed intermediate lobe of pituitary.
- In humans, MSH is released from *pars intermedia* of anterior pituitary in extremely low quantities.

Adenohypophysis: Non-Tropic Hormones

Melanocyte Stimulating Hormone (MSH)

- MSH regulates the activity of pigment-containing cells (melanocytes) in the skin of many vertebrates.
- It stimulates synthesis of black pigment melanin and its dispersal in melanocytes, leading to darkening of the skin.

Adenohypophysis: Non-Tropic Hormones

Melanocyte Stimulating Hormone (MSH)

- In humans most of skin pigmentation is controlled by ACTH.
- In mammals, MSH also inhibits hunger by acting on neurons in the brain.
- It is also involved in fat metabolism in mammals.

Adenohypophysis: Non-Tropic Hormones

Prolactin (PRL)

- PRL has diverse effects in different vertebrates.
- In birds, it regulates fat metabolism and reproduction.
- In amphibians, it delays metamorphosis.
- In freshwater fishes, it regulates salts and water.
- In mammals, it stimulates mammary gland growth and milk production.

Adenohypophysis: Non-Tropic Hormones

Growth Hormone (GH)

- Also called somatotropin & somatotropic hormone
- It is a protein that contains 191 amino acids in a single chain.

Control of GH Secretion

- Production and release of GH is under the control of GH releasing hormone (GHRH) and somatostatin (GH-inhibiting hormone GIH).

Adenohypophysis: Non-Tropic Hormones

Effects of GH

- Growth hormone exerts both metabolic and developmental effects.

Metabolic Effects of GH

- Induces the mobilization of stored fat for energy metabolism.
- Stimulates fatty acid uptake in muscles.
- Decreases rate of glucose utilization causing an elevation of blood glucose.

Adenohypophysis: Non-Tropic Hormones

Metabolic Effects of GH

- Increases rate of protein synthesis in most cells of the body .
- Stimulates insulin secretion both directly, through its action on the pancreatic beta cells (tropic effect of GH), and indirectly, through its effect in elevating plasma glucose levels.

Adenohypophysis: Non-Tropic Hormones

Metabolic Effects of GH

- Overall, GH enhances body proteins, uses up fat stores, and conserves carbohydrates.

Developmental Effects of GH

- Causes growth of almost all tissues of the body.
- Stimulates RNA and protein synthesis, promoting the growth of tissues, particularly cartilage and bone.

Adenohypophysis: Non-Tropic Hormones

Developmental Effects of GH

- Tissue growth due to GH occurs by an increase in cell number (i.e., cell proliferation) rather than an increase in cell size.
- GH works synergistically with thyroid hormones to promote tissue growth during development.

Adenohypophysis: Non-Tropic Hormones

Developmental Effects of GH

- GH also stimulates liver to produce growth-promoting factors e.g. insulin-like growth factors (IGFs) or somatomedins that act directly on cells to promote growth.

Adenohypophysis: Non-Tropic Hormones

Developmental Abnormalities Due to GH

- Disturbances in the secretion of GH lead to abnormal growth and development in humans:

Gigantism

- Hypersecretion of GH before puberty causes excessive size and stature.

Adenohypophysis: Non-Tropic Hormones

Developmental Abnormalities Due to GH

Acromegaly

- Hypersecretion of GH after maturity causes enlargement of bones of the head and limbs.

Dwarfism

- Insufficient secretion of GH during childhood and adolescence causes underdevelopment of the body (short stature).

END

Endocrine Physiology

Hormones of Neurohypophysis

LECTURE NO 78

Hormones of Neurohypophysis

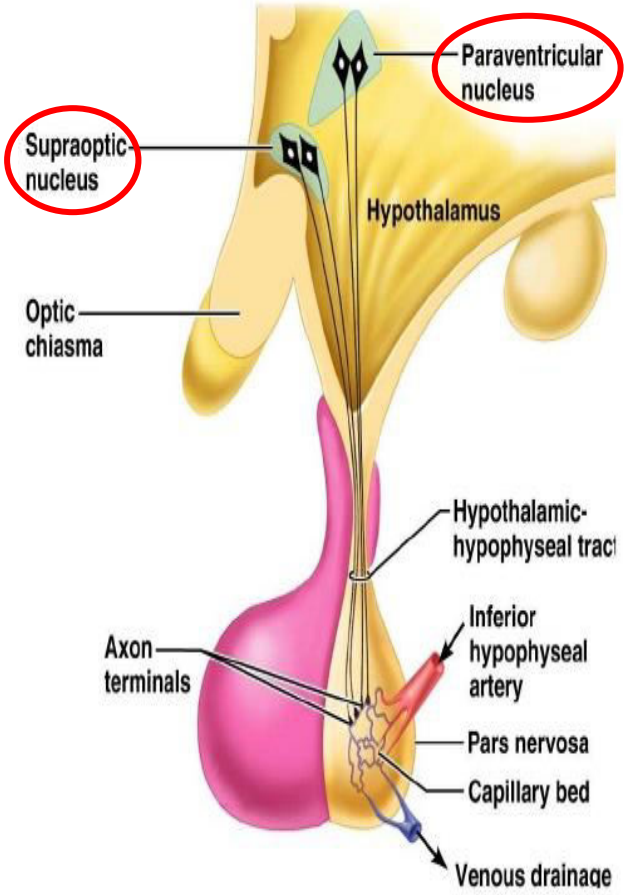
Neurohypophysis

- The posterior lobe of pituitary is also called as neurohypophysis or *pars nervosa*.
- It stores and releases two neurohormones:

1. Oxytocin
2. Antidiuretic hormone (ADH) or vasopressin

- Both hormones are peptides, containing nine amino acid residues.

Hormones of Neurohypophysis



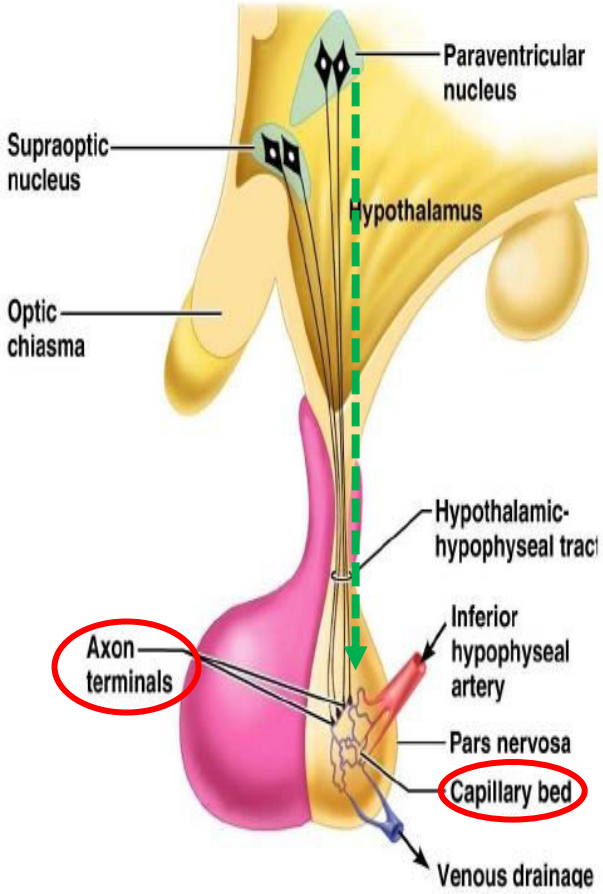
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Synthesis of ADH and Oxytocin

- Neurohypophyseal hormones are synthesized and packaged in the cell bodies of two groups of neurosecretory cells in the anterior portion of the hypothalamus:

1. Supraoptic nuclei
2. Paraventricular nuclei

Hormones of Neurohypophysis



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Release of ADH and Oxytocin

- After their synthesis, these hormones are transported within the axons of the hypothalamo-hypophyseal tract to nerve terminals in the neurohypophysis, where they are released into a capillary bed.

Hormones of Neurohypophysis

Functions of ADH

- It promotes water reabsorption in the kidneys and decreases urine volume, helping to regulate blood osmolarity.
- It also acts as a vasoconstrictor, increasing the arterial pressure. For this reason, it is named vasopressin.

Hormones of Neurohypophysis

END

Functions of Oxytocin

- In mammals, oxytocin stimulates uterine contractions during parturition.
- It stimulates release of milk from the mammary glands during nursing.
- It functions in regulating mood and sexual arousal in both males and females.
- In birds, it stimulates motility of the oviduct.

Endocrine Physiology

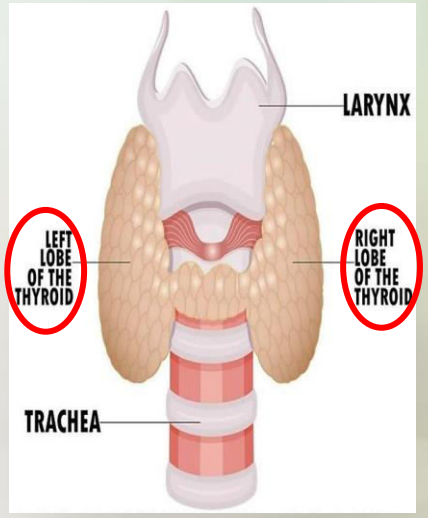
Thyroid Gland and Its Hormones

LECTURE NO 79

Thyroid Gland and Its Hormones

Thyroid Gland

- Thyroid gland lies on the ventral surface of trachea, just below the Adam's apple.
- It consists of two lobes: right and left.



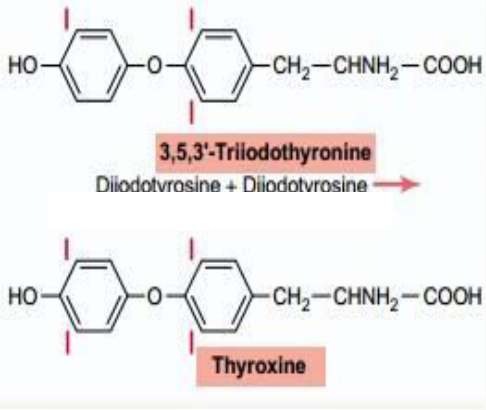
Thyroid Gland and Its Hormones

Glandular Cells of Thyroid

- Thyroid has two types of glandular cells:

1. Follicular cells which secrete two hormones:
   - Triiodothyronine (T_3)
   - Tetraiodothyronine (T_4) also known as thyroxine
2. Parafollicular cells which secrete one hormone:
   - Calcitonin

Thyroid Gland and Its Hormones



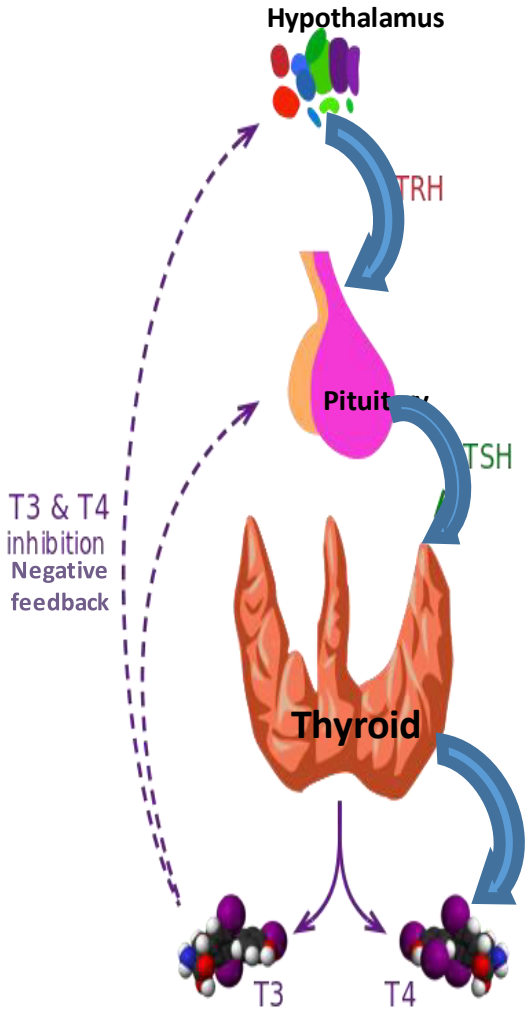
Secretion of T_3 and T_4

- The secretion of T_4 predominates T_3.
- T_3 is more active form and carries out the major functional roles.
- So, the target cells convert most of T_4 to T_3.

Thyroid Gland and Its Hormones

Secretion of T_3 and T_4

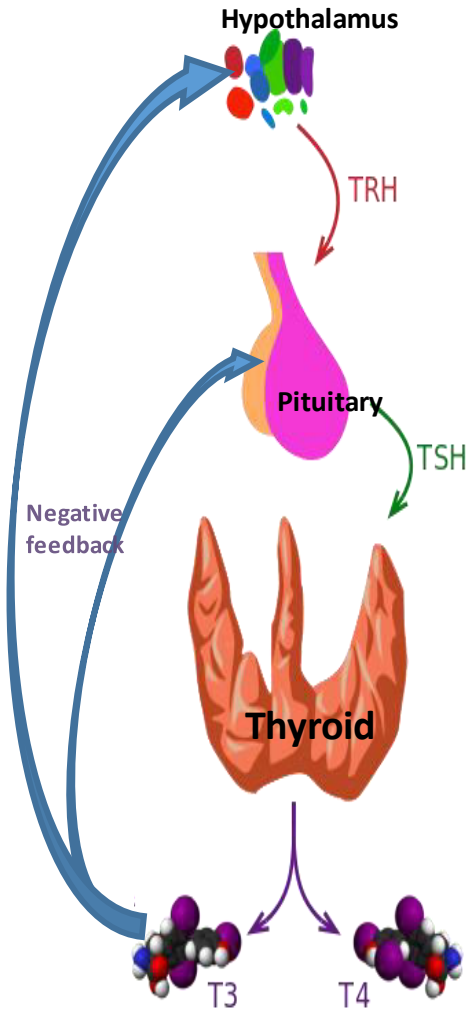
- Stimulus for the release of thyroid hormones is the TSH from anterior pituitary.
- TSH is released on the stimulus from hypothalamic TRH.
- Stimuli for this +ve feedback loop are stress, cold, low skin temp. and low metabolic rate.



Thyroid Gland and Its Hormones

Feedback Regulation of T_3 and T_4

- High conc. of T_3 & T_4 feedback negatively to the hypothalamus and pituitary.
- This causes reduction in secretion of TRH and TSH.
- As an effect, release of thyroid hormones is reduced.



Thyroid Gland and Its Hormones

Functions of T_3 and T_4

- These hormones have diverse effects on the physiology of virtually all tissues of the body.
- Their roles can be grouped into two categories:
  - Metabolic roles
  - Developmental roles

Thyroid Gland and Its Hormones

Metabolic Roles of T_3 & T_4

- They stimulate cellular respiration, oxygen consumption and metabolic rate and, in turn, heat production. (important in thermo-regulation).
- They sensitize some tissues to epinephrine that helps to maintain normal blood pressure, heart rate and muscle tone.

Thyroid Gland and Its Hormones

Metabolic Roles of T_3 & T_4

- They also promote normal motility of gastrointestinal tract.
- They also regulate reproductive functions.

Thyroid Gland and Its Hormones

Developmental Roles of T_3 & T_4

- They significantly affect the development and maturation of vertebrate animals.
- They are involved in the normal functioning of bone-forming cells and the branching of nerve cells during embryonic development of the brain.

Thyroid Gland and Its Hormones

Developmental Roles of T_3 & T_4

- They control metamor-phosis of a tadpole larva into adult frog.
- Developmental effects of growth hormone occur only in the presence of thyroid hormones.

Thyroid Gland and Its Hormones

Hyperthyroidism

- Excessive secretion of thyroid hormones as in Graves' disease.
- It is an autoimmune disorder in which antibodies that mimic TSH bind to the receptors for TSH and cause sustained thyroxine production.

Thyroid Gland and Its Hormones

Hyperthyroidism

- It leads to:
  - high body temp.
  - profuse sweating
  - weight loss
  - Irritability
  - high blood pressure
  - protruding eyes (exophthalmia)

Thyroid Gland and Its Hormones

Hypothyroidism

- Characterized by low production of throxine.
- It results from the lack of dietary iodine.
- It is represented by two types of diseases:
  - Cretinism
  - Goiter
- Cretinism results from iodine deficiency during early stages of development.

Thyroid Gland and Its Hormones

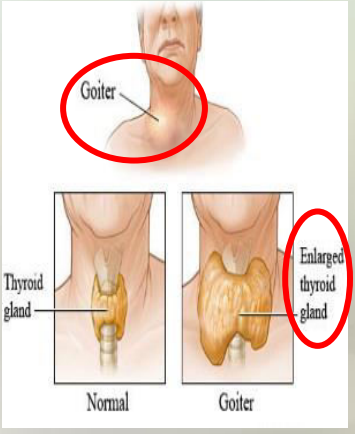
Hypothyroidism

- In cretinism somatic, neural and sexual development is severely retarded, metabolic rate and resistance to infection is also reduced.
- Inadequate production of thyroxine in adults leads to goiter.

Thyroid Gland and Its Hormones

Hypothyroidism

- In goiter, TSH is produced excessively that causes over-stimulation of thyroid gland resulting in its enlargement (hypertrophy).



Thyroid Gland and Its Hormones

Calcitonin

- A protein hormone secreted by the parafollicular cells of thyroid gland.
- It is released in response to hypercalcemia.
- It is an important hormone for calcium metabolism and calcium homeostasis.

Thyroid Gland and Its Hormones

Calcitonin

- It promotes calcium deposition in bone matrix.
- It suppresses Ca^{2+} loss from bones.
- It enhances Ca^{2+} excretion by the kidneys.

END

Endocrine Physiology

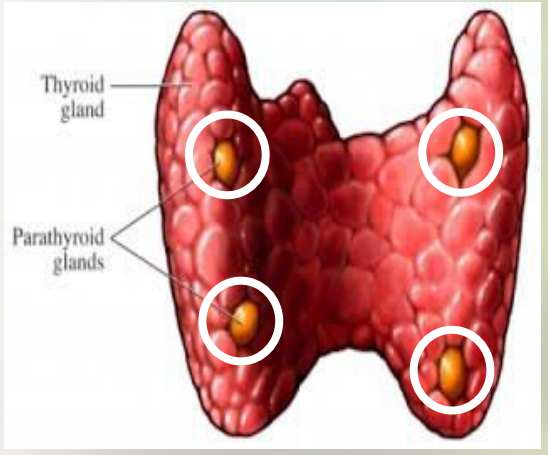
Parathyroid Gland and Its Hormones

LECTURE NO 80

Parathyroid Gland and Its Hormones

Parathyroid Glands

- The parathyroid glands are tiny, pea-sized glands embedded in the thyroid lobes.
- In each lobe, two glands are embedded.



Parathyroid Gland and Its Hormones

Parathormone (PTH)

- The parathyroid glands secrete parathormone (PTH).
- It is released in response to a drop in plasma Ca^{2+} levels.
- It regulates the conc. of calcium and phosphate ions in the blood.
- It increases plasma Ca^{2+} by promoting Ca^{2+} mobilization from bone.

Parathyroid Gland and Its Hormones

Parathormone (PTH)

- It promotes calcium reabsorption by kidney tubules and decrease the amount of calcium excreted in the urine.
- It enhances calcium absorption from small intestine into the blood.
- It regulates phosphate ions in the blood by their absorption into the bones and enhances their renal excretion.

Parathyroid Gland and Its Hormones

Role of Calcitriol in the Action of PTH

- PTH stimulates 1α,25-hydroxylase activity in kidney that stimulates production of calcitriol, the active form of vit. D.
- Calcitriol works with PTH to stimulate Ca^{2+} reab-sorption and phosphate excretion through kidney
- It also helps in release of Ca^{2+} from bone and its absorption from the gut.

Parathyroid Gland and Its Hormones

PTH Over Secretion

- Bones become soft and deformed and prone to fracture due to release of calcium from bones.
- Elevation of blood Ca^{2+} levels (hypercalcemia).
- Suppression of nervous system and weakness of muscles due to hypercalcemia.
- Stone formation in kidneys due to excess Ca^{2+} salts precipitation.

Parathyroid Gland and Its Hormones

PTH Under Secretion

- Hypocalcemia.
- Increased excitability of neurons.
- Muscle tetany due to which muscles remain in contracted state.

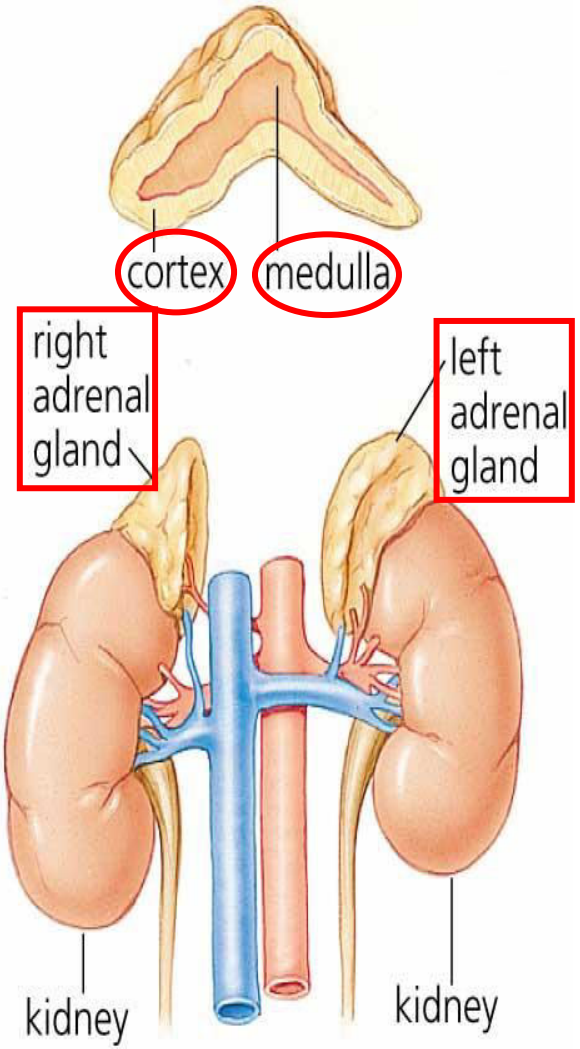
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Endocrine Physiology

Adrenal Glands

LECTURE NO 81

Adrenal Glands



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Adrenal Glands

- Mammals have two adrenal glands, one attached to the upper end of each kidney.
- Each adrenal gland is in fact two glands in one:
- Adrenal cortex which forms an outer layer.
- Adrenal medulla which forms inner, central portion.

Adrenal Glands

Adrenal Cortex and Medulla

- The two portions of the adrenals have different cell types, functions, and embryonic origins.
- The cells of cortex are true endocrine cells, derived from non-neural, mesodermal tissue.

Adrenal Glands

Adrenal Cortex and Medulla

- Cells of medulla are derived from epidermal, neural tissue during embryonic development.
- They are functionally related to the sympathetic nervous system.
- Thus, like pituitary gland, each adrenal gland is a fused endocrine and neuroendocrine gland.

Adrenal Glands

Hormones of Adrenal Cortex

- Adrenal cortex produces two major types of steroid hormones:
  - Mineralocorticoids
  - Glucocorticoids
- They are collectively called as corticosteroids.
- These hormones are involved in blood ion and glucose regulation and anti-inflammatory reactions.

Adrenal Glands

Hormones of Adrenal Cortex

- Adrenal cortex also secretes small amounts of sex hormones, especially androgenic hormones.
- These androgens exhibit about the same effects in the body as the male sex hormone testosterone.

Adrenal Glands

Hormones of Adrenal Medulla

- The cells of the adrenal medulla synthesize and secrete catecholamines, i.e. epinephrine and norepinephrine.
- These hormones are released under visceral motor stimulation.

Adrenal Glands

Chromaffin Cells

- The cells of adrenal medulla are modified postganglionic sympathetic neurons.
- They are known as chromaffin cells because they stain easily with chromium salts.

Adrenal Glands

Chromaffin Cells

- Chromaffin cells that produce norepinephrine have dark staining irregular granules.
- Chromaffin cells that produce epinephrine have light-staining, spherical granules.

Effort by: AARISH ALI

END

Endocrine Physiology

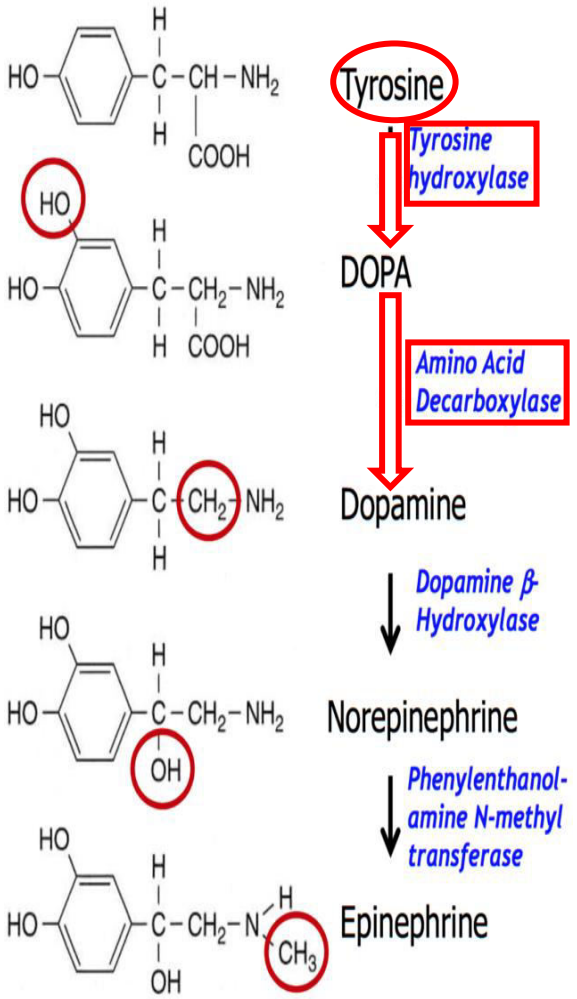
Catecholamines: Synthesis and Release

LECTURE NO 82

Catecholamines: Synthesis and Release

Synthesis of Catecholamines

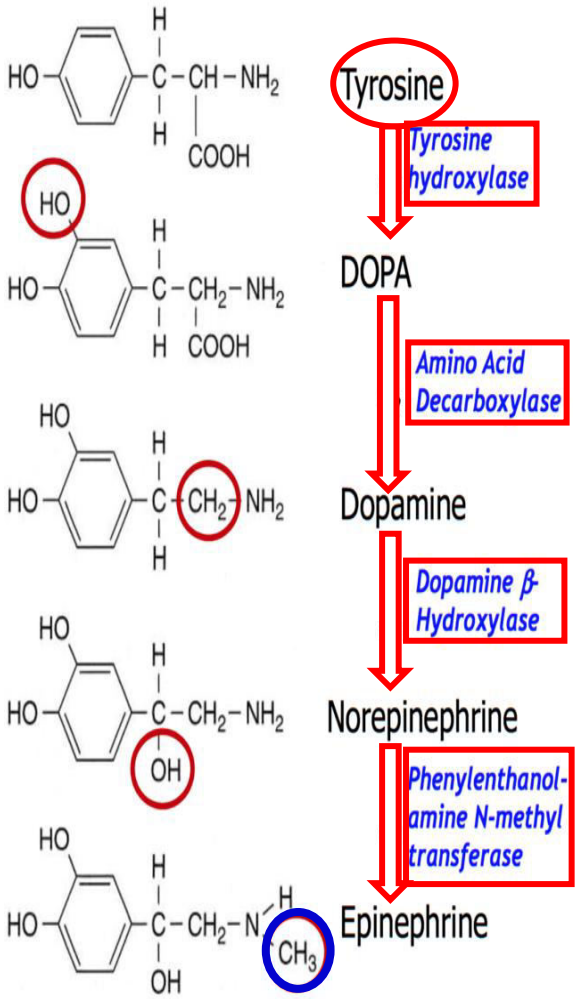
- Norepinephrine is synthesized from tyrosine.
- Dopa and dopamine are formed as intermediate compounds.
- Conversion of tyrosine to dopa and dopamine occurs in the cytosol.
- The reactions are catalyzed by tyrosine hydroxylase and dopa decarboxylase enzymes.



Catecholamines: Synthesis and Release

Synthesis of Catecholamines

- Dopamine is incorporated into the granules where it is converted to norepinephrine
- This reaction is catalyzed by dopamine β-hydroxylase in the secretory granules.
- Norepinephrine is methylated to epinephrine.
- This reaction is catalyzed by phenylethanolamine N-methyl transferase.



Catecholamines: Synthesis and Release

Synthesis of Catecholamines

- Phenylethanolamine N-methyl transferase, is found in the cytosol.
- Thus, norepinephrine must come out of the secretory granules to be converted to epinephrine.
- Epinephrine re-enters the granules.

Catecholamines: Synthesis and Release

Release of Catecholamines

- Catecholamines are released as secretory granules by exocytosis from chromaffin cells .
- Each chromaffin cell secretes either norepi-nephrine or epinephrine.

Catecholamines: Synthesis and Release

Release of Catecholamines

- The granules also contain enkephalin, ATP, and acidic proteins chromogranins to which the catecholamines are bound.
- Once a pore is opened in the vesicle, the catecholamines diffuse out, liberating from the chromogranins.

Catecholamines: Synthesis and Release

Catecholamine Release Mechanism

- Release of epinephrine and norepinephrine is controlled by the action of preganglionic sympathetic nerves.
- These nerves release acetylcholine as neurotransmitter.
- ACh increases chromaffin cells' conductance for Ca^{2+}.

Catecholamines: Synthesis and Release

Catecholamine Release Mechanism

- Increased conductance results in influx of Ca^{2+} and elevation of intracellular Ca^{2+} levels.
- Rise in intracellular Ca^{2+} causes exocytosis of both epinephrine and norepinephrine.

Catecholamines: Synthesis and Release

Positive Feedback of Catecholamine Release

- Release of catecholamines causes an increase in blood flow to the adrenals which enhances catecholamine release.
- Thus catecholamine releases has a positive feedback on further catecholamine release.

Catecholamines: Synthesis and Release

END

Negative Feedback Controls Release

- ATP is released in granules containing catecholamines.
- ATP and its breakdown product adenosine, reduces calcium influx, which inhibits release of catecholamines
- This provides negative-feedback control on catecholamine release.

Endocrine Physiology

Catecholamines: Effects and Mode of Action

LECTURE NO 83

Catecholamines: Effects and Mode of Action

Effects of Catecholamines

- They have numerous cardiovascular and metabolic effects.
- They affect contraction of smooth muscles and induce vasoconstriction.
- They stimulate glycolysis and lipolysis.

Catecholamines: Effects and Mode of Action

Effects of Catecholamines

- They stimulate the sympathetic nervous system for fight-or-flight response.
- In this response, various tissues are activated and the body is prepared for emergency to either attack or flee from the objects of stress.

Catecholamines: Effects and Mode of Action

Effects of Catecholamines

- The fight-or-flight effects include:
  - dilating the pupil
  - increasing heart rate
  - mobilizing energy
  - diverting blood flow to skeletal muscles.

Catecholamines: Effects and Mode of Action

Adrenergic Receptors (Adrenoreceptors)

- These hormones bind to a class of G protein-coupled receptors, which are present on cell membranes of various tissues of the body.
- There are two groups of adrenoreceptors: α & β.
- Each group has two subtypes in different tissues: α_1, α_2, and β_1, β_2.

Catecholamines: Effects and Mode of Action

Mode of Action Through α_1-Adrenoreceptors

- α_1-adrenoreceptors are coupled with an inhibitory G protein.
- They mediate smooth muscle contraction in many tissues.
- Stimulation of these receptors results in a decrease in cAMP and activation of inositol trisphosphate ($InsP_3$) pathway.

Catecholamines: Effects and Mode of Action

Mode of Action Through α_1-Adrenoreceptors

- Activation of $InsP_3$ pathway leads to elevation of intracellular $InsP_3$.
- Elevated $InsP_3$ causes release of Ca^{2+} from stores within the cell.
- The resulting rise in cytosolic Ca^{2+} causes muscle contraction.

Catecholamines: Effects and Mode of Action

Mode of Action Through α_2-Adrenoreceptors

- α_2-adrenoreceptors are located in presynaptic cells at noradrenergic synapses.
- They are also located on some postsynaptic sites in liver, brain, and smooth muscles.

Catecholamines: Effects and Mode of Action

Mode of Action Through α_2-Adrenoreceptors

- Their stimulation causes autoinhibition of norepinephrine release through –ve feedback.
- This action is mediated by an inhibitory effect of these receptors on adenylate cyclase.

Catecholamines: Effects and Mode of Action

Mode of Action Through β_1-Adrenoreceptors

- β_1-adrenoreceptors are stimulated due to neuronal release of norepinephrine.
- These receptors are coupled to a stimulatory G protein.
- Through these receptors, catecholamines act by activating adenylate cyclase, leading to an increase in cAMP.

Catecholamines: Effects and Mode of Action

Mode of Action Through β_1-Adrenoreceptors

- The elevation of cAMP increases calcium conductance, thereby raising the intracellular calcium level.
- This results in increased contraction of cardiac muscles and release of fatty acids from adipose tissue.

Catecholamines: Effects and Mode of Action

Mode of Action Through β_2-Adrenoreceptors

- β_2-adrenoreceptors are stimulated due to higher levels of circulating catecholamines.
- These receptors are also coupled to a stimulatory G protein that causes elevation of cAMP.
- Here, the cAMP causes activation of Ca^{2+} pump rather than an increase in Ca^{2+} conductance.

Catecholamines: Effects and Mode of Action

Mode of Action Through β_2-Adrenoreceptors

- Ca^{2+} pump stimulates sequestering of Ca^{2+} ions by mitochondria and ER.
- It also stimulates Ca^{2+} extrusion from the cell.
- As a result, cytosolic calcium levels fall, promoting smooth muscle relaxation.
- This results in broncho-dilation and vasodilation.

END

Endocrine Physiology

Adrenal Cortex: Corticosteroids

LECTURE NO 84

Adrenal Cortex: Corticosteroids

Hormones of Adrenal Cortex

- Adrenal cortex synthesizes and secretes a family of steroid hormones collectively called corticosteroids.
- Corticosteroids belong to three functional categories:
  - Glucocorticoids
  - Mineralocorticoids
  - Reproductive hormones

Adrenal Cortex: Corticosteroids

Glucocorticoids

- Glucocorticoids include: cortisol, cortisone and corticosterone.
- They have a primary effect on glucose metabolism.
- They are involved in gluconeogenesis under stress, starvation and disease.

Adrenal Cortex: Corticosteroids

Glucocorticoids' Role in Gluconeogenesis

- Glucocorticoids act on liver and increase the synthesis of enzymes that promote gluconeogenesis.
- The glucose thus produced causes a rise in blood glucose levels.
- They reduce uptake of glucose by peripheral tissues such as muscles.

Adrenal Cortex: Corticosteroids

Glucocorticoids' Role in Gluconeogenesis

- They also reduce uptake of amino acids by muscles, increasing the available amino acids for deamination and conversion to glucose.
- They also stimulate mobilization of fatty acids from stores of fat in adipose tissue, which can be used for gluconeogenesis.

Adrenal Cortex: Corticosteroids

Glucocorticoids' Role in Gluconeogenesis

- All these mechanisms of gluconeogenesis are important during starvation and stress as they make quick energy available to muscles and critical tissues e.g. brain.
- The end result of this process is the degradation of tissue proteins and stored fat deposits.

Adrenal Cortex: Corticosteroids

Other Roles of Glucocorticoids

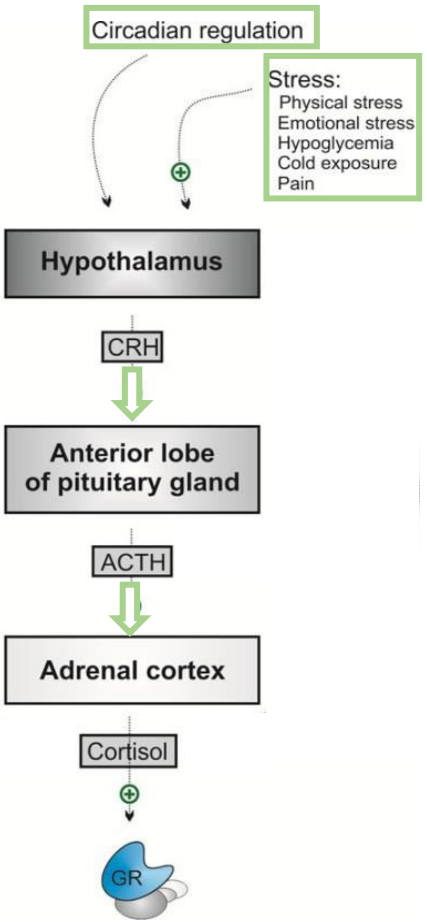
- They stimulate gastric secretions.
- They suppress certain components of immune system and act as anti-inflammatory agents.

Adrenal Cortex: Corticosteroids

Mode of Action of Glucocorticoids

- The glucocorticoids, like other steroid hormones, bind to specific receptors in the cytosol, forming hormone-receptor complexes.
- These complexes enter the nucleus and regulate the transcription of specific genes.

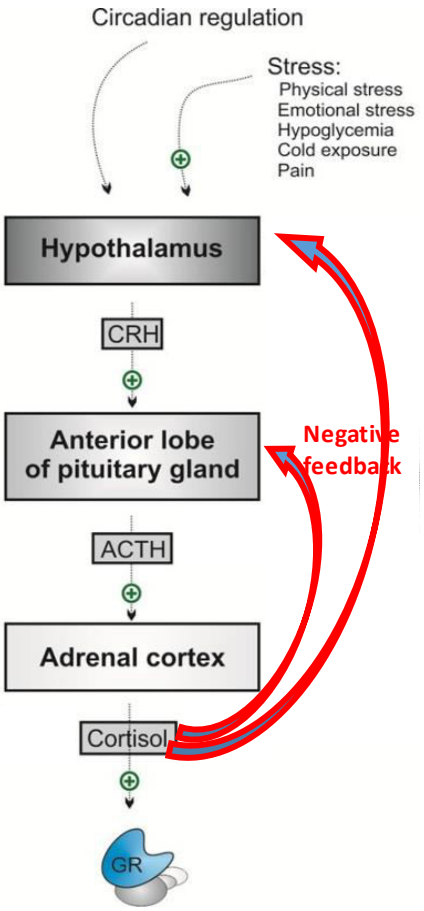
Adrenal Cortex: Corticosteroids



Secretion of Glucocorticoids

- Glucocorticoid secretion is stimulated by CRH and ACTH release which are secreted in response to stress and starvation.
- An endogenous biological clock in a diurnal rhythm causes their maximal release just prior to waking to activate energy-mobilizing actions.

Adrenal Cortex: Corticosteroids



Regulation of Glucocorticoids

- Glucocorticoid secretion is regulated by negative feedback by the hormones themselves on:
- CRH-secreting neurons of the hypothalamus
- ACTH-secreting cells of the anterior pituitary.

Adrenal Cortex: Corticosteroids

Mineralocorticoids

- Mineralocorticoids have effects on mineral metabolism.
- They act principally in maintaining salt and water balance in the body.
- The primary mineralocorticoid hormone is aldosterone.

Adrenal Cortex: Corticosteroids

Aldosterone

- It is secreted under the stimulation of ACTH and angiotensin II (produced during low blood pressure or volume).
- High blood K^+ also stimulates aldosterone production.

Adrenal Cortex: Corticosteroids

Effects of Aldosterone

- It functions in ion and water homeostasis .
- It enhances reabsorption of water and Na^+ and Cl^- ions from the filtrate in the kidneys.
- This contributes to raise blood pressure and volume.
- It slso functions in body's response to severe stress.

Adrenal Cortex: Corticosteroids

END

Adrenal Reproductive Hormones

- Adrenal cortex produces small amounts male and female sex hormones i.e. androgens, estrogens and progestins.
- Androgens secreted by adrenal cortex include:

1. Testosterone
2. Dihydrotestosterone
3. Androstenedione
4. Dehydroepiandrosterone (DHEA).

Endocrine Physiology

Pancreas: Insulin and Glucagon Hormones

LECTURE NO 85

Pancreas: Insulin and Glucagon Hormones

Pancreas

- Pancreas is a dual gland having exocrine as well as endocrine roles.
- It is composed of two types of tissues:
  - Acini which are exocrine.
  - Islets of Langerhans which are endocrine.

Pancreas: Insulin and Glucagon Hormones

Islets of Langerhans

- The islets contain four types of cells:

1. Beta Cells

- constitute 60% of the Islets.
- They secrete insulin

2. Alpha Cells

- constitute 25% of the islets
- They secrete glucagon

Pancreas: Insulin and Glucagon Hormones

Islets of Langerhans

3. Delta Cells

- constitute 10% of islets.
- Secrete somatostatin that inhibits the secretion of insulin and glucagon.

4. PP Cells

- Present in small number.
- Secrete a hormone of uncertain function called pancreatic polypeptide.

Pancreas: Insulin and Glucagon Hormones

Insulin Releasing Stimuli

- Major stimulus for the beta cells to secrete insulin is high blood glucose level.
- Release of insulin is also stimulated by glucagon, GH, gastric inhibitory peptide (GIP, also known as glucose-dependent insulin-releasing peptide), epinephrine, and elevated levels of amino acids.

Pancreas: Insulin and Glucagon Hormones

Effects of Insulin

- Insulin has major effects on carbohydrate, fat and protein metabolism.
- Insulin has two major actions on carbohydrate metabolism which reduce glucose level in blood.

1. Increasing rate of uptake of glucose by the cells of liver, muscles and adipose tissue.
2. Stimulating glycogenesis

Pancreas: Insulin and Glucagon Hormones

Effects of Insulin

- Insulin affects lipid metabolism by stimulating lipogenesis in liver and adipose tissue.
- It affects protein metabolism by stimulating the uptake of amino acids into liver and muscles.

Pancreas: Insulin and Glucagon Hormones

Deficiency of Insulin: *Diabetes mellitus*

- An absolute deficiency of insulin due to loss of pancreatic beta-cells causes *Diabetes mellitus* Type-I.
- A relative deficiency of insulin associated with defective insulin receptors causes *Diabetes mellitus* Type-II.

Pancreas: Insulin and Glucagon Hormones

Symptoms of Diabetes

- Diabetes lead to:
- Hyperglycemia
- Glycosuria (spillover of glucose into the urine)
- Reduced ability to utilize glucose by the cells.
- Reduced ability to synthesize lipids and proteins.
- Accumulation of non-metabolized fat particles in the blood as ketone bodies.

Pancreas: Insulin and Glucagon Hormones

Symptoms of Diabetes

- The disturbances in carbohydrate, lipid, and protein metabolism produce a large number of complications in various organs. e.g.,
  - Cataract
  - Cardiovascular diseases
  - Renal diseases

Pancreas: Insulin and Glucagon Hormones

Glucagon

- Glucagon is secreted in response to hypoglycemia.
- It increases glucose level in blood by:
  - Stimulating glycogenolysis in the liver
  - Stimulating lipolysis, providing lipids for gluconeogenesis

Pancreas: Insulin and Glucagon Hormones

Antagonistic Actions of Insulin and Glucagon

- The actions of insulin and glucagon are antagonistic to each other.
- This is important in maintaining an appropriate glucose level in blood.

END

Endocrine Physiology

Role of Testes as Endocrine Tissue

LECTURE NO 86

Role of Testes as Endocrine Tissue

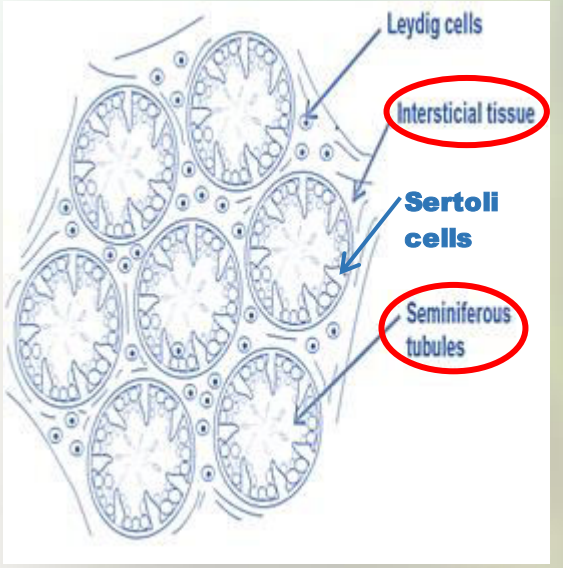
Roles of Testes

- Male gonads, primarily involved in spermatogenesis.
- Also act as an endocrine tissue as certain cells in them secrete hormones which include:
  - Inhibin
  - Androgens (male reproductive hormones)

Role of Testes as Endocrine Tissue

Types of Cells in Testes

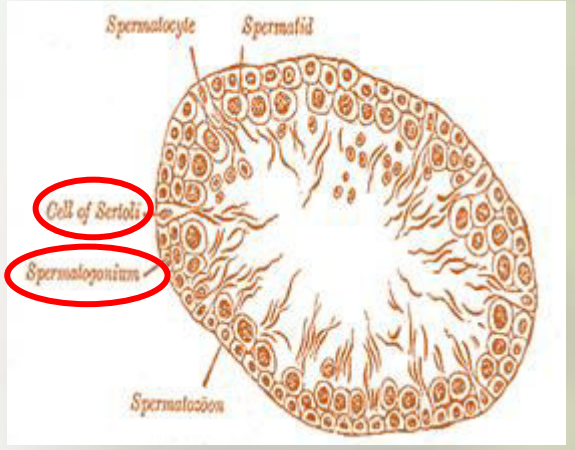
- In a cross section, testes have two major parts:
  - Seminiferous tubules
  - Interstitial tissue



Role of Testes as Endocrine Tissue

Seminiferous Tubules

- These tubules are lined with two major types of cells:
  - Germ cells
  - Sertoli cells



Role of Testes as Endocrine Tissue

Role of Germ Cells

- Germ cells are involved in spermatogenesis.

Role of Sertoli Cells

- They support germ cells in spermatogenesis.
- They also synthesize the hormone inhibin, on stimulation from androgens.

Role of Testes as Endocrine Tissue

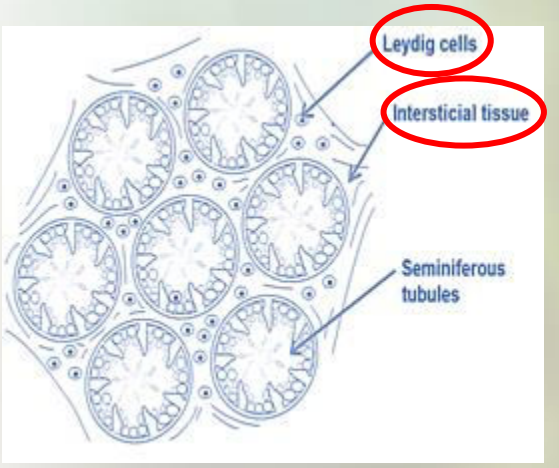
Role of Inhibin

- Inhibin locally regulates spermatogenesis.
- It also down-regulates FSH synthesis and inhibits its secretion.

Role of Testes as Endocrine Tissue

Interstitial Tissue

- Interstitial tissue lies b/w seminiferous tubules.
- It constitutes 20% of the mass of adult testes.
- Most important cells of interstitial tissue are the Leydig cells.



Role of Testes as Endocrine Tissue

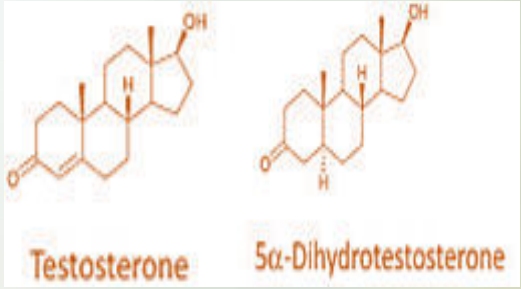
Endocrine Role of Leydig Cells

- Leydig cells produce and secrete several male sex hormones, collectively called as androgens.
- Androgens include:
  - Testosterone
  - Dihydrotestosterone
  - Androstenedione

Role of Testes as Endocrine Tissue

Androgens

- Testosterone is the primary male sex hormone and is more abundant than others.



- Dihydrotestosterone is more active hormone in the target tissues. Most of the testosterone is eventually converted into this form.

Role of Testes as Endocrine Tissue

Androgens

- Androstenedione is a weak androgen. It is formed as an intermediate in the biosynthesis of testosterone.



Role of Testes as Endocrine Tissue

Regulation of Testosterone Secretion

- A decrease in blood testosterone stimulates secretion of GnRH.
- GnRH promotes release of FSH and LH.
- FSH and LH stimulate the production and release of testosterone from Leydig cells.

Role of Testes as Endocrine Tissue

Regulation of Testosterone Secretion

- When testosterone level is increased, it causes secretion of inhibin from sertoli cells.
- Testosterone and inhibin provide -ve feedback to the hypothalamus which lowers GnRH production.
- It diminishes release of FSH & LH from pituitary which result in reducing testosterone secretion.

END

Endocrine Physiology

Role of Ovaries as Endocrine Tissue

LECTURE NO 87

Role of Ovaries as Endocrine Tissue

Ovarian Hormones

- Ovaries produce and release mainly two groups of female sex hormones:
  - Progesterone
  - Estrogens (Estradiol, Estrone and Estriol)
- Ovaries also secrete:
  - Relaxin prior to parturition.
  - Inhibin which signals the pituitary to inhibit FSH.

Role of Ovaries as Endocrine Tissue

Role of Progesterone and Estrogens

- These hormones work together to:
  - Promote the development of female secondary sexual characters.
  - Maintain the uterine and ovarian cycles
  - Help in fertility

Role of Ovaries as Endocrine Tissue

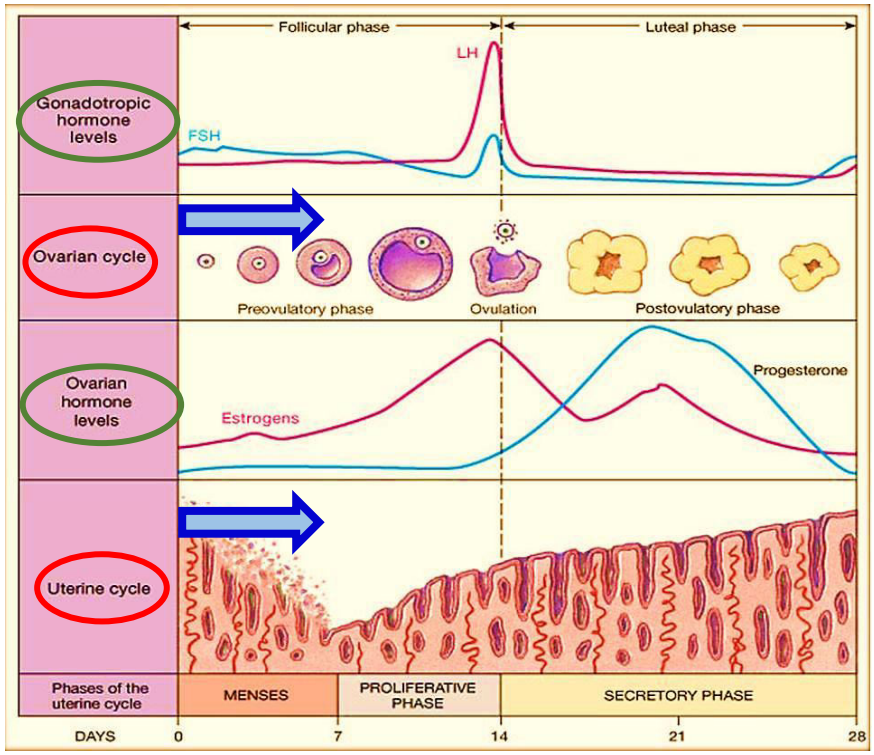
Uterine and Ovarian Cycles

- These are two closely linked reproductive cycles in human females.
- Uterine cycle (menstrual cycle) involves cyclic changes in the uterus. This cycle averages 28 days.
- Ovarian cycle involves cyclic events in ovaries and involving follicle growth and ovulation.

Role of Ovaries as Endocrine Tissue

Uterine and Ovarian Cycles

- Hormonal activity links the two cycles.
- Hormones synchronize follicle growth and ovulation with the establishment of uterine lining.



http://images.slideplayer.com/34/8319917/slides/slide_3.jpg

Role of Ovaries as Endocrine Tissue

Secretion of Estradiol in Ovarian Cycle

- Ovarian cycle begins when GnRH, released from the hypothalamus, stimulates pituitary to secrete FSH and LH.
- These hormones stimulate follicle growth.
- Theca interna cells of growing follicles start to secrete an estrogen, estradiol.

Role of Ovaries as Endocrine Tissue

Secretion of Estradiol in Ovarian Cycle

- Amount of estradiol rises slowly during the follicular phase till the maturation of oocyte.
- Prior to ovulation, higher estrogen levels feedback positively to the pituitary and hypothalamus, producing a surge in release of FSH and LH.

Role of Ovaries as Endocrine Tissue

Secretion of Estradiol in Ovarian Cycle

- This FSH accelerates maturation of the developing follicles.
- As the follicle completes its maturation, it ruptures under the influence of LH, releasing the ovum.

Role of Ovaries as Endocrine Tissue

Secretion of Inhibin

- FSH stimulates secretion of inhibin from the granulosa cells of the ovarian follicles.
- Its secretion reaches a peak near ovulation.
- It feeds back the anterior pituitary to suppress the release of FSH (but not LH).

Role of Ovaries as Endocrine Tissue

Corpus Luteum: Secretion of Progesterone

- After release of ovum, LH transforms the ruptured follicle into a temporary endocrine tissue, corpus luteum.
- Corpus luteum secretes progesterone and estrogens.
- They exert -ive feedback on hypothalamic GnRH which causes decrease in secretion of FSH and LH.

Role of Ovaries as Endocrine Tissue

Synchrony of Ovarian and Uterine Cycles

- Increase in estrogens during the follicular phase simultaneously stimulates proliferation of the endometrium tissue lining the uterus.
- Later, progesterone stimulates secretion of endometrial fluid that prepares uterus for implantation of a fertilized ovum.

Role of Ovaries as Endocrine Tissue

Synchrony of Ovarian and Uterine Cycles

- If fertilization does not happen, corpus luteum degenerates after about 14 days in humans, and secretion of estrogen and progesterone drops.
- In humans and some other primates, this triggers shedding of the uterine lining commonly called menstruation.

Role of Ovaries as Endocrine Tissue

Role of Corpus Luteum in Pregnancy

- If the egg is fertilized and implants, active corpus luteum is maintained.
- So progesterone and estrogen secretion continues until placenta fully develops to take over the production of these hormones.
- At this stage, corpus luteum degenerates.

Role of Ovaries as Endocrine Tissue

Role of Corpus Luteum in Pregnancy

- In many mammals, (e.g. rat) corpus luteum is stimulated by prolactin and continues to grow and secrete estrogen and progesterone throughout the gestation period.

END

Endocrine Physiology

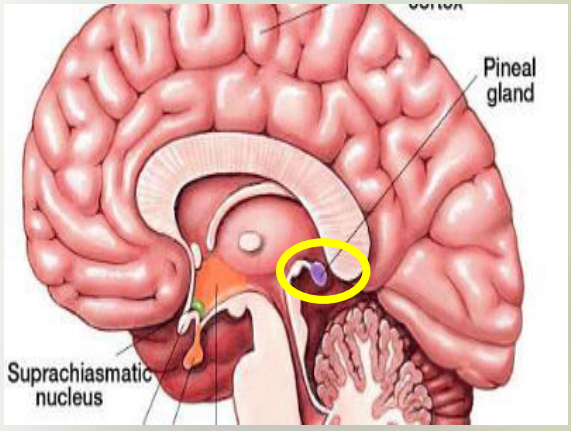
Pineal Gland

LECTURE NO 88

Pineal Gland

Pineal Gland

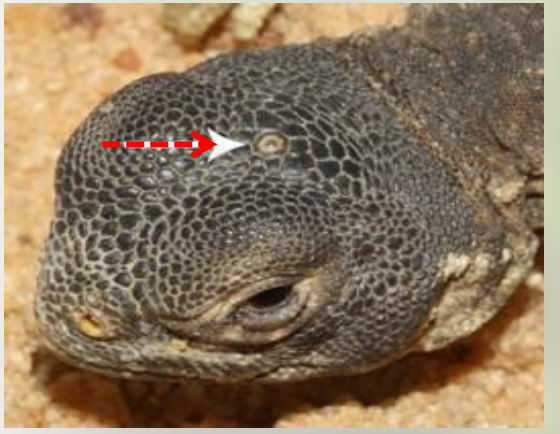
- A small reddish-gray gland (5–8 mm) found on the dorsal surface of forebrain in vertebrates .
- Its shape resembles a pine cone, hence its name.



Pineal Gland

Pineal Gland: A Modified Photoreceptor

- Pineal gland is actually a modified photoreceptor.
- In some amphibians and reptiles, it is linked to the light-sensing organ, the parietal eye or third eye.



Pineal Gland

Pineal Gland: A Modified Photoreceptor

- In birds and mammals, it contains light sensitive cells.
- It also has nervous connections with the eyes.
- Light controls its secretory activity.

Pineal Gland

Hormone Melatonin

- Pineal gland synthesizes and secretes the hormone melatonin.
- It is synthesized from serotonin, which itself is synthesized from the amino acid tryptophan.

Pineal Gland

Role of Melatonin in Regulating Biorhythms

- Melatonin is involved in the regulation of rhythmic activities related to light and seasons marked by changes in day length.

Pineal Gland

Role of Melatonin in Regulating Biorhythms

- Melatonin production is stimulated by darkness and inhibited by light.
- So, it is secreted at night, and the amount released depends on the length of the night.
- In winter days are short and nights are long, more melatonin is secreted.

Pineal Gland

Role of Melatonin in Regulating Biorhythms

- Main target of melatonin is the supra-chiasmatic nucleus (SCN) in the hypothalamus.
- SCN functions as a biological clock.
- Melatonin decreases the activity of SCN, and this effect is related to its role in mediating biorhythms.

Pineal Gland

Role of Melatonin in Sleep-Wake Cycles

- All birds and mammals show characteristic sleep/wake cycles.
- Melatonin plays an important role in these cycles.
- It modulates the brainstem circuits that govern the sleep–wake cycle.

Pineal Gland

Role of Melatonin in Sleep-Wake Cycles

- Melatonin synthesis increases as the light decreases and reaches a maximum between 2 to 4 A.M.
- In elder people, less melatonin is produced, that is why older people often sleep less at night and develop insomnia.

Pineal Gland

Melatonin Regulates Seasonal Fertility

- Pineal gland plays a regulatory role in seasonal sexual and reproductive activities in many animals.
- Melatonin causes a decrease in the secretion of gonadotropic hormones from pituitary, causing inhibition of gonads and partly involutes them.

Pineal Gland

Melatonin Regulates Seasonal Fertility

- This occurs during the early winter when nights are longer.
- Before spring, secretion of gonadotropins resumes and the gonads become functional and get ready for springtime reproductive activities.

Pineal Gland

Melatonin and Sexual Development in Humans

- Children produce high levels of melatonin that is believed to inhibit sexual development.
- When puberty arrives, melatonin production is reduced.
- It also plays some role in controlling sexual drive and reproduction in humans.

END

Endocrine Physiology

Placental Hormones

LECTURE NO 89

Placental Hormones

Placental Hormones

- In pregnancy, placenta secretes large quantities of hormones, essential for maintenance of pregnancy.
- These hormones include:
  - Human chorionic gonadotropin
  - Estrogens
  - Progesterone
  - Human chorionic somatomammotropin

Placental Hormones

Human Chorionic Gonadotropin

- It is a glycoprotein.
- Its molecular structure and functions resemble LH secreted by pituitary.
- Its secretion starts shortly after embryo implants in the endometrium.

Placental Hormones

Human Chorionic Gonadotropin

- Rate of secretion rises rapidly and reaches a maximum at about 10 to 12 weeks of pregnancy.
- Secretion decreases by 16 to 20 weeks and continues at this level for the remainder of pregnancy.

Placental Hormones

Functions of Human Chorionic Gonadotropin

- It causes persistence and growth of corpus luteum
- It causes corpus luteum to secrete larger amount of progesterone and estrogens up to the 12th week, till the placenta starts to secrete sufficient quantities of these hormones.

Placental Hormones

Functions of Human Chorionic Gonadotropin

- It also helps to prevent menstruation.
- It also causes the endometrium to grow and store large amounts of nutrients.

Effect on Fetal Testes

- It exerts an interstitial cell stimulating effect on the testes of male fetus.

Placental Hormones

Effect on Fetal Testes

- It causes production of testosterone by testes of male fetus until birth.
- Testosterone during gestation causes the fetus to grow male sex organs instead of female organs.
- Human chorionic gonadotropin also causes the testes to descend into the scrotum.

Placental Hormones

Estrogen Secretion by Placenta

- Placenta produces enormous amounts of estrogens.
- Near the end of pregnancy, placental estrogens are 30 times more than the normal levels.

Placental Hormones

Functions of Estrogens in Pregnancy

- During pregnancy, estrogens exert mainly a proliferative function on reproductive and associated organs of the mother.

Placental Hormones

Functions of Estrogens in Pregnancy

They cause:

- enlargement of uterus
- enlargement of breasts and growth of breast ductal structure
- enlargement of external genitalia
- Relaxation of pelvic ligaments to make pubic symphysis elastic for easy passage of the fetus through birth canal

Placental Hormones

Progesterone Secretion by Placenta

- Progesterone is secreted in tremendous quantities by placenta.
- This secretion is about 10-fold more than normal levels.

Placental Hormones

Progesterone Functions During Pregnancy

- Causes development of decidual cells in endometrium. These cells play an important role in the nutrition of the early embryo.
- It decreases contractility of the uterus to prevent spontaneous abortion.

Placental Hormones

Progesterone Functions During Pregnancy

- Contributes to the development of conceptus by increasing the secretions of fallopian tubes and uterus to provide nutrition to the developing morula and blastocyst.

Placental Hormones

Human Chorionic Somatomammotropin

- It is a protein hormone
- Its secretion starts during the fifth week of pregnancy.
- Its amount increases progressively throughout the remainder of pregnancy in direct proportion to the weight of placenta.

Placental Hormones

Functions of Chorionic Somatomammotropin

- It is involved partially in the development of breasts and lactation.
- It promotes the formation of protein tissues just like the growth hormone.

Placental Hormones

Functions of Chorionic Somatomammotropin

- It causes decreased insulin sensitivity and decreased utilization of glucose in the mother, making more glucose available to the fetus.
- It also promotes release of free fatty acids from the fat stores of the mother, providing this alternative source of energy.

Placental Hormones

END

Functions of Chorionic Somatomammotropin

- This hormone acts like a general metabolic hormone that has specific nutritional implications for both the mother and the fetus.

Endocrine Physiology

Metabolic and Developmental Hormones

LECTURE NO 90

Metabolic and Developmental Hormones

Hormones Involved in Metabolism and Development

- Several hormones regulate metabolism and affect developmental processes. These include:
  - Insulin
  - Glucagon
  - Glucocorticoids
  - Catecholamines
  - Thyroxine
  - Growth hormone

Metabolic and Developmental Hormones

Hormones Involved in Metabolism and Development

- These hormones are secreted by different endocrine glands.
- They belong to different chemical categories i.e. proteins, steroids and amines.

Metabolic and Developmental Hormones

Insulin	
Tissue of Origin	Pancreas (beta cells)
Chemical Category	Protein
Target Tissue	All tissues, except most neural tissues
Primary Actions	Increases glucose and amino acid uptake by cells
Regulation	Secreted in response to high plasma glucose and ammo acid levels. Secretion is lowered with low glucose level and inhibited by somatostatin.

Metabolic and Developmental Hormones

Glucagon	
Tissue of Origin	Pancreas (alpha cells)
Chemical Category	Protein
Target Tissue	Liver, Adipose tissue
Primary Actions	Release of glucose from liver by stimulating glycogenolysis. Also promotes lipolysis
Regulation	Low serum glucose increases secretion, somatostatin inhibits release

Metabolic and Developmental Hormones

Glucocorticoids (Cortisol, Cortisone, Corticosterone)	
Tissue of Origin	Adrenal cortex
Chemical Category	Steroid
Target Tissue	Liver, Adipose tissue
Primary Actions	Mobilize amino acids from muscles and fatty acids from adipose tissue. Stimulate gluconeogenesis by liver to raise blood glucose. Exhibit anti-inflammatory actions.
Regulation	Secretion increases during stress. -ve feedback via CRH and ACTH and biological clock regulate secretion .

Metabolic and Developmental Hormones

Catecholamines (Epinephrine, Norepinephrine)	
Tissue of Origin	Adrenal medulla (chromaffin cells)
Chemical Category	Amine
Target Tissue	Most tissues
Primary Actions	Increase cardiac activity. Induce vasoconstriction. Increase glycolysis and lipolysis causing hyperglycemia.
Regulation	Secreted by sympathetic stimulation. Control through simultaneous release of ATP.

Metabolic and Developmental Hormones

Thyroxine	
Tissue of Origin	Thyroid gland
Chemical Category	Amine
Target Tissue	Most cells, specially muscles, heart, Liver and kidney
Primary Actions	Increases metabolic rate and thermogenesis. Promotes growth and development. Promotes amphibian metamorphosis
Regulation	TSH induces release. -ve feedback to pituitary and hypothalamus decrease TSH & TRH secretion that decrease its secretion.

Metabolic and Developmental Hormones

Growth Hormone	
Tissue of Origin	Adenohypophysis
Chemical Category	Protein
Target Tissue	All tissues
Primary Actions	Stimulates RNA and protein synthesis. Promotes tissue growth. Increases uptake of glucose and amino acids by the cells. Increases lipolysis. Stimulates antibody formation.
Regulation	Reduced plasma glucose and increased amino acid levels stimulate release. Somatostatin inhibits release.

Metabolic and Developmental Hormones

Integration of Metabolism and Development

- The processes of growth and development of animals are related to the rate of metabolism.
- These processes are controlled and integrated by coordinated activity of these hormones.

END

Endocrine Physiology

Hormones for Water Regulation and Ion Balance

LECTURE NO 91

Hormones for Water Regulation and Ion Balance

Regulation of Water and Ions

- The organs of vertebrates which regulate water and ions are: kidneys, skin, gills, intestine and bone.
- The absorption or excretion of water and ions is regulated through these organs by the coordinated activity of many hormones.

Hormones for Water Regulation and Ion Balance

Hormones in Regulation of Water and Ions

- These hormones include:
  - Antidiuretic hormone (ADH)
  - Aldosterone
  - Atrial natriuretic Hormone
  - Calcitonin
  - Parathormone

Hormones for Water Regulation and Ion Balance

Vasopressin or Antidiuretic hormone (ADH)	
Tissue of Origin	Neurohypophysis
Chemical Category	Peptide (Nonapeptide)
Target Tissue	Kidneys
Primary Actions	Increases water reabsorption
Regulation	Increased plasma osmotic pressure or decreased blood volume stimulates release.

Hormones for Water Regulation and Ion Balance

Aldosterone (Mineralocorticoid)	
Tissue of Origin	Adrenal cortex
Chemical Category	Steroid
Target Tissue	Distal kidney tubules
Primary Actions	Promotes reabsorption of Na^+ from urinary filtrate
Regulation	Angiotensin II stimulates secretion

Hormones for Water Regulation and Ion Balance

Atrial Natriuretic Hormone	
Tissue of Origin	Heart (atrium)
Chemical Category	Peptide
Target Tissue	Kidneys
Primary Actions	Reduces Na^+ and water reabsorption
Regulation	Increased venous pressure stimulates release

Hormones for Water Regulation and Ion Balance

Calcitonin	
Tissue of Origin	Thyroid (parafollicular cells)
Chemical Category	Peptide
Target Tissue	Bones, kidneys
Primary Actions	Decreases release of Ca^{2+} from bone. Increases renal Ca^{2+} and PO_4^{3-} excretion
Regulation	Increased plasma Ca^{2+} stimulates secretion

Hormones for Water Regulation and Ion Balance

Parathormone (PTH)	
Tissue of Origin	Parathyroid gland
Chemical Category	Peptide
Target Tissue	Bones, kidneys, intestine
Primary Actions	Increases release of Ca^{2+} from bone. Increases intestinal Ca^{2+} absorption with calcitriol. Decreases renal Ca^{2+} excretion.
Regulation	Decreased plasma Ca^{2+} stimulates secretion.

Hormones for Water Regulation and Ion Balance

Regulation of Water and Ions

- As epithelial cells are responsible for the uptake or excretion of water and ions, these hormones act primarily on the epithelial tissues of the organs involved.

END

Endocrine Physiology

Reproductive Hormones

LECTURE NO 92

Reproductive Hormones

Reproductive Hormones

- Reproductive hormones of vertebrates belong to two chemical categories:
  - Steroid Sex Hormones
  - Peptide Hormones
- These hormones:
  - Develop primary and secondary sexual characters
  - Control and regulate reproductive activities.

Reproductive Hormones

Steroid Sex Hormones

- These include:
  - Progesterone
  - Estrogens
  - Androgens
- Androgens predominate in males.
- Estrogens and Progesterone predominate in females.

Reproductive Hormones

Steroid Sex Hormones

- They are produced by the gonads i.e. ovaries and testes.
- They are also produced by adrenal cortex of both sexes in varying quantities.

Reproductive Hormones

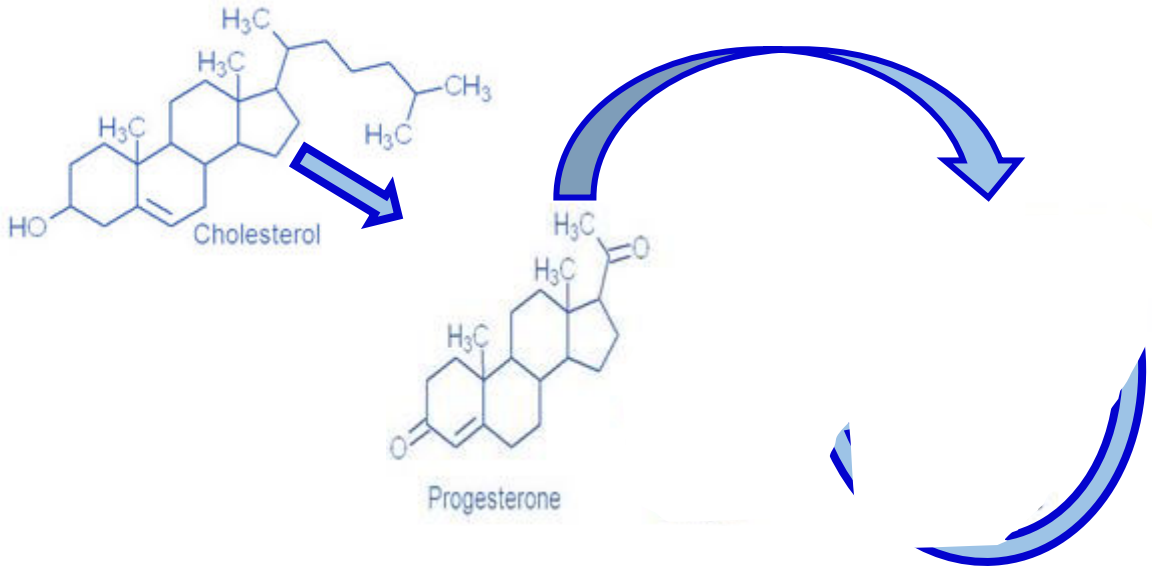
Peptide Hormones

- Two peptide hormones released by pituitary function in parturition (child birth) and lactation.
- These hormones are:
  - Prolactin
  - Oxytocin

Reproductive Hormones

Synthesis of Steroid Sex Hormones

- All steroid hormones are synthesized from cholesterol.
- Cholesterol is first converted to progesterone.
- Progesterone is then transformed into androgens (androstenedione and testosterone).
- Estrogens are formed from androgens.



Reproductive Hormones

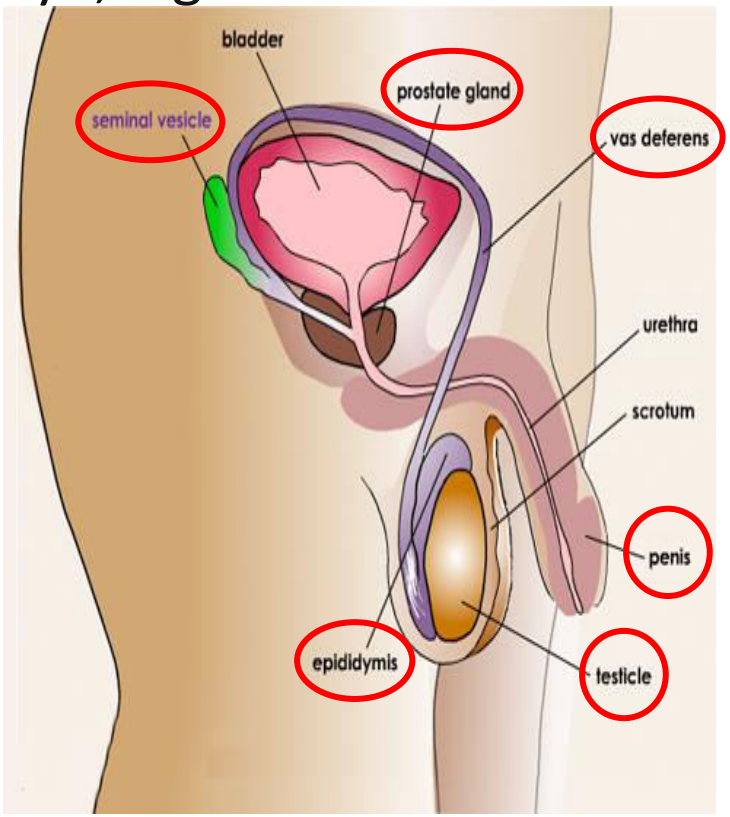
Importance of Steroid Sex Hormones

- Estrogens and androgens are important in both sexes in various aspects of:
  - Growth and development
  - Morphologic differentiation
  - Regulation of reproductive cycles
  - Development of sexual behaviors

Reproductive Hormones

Roles of Androgens in Males

- Androgens trigger development of primary male sexual characters in the embryo; e.g.
  - Testes
  - Penis
  - vas deferens
  - Epididymis
  - prostate gland
  - seminal vesicles

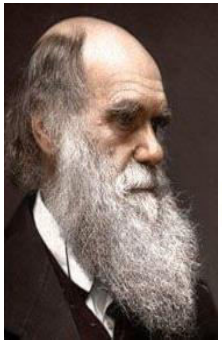


Reproductive Hormones

Roles of Androgens in Males

- They also produce male secondary sexual characters at the time of puberty e.g. (e.g.,, the rooster's comb and plumage, and facial hair in men, voice coarseness and body musculature).
- Lion's mane
- Rooster's comb
- Plumage
- Facial hair in men
- body musculature
- Voice coarseness




Reproductive Hormones

Roles of Androgens in Males

- Androgens stimulate spermatogenesis by the germinal cells of testes.
- They also contribute to growth and protein synthesis; particularly synthesis of myofibrillar proteins in muscles of males, adding greater muscularity to males.

Reproductive Hormones

Roles of Steroid Sex Hormones in Females

- Estrogens and progesterone are the primary steroid sex hormones in females.
- Estrogens stimulate development of female primary sexual characteristics e.g. uterus, ovaries and vagina.

Reproductive Hormones

Roles of Steroid Sex Hormones in Females

- They also develop female secondary sexual characters e.g. breasts.
- They regulate the reproductive cycles: menstrual cycle in human and some other primate females and estrous cycle in other mammalian females.

Reproductive Hormones

Reproductive Role of Prolactin and Oxytocin

- Prolactin stimulates mammary gland growth and milk synthesis in mammals.
- Oxytocin is specialized in stimulating uterine contractions during parturition (child birth).
- It also stimulates ejection of milk from the mammary glands.

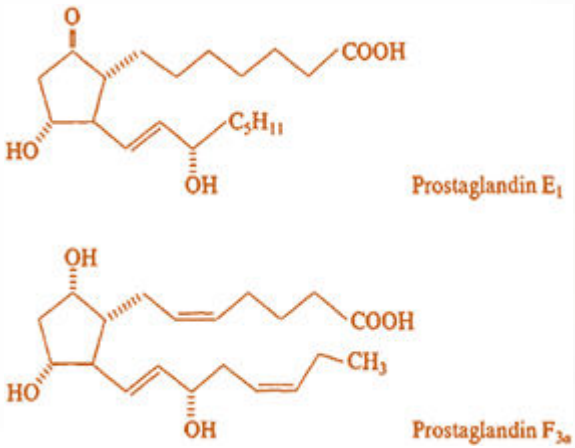
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Endocrine Physiology

Prostaglandins

LECTURE NO 93

Prostaglandins



Prostaglandins

- Prostaglandins are about 16 hormone-like local regulators substances .
- They are derived from cyclic, long-chain, unsaturated, hydroxy fatty acids.

Prostaglandins

Synthesis

- Prostaglandins are produced by all or nearly all tissues.
- They are synthesized in membranes from arachidonic acid, which is produced by cleavage of membrane phospholipids by phospholipases.

Prostaglandins

Mode of Action

- In some cases, they act locally as paracrine agents.
- In other cases they act on distant target tissues in endocrine fashion.
- They have a rapid, short-lasting effect, similar to that of lipid-insoluble hormones.
- They bind to cell-surface receptors linked to the cAMP pathway.

Prostaglandins

Functions

- Prostaglandins have diverse actions on a variety of tissues, particularly involving smooth muscles.

Prostaglandins

Prostaglandins Role in Fertilization

- Prostaglandins present in semen aid in fertilization by reacting with the female cervical mucus to make it more receptive to sperm movement.
- They also stimulate the smooth muscles of the uterine wall to contract, helping sperms reach an egg.

Prostaglandins

Prostaglandins Aid in Labor

- Certain prostaglandins are secreted by the placenta at the onset of childbirth.
- They make the muscles of uterus more excitable, enhancing uterine contractions during labor.

Prostaglandins

Prostaglandins Role in Immune Response

- Damaged tissues produce prostaglandins during immune response.
- Some prostaglandins act as local regulators of inflammation.
- They promote fever and inflammation and also intensify the sensation of pain.

Prostaglandins

Prostaglandins Role in Immune Response

- They worsen pain by increasing nociceptor sensitivity.
- Prostaglandin E_2, formed from interleukin-1, produced in immune response, causes fever by acting on the hypothalamus to elicit fever reaction.

Prostaglandins

Prostaglandins Role in Immune Response

- The anti-pyretic, anti-inflammatory and pain-relieving effects of aspirin and ibuprofen are actually due to the inhibition of prostaglandin synthesis by these drugs.

Prostaglandins

Prostaglandins Role in Blood Clot Formation

- Prostaglandins help platelets to aggregate and form blood clots.
- This is the basis of use of aspirin by patients at risk for a heart attack due to formation of clots.

Prostaglandins

Maintenance of Stomach Lining

- Prostaglandins help to maintain a protective lining in the stomach.
- As aspirin interferes with prostaglandin synthesis, long-term aspirin therapy can result in damage to this protective lining, causing stomach irritation and ulcer.

Prostaglandins

Production of Erythropoietin

- Several prostaglandins stimulate the production of erythropoietin by kidneys.
- Erythropoietin is a hormone that stimulates the production of erythrocytes.

Prostaglandins

Blood Pressure Regulation

- Some prostaglandins act on the smooth muscles of blood vessels.
- They regulate blood pressure by vasodilation and vasoconstriction.

END

Endocrine Physiology

Feedback Mechanisms

LECTURE NO 94

Feedback Mechanisms

Feedback Loops

- A feedback mechanism connects the response to the initial stimulus through a feedback loop or circuit.
- Feedback loops are characteristic of controlled pathways of hormones.
- Through these loops, secretion of hormone is regulated.

Feedback Mechanisms

Types of Feedback Loops

- Feedback loops may follow negative or positive pathways.

Negative Feedback

- Secretory activities of most endocrine tissues, are modulated by negative feedback, especially those involved in maintaining homeostasis.

Feedback Mechanisms

Negative Feedback

- In this type of feedback, the hormone itself or its products tend to suppress its further release.
- This mechanism ensure a proper level of hormone activity at the target tissue and prevents its over-activity by over-secretion.

Feedback Mechanisms

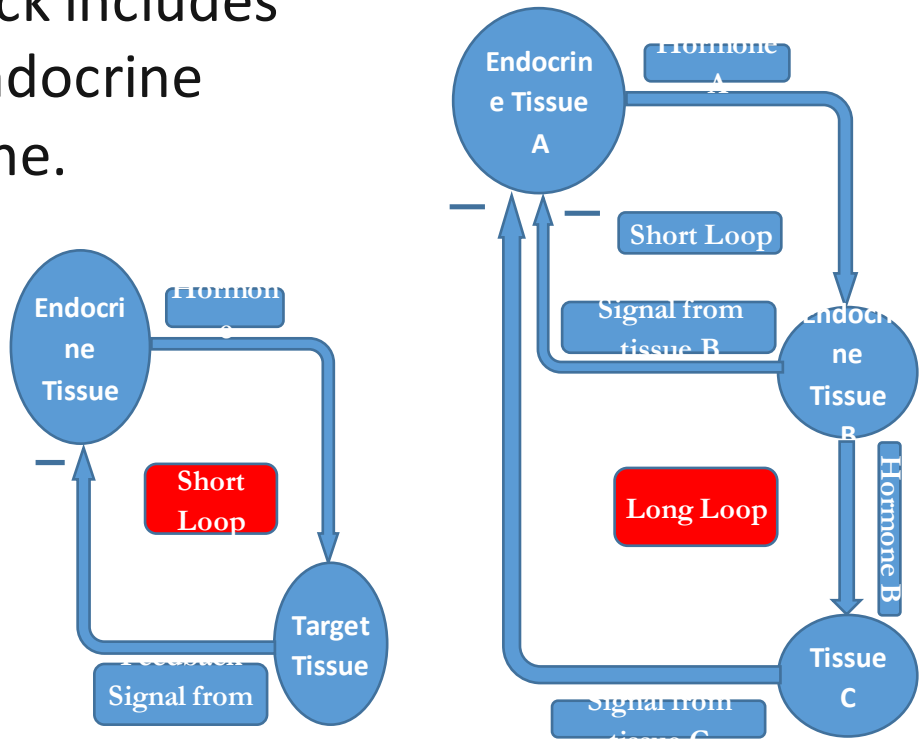
How Negative Feedback Operates

- Feedback may cause regulation of hormone at any level e.g.,
- Synthesis may be stopped by regulating gene transcription and translation steps.
- Steps involved in the activation of hormone may be regulated.
- Releasing of hormone may be blocked.

Feedback Mechanisms

Loops of Negative Feedback

- -ve feedback involves either a short loop or long loop.
- In short-loop, hormone or a byproduct of its activity, acts back directly on the secreting endocrine tissue.
- Long-loop feedback includes more than one endocrine gland and hormone.



Feedback Mechanisms

Positive Feedback

- In +ve feedback, the secretion of a hormone directly or indirectly leads to its increased secretion.
- This happens when an extremely rapid or strong response is required.
- +ve feedback is also common in the early phases of response in most cases of hormonal action.

Feedback Mechanisms

Example of Positive Feedback

- Increase in the level of LH before ovulation due to stimulatory effect of estrogen on the anterior pituitary.
- The secreted LH acts on the ovaries to stimulate additional secretion of estrogen.
- This estrogen, in turn, causes more secretion of LH.

Feedback Mechanisms

Example of Positive Feedback

- The +ve feedback is ultimately countered by a –ve feedback that ends the rapid increase.
- e.g. when LH reaches a certain conc., typical -ve feedback control is exerted to lower LH and estrogen concentrations.

END

Endocrine Physiology

Mechanisms of Hormone Action

LECTURE NO 95

Mechanisms of Hormone Action

Phases of Hormone Action

- Action of hormone is divided into two phases:
  - Forming the hormone-receptor complex
  - Producing the effect or response

Mechanisms of Hormone Action

Hormone-Receptor Complex

- First step of a hormone's action is to bind to specific receptors at the target cell.
- This binding results in the formation of a hormone-receptor complex.

Mechanisms of Hormone Action

Hormone Receptors

- Hormone receptors are large proteins.
- The target cells usually have 2000 to 100,000 receptors.
- Cells that lack receptors for a hormone do not respond to that hormone.

Mechanisms of Hormone Action

Hormone Receptor Locations

- Receptors for lipid-insoluble, hydrophilic hormones (catecholamines, protein and peptide hormones) are present in the plasma membrane.
- These hormones, being lipid-insoluble, cannot penetrate the plasma membrane, so bind to the surface receptors.

Mechanisms of Hormone Action

Hormone Receptor Locations

- Steroid hormones have two receptors in a cell.
- Their primary receptors are present in the cytoplasm.
- Their secondary receptors are located in the nucleus.

Mechanisms of Hormone Action

Hormone Receptor Locations

- Steroid hormones, being lipid soluble, can readily penetrate the plasma membrane and bind to receptors inside the cell.
- Receptors for thyroid hormones are found in the nucleus.

Mechanisms of Hormone Action

Producing the Effect or Response

- Formation of hormone-receptor complex initiates a cascade of reactions in the cell.
- These intracellular reactions and mechanisms of action vary for the hormones binding to cytoplasmic, nuclear and cell-surface receptors.

Mechanisms of Hormone Action

Mechanism of Action of Lipid-Soluble Hormones

- Lipid-soluble thyroid hormones and steroids bind to cytoplasmic or nuclear receptors.
- Hormone-receptor complex translocates to the nucleus and acts directly on the DNA causing changes in gene expression.
- Effects are long-term and last for hours or days.

Mechanisms of Hormone Action

Mechanism of Action of Lipid-Insoluble Hormones

- Their binding to cell-surface receptors leads to production of one or more 2nd messengers.
- 2nd messengers amplify the signal and mediate rapid, short-lived responses via various effector proteins.

Mechanisms of Hormone Action

Mechanism of Action of Prostaglandins

- Prostaglandins are lipid-soluble.
- However, they bind to cell-surface receptors and produce a rapid, short-lasting effect, similar to that of lipid-insoluble hormones.

END

Endocrine Physiology

Lipid Soluble Hormones: Mechanism of Action

LECTURE NO 96

Lipid Soluble Hormones: Mechanism of Action

Transport of Lipid Soluble Hormones

- Lipid-soluble steroid and thyroid hormones are carried through blood by forming complexes with carrier proteins.
- The carrier proteins are necessary because blood is an aqueous solution and lipid-soluble hormones can't dissolve in it.

Lipid Soluble Hormones: Mechanism of Action

Transport of Lipid Soluble Hormones

- In such un-dissolved form, these hormones would be taken up by any lipids in circulation.
- The binding to carrier proteins ensures delivery of the hormones to their target tissues.

Lipid Soluble Hormones: Mechanism of Action

Entry into the Target Cells

- When these hormones reach their target tissues, they dissociate from carrier proteins.
- They readily enter the cells by diffusing across the plasma membrane.

Lipid Soluble Hormones: Mechanism of Action

Receptors for Lipid-Soluble Hormones

- The receptors for steroid hormones are present in the target cell cytoplasm.
- Here hormone-receptor complexes are formed.
- These complexes then move into the nucleus.

Lipid Soluble Hormones: Mechanism of Action

Receptors for Lipid-Soluble Hormones

- The receptors for non-steroid lipid soluble thyroid hormones are present in the nucleus.
- The hormone-receptor complexes are formed inside the nucleus.

Lipid Soluble Hormones: Mechanism of Action

Receptors for Lipid-Soluble Hormones

- All receptors that bind lipid-soluble hormones share a highly conserved DNA-binding domain.
- In the absence of hormone, an inhibitor protein is bound to this domain, making the receptor inactive.

Lipid Soluble Hormones: Mechanism of Action

Receptors for Lipid-Soluble Hormones

- Binding of hormone to the receptor causes the inhibitor protein to dissociate.
- This exposes its DNA-binding site and the receptor is activated.

Lipid Soluble Hormones: Mechanism of Action

Action of Hormone in the Nucleus

- Inside the nucleus, the DNA-binding domain of the receptor binds to the specific regulatory sequences of DNA.
- This induces the transcription of specific genes causing synthesis of specific proteins.

Lipid Soluble Hormones: Mechanism of Action

END

Long-Lasting Effect of Lipid-Soluble Hormones

- Since the lipid-soluble hormones act on cell's DNA to stimulate or inhibit production of particular proteins, their effects persist for hours to days.
- In comparison, the effects of lipid-insoluble hormones usually last only minutes to hours.

Endocrine Physiology

Lipid Insoluble Hormones: Intracellular Signaling

LECTURE NO 97

Lipid Insoluble Hormones: Intracellular Signaling

Hormone-Receptor Complex

- The water soluble hormones do not penetrate the cell.
- Their receptors are present in the plasma membrane.
- So, the hormone-receptor complexes are formed on the cell membrane of target cell.

Lipid Insoluble Hormones: Intracellular Signaling

Signaling Through Second Messengers

- Binding of hormone to its receptor results in the activation of many cellular proteins.
- This results in a cascade of enzymatic reactions in the cell which produce a second messenger.

Lipid Insoluble Hormones: Intracellular Signaling

Signaling Through Second Messengers

- The second messenger causes subsequent intracellular effects that transduce the extracellular hormonal signal into a specific intracellular response.

Lipid Insoluble Hormones: Intracellular Signaling

Responses of Water-Soluble Hormones

- The responses of water-soluble hormones are diverse. e.g.
- Activation of an enzyme.
- A change in the uptake or secretion of specific molecules.
- Rearrangement of cytoskeleton.
- Movement of certain proteins into the nucleus which alter transcription of genes.

Lipid Insoluble Hormones: Intracellular Signaling

Second Messenger Types

- The second messengers involved in signal transduction fall into three distinct groups:

1. Cyclic nucleotide monophosphates e.g. cAMP and cGMP.
2. Inositol phospholipids e.g. inositol trisphosphate ($InsP_3$) and diacylglycerol (DAG)
3. Ca^{2+} ions and associated calmodulin

Effort by: AARISH ALI

END

Endocrine Physiology

Cyclic Nucleotide Signaling Systems

LECTURE NO 98

Cyclic Nucleotide Signaling Systems

Cyclic Nucleotide Second Messengers

- Water-soluble hormones exert intracellular actions by stimulating the formation of a 2nd messenger in the cell.
- The most common 2nd messengers are the cyclic nucleotide monophosphates cAMP and cGMP which cause subsequent intracellular effects of hormones.

Cyclic Nucleotide Signaling Systems

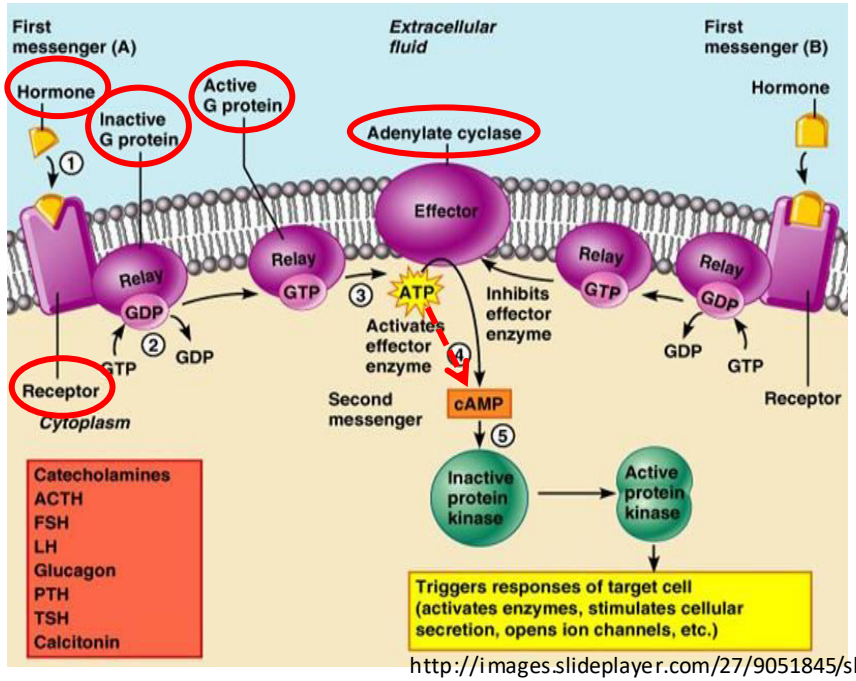
Signaling System of cAMP

- Many hormones use the Adenylate Cyclase–cAMP second messenger signaling system to stimulate their target tissues.

Cyclic Nucleotide Signaling Systems

Signaling System of cAMP

- The hormone binds to the receptor which is coupled to a G protein.
- G protein stimulates adenylate cyclase in the membrane.
- Activated AC catalyzes the conversion of ATP into cAMP in cytoplasm.

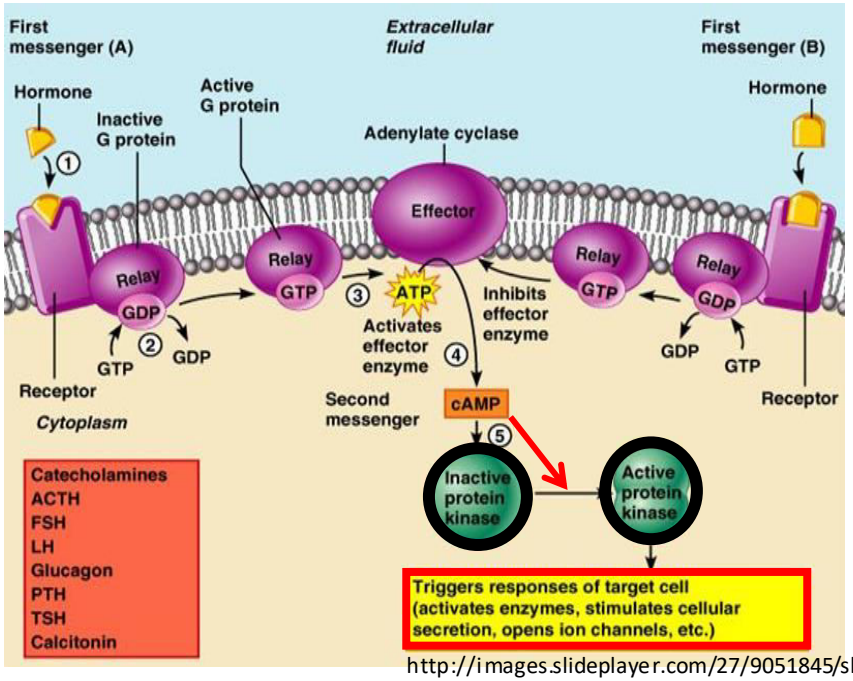


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Cyclic Nucleotide Signaling Systems

Signaling System of cAMP

- cAMP then activates an enzyme called cAMP-dependent protein kinase.
- Protein kinase phosphorylates specific proteins in the cell, triggering a cascade of biochemical reactions that ultimately lead to cell's response to the hormone.



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Cyclic Nucleotide Signaling Systems

Action of cAMP Varies with Cell-Type

- Specific action in response to cAMP in target cell depends on the nature of the intracellular machinery.
- Different types of cells have different sets of enzymes.
- Therefore, different functions are elicited in different target cells.

Cyclic Nucleotide Signaling Systems

Action of cAMP Varies with Cell-Type

Response to cAMP in different cell types include:

- Initiating synthesis of specific chemicals.
- Causing muscle contraction or relaxation.
- Initiating secretion by the cells
- Altering cell permeability

Cyclic Nucleotide Signaling Systems

cGMP as a Second Messenger

- Many animal cells also use cyclic GMP as a second messenger.
- The pattern of cGMP activities is similar to that of cAMP.
- However, cGMP signaling pathway uses different enzymes and factors stimulating these enzymes.

Cyclic Nucleotide Signaling Systems

Differences b/w cGMP and cAMP Pathways

- Production of cGMP is catalyzed by guanylate cyclase enzyme from GTP.
- Guanylate cyclase occurs in two forms: one bound to the membrane and one free in cytoplasm. In contrast, adenylate cyclase is always bound to the membrane.

Cyclic Nucleotide Signaling Systems

Differences b/w cGMP and cAMP Pathways

- Guanylate cyclase becomes active as the Ca^{2+} concentration is increased within the cell, while adenylate cyclase activity is increased when Ca^{2+} conc. is low.
- cGMP activates a specific protein kinase G instead of protein kinase A.

Cyclic Nucleotide Signaling Systems

Amplification of the Effect

- The mechanism of cyclic nucleotide signaling system is important b/c it involves a cascade of biochemical reactions in which each activated enzyme stimulates many more molecules of the next enzyme.

Cyclic Nucleotide Signaling Systems

Amplification of the Effect

- This is like a chain reaction that can initiate a powerful activity in the cell even in the presence of slightest amount of the hormone.

END

Endocrine Physiology

Inositol Phospholipid and Ca^{+2} Signaling Systems

LECTURE NO 99

Inositol Phospholipid and Ca^{+2} Signaling Systems

Inositol Phospholipid Signaling System

- Some hormones use the inositol phospholipid pathway to produce their effect.
- These hormones include:
  - Angiotensin II
  - Catecholamines
  - GnRH
  - GHRH
  - Oxytocin
  - TRH
  - Vasopressin

Inositol Phospholipid and Ca^{+2} Signaling Systems

Mechanism of Inositol Phospholipid Signaling System

- The hormone binds to the transmembrane receptors that are linked to G proteins.
- G proteins activate the enzyme phospholipase C.

Inositol Phospholipid and Ca^{+2} Signaling Systems

Mechanism of Inositol Phospholipid Signaling System

- phospholipase C catalyzes the breakdown of some phospholipids in the cell membrane, e.g. phosphatidylinositol biphosphate (PIP_2) into two different second messengers:
  - inositol trisphosphate (InsP3)
  - diacylglycerol (DAG)

Inositol Phospholipid and Ca^{+2} Signaling Systems

Role of $InsP_3$

- $InsP_3$ mobilizes Ca^{+2} ions from intracellular calcium stores (mitochondria and endoplasmic reticulum).
- These calcium ions have their own second messenger effects on smooth muscle contraction and changes in cell secretion.

Inositol Phospholipid and Ca^{+2} Signaling Systems

Role of DAG

- Diacylglycerol activates the enzyme protein kinase C (PKC).
- PKC phosphorylates a large number of proteins, leading to the cell's response.

Inositol Phospholipid and Ca^{+2} Signaling Systems

Role of DAG

- DAG also acts as a precursor for the synthesis of prostaglandins and other local hormones.
- These hormones cause multiple effects in tissues throughout the body.

Inositol Phospholipid and Ca^{+2} Signaling Systems

Ca^{+2} Signaling Systems

- Some hormones interact with membrane receptors that open calcium channels.
- Calcium entering through these channels acts as a second messenger.
- On entering a cell, calcium ions bind with the protein calmodulin.

Inositol Phospholipid and Ca^{+2} Signaling Systems

Ca^{+2} Signaling Systems

- Calmodulin has four calcium sites.
- When calcium ions bind to these sites, calmodulin becomes activated.
- Activated calmodulin activates calmodulin-dependent protein kinase enzyme.

Inositol Phospholipid and Ca^{+2} Signaling Systems

Ca^{+2} Signaling Systems

- Calmodulin-dependent protein kinase phosphorylates certain proteins which induce cell's response to the hormone.

Inositol Phospholipid and Ca^{+2} Signaling Systems

END

Ca^{+2} Signaling Systems

Example

- This pathway is used in the contraction of smooth muscles.
- It involves activation of myosin kinase, which acts on the myosin of smooth muscle and phosphorylates it.
- Phosphorylation of myosin causes contraction .

Muscle Physiology

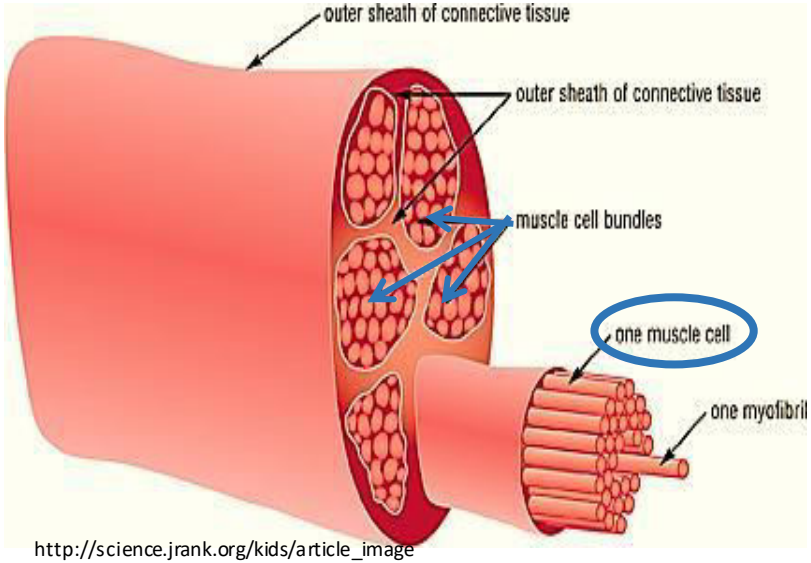
The Structure of Muscle

LECTURE NO 100

The Structure of Muscle

Myofibers—The Muscle Cells

- Each muscle is composed of thousands of myofibers which are thin, long, cylindrical and multinucleated cells, arranged in bundles called fascicles.
- Myofibers are 5 to 100 µm in diameter, and up to many centimeters in length.

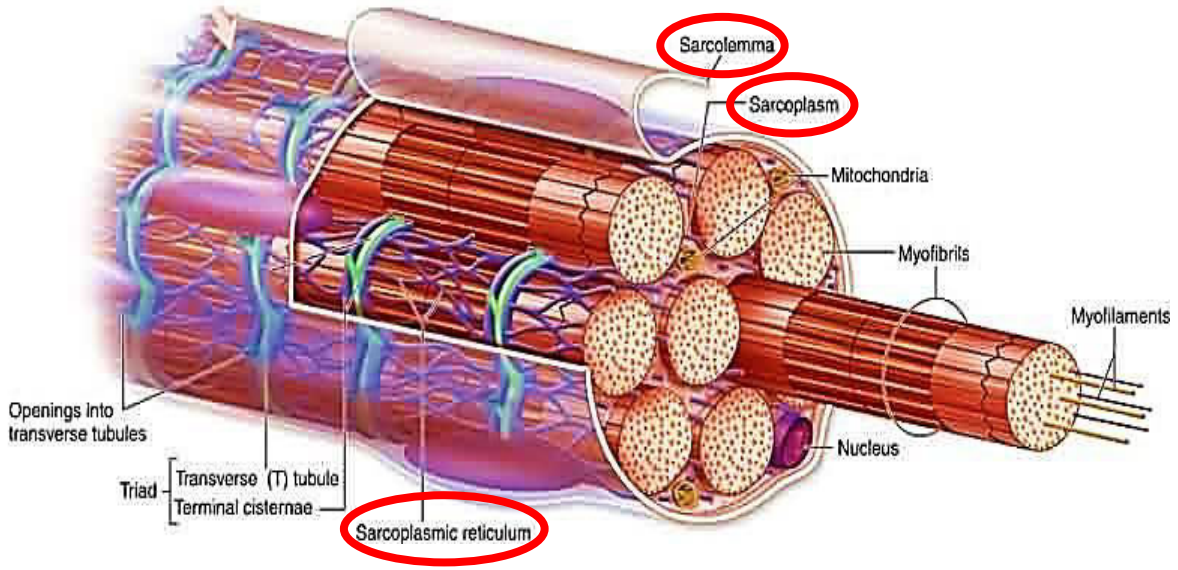


http://science.jrank.org/kids/article_images/cells_p15.jpg

The Structure of Muscle

Myofibers—The Muscle Cells

- The membrane of the cell is called sarcolemma.
- Its cytoplasm is called sarcoplasm.
- Its endoplasmic reticulum is known as sarcoplasmic reticulum.

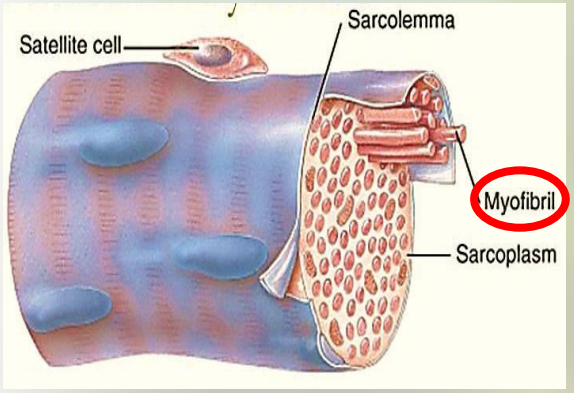


https://cdn.thinglink.me/api/image/578290998078078977/1240/10/scaletowidth

The Structure of Muscle

Myofibrils

- Within each muscle fiber, there are many myofibrils running in parallel fashion.
- Myofibrils are 1-2 µm in diameter and extend the entire length of the cell.

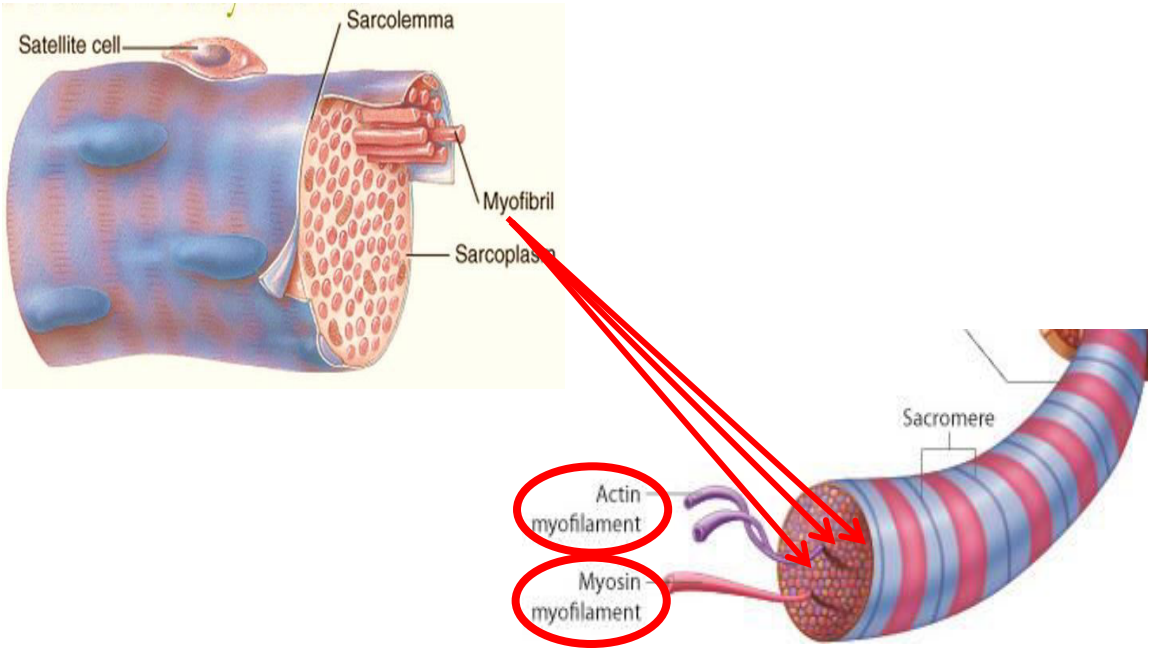


http://slideplayer.com/6381982/22/images/3/Muscle+Fiber+or+Myofibers.jpg

The Structure of Muscle

Myofilaments

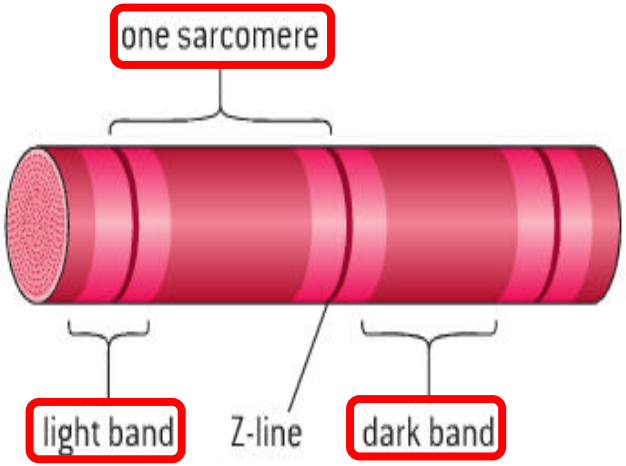
- A myofibril is composed of myofilaments.
- Myofilaments are of two types:
  - Thin filaments composed of actin
  - Thick filaments composed of myosin



The Structure of Muscle

Sarcomere

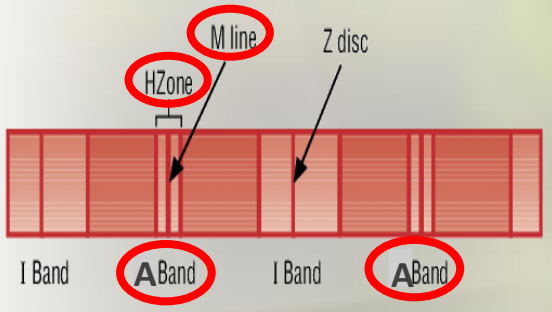
- The arrangement of thick and thin filaments creates a pattern of repeating light and dark bands.
- This pattern gives a striped appearance to the muscle cell.
- Each repeating unit is called a sarcomere and is the basic functional contractile unit of the muscle.



The Structure of Muscle

Structure of Sarcomere

- Each dark band in the sarcomere is called A band.
- It is anisotropic i.e. it polarizes visible light.
- It has a lighter stripe in its center called H-zone.
- H-zone is bisected by a dark line called M-line.

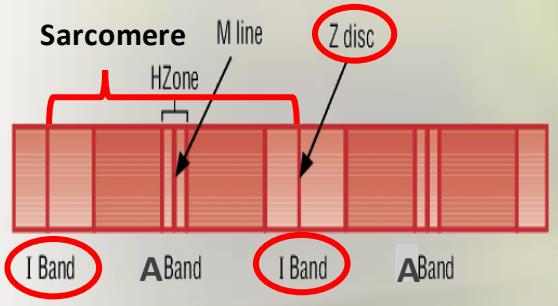


The Structure of Muscle

Structure of Sarcomere

- The light band is called I band which is isotropic i.e. non-polarizing.
- It has a mid-line called Z-line.
- A sarcomere is the region of a myofibril between two successive Z-lines.

END



Muscle Physiology

Myofilament Substructure

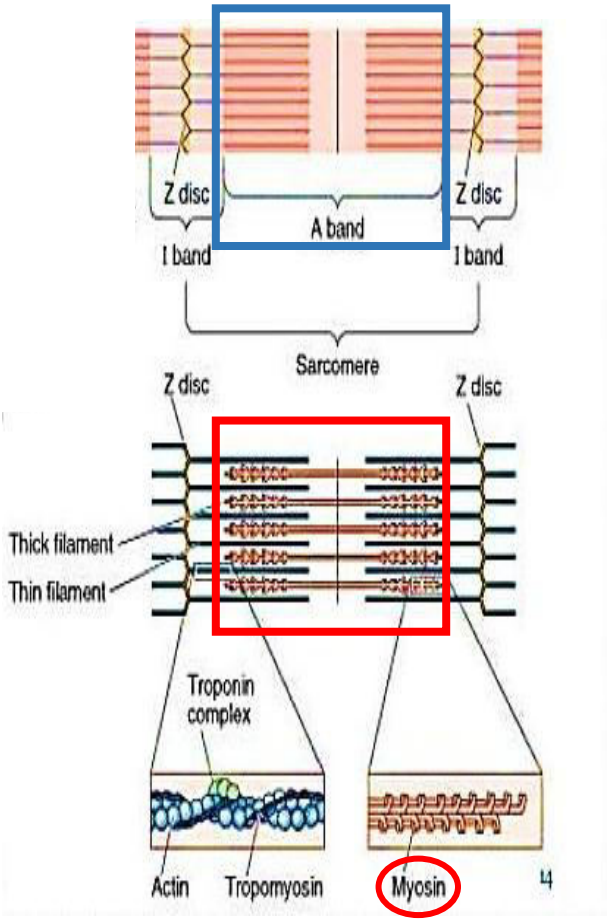
LECTURE NO 101

Myofilament Substructure

Myofilament

- A myofilament is made up of thick and thin filaments.

Myofilament Substructure



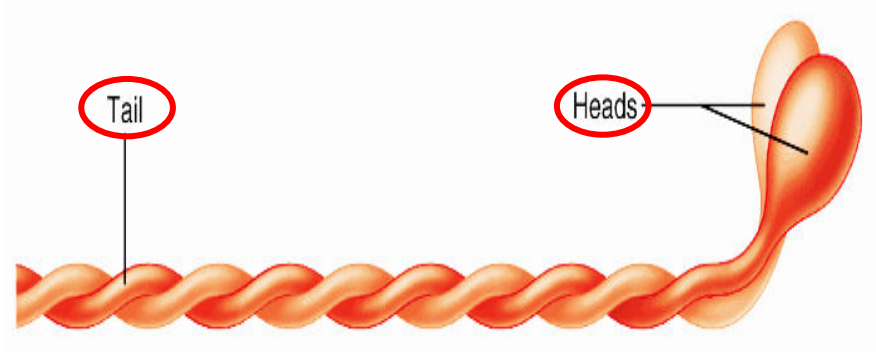
Thick Filaments

- Thick filaments extend the entire length of A band.
- These filaments are about 16 nm in diameter.
- They are composed of about 300 myosin molecules.

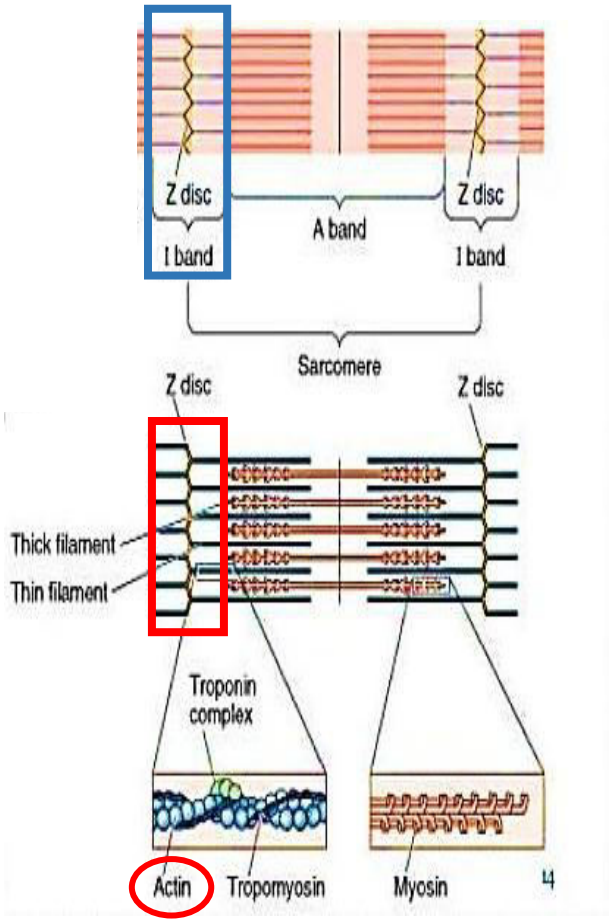
Myofilament Substructure

Myosin Molecule

- A myosin molecule consists of two identical heavy chains.
- These chains are coiled together to form a long tail.
- Myosin also has two globular heads, made from two heavy chains and three or four calcium-binding light chains.
- The heads form cross bridges between the thick and the thin myofilaments during contraction.



Myofilament Substructure



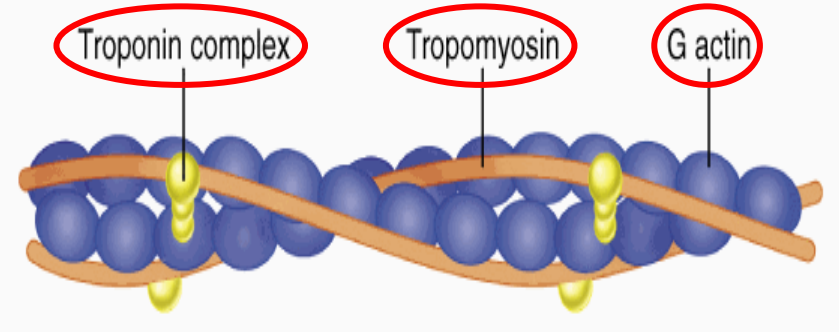
Thin Filaments

- Thin filaments are 7-8 nm thick and extend across the I band.
- They also overlap myosin filaments in the darker regions of A band.
- They are composed chiefly of actin molecules.

Myofilament Substructure

Thin Filaments

- In thin filaments, actin molecules are arranged in two chains which twist around each other.
- Two strands of another protein tropomyosin twist around the actin.
- In a relaxed muscle fiber, tropomyosin blocks cross-bridging b/w myosin and actin filaments.
- Thin filaments also have a three polypeptide complex troponin at intervals of about 40nm along the thin filament.



Myofilament Substructure

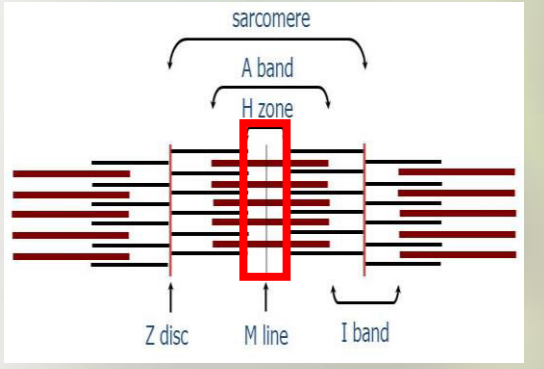
Thin Filaments

- One troponin polypeptide (TnI) is inhibitory and binds to actin.
- Other, TnT binds to tropomyosin and helps position it on actin.
- Third (TnC) binds the calcium ions.
- Both troponin and tropomyosin help control myosin-actin interactions involved in contractions.

Myofilament Substructure

The H Zone

- The center of A band appears lighter than the other regions in a relaxed sarcomere.
- This region is called H zone.
- This zone contains only thick filaments.

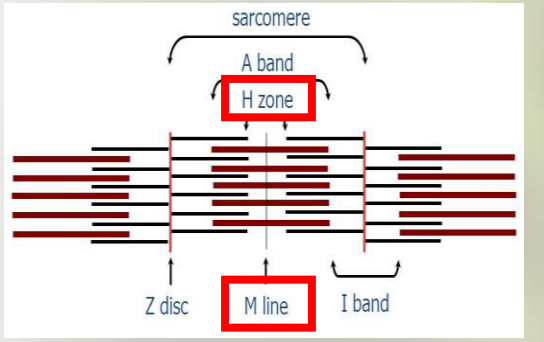


Myofilament Substructure

The H Zone

- There are no overlaps between the actin and myosin in this region.
- The H zone is bisected by a dark line, the M line.
- M line contains enzymes important in energy metabolism.

END



Muscle Physiology

Contraction of Muscle: Sliding Filament Theory

LECTURE NO 102

Contraction of Muscle: Sliding Filament Theory

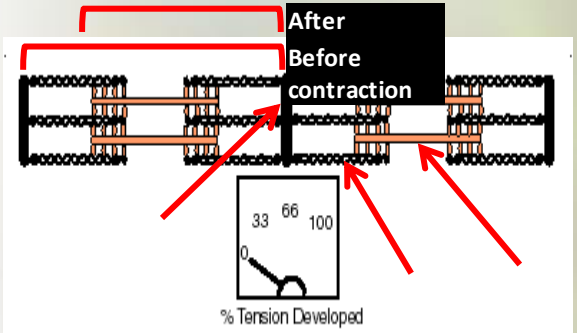
Sliding Filament Theory

- H. E. Huxley and A. F. Huxley proposed this theory in 1954 to explain muscle contraction.
- It states that during muscle contraction the thin and thick filaments in sarcomeres slide and undergo shifting.

Contraction of Muscle: Sliding Filament Theory

Sliding Filament Theory

- When a muscle contracts, the thin actin filaments slide b/w the thick myosin filaments and move closer to the center of sarcomere.
- As a result, sarcomere becomes shorter.



http://www.apsubiology.org/anatomy/2010/2010_Exam_Reviews/Exam_3_Review/slidingfilaments.gif

Contraction of Muscle: Sliding Filament Theory

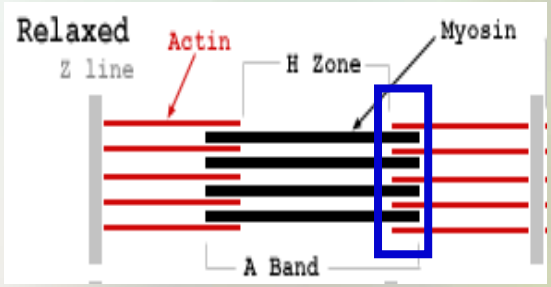
Sliding Filament Theory

- When a muscle relaxes or is stretched, the overlap between thin and thick filaments is reduced, and the sarcomere elongates.
- Changes in sarcomere length during stretch and contraction, correspond to changes in muscle length.

Contraction of Muscle: Sliding Filament Theory

Sliding Filament Theory: Explanation

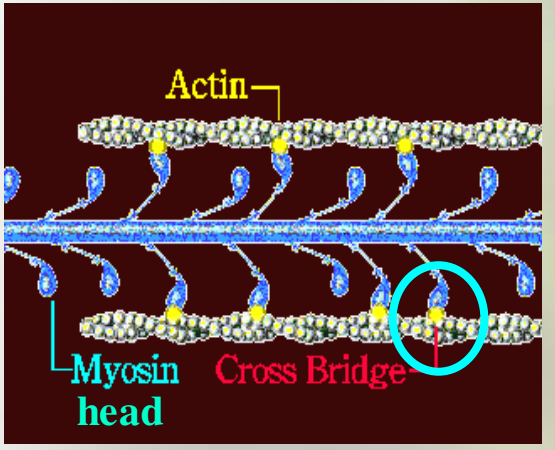
- In a relaxed muscle fiber, thick and thin filaments overlap only at the ends of A band.



Contraction of Muscle: Sliding Filament Theory

Sliding Filament Theory: Explanation

- Sliding begins when myosin heads attach to the binding sites on actin thin filaments, i.e. cross bridges are formed.

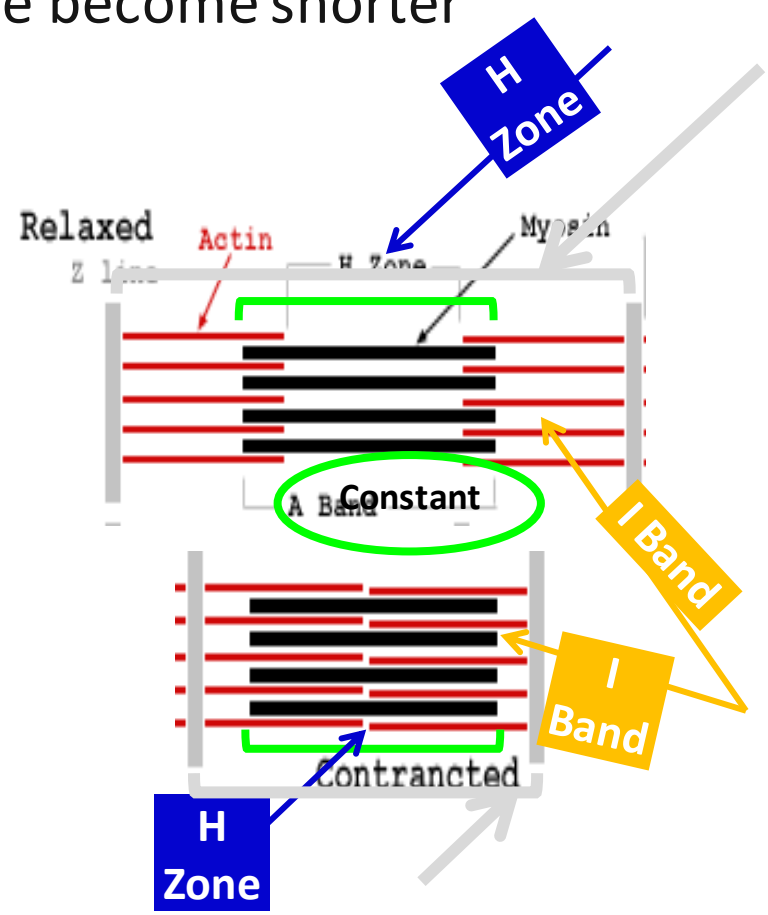


https://kristindockter.wikispaces.com/file/view/HEM21.gif/133853275/HEM21.gif

Contraction of Muscle: Sliding Filament Theory

Sliding Filament Theory: Explanation

- During contraction:
  - A bands maintain a constant length.
  - I bands and H zone become shorter
  - Z lines get closer.



Contraction of Muscle: Sliding Filament Theory

Sliding Filament Theory: Explanation

- When muscle is stretched, A band again maintains a constant length, but the I bands and H zone become longer.

Contraction of Muscle: Sliding Filament Theory

Sliding Filament Theory: Explanation

- Neither the myosin thick filaments nor the actin thin filaments change their lengths during shortening or stretching.
- It is the extent of overlap between actin and myosin filaments that changes.

Contraction of Muscle: Sliding Filament Theory

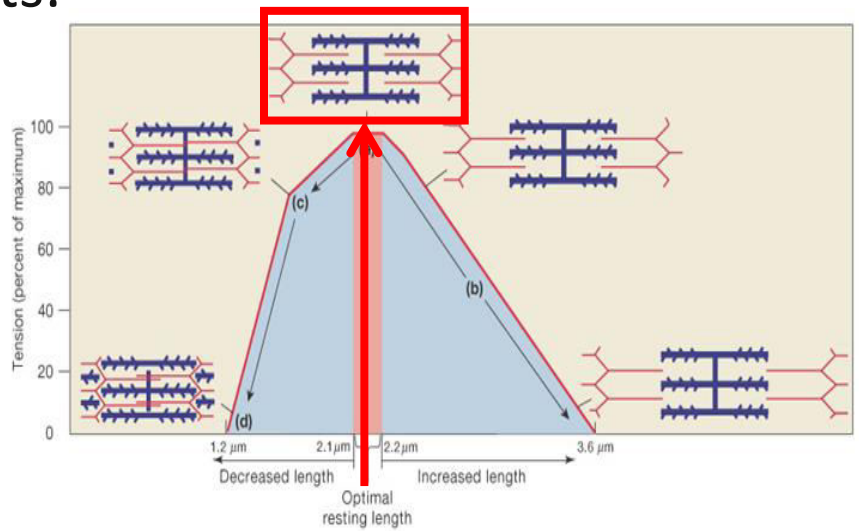
Length-Tension Curve

- Strongest evidence in support of sliding filament model comes from the length-tension relation of a sarcomere.
- Measurement of the length of sarcomere during contraction and resulting force generates a length-tension curve.
- This curve explains the assumptions of sliding-filament theory.

Contraction of Muscle: Sliding Filament Theory

Explanation of the Curve

- The tension produced by the muscle is maximum when largest number of cross-bridges are formed between actin and myosin.
- These cross bridges form due to the overlap of thick and thin filaments.

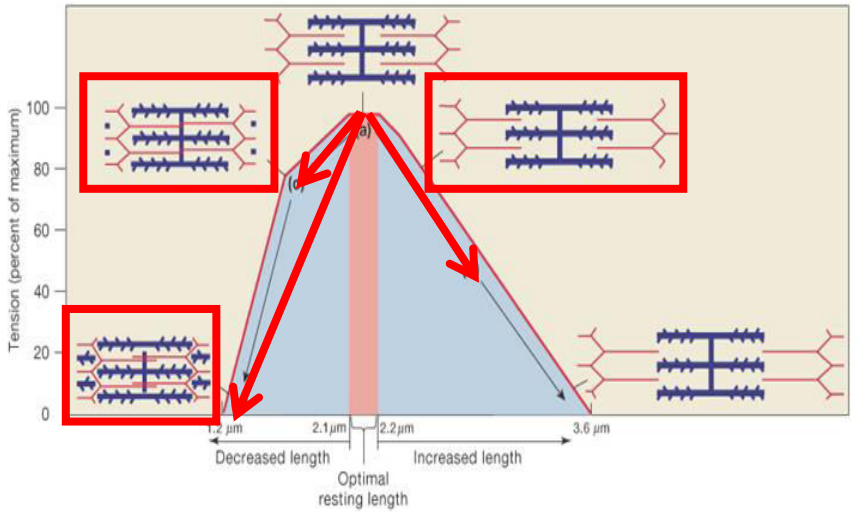


http://faculty.pasadena.edu/dkwon/chapt_11/images/image78.png

Contraction of Muscle: Sliding Filament Theory

Explanation of the Curve

- Tension is reduced with increased length of sarcomere, because the thick and thin filaments overlap less and fewer crossbridges can be formed.
- It also reduces with decreased length, as thin filaments begin to collide with one another, preventing further shortening.

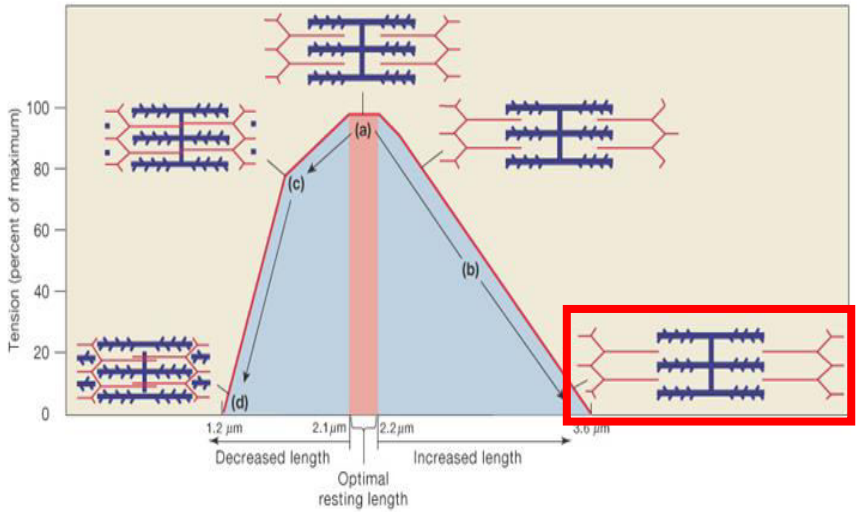


http://faculty.pasadena.edu/dkwon/chapt_11/images/image78.png

Contraction of Muscle: Sliding Filament Theory

Explanation of the Curve

- The curve also predicts the consequence of over stretching a sarcomere so far that there remains no overlap between actin and myosin filaments, making it impossible to develop any crossbidges.
- As a result no active tension will develop.



http://faculty.pasadena.edu/dkwon/chapt_11/images/image78.png

Contraction of Muscle: Sliding Filament Theory

Conclusion

- This curve shows that tension produced by contraction of sarcomere is proportional to its shortening which, as proposed by sliding filament theory, is due to sliding of thick and thin filaments and formation of cross bridges in the sarcomere during contraction.

END

Muscle Physiology

Role of ATP in Cross Bridge Working

LECTURE NO 103

Role of ATP in Cross Bridge Working

Cross-Bridge Attachment and Detachment

- Myosin cross-bridges must attach to binding sites on actin filaments in order to generate force.
- Cross-bridges must also be able to detach b/c attached cross-bridges would prevent filaments from sliding past one another, locking the muscle at one length.

Role of ATP in Cross Bridge Working

Cross-Bridge Attachment and Detachment

- Detachment of cross-bridges from actin is also necessary for the muscle to relax.
- So, during contraction, cross-bridges must attach to and detach from thin filaments in a cyclic manner.
- In this cycle, ATP plays a crucial role.

Role of ATP in Cross Bridge Working

Cross-Bridge Attachment

- Cross bridges are formed when actin (A) and myosin (M) bind and form a stable complex actomyosin (AM).

A + M = AM

1

- This happens in the absence of ATP.

Role of ATP in Cross Bridge Working

Role of ATP in Cross-Bridge Detachment

- Cross-bridge detachment occurs in the presence of ATP.
- ATP causes the AM complex to rapidly dissociate into actin and myosin-ATP.

AM + ATP = A + M-ATP

Role of ATP in Cross Bridge Working

$$\text{M-ATP} \longrightarrow \text{M-ADP-P}_i$$

$$\text{M-ADP-P}_i \xrightarrow{\text{Very slow}} \text{M + ADP + P}_i$$

Role of ATP in Cross-Bridge Detachment

- The ATP in Myosin-ATP complex hydrolyzes to form myosin-ADP-P_i complex.
- ADP and P_i unbind from myosin very slowly.

Role of ATP in Cross Bridge Working

Cyclic Activity of AM Complex

- The release of ADP and P_i is greatly speeded up when actin binds to myosin in the myosin-ADP-P_i complex.

$$A + M\text{-}ADP\text{-}P_i \xrightarrow{\text{Fast}} AM + ADP + P_i$$

- Binding of actin forms another actomyosin complex.
- This reaction is kinetically favored as it releases energy.
- Combining these reactions produces a cycle of binding and unbinding of myosin with actin; with a net use of one molecule of ATP per cycle.

Role of ATP in Cross Bridge Working

Rigor Mortis

- After death, human and other animal's bodies gradually become rigid.
- This condition is called rigor mortis.
- This rigidity happens because ATP's are not available in dead body for detachment of actin and myosin, so muscles cannot relax.

END

Muscle Physiology

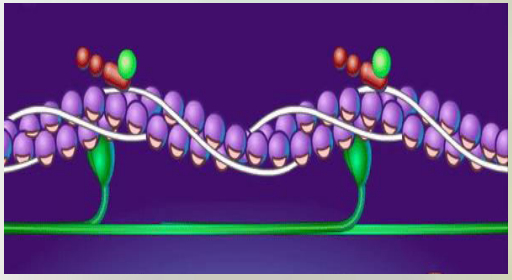
Production of Force for Sliding of Filaments

LECTURE NO 104

Production of Force for Sliding of Filaments

Myosin Head Rotation Produces Force

- Cross-bridges b/w myosin heads & actin pull the thin filament toward the center of sarcomere.

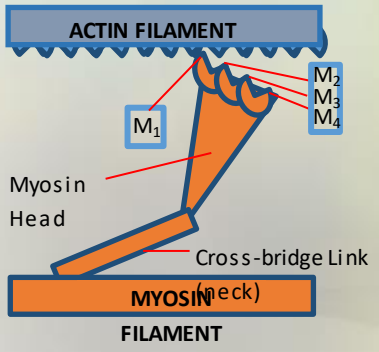


- The force for pulling is produced by the partial rotation of myosin heads

Production of Force for Sliding of Filaments

How Rotation is Produced?

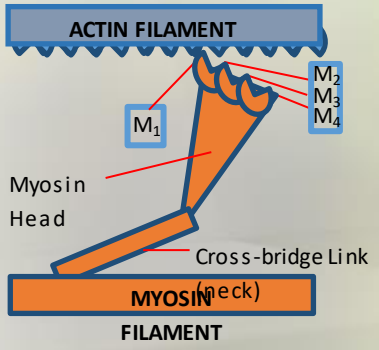
- Rotation is produced b/c of sequential binding of four sites (M_1 to M_4) of myosin head with the binding sites of actin filament.



Production of Force for Sliding of Filaments

Energy Storage in the Link

- As the myosin head rotates against the actin filament, the link is stretched elastically and stores mechanical energy due to tension developed in it.



Production of Force for Sliding of Filaments

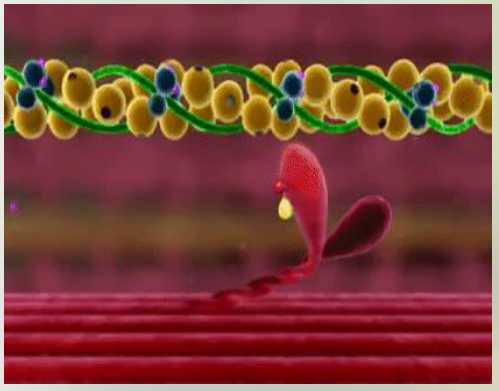
Transmission of Force to Thick Filament

- The tension produced in the link is transmitted to the thick myosin filament.
- This tension provides force to shorten the sarcomere.

Production of Force for Sliding of Filaments

Detachment of Myosin Head

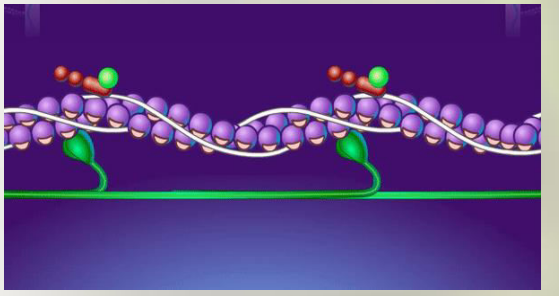
- When rotation of the head is complete, it dissociates from the actin filament and rotates back to its relaxed position.



Production of Force for Sliding of Filaments

Detachment of Myosin Head

- Dissociation occurs when Mg^{2+} & ATP bind to head.
- ATP is then hydrolyzed, and myosin head changes conformation.
- Head rebind to a little farther site on the actin filament.

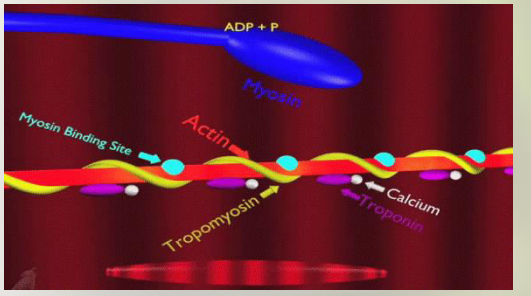


Production of Force for Sliding of Filaments

Cyclic Repetition

- Attachment, rotation, and detachment of myosin heads is repeated over and over in cyclic manner, causing filaments to slide past one another in small steps, resulting in sarcomere contraction.

END



Muscle Physiology

Role of Calcium in Contraction

LECTURE NO 105

Role of Calcium in Contraction

Role of Ca^{2+} in Cross-Bridge Attachment

- Ca^{2+} ions play a crucial role in regulating the contractile activity of muscles.
- Calcium helps to expose the active sites of actin that bind myosin heads to form cross-bridges, necessary for contraction.

Role of Calcium in Contraction

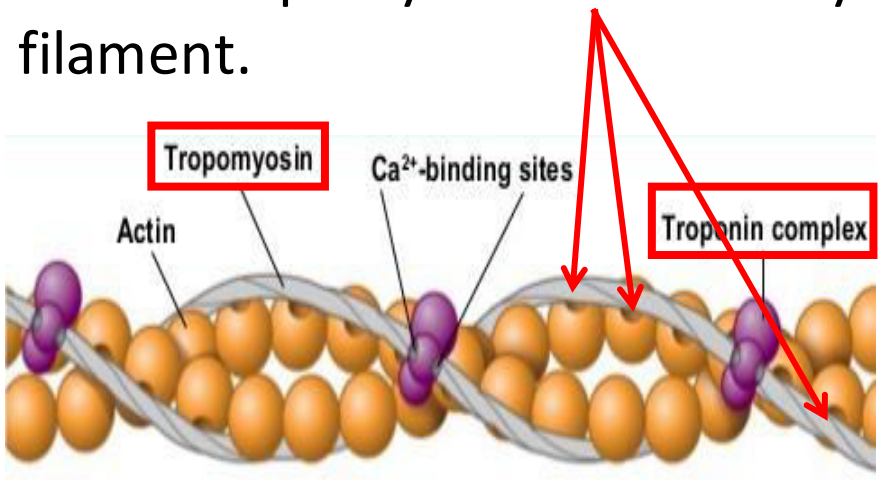
Ca^{2+} Acts Through Regulatory Proteins

- Ca^{2+} induces contraction with the help of two regulatory proteins associated with actin filaments:
  - Troponin
  - Tropomyosin

Role of Calcium in Contraction

Tropomyosin Masks Active Sites of Actin

- In a relaxed myofibril, tropomyosin coils around the actin filaments.
- It sterically (physically) covers the myosin-binding sites of actin, preventing actin and myosin from interacting.
- Troponin complex binds to tropomyosin about every 40 nm along the thin actin filament.

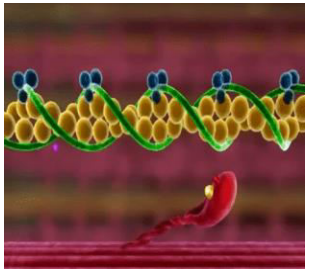


Biology. (8th ed) 2008. By Neil A. Campbell and Jane B. Reece

Role of Calcium in Contraction

Binding of Ca^{2+} to Troponin

- Troponin has high affinity for Ca^{2+}.
- Each troponin complex can bind four Ca^{2+} ions.
- When Ca^{2+} bind to troponin molecule, it undergoes a change in conformation.
- Conformational change in troponin causes a shifting in the position of tropomyosin.
- Tropomyosin movement exposes myosin binding sites on the thin actin filament.



Role of Calcium in Contraction

Ca^{2+} Binding to Troponin Results in Contraction

- Ca^{2+} binding to troponin, removes inhibition for attachment b/w myosin heads and thin filaments.
- So thin and thick filaments can make cross-bridges & slide past each other causing contraction of muscle fiber.

Role of Calcium in Contraction

Concentration of Ca^{2+}

- The concentration of Ca^{2+} ions in cytosol required for binding of cross-bridges to actin is above 10^{-7} M.

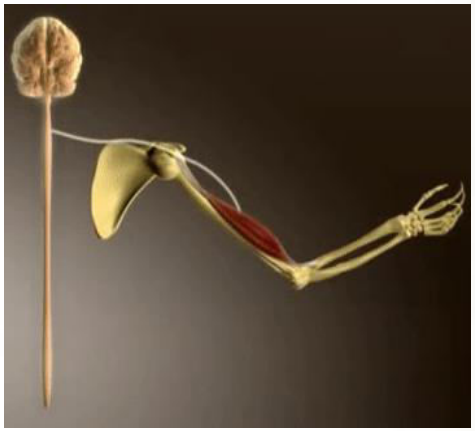
END

Muscle Physiology

Excitation Contraction Coupling

LECTURE NO 106

Excitation Contraction Coupling



Nerve Impulse Triggers Muscle Contraction

- The skeletal muscles contract in response to a nerve impulse from brain that arrives through motor neurons at the neuromuscular junction.

Excitation Contraction Coupling

Release of Acetylcholine

- At the neuromuscular junction, motor neurons release acetylcholine as neurotransmitter.
- ACh binds to receptors (ligand-gated ion channels) in postsynaptic muscle fibers.
- Opening of these channels causes movement of sodium (Na^+) and potassium (K^+) ions.

Excitation Contraction Coupling

Membrane Excitation

- Movement of ions causes change of potential in muscle cell membrane resulting in membrane excitation.
- Membrane potential of excited muscle fiber is known as end-plate potential.

Excitation Contraction Coupling

Generation of Action Potential

- Membrane excitation results in triggering an all-or-none AP in muscle fiber membrane.
- The AP propagates away, exciting the entire membrane of the muscle fiber.
- This sets in motion the sequence of events leading to contraction.

Excitation Contraction Coupling

Excitation-Contraction Coupling

- The sequence of events that convert an action potential to muscle contraction is known as excitation-contraction coupling.

Excitation Contraction Coupling

Latency Period in Coupling

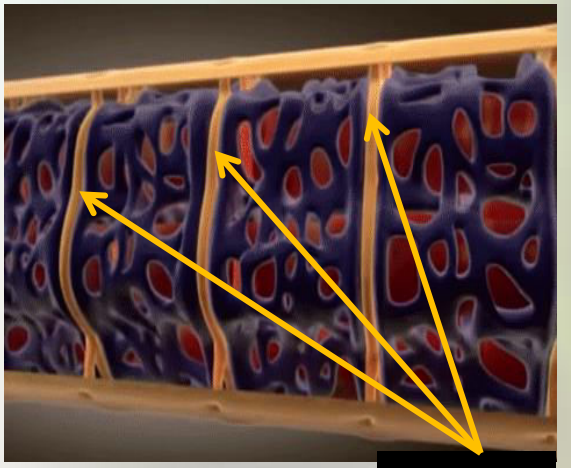
- After the arrival of an AP, it takes several milliseconds to begin contraction.
- This latency is because of large size of skeletal muscle fibers which cannot contract unless AP spreads deep into the fiber to the vicinity of each myofibril.

Excitation Contraction Coupling

Latency Period in Coupling

- During the latent period, AP is transmitted along the transverse tubules deep within the fiber.

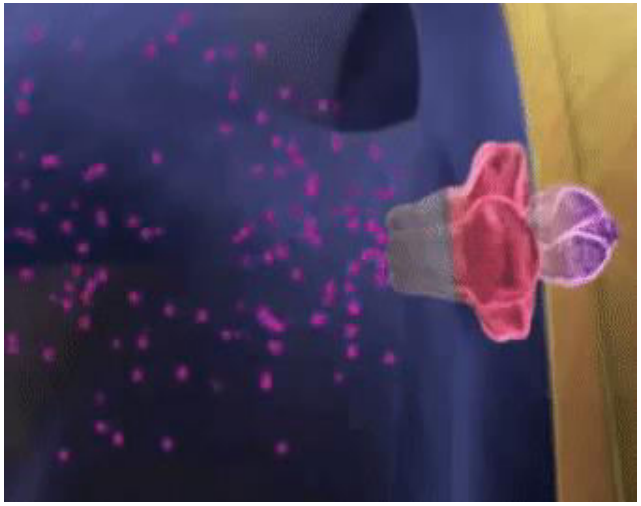
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Excitation Contraction Coupling

Release of Ca^{2+} and Contraction

- Transmission of AP through T tubules results in the release of Ca^{2+} ions form stores of sarcoplasmic reticulum.
- This increases Ca^{2+} conc. inside the muscle fiber in the immediate vicinity of myofibrils.
- These calcium ions cause the contraction to begin.



Excitation Contraction Coupling

Release of Ca^{2+} and Contraction

- So, the net effect of excitation-contraction coupling is to link an AP in the plasma membrane of the muscle fiber to the concentration of free Ca^{2+} in the cytosol that initiate contraction.

END

Muscle Physiology

T-Tubules: Propagation of AP into the Myofibril

LECTURE NO 107

T-Tubules: Propagation of AP into the Myofibril

Muscle Fiber Size: A Problem in Propagation of AP

- An AP arriving at the neuromuscular junction causes a potential difference across the surface membrane of muscle cell.
- This potential difference can directly affect less than a micrometer area of the membrane.

T-Tubules: Propagation of AP into the Myofibril

Muscle Fiber Size: A Problem in Propagation of AP

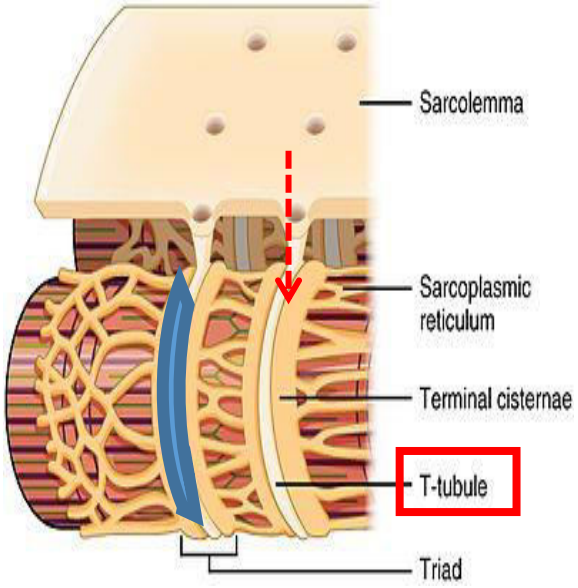
- The skeletal muscle fibers are 50-100 µm in diameter.
- Small APs spreading along the surface membrane cannot cause current flow deep within these huge muscle fibers.

T-Tubules: Propagation of AP into the Myofibril

Spread of Depolarization in Muscle Fiber

- Muscle fibers have a specialized mechanism to spread depolarization deep into the myofibrils.
- This mechanism involves transmission of APs along transverse tubules.
- T tubules couple depolarization of surface membrane to the myofibrils.

T-Tubules: Propagation of AP into the Myofibril



https://upload.wikimedia.org/wikipedia/commons/thumb/9/94/1023_T-tubule.jpg/400px-1023_T-tubule.jpg

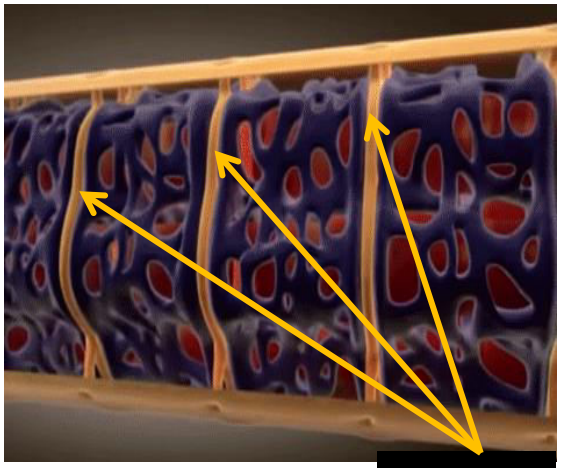
T Tubules

- T tubules are thin internal extensions of the cell membrane.
- They are less than 0.1 µm in diameter.
- They innervate the cell at the level of Z disk.
- Make branching networks around the perimeter of each myofibril.

T-Tubules: Propagation of AP into the Myofibril

Role of T Tubules

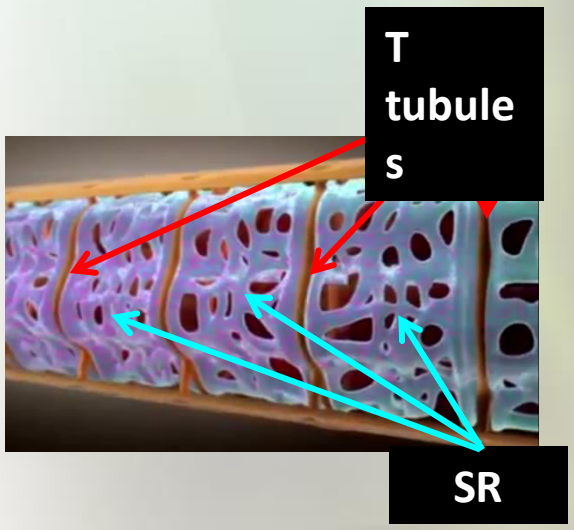
- T tubule system provides the anatomic link between the surface membrane and the myofibrils deep inside the muscle fiber.
- When an action potential spreads over a muscle fiber membrane, a potential change also spreads along the T tubules to the deep interior of the muscle fiber.



T-Tubules: Propagation of AP into the Myofibril

Role of T Tubules

- In the cell interior, T tubes are linked to SR.
- SR releases Ca^{2+} that permit myosin cross-bridges to attach and generate force for contraction.



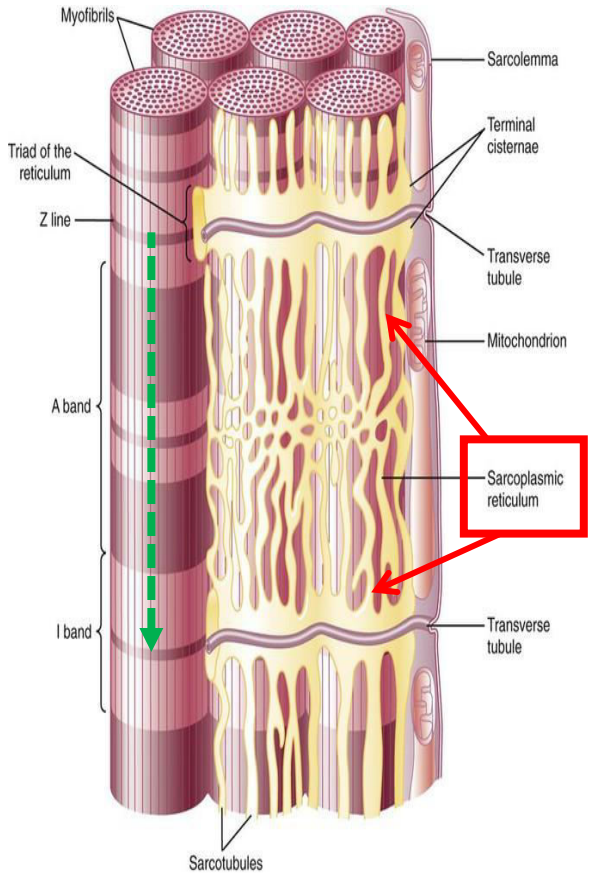
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Muscle Physiology

Sarcoplasmic Reticulum

LECTURE NO 108

Sarcoplasmic Reticulum



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Sarcoplasmic Reticulum

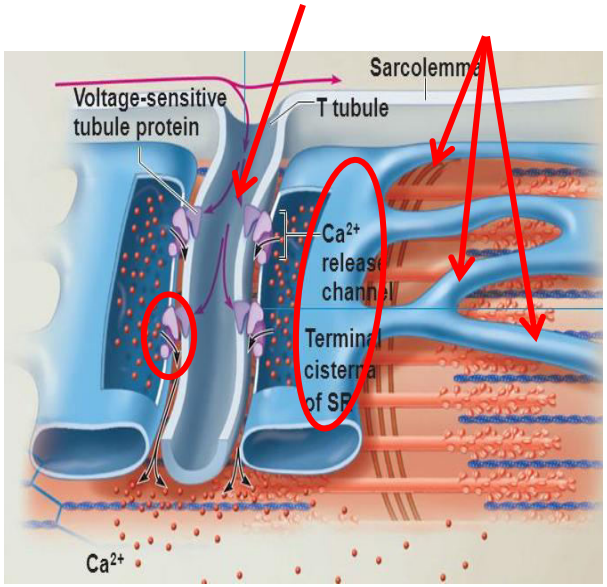
- SR is a network of membrane-bound tubules extending throughout muscle cells on either side of a Z disk and extends from one Z disk to the next as well.
- In many features, it is similar to the ER of other cells.

Sarcoplasmic Reticulum

Sarcoplasmic Reticulum

- The SR has a special organization that is extremely important in controlling muscle contraction.
- Rapidly contracting types of muscle fibers have extensive network of sarcoplasmic reticulum.

Sarcoplasmic Reticulum



http://images.slideplayer.com/34/10164988/slides/slide_14.jpg

Structure of SR

- Composed of two major parts:
  - Terminal cisternae
  - Longitudinal tubules
- Terminal cisternae are larger chambers that are closely associated with T-tubules (12 nm apart).
- These are the primary site of calcium release as they have channel proteins that open with AP in T tubules.

Sarcoplasmic Reticulum

Structure of SR

- Long longitudinal tubules run between the terminal cisternae and surround the myofibrils.
- These are the locations where ion channels for calcium ion absorption (calcium pumps) are most abundant.
- So, they are involved in Ca^{2+} sequestering activity.

Sarcoplasmic Reticulum

Calcium Sequestering by SR

- The main function of SR is to sequester and store calcium (Ca^{2+}) ions.
- It takes up Ca^{2+} ions due to the activity of calcium pumps in its membrane.
- These pumps actively transport Ca^{2+} ions from sarcoplasm and concentrate it inside the reticular tubules.

Sarcoplasmic Reticulum

Role of Calsequestrin

- Inside the SR, Ca^{2+} is stored in bound form to a protein “calsequestrin”.
- Each molecule of this protein can bind around 50 Ca^{2+}.
- This decreases the conc. of free Ca^{2+} within the SR and enhances its capacity to store more calcium ions.

Sarcoplasmic Reticulum

Importance of Ca^{2+} Sequestering by SR

- Due to sequestering by SR, cytosolic Ca^{2+} level is kept below 10^{-7} M.
- This conc. is required to remove Ca^{2+} bound to troponin and preventing contraction.

Sarcoplasmic Reticulum

END

Importance of Ca^{2+} Sequestering by SR

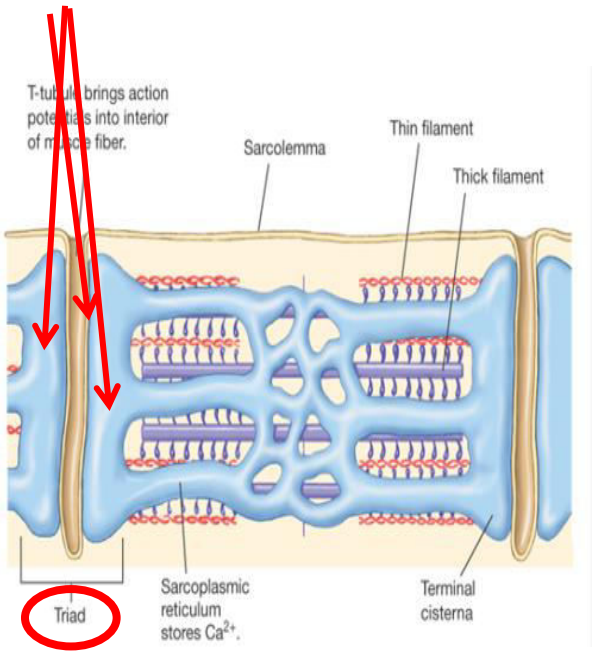
- When an AP arrives through T tubules, it causes opening of calcium channels in the terminal cisternae of SR and large quantities of stored Ca^{2+} ions are released.
- When Ca^{2+} ions are released, contraction is activated.

Muscle Physiology

Membrane Receptors in Triads

LECTURE NO 109

Membrane Receptors in Triads

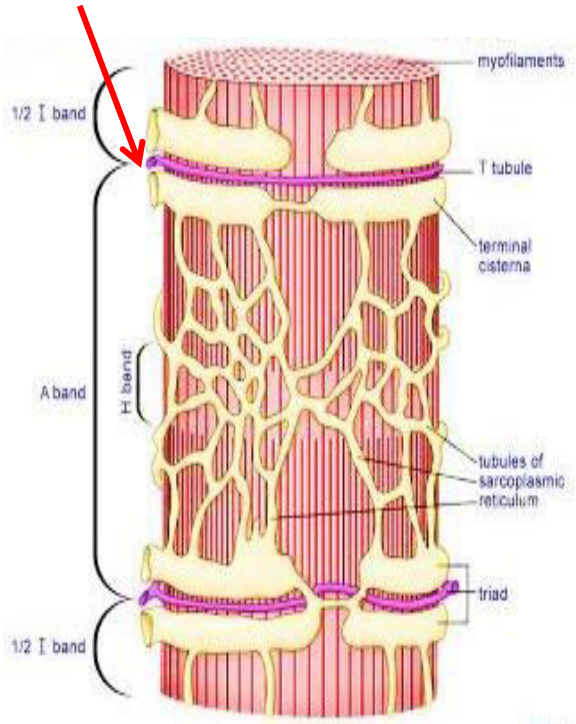


http://faculty.pasadena.edu/dkwon/chapt_11/images/image16.png

Triads

- In the skeletal muscle fiber, a T tubule is associated with two terminal cisternae of sarcoplasmic reticulum on its both sides.
- This arrangement of three associated tubes or sacks forms a structure that is called a triad.

Membrane Receptors in Triads



http://histology.kasralainy.edu.eg/_/rsrc/1428343080738/home/first-year/faqs/muscular-tissue/392117_143254292452214_41220315_n.jpg

Triads

- Triads are located at the junction between the A and I bands of the sarcomere.
- Each skeletal muscle fiber has many thousands of triads which are visible in longitudinal sections of muscle fiber.

Membrane Receptors in Triads

Function of Triads

- Triads form anatomical basis of excitation-contraction coupling, due to which a stimulus excites the muscle and causes its contraction.
- Due to the close proximity of T tubules and SR at the triads, an AP in T tubule causes the SR to release Ca^{2+}.

Membrane Receptors in Triads

Receptors in Triads and Release of Ca^{2+}

- The membrane of SR in the triad region has specialized Ca^{2+} ion channel proteins called ryanodine receptors.
- When they open, Ca^{2+} ions are released from the SR.

Membrane Receptors in Triads

Opening of Ryanodine Receptors

- They open when dihydropyridine receptors (a cluster of proteins) in the T-tubule membrane are activated.
- Dihydropyridine receptors are voltage-sensitive receptors which are activated by AP in the T tubule.

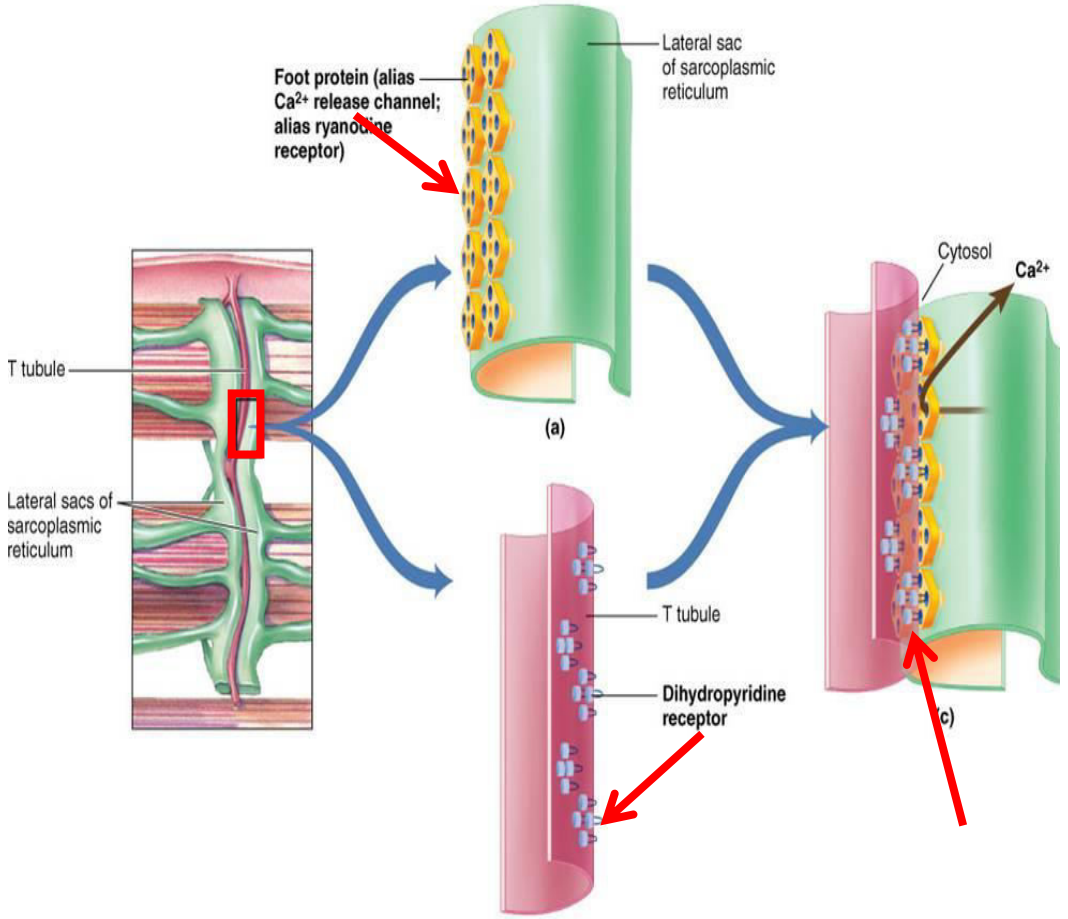
Membrane Receptors in Triads

Opening of Ryanodine Receptors

- Activation of Dihydropyridine receptors causes a change in their conformation.
- In the activated state, these receptors mechanically interact with ryanodine receptors causing conformational change in them too.

Membrane Receptors in Triads

Ryanodine and Dihydropyridine Receptors



http://www.austincc.edu/apreview/NursingPics/MusclePics/Picture11.jpg

Membrane Receptors in Triads

Opening of Ryanodine Receptors

- Opening of ryanodine receptors allows release of calcium ions from SR.
- Release of calcium ions causes contraction to happen.

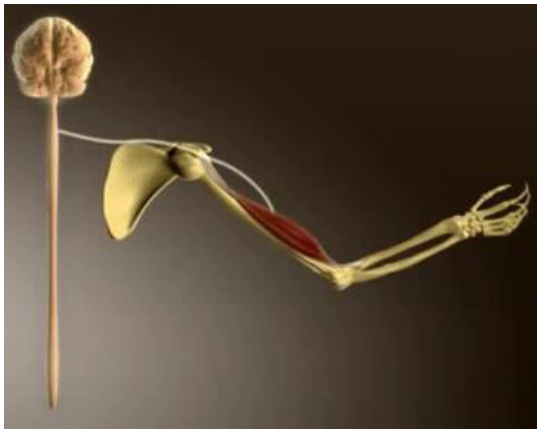
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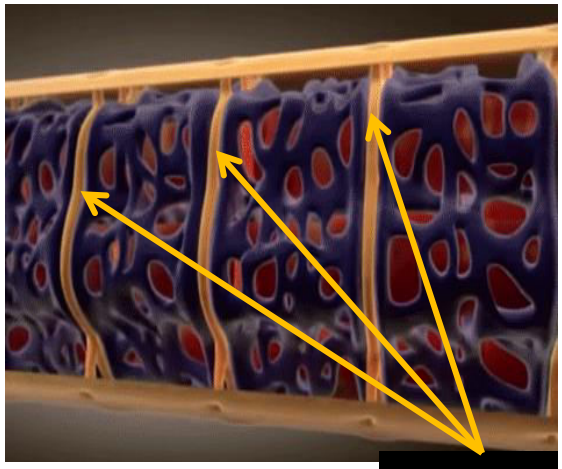
Muscle Physiology

Summary of Muscle Contraction Mechanism

LECTURE NO 110

Summary of Muscle Contraction Mechanism





Contraction-Relaxation Cycle

- Sequence of events that occur in a relaxed skeletal muscle, which lead to contraction and then relaxation:

1. An AP due to neuronal input causes the depolarization of surface membrane.
2. AP is conducted deep into the muscle fiber along the T tubules.

Summary of Muscle Contraction Mechanism

Contraction-Relaxation Cycle

3. Depolarization of T-tubule membrane, causes voltage-sensitive dihydropyridine receptors in T-tubule membrane to undergo a conformational change.
- These receptors make direct mechanical link to ryanodine receptors in SR membrane, opening Ca^{2+} channels in it.

Summary of Muscle Contraction Mechanism

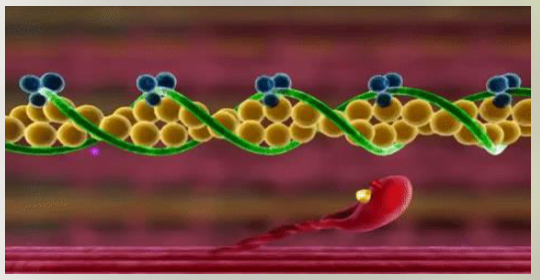
Contraction-Relaxation Cycle

4. Ca^{2+} flow out of lumen of SR increases free Ca^{2+} conc. in myoplasm from a resting value of below 10^{-7} M to an active level of about 10^{-6} M.

Summary of Muscle Contraction Mechanism

Contraction-Relaxation Cycle

5. Ca^{2+} bind to troponin, inducing conformational change in it.
- Troponin causes change in the position of tropo-myosin, exposing myosin binding sites on actin to form cross-bridges.



Summary of Muscle Contraction Mechanism

Contraction-Relaxation Cycle

6. When cross-bridges form, myosin heads rotate, producing force that pulls thin filaments.

- This causes the sarcomere to shorten.

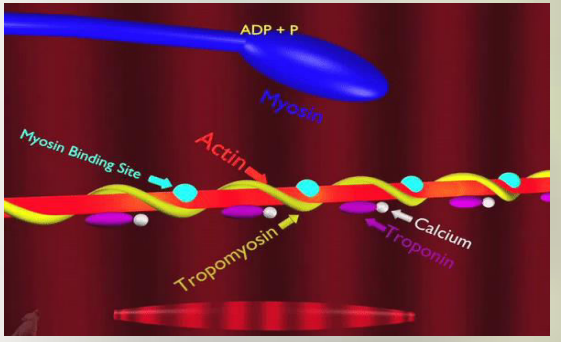
7. Now, ATP binds to the myosin head causing the myosin head to detach from the thin filament.

Effort by: AARISH ALI

Summary of Muscle Contraction Mechanism

Contraction-Relaxation Cycle

8. Hydrolysis of ATP produces energy that causes conformational change in the myosin, which reattaches to the next site along the actin filament. (cycle repeats)



Summary of Muscle Contraction Mechanism

Contraction-Relaxation Cycle

9. Finally, calcium pumps in the SR membrane actively transport Ca^{2+} from myoplasm back into the SR.

- As the conc. of free Ca^{2+} in the myoplasm drops, Ca^{2+} bound to troponin is released.

Summary of Muscle Contraction Mechanism

END

Contraction-Relaxation Cycle

- Tropomyosin again masks the cross-bridge binding sites on actin filament, inhibiting cross-bridge attachment.
- This causes muscle to relax.
- The muscle remains relaxed until the next depolarization/nerve impulse.

Muscle Physiology

Isometric and Isotonic Contractions

LECTURE NO 111

Isometric and Isotonic Contractions

Muscle Contractions

- Based on variables of:
  - Force (tension)
  - Length (shortening or lengthening)
- Muscle contractions are categorized into two types:
  - Isotonic contraction
  - Isometric contraction

Isometric and Isotonic Contractions

Isotonic Contraction (Constant Tension)

- A contraction during which muscle length changes while muscle tension remains constant.
- Isotonic contractions occur because the force exerted by muscle contraction is greater than the external force against it.

Isometric and Isotonic Contractions

Isotonic Contraction

- The change in length of the muscle results in the movement of a body part.
- So, such contractions are produced during locomotion.

Isometric and Isotonic Contractions

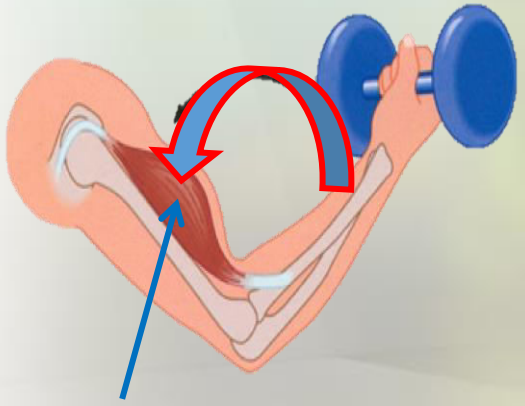
Types of Isotonic Contractions

- Based on how the length changes, isotonic contractions are classified into two types:
  - Concentric contractions
  - Eccentric contractions

Isometric and Isotonic Contractions

Concentric Contractions

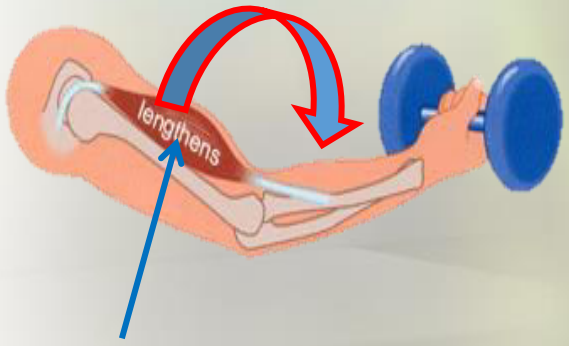
- A muscle generates tension and shortens.
- During Lifting a weight, the bicep muscle undergo a concentric contraction.



Isometric and Isotonic Contractions

Eccentric Contractions

- A muscle generates tension and lengthens.
- During lowering the weight, the bicep muscles generate force but the muscle lengthens.



Isometric and Isotonic Contractions

Isometric Contraction

- During contraction, tension in a muscle increases without a corresponding change in length (iso = same, metric=length).
- Isometric contractions occur when the force exerted by the muscle contraction is equal to the opposing external force.

Isometric and Isotonic Contractions

END

Isometric Contraction

- Isometric contractions are important in maintaining posture or stabilizing a joint.
- When one grips something hard, there is no movement of arm, but the muscles in the arm contract to provide a force to keep the object in place against gravity.

Muscle Physiology

Muscle Twitch and Tetanus

LECTURE NO 112

Muscle Twitch and Tetanus

Muscle Twitch

- A muscle twitch is a single contraction in response to a single action potential.

Three Components of a Single Muscle Twitch

- Latent period, or lag phase
- Contraction phase
- Relaxation phase

Muscle Twitch and Tetanus

The Latent Period

- A short delay from the time when the AP reaches the muscle until tension is observed in the muscle.
- This is the time required for Ca^{2+} to diffuse out of the SR and bind to troponin, and to move tropomyosin off the active sites till cross bridges are formed.

Muscle Twitch and Tetanus

The Contraction Phase

- The contraction phase is when the muscle is generating tension.
- It is associated with cycling of the cross bridges that result in the shortening of sarcomeres.

Muscle Twitch and Tetanus

The Relaxation Phase

- Relaxation phase is the time when the muscle returns to its normal length.

The Length of a Twitch

- The length of a twitch varies among different muscle types.
- It may be as short as 10 milliseconds or as long as 100 ms.

Muscle Twitch and Tetanus

Tetanus

- A sustained muscle contraction evoked by stimulation from simultaneous multiple impulses is known as tetanic contraction.
- During tetanized state, the contracting tension in the muscle remains constant in a steady state.

Muscle Twitch and Tetanus

Types of Tetanus

- A tetanic contraction can be either:
- Unfused (incomplete)
- Fused (complete).

Unfused Tetanus

- An unfused tetanus occurs when the muscles are being stimulated at a faster rate.
- The fibers do not completely relax before the next stimulus.

Muscle Twitch and Tetanus

END

Fused Tetanus

- When there is no relaxation of the muscle fibers between stimuli and the twitches overlap.
- It occurs during a high frequency of stimulation.
- A fused tetanic contraction is the maximal possible contraction.

Muscle Physiology

Neural Control of Muscle Contraction

LECTURE NO 113

Neural Control of Muscle Contraction

Coordination of Muscle Contraction

- Animal movement involves simultaneous contraction of many muscles and their fibers.
- These contractions need to be correctly timed and coordinated with respect to one another.

Neural Control of Muscle Contraction

Coordination of Muscle Contraction

- Muscle coordination is controlled by nervous system.
- Neuronal impulses conducted through α motor neurons.
- Reaching neuromuscular junctions.

Neural Control of Muscle Contraction

Coordination of Muscle Contraction

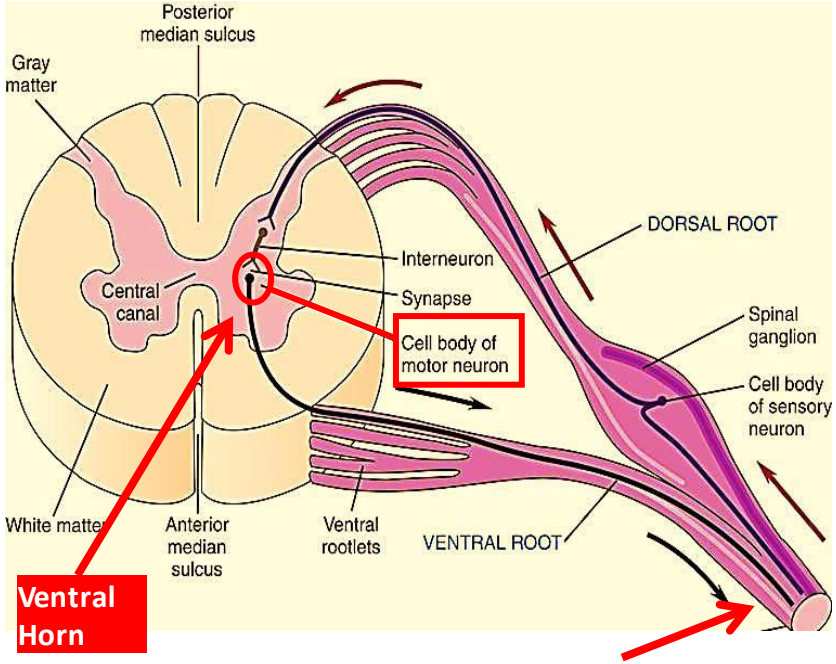
Nervous system also :

- Regulates strength of contractions
- Determines number and type of fibers activated at a time.

Neural Control of Muscle Contraction

Spinal Motor Neurons

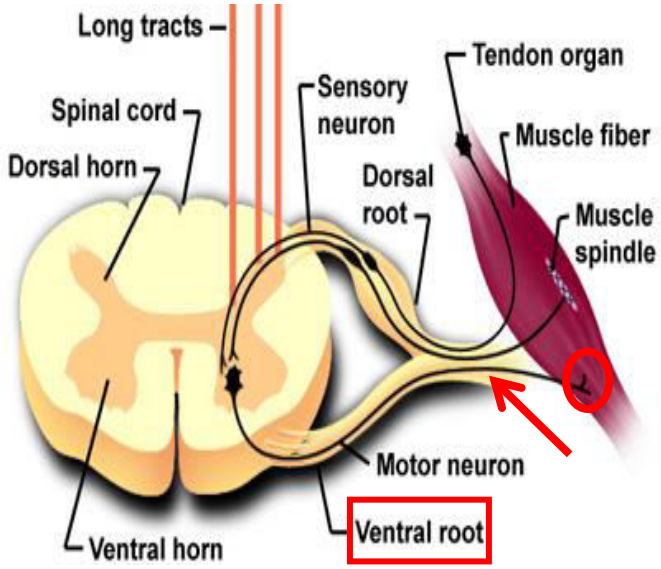
- All vertebrate skeletal muscles are innervated by spinal motor neurons.
- Cell bodies are located in ventral horn of gray matter of spinal cord.



Neural Control of Muscle Contraction

Spinal Motor Neurons

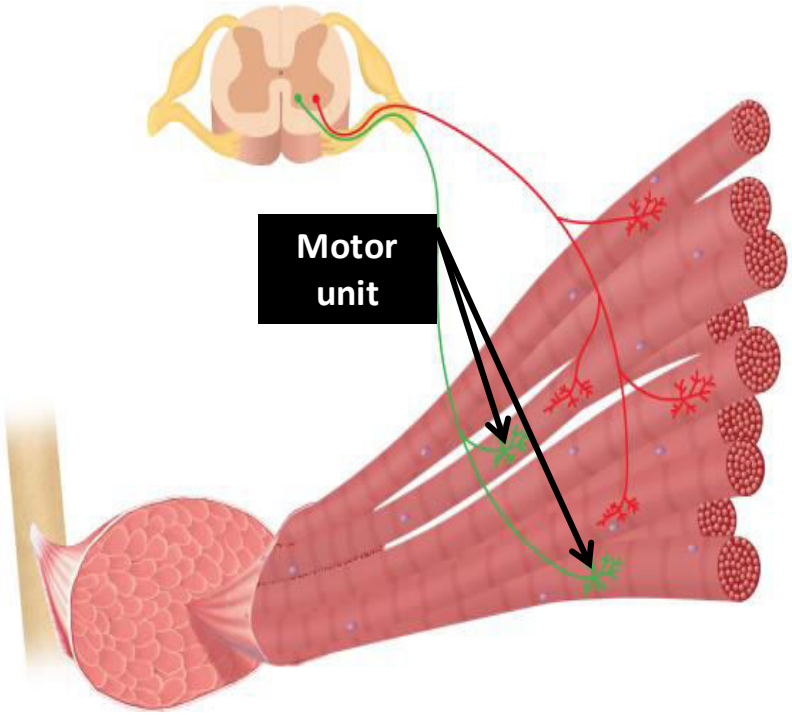
- Axons leave spinal cord by a ventral root.
- Axons reach muscles through peripheral nerves.
- In muscles, they branch off repeatedly.
- Branches innervate hundreds of skeletal muscle fibers.



Neural Control of Muscle Contraction

Spinal Motor Neurons

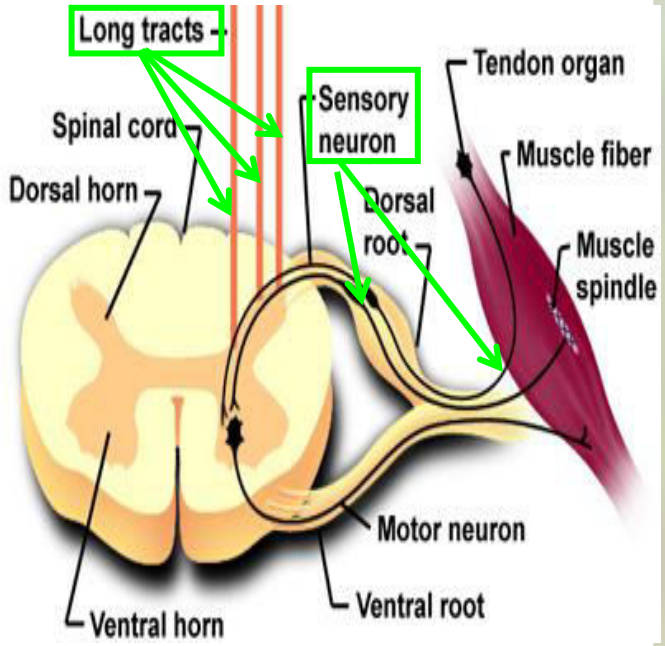
- A motor neuron and the muscle fibers that it innervates form a motor unit.



Neural Control of Muscle Contraction

Spinal Motor Neurons

- Receive synaptic inputs from:
  - Sensory motor neurons. innervating muscle fibers
  - Interneurons coming from brain
- These are the only means for controlling contraction of muscles.
- Known as "the final common pathway" of neuronal output.



Neural Control of Muscle Contraction

Fine Motor Control of Muscle Contraction

- Synaptic inputs initiate AP in a motor neuron.
- AP spreads into all of its terminal branches.
- Activates alll of its endplates.

Neural Control of Muscle Contraction

Fine Motor Control of Muscle Contraction

- All spinal α motor neurons produce neurotransmitter acetylcholine (ACh).
- ACh is released onto all of the fibers in the neuron's motor unit.

Neural Control of Muscle Contraction

END

Fine Motor Control of Muscle Contraction

- All muscle fibers in a motor unit contract with an AP.
- Frequency of APs determines whether a single twitch or a sustained tetanic contraction is produced.

Muscle Physiology

Muscle Fatigue

LECTURE NO 114

Muscle Fatigue

Muscle Fatigue

- Prolonged and sustained strong contractions of a muscle lead to muscle fatigue.
- Skeletal muscle is no longer able to contract optimally.

Muscle Fatigue

Physiological Causes of Muscle Fatigue

- Muscle fatigue has physiological causes.
- Can occur in neuromuscular junction or any other contractile element of the muscle.

Muscle Fatigue

Physiological Causes of Muscle Fatigue

- Transmission of nerve signals through neuromuscular junction diminishes after intense prolonged muscle activity.
- Happens because of impaired membrane excitability due to imbalances of ions.

Muscle Fatigue

Physiological Causes of Muscle Fatigue

- Ion imbalances occur due to inadequate functioning of the Na^+/K^+ pump.
- Inactivation or insensitivity of Na^+ and K^+ channels.
- Quick recovery (within 30 minutes).

Muscle Fatigue

Physiological Causes of Muscle Fatigue

- Interruption in excitation-contraction coupling.
- Impaired Ca^{2+} release from internal cell sources.
- Recovery time: 24 to 72 hours.

Muscle Fatigue

Physiological Causes of Muscle Fatigue

- Interruption of blood flow through a contracting muscle.
- Almost complete muscle fatigue.
- Cause: loss of nutrients and O_2 supply.

Muscle Fatigue

Other potential fatigue contributors

- Accumulation of inorganic phosphates.
- Accumulation of H^+ ions (metabolic acidosis) disrupting tissue metabolism.
- Glycogen depletion.
- Depletion of ATP.

Muscle Fatigue

Physiological Causes of Muscle Fatigue

- Lactic acid accumulation.
- Recent researches have discarded this factor as contributor to fatigue.

END

Muscle Physiology

Lever System of the Body

LECTURE NO 115

Lever System of the Body

Lever System of the Body

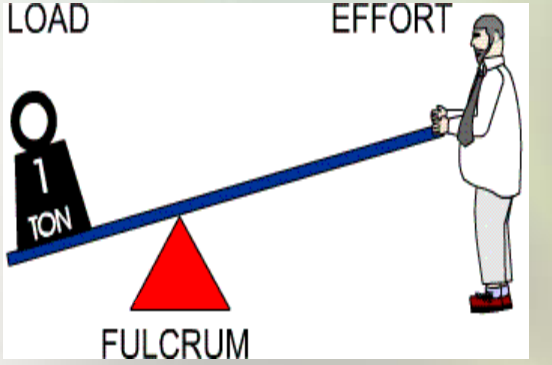
- Lever system helps in movement of body parts.
- Bones, ligaments, and muscles form levers in the body.
- Muscles apply tension to their points of insertion on bones.
- Bones, in turn, cause movement.

Lever System of the Body

Lever System

- In physical terms, a lever system has three components:
  - Axis or fulcrum
  - Force or effort
  - Load or resistance

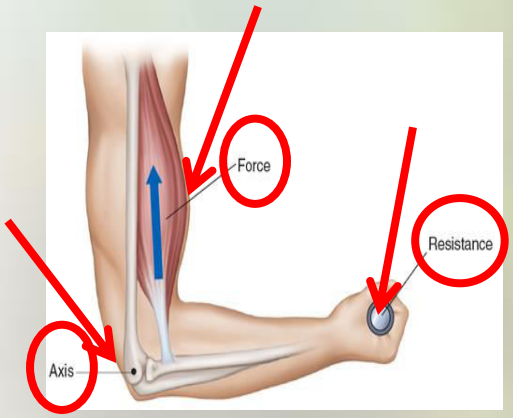
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Lever System of the Body

Lever System of the Body

- In lever system of body:
  - A joint forms the axis
  - Muscles attached to the joint provide the force
  - Weight or load is moved.



Lever System of the Body

Kinesiology

- Study of muscles, lever systems, and their working.

Kini zi
ology

Lever System of the Body

Types of Lever Systems of the Body

Levers may be:

- First class
- Second class
- Third class

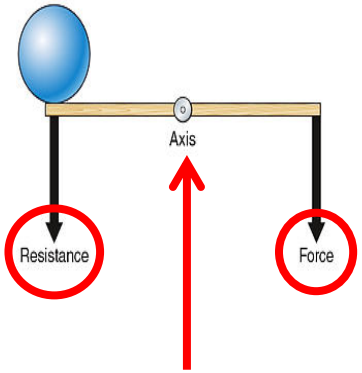
Classification depends on:

- Point of muscle insertion
- Its distance from axis (fulcrum)
- Length of the lever arm

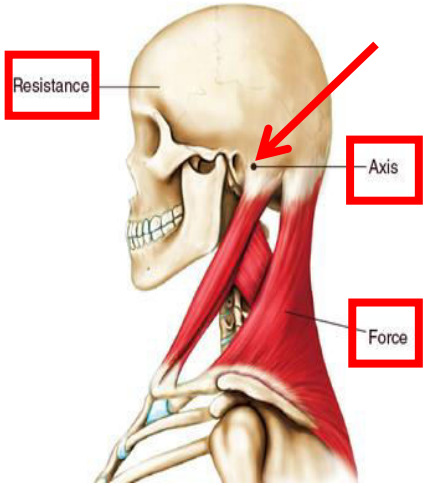
Lever System of the Body

First-Class Lever

- Axis (fulcrum) is located between the weight (resistance) and force. (e.g scissors)
- Few first-class levers in human body.
- Example: Joint b/w head and first vertebra.
- Weight (resistance) = head, axis = joint, force comes from posterior muscles attached to the skull.



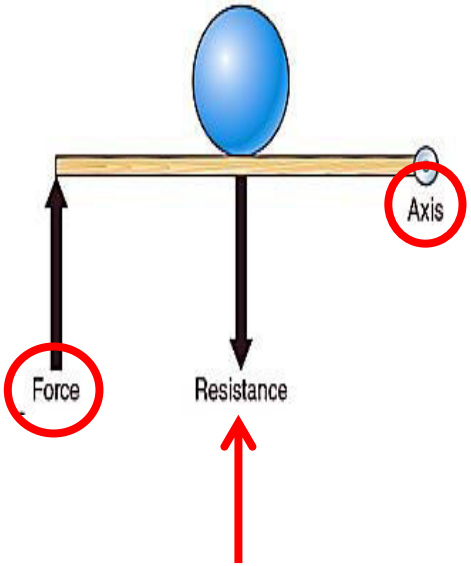




Lever System of the Body

Second-Class Lever

- Weight (resistance) is located between axis (fulcrum) and force.
- Example: a wheelbarrow



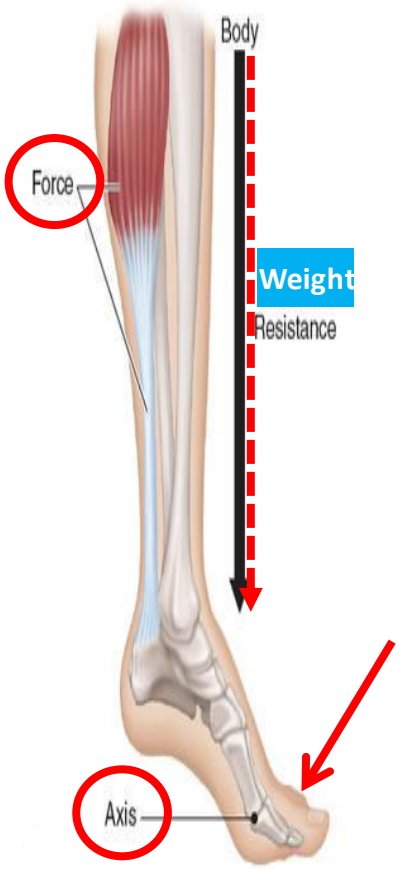


Wheelbar row

Lever System of the Body

Second-Class Lever

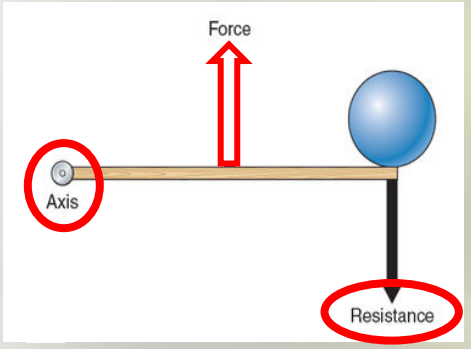
- Example: the lower leg when someone stands on tiptoes.
- Axis: formed by metatarso-phalangeal (tiptoe) joints.
- Resistance: weight of body.
- Force: applied by calf muscles.



Lever System of the Body

Third-Class Lever

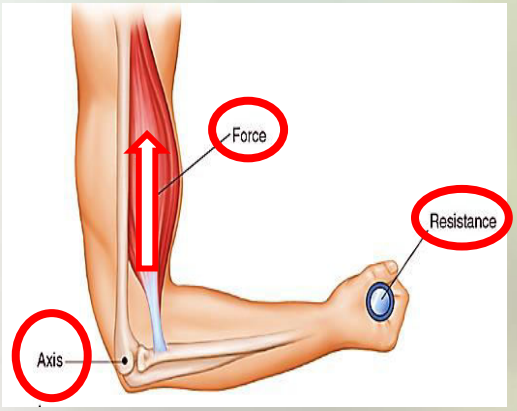
- Force is applied between resistance (weight) and axis (fulcrum).
- Most common in human body.



Lever System of the Body

Third-Class Lever

- Example: elbow joint.
- Joint = axis (fulcrum).
- Resistance (weight) is forearm, wrist, & hand.
- Force is provided by the biceps muscle when the elbow is flexed.



3

END

Muscle Physiology

Cardiac Muscles

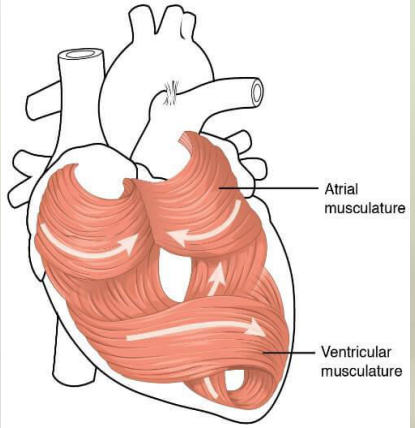
LECTURE NO 116

Cardiac Muscles

Cardiac Muscles

- Found only in the heart.
- Striated muscles
- Share many characteristics with skeletal muscle but differ in several important ways.

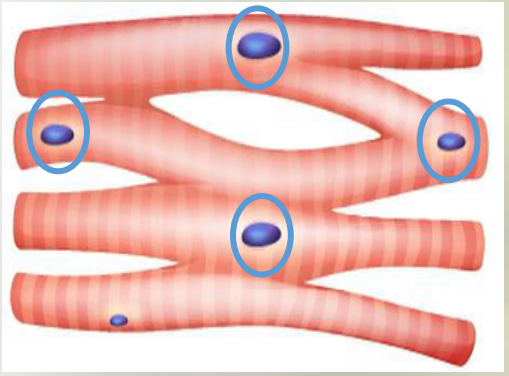
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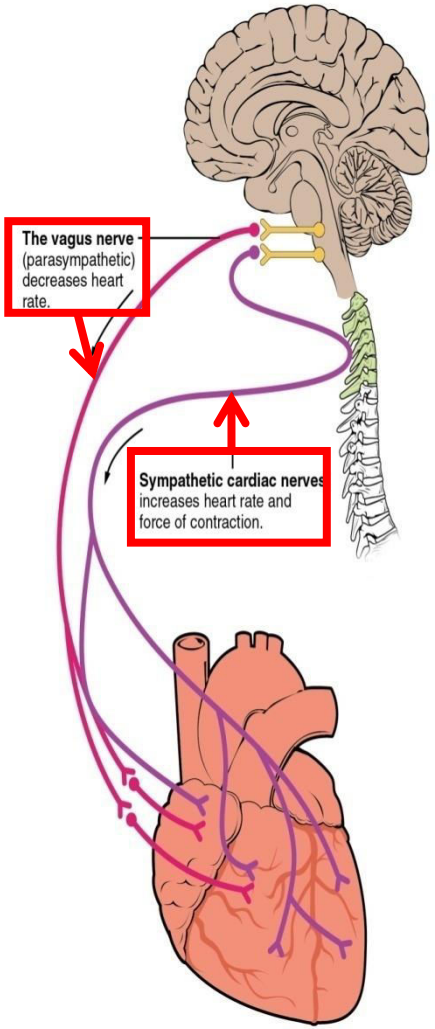
Cardiac Muscles

Features of Cardiac Muscles

- Cardiac muscle cells (myocyte) are mononucleate.
- Skeletal muscle cells are multinucleate.



Cardiac Muscles



Features of Cardiac Muscles

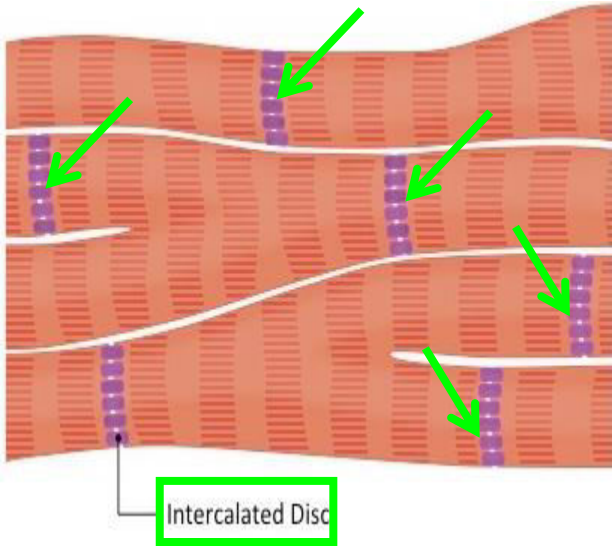
- Cardiac fibers innervated only diffusely by both:
  - Excitatory sympathetic neurons
  - Inhibitory parasympathetic neurons
- Skeletal muscle fibers are individually innervated by an excitatory motor axon.

Cardiac Muscles

Features of Cardiac Muscles

- Cardiac neural innervations are modulatory only.
- Contractions are induced by electrical activity of pacemaker.
- They don’t produce discrete PSP.
- They only increase or decrease the strength of myogenic contractions.

Cardiac Muscles



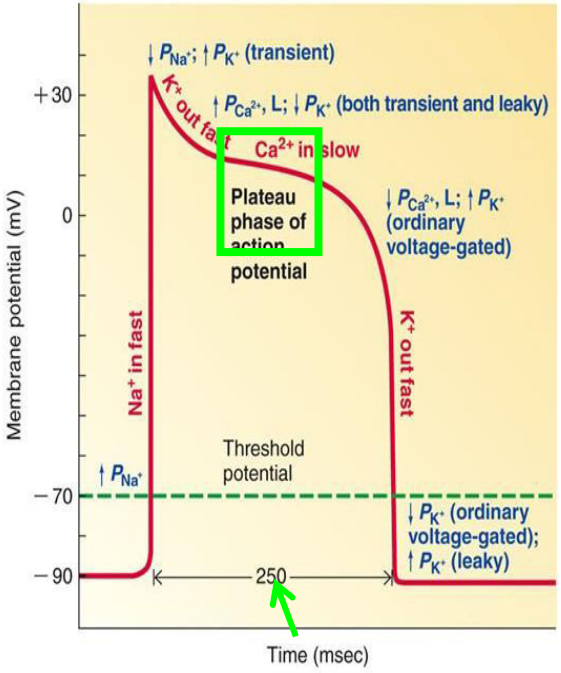
Features of Cardiac Muscles

- Cardiac muscle cells are connected electrically by intercalated disks.
- AP initiated in the pacemaker region spreads rapidly from cell to cell, through fast-conducting pathways to all muscle cells.

Cardiac Muscles

Contraction Mechanism of Cardiac Muscles

- Fundamentally resembles that of skeletal twitch muscles
- Activated by an increase in cytosolic Ca^{2+} conc.
  - However APs differ in length.
  - Skeletal muscle APs are of very short duration.
  - AP in cardiac muscle has a plateau phase of hundreds ms long.



Cardiac Muscles

Contractile Mechanism of Cardiac Muscles

- Long duration of AP and long refractory period prevent tetanic contractions of cardiac muscle fibers.
- Also permit the muscle to relax, allowing ventricle to fill with blood between APs.

Cardiac Muscles

Contractile Mechanism of Cardiac Muscles

- Regularly paced, prolonged APs, cause the heart to beat at a rate suitable for its function as a pump.

Cardiac Muscles

Contractile Mechanism of Cardiac Muscles

- Contraction is activated by raising cytosolic Ca^{2+} conc.
- Ca^{2+} influx across the plasma membrane and release from the SR.
- Mammalian cardiac cells possess an elaborate system of SR and T-tubules.

END

Muscle Physiology

Smooth Muscles

LECTURE NO 117

Smooth Muscles

General Features

- Least specialized muscle fibers.
- Have myosin similar to that found in contractile nonmuscle cells.
- Non-striated.
- Involuntary: under autonomic control.
- Contract and relax slowly.
- Capable of more sustained contractions.

Smooth Muscles

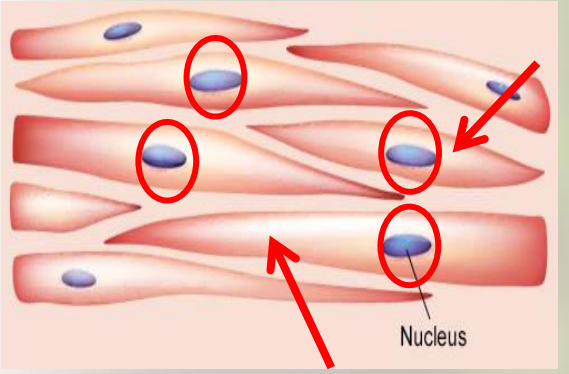
Types

Two types:

- Single-unit smooth muscles
- Multi-unit muscles

Single-Unit Muscles

- Cells small, spindle shaped & mononucleate.



Smooth Muscles

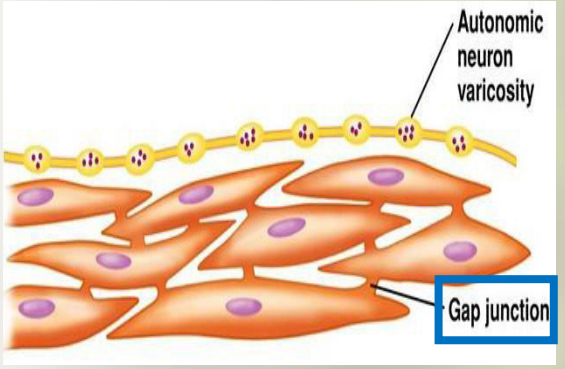
Single-Unit Muscles

- Found in the walls of visceral organs:
  - Alimentary canal
  - Urinary bladder
  - Ureters
  - Uterus
- Neurons synapse onto few cells.
- Modulate rate and strength of contraction.

Smooth Muscles

Single-Unit Muscles

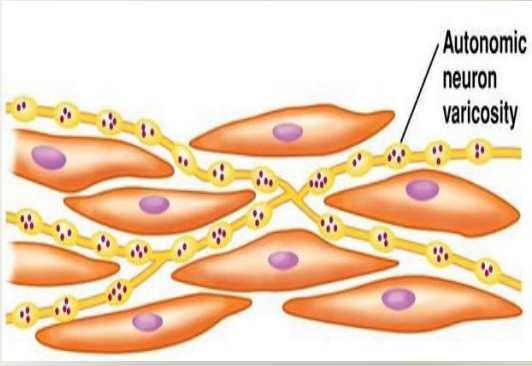
- Cells coupled electrically through gap junctions.
- Entire muscle mass contracts if only a few cells are excited or contracted.



Smooth Muscles

Multi-Unit Muscles

- Each cell innervated.
- Acts independently
- Contracts only through synaptic input.
- Muscles in the iris and walls of blood vessels.



Smooth Muscles

Excitation-Contraction Coupling

- A different mechanism.
- Have a poorly developed SR.
- Calcium-regulating functions performed by plasma membrane.

Smooth Muscles

Excitation-Contraction Coupling

- Also lack troponin and tropomyosin.
- An elongated protein caldesmon binds to actin filaments.
- Restricts myosin-actin interactions, inhibiting muscle contraction.

Smooth Muscles

Excitation-Contraction Coupling

- On excitation Ca^{2+} binds to protein calmodulin.
- A Ca^{2+}/ calmodulin complex formed.
- This complex binds to caldesmon.
- Removes inhibition on myosin-actin interactions.
- Contraction begins.

END

Cardiovascular Physiology

Excitatory and Conductive System of Heart

LECTURE NO 118

Excitatory and Conductive System of Heart

Excitatory and Conductive System

Heart has:

- A specialized system for generating rhythmic electrical impulses.
- A system to conduct these impulses rapidly throughout its muscles.
- These impulses cause and control rhythmic contraction of cardiac chambers.

Excitatory and Conductive System of Heart

Effects of Rhythmic Electrical Impulses

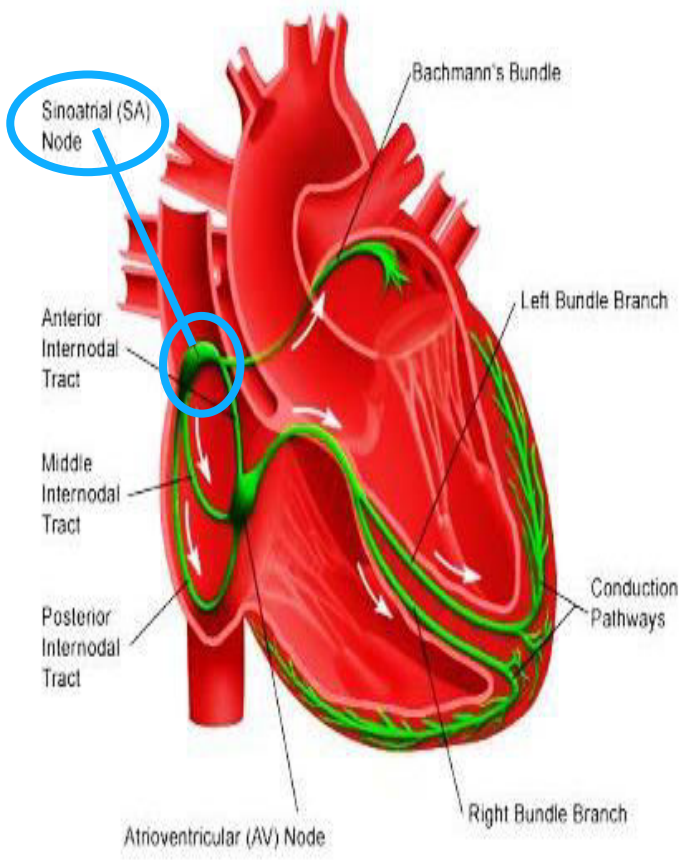
Rhythmicity of impulses cause:

- Contraction of atria about 1/6th sec. ahead of ventricular contraction.
- Filling of ventricles before they contract.
- Contraction of all portions of ventricles simultaneously.

Excitatory and Conductive System of Heart

Components of Excitatory System

- Comprises of pacemaker region (sinoatrial or S-A node).
- Rhythmical impulses are generated in the pacemaker.



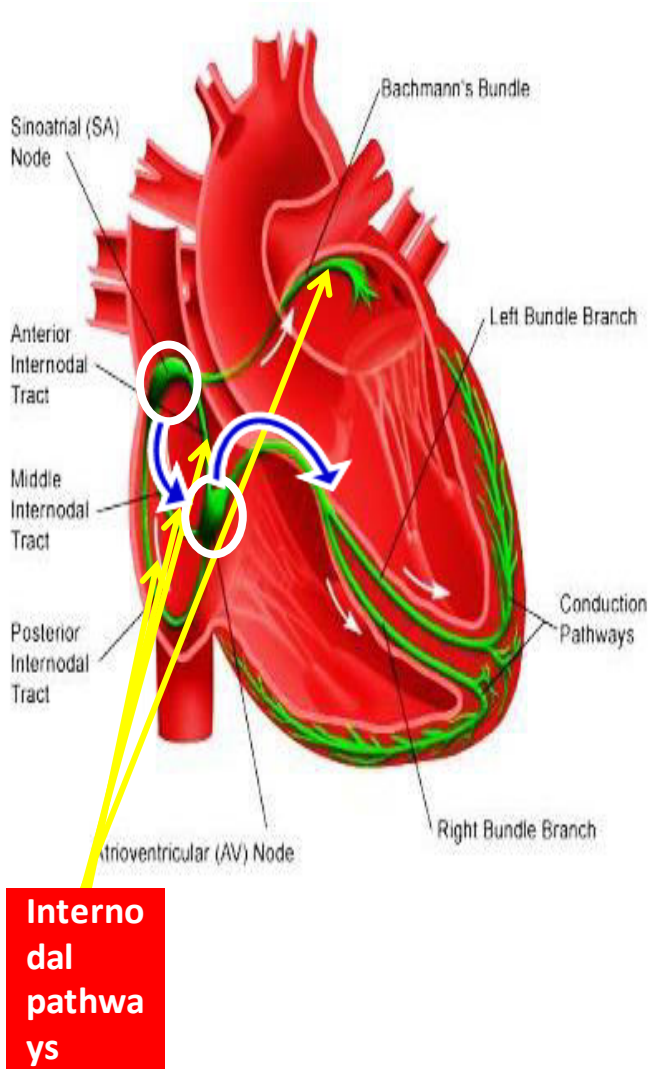
http://2.bp.blogspot.com/_7zQULPNQ7FQ/SohMP3pMV5I/AAAAAAAAATQ/y2VYsUGRUyQ/s400/cardiac-conduction-system.jpg

Excitatory and Conductive System of Heart

Components of Conductive System

Include:

1. Atrial internodal pathways (conduct impulse from S-A node to the A-V node).
2. A-V node (conducts impulses into the ventricles).

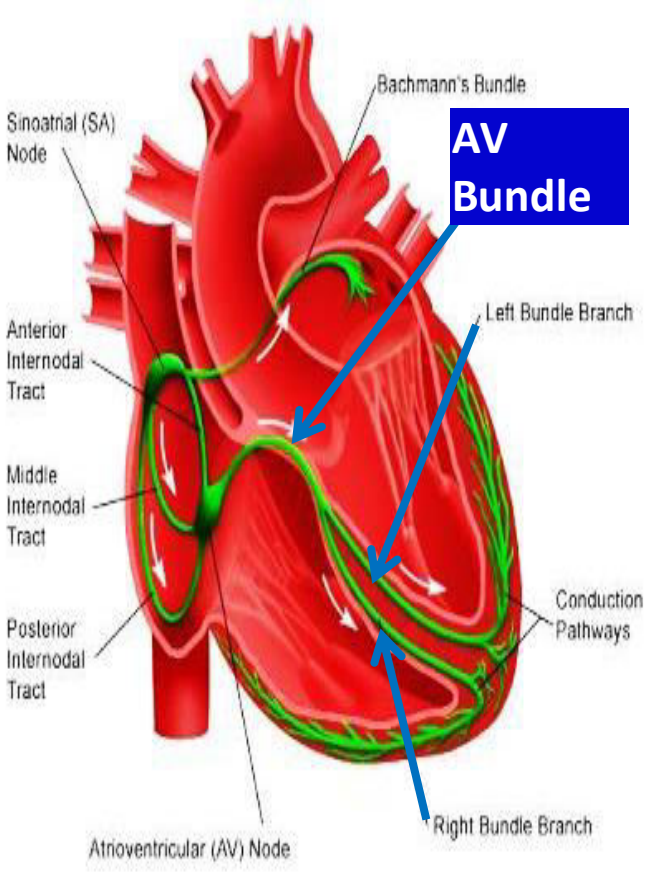


Excitatory and Conductive System of Heart

Components of Conductive System

3. A-V bundle (bundle of His), conducts impulses from atria into the ventricles.

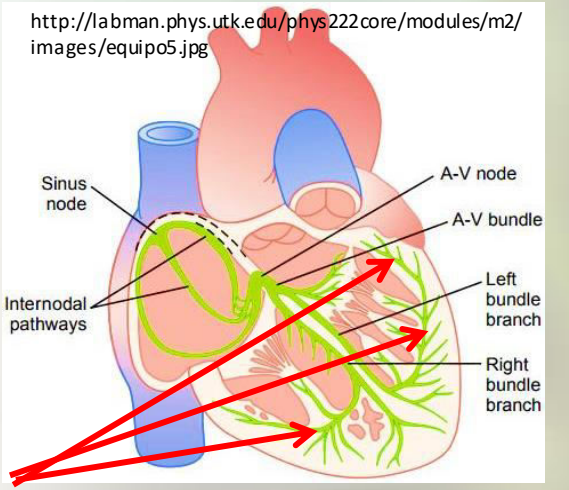
4. Left and right branches of A-V bundle.



Excitatory and Conductive System of Heart

Components of Conductive System

5. Purkinje fibers, branch off from bundle branches (conduct impulse to all parts of the ventricles).



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END

Cardiovascular Physiology

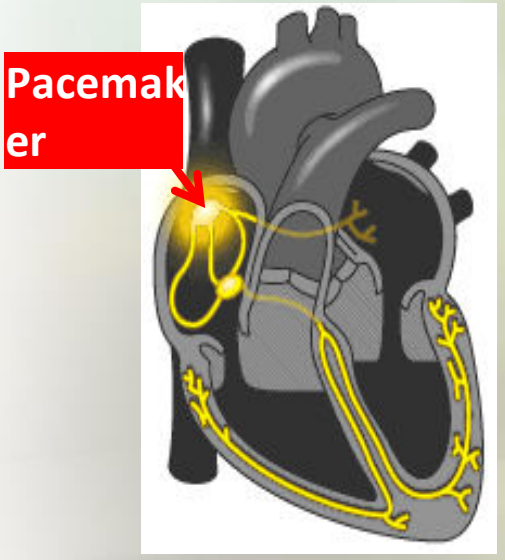
Pacemakers

LECTURE NO 119

Pacemakers

Pacemaker

- The excitatory region.
- Generates rhythmical impulses.
- Control the rhythmicity of cardiac chambers.



Pacemakers

Pacemaker Types

- Two basic types of pacemakers in animals with pumping hearts:
  - Neurogenic pacemakers
  - Myogenic Pacemakers

Pacemakers

Neurogenic Pacemakers

- Consists of neurons.
- Found in many invertebrate hearts.
- Such hearts are neurogenic hearts.

Pacemakers

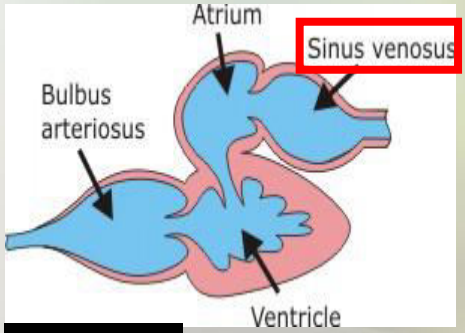
Myogenic Pacemakers

- Consists of specialized self-excitatory muscle cells.
- Found in some invertebrate & all vertebrate hearts.
- Such hearts are known as myogenic hearts.

Pacemakers

Vertebrate Myogenic Pacemaker

- In fishes and amphibians, situated in sinus venosus (a chamber of heart).

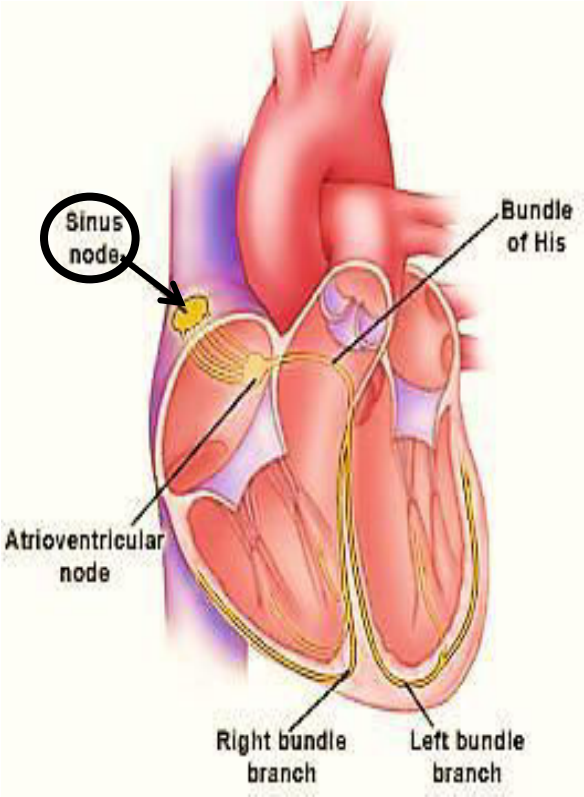


Fish heart

Pacemakers

Vertebrate Myogenic Pacemaker

- In amniotes, located in superior posterolateral wall of the right atrium (a vestigial remnant of sinus venosus).
- Named as sinus node or sinoatrial node (SA node).



Pacemakers

Characteristics of S-A Node

Consists of cells that are:

- Small
- Flattened
- Weakly contractile
- Specialized muscle cells
- Capable of self-excitation.

- Human S-A node is:
  3 millimeters wide
  15 millimeters long
  1 millimeter thick

Pacemakers

Characteristics of S-A Node

- S-A nodal fibers connect directly with atrial muscle fibers.
- An AP in sinus node spreads immediately into atrial muscle wall.

END

Cardiovascular Physiology

Autorhythmicity of Pacemaker

LECTURE NO 120

Autorhythmicity of Pacemaker

Autorhythmicity

- Ability of certain cardiac cells to spontaneously and repetitively generate an electrical impulse without a stimulus from the nervous system.

Autorhythmicity of Pacemaker

Pacemaker

- Any cardiac cells that can generate an impulse and can maintain heart rate.
- Include:
  - Sinoatrial node
  - Cells of the AV node
  - Bundle of His
  - Purkinje cells

Autorhythmicity of Pacemaker

S-A Node as Pacemaker

- Rhythmical discharge of sinoatrial node is:
  - 70 to 80 times per minute
  - It is faster than other impulse-generating parts.
- S-A node is always the pacemaker of a normal heart.

Autorhythmicity of Pacemaker

Ectopic Pacemakers

- If SA node stops functioning:
  - Other autorhythmic components may take control of rhythmicity.
  - But at a slower rate.
- Such pacemakers are ectopic pacemakers.

Autorhythmicity of Pacemaker

Ectopic Pacemakers

A-V node

- Rhythmic discharge rate : 40 to 60 times per minute.

Purkinje fibers

- Discharge rate: 15 to 40 times per minute.

Autorhythmicity of Pacemaker

Autorhythmicity of Pacemaker: Cause

- Absence of a stable resting potential.
- “Resting membrane potential” of S-A nodal fibers: -55 to -60 mV.
- Comparison: -85 to -90 mV of ventricular muscle fibers.

Autorhythmicity of Pacemaker

Autorhythmicity of Pacemaker: Cause

- Cause of less –ve RMP:
  - Cell membranes are naturally leaky to sodium and calcium ions.
- Effect:
  - S-A nodal fibers undergo a steady depolarization, termed as pacemaker potential.

Autorhythmicity of Pacemaker

END

Pacemaker Potentials and Autorhythmicity

- As pacemaker potential reaches threshold, an all-or-none cardiac AP is generated.
- AP spreads to whole cardiac tissue.
- Contraction is caused.

Cardiovascular Physiology

Role of Ion Channels in Self Excitation

LECTURE NO 121

Role of Ion Channels in Self Excitation

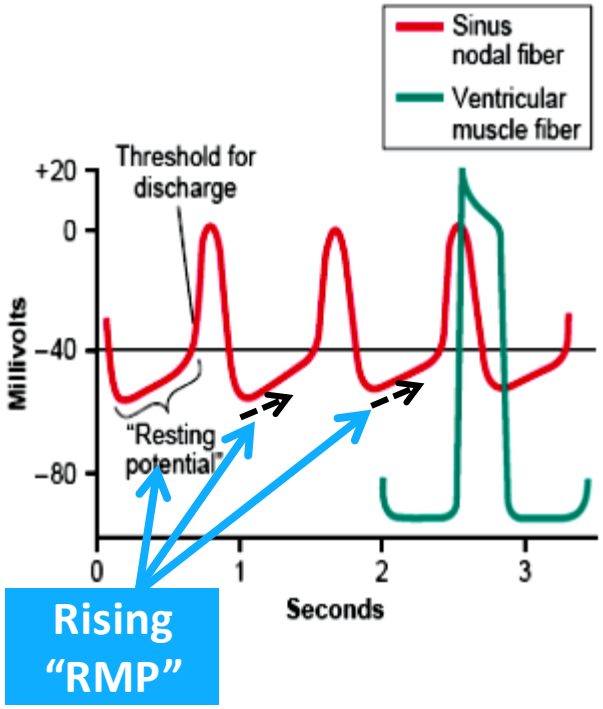
Basis of Self-Excitation

- Inherent leakiness of S-A nodal fibers to sodium and calcium ions.
- Leakiness is due to three types of membrane ion channels:
  - Fast sodium channels
  - Slow sodium-calcium channels
  - Potassium channels

Role of Ion Channels in Self Excitation

Activity of Ion Channels

- These channels activate and deactivate at a pace that keeps:
- The RMP much less negative (only -55 millivolts)
- An unstable and rising RMP (due to continuous influx of Na^+ and Ca^{2+} ions.



Role of Ion Channels in Self Excitation

Role of Fast Sodium Channels

- Open immedialtely as the V_m goes less -ve than -55 mV after an AP.
- Remain open for only a few milli-sec.
- Let Na^+ ions immediately move inside the cell.

Role of Ion Channels in Self Excitation

Role of Fast Sodium Channels

- This Na^+ influx prevents developing a stable, more negative RMP.
- These channels become inactivated and blocked above -55 mV.

Role of Ion Channels in Self Excitation

Role of Slow Sodium-Calcium Channels

- At -55 mV and above, slow sodium-calcium channels start to open.
- Na^+ ions tend to leak inside.
- Membrane potential gradually rises.
- Preventing to establish a stable RMP.

Role of Ion Channels in Self Excitation

Role of Slow Sodium-Calcium Channels

- As V_m reaches threshold of -40 mV, sodium-calcium channels become activated maximally.
- Influx of sodium and calcium ions starts.
- This trigger AP.

Role of Ion Channels in Self Excitation

Role of Slow Sodium-Calcium Channels

- Rise in membrane potential is slower in the range of -55 to -40 mV.
- So, the sinoatrial nodal AP is slower to develop than AP of ventricular muscle fibers.

Role of Ion Channels in Self Excitation

END

Role of Potassium Channels

- After the action potential, the K^+ channels open slowly.
- Their opening causes the return of membrane potential to its negative state.
- Owing to their slow activation, duration of S-A nodal AP is longer.

Cardiovascular Physiology

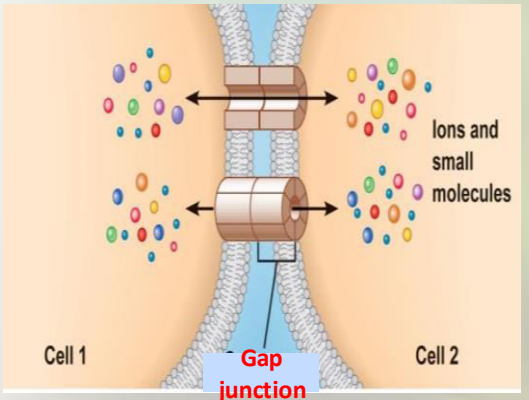
Transmission of Excitation Over the Heart

LECTURE NO 122

Transmission of Excitation Over the Heart

Gap Junctions

- Cardiac cells coupled electrically through gap junctions.
- Regions of low resistance between cells.



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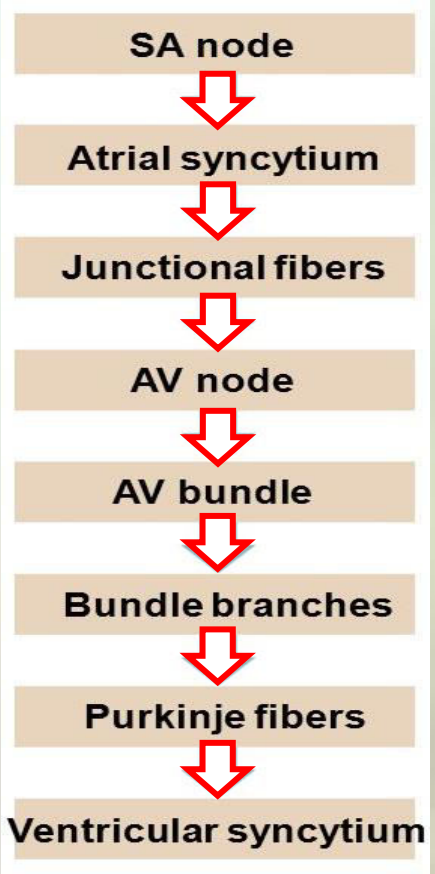
Transmission of Excitation Over the Heart

Role of Gap Junctions

- Allow current flow from one cell to the next.
- Pacemaker's electrical activity spreads over the entire heart via gap junctions.

Transmission of Excitation Over the Heart

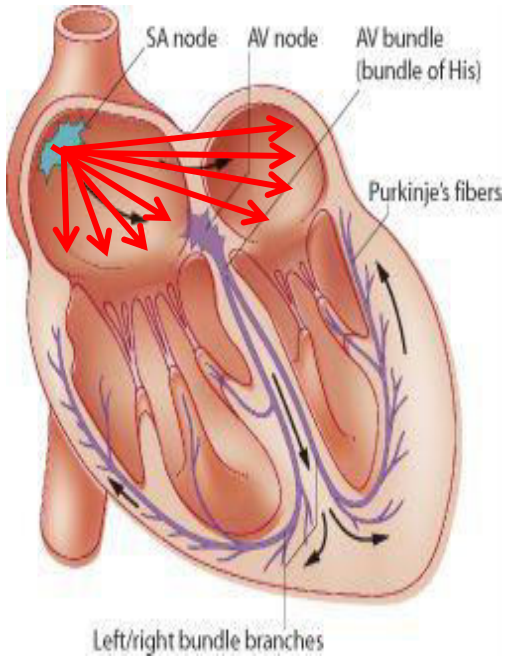
Path of Transmission of Excitation Over the Heart



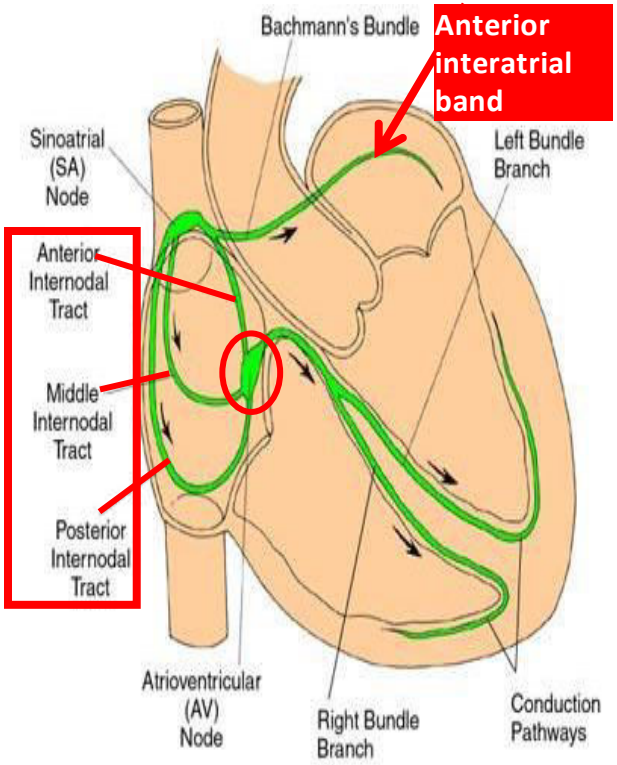
Transmission of Excitation Over the Heart

Transmission of Cardiac Impulse Through Atria

- S-A nodal fibers connect directly with atrial muscle fibers.
- Wave of excitation spreads over both atria.
- Velocity of conduction in most atrial fibers: ~0.3 m/s. Conduction is rapid in smaller bands of atrial fibers (~1 m/s).



Transmission of Excitation Over the Heart



Internodal Pathways

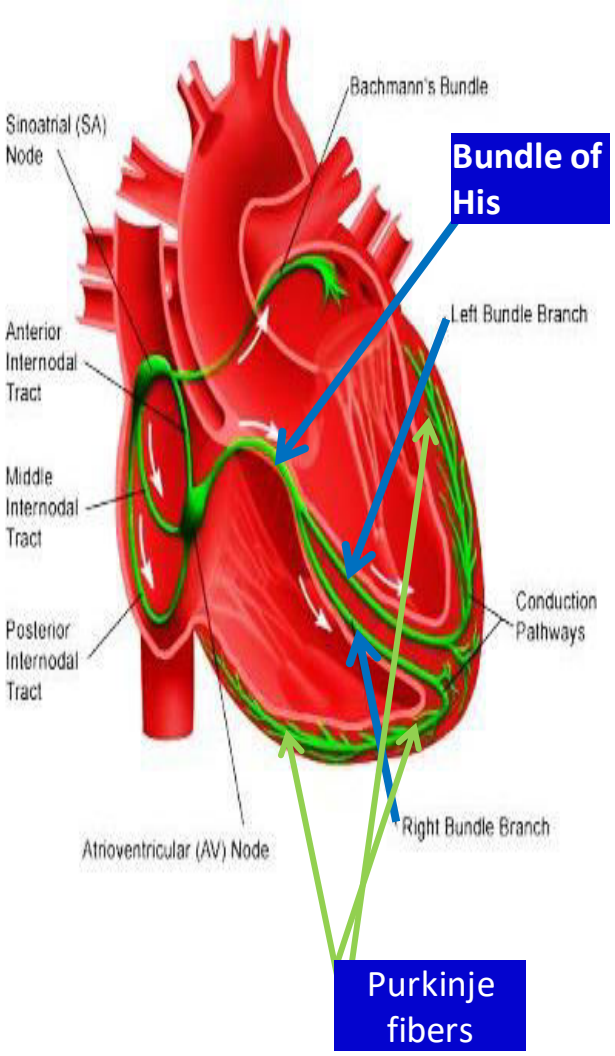
- Spread of excitation from atrial musculature to A-V node through small junctional fibers.
- Junctional fibers form internodal pathways, including:
- Anterior interatrial band
- Three small bands (curve through anterior, middle, and posterior atrial walls)

Transmission of Excitation Over the Heart

Delay in Impulse Conduction

- Velocity of wave of excitation in junctional fibers: ~0.05 m/s.
- Slowness causes delay in conduction of impulse to ventricles.
- This delay allows:
- atrial contractions to precede ventricular contractions.
- time for blood to move from atria to ventricles.

Transmission of Excitation Over the Heart



Conduction Through Bundle of His and Purkinje Fibers

- Bundle of His, its branches and Purkinje fibers deliver wave of excitation to all regions of ventricular myocardium.
- All the ventricular muscle fibers contract together.

Transmission of Excitation Over the Heart

Conduction Through Bundle of His and Purkinje Fibers

- Conduction is rapid through the bundle of His and Purkinje fibers (4-5 m/s).

END

Cardiovascular Physiology

Effect of ACh and Catecholamines on Excitation

LECTURE NO 123

Effect of ACh and Catecholamines on Excitation

Effect of Acetylcholine on Pacemaker Potentials

- Parasympathetic cholinergic fibers of Vagus nerve (10th cranial nerve) innervate S-A and A-V nodes.
- ACh is released from terminals of these nerve fibers
- ACh slows heart rate (-ve chronotropic effect).

Effect of ACh and Catecholamines on Excitation

Effect of Acetylcholine on Pacemaker Potentials

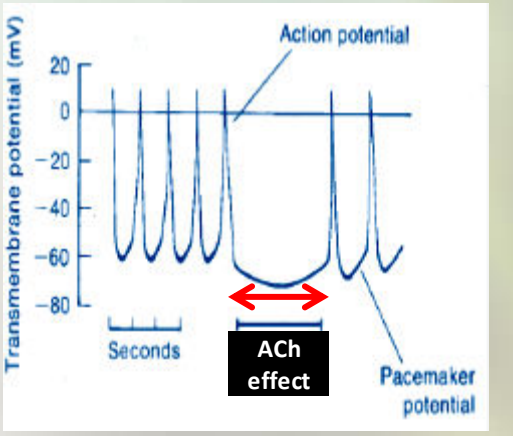
- ACh increases K^+ ion conductance of pacemaker cells.
- Flow of K^+ keeps V_m near potassium equilibrium potential for a longer time.

Effort by: AARISH ALI

Effect of ACh and Catecholamines on Excitation

Effect of Acetylcholine on Pacemaker Potentials

- Actions of ACh slow pacemaker depolarization.
- So the interval between APs is increased.



Effect of ACh and Catecholamines on Excitation

Effect of Acetylcholine on Pacemaker Potentials

- ACh also reduces velocity of conduction of excitation from:
  - Atria to A-V node
  - A-V node to ventricles

Effect of ACh and Catecholamines on Excitation

Effect of Catecholamines on Pacemaker Potentials

- Adrenergic nerve fibers innervate S-A node, atria, A-V node and ventricles.
- Norepinephrine is released from them.

Effect of ACh and Catecholamines on Excitation

Effect of Catecholamines on Pacemaker Potentials

- Norepinephrine increases the rate of contraction of myocardium.
- It increases Na^+ and Ca^{2+} conductance.
- Na^+ and Ca^{2+} accelerate pacemaker depolarization.

Effect of ACh and Catecholamines on Excitation

Effect of Catecholamines on Pacemaker Potentials

- Norepinephrine also increases:
  - Force of contraction of myocardium (positive inotropic effect).
  - Speed of conduction of wave of excitation over the heart (positive dromotropic effect).

END

Cardiovascular Physiology

Cardiac Output and Stroke Volume

LECTURE NO 124

Cardiac Output and Stroke Volume

Cardiac Output

- “Quantity of blood pumped into the aorta each minute by the heart”.
- Average cardiac output for resting adult human: 5 L/min.

Cardiac Output and Stroke Volume

Factors Affecting Cardiac Output

Cardiac output varies widely with:

- Basic level of body metabolism
- Physical activity of the body
- Size of the body
- Age
- Gender

Cardiac Output and Stroke Volume

Stroke Volume

- "Volume of the blood ejected by each beat of heart".
- It is the difference b/w volume of the ventricle just before contraction (end-diastolic volume) and volume of ventricle at the end of a contraction (end-systolic volume).

Cardiac Output and Stroke Volume

Measurement of Stroke Volume

- Determined by dividing cardiac output by heart rate.
- Changes in stroke volume result from changes in end-diastolic or end-systolic volume.

Cardiac Output and Stroke Volume

Determining the End-Diastolic Volume

End-diastolic volume depends on four parameters:

- Venous filling pressure
- Pressures generated during atrial contraction
- Distensibility of the ventricular wall
- Time available for filling the ventricle

Cardiac Output and Stroke Volume

END

Determining the End-Systolic Volume

End-systolic volume depends on two parameters:

- Pressures generated during ventricular systole
- Pressure in the outflow channels i.e. aortic and pulmonary arteries.

Cardiovascular Physiology

Changes in Pressure and Flow During One Beat

LECTURE NO 125

Changes in Pressure and Flow During One Beat

Fluctuations in Pressure and Flow

- Chambers of heart contract during each heart beat
- Contraction rhythm result in sequential fluctuations of pressure and volume

Changes in Pressure and Flow During One Beat

Diastole

During diastole:

- Aortic valves are closed
- Ventricles and atria are relaxed
- Pressure difference b/w relaxed chambers and systemic and pulmonary arteries is large

Changes in Pressure and Flow During One Beat

Diastole

- Atrioventricular valves remain open
- Blood flows from venous system directly into the ventricles by venous filling pressure (passing through atria)

Changes in Pressure and Flow During One Beat

Atrial Contraction

- Contraction causes pressure rise in them
- Blood is ejected into the ventricles

Changes in Pressure and Flow During One Beat

Evident Role of Venous Filling Pressure

- Atrial contraction provides only 30% volume of total mammalian ventricular output
- Ventricular filling is largely determined by venous filling pressure
- Atrial contraction simply tops up nearly full ventricles

Changes in Pressure and Flow During One Beat

Ventricular Contraction

- When ventricular muscles begin to contract, pressure rises in chambers
- Atrioventricular valves close—to prevent backflow of blood

Changes in Pressure and Flow During One Beat

Ventricular Contraction

- Aortic valves are closed in the beginning
- Ventricles become sealed chambers
- Pressure rises in them (due to contracting muscles) without a volume change
- i.e. ventricular contraction is isometric

Changes in Pressure and Flow During One Beat

Ventricular Contraction

- When pressure within ventricles exceeds that in aorta and pulmonary arteries, aortic valves open
- Blood is ejected into the aorta and pulmonary arteries
- Result: ventricular volume decreases

Changes in Pressure and Flow During One Beat

END

Ventricular Relaxation

After Pumping:

- Ventricles begin to relax
- Intraventricular pressure falls
- Aortic valves close
- Atrioventricular valves open
- Ventricular filling starts again—by venous filling pressure
- The cycle is repeated

Cardiovascular Physiology

Work Done by the Heart

LECTURE NO 126

Work Done by the Heart

Work

In physics, work is defined as:

- The product of force and distance

W = F x s

Work = Force x Distance

Work Done by the Heart

Work Done by the Heart

- For the heart, work is to move a volume of blood

So, work is equal to:

- Product of volume of blood moved

And

- Pressure required to move it

W = V x P

Work = Volume x Pressure

Work Done by the Heart

Calculating the Work Done by the Heart

- Volume is taken as the stroke volume (blood volume ejected during ventricular contraction)
- Ventricular pressure is taken as mean arterial pressure

Work Done by the Heart

Calculating the Work Done by the Heart

- Work done by the heart during one stroke (SW) is the product of:
- Stroke volume (SV)

and

- Mean arterial pressure (MAP).

$$SW \cong SV \times MAP$$

Work Done by the Heart

Work Done By the Two Ventricles

- Work done differs for right and left ventricles of a mammalian heart
- Both ventricles eject equal volumes of blood
- Right ventricle pumps blood to shorter pulmonary circuit
- Left ventricle pumps blood to extensive systemic circuit

Work Done by the Heart

Work Done By Right Ventricle

- Pressure generated by right ventricle is much lower
- External work done by it is much less

Work Done by the Heart

Work Done By Left Ventricle

- Pressure generated by left ventricle is much higher
- Work done by this ventricle is higher

END

Cardiovascular Physiology

Electrocardiogram (ECG)

LECTURE NO 127

Electrocardiogram (ECG)

Basis of ECG

Cardiac impulses:

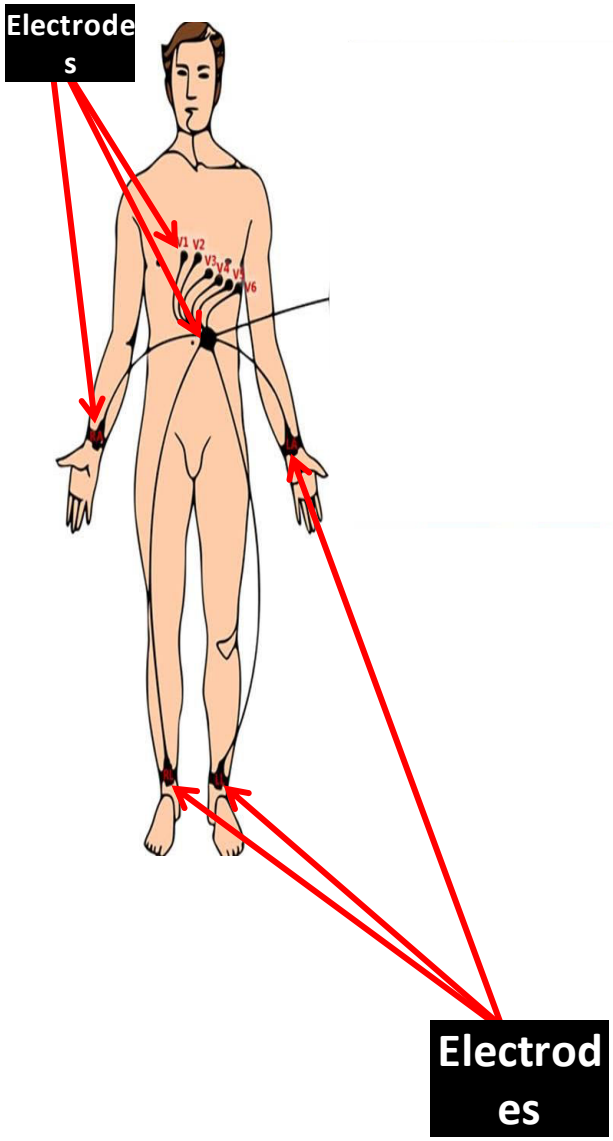
- Are due to depolarization and repolarization of cardiac muscle fibers
- Cause electrical changes while passing through the heart muscles

Electrocardiogram (ECG)

Basis of ECG

- Electrical current in heart spreads to the adjacent tissues
- A small portion spreads all the way to surface of the body

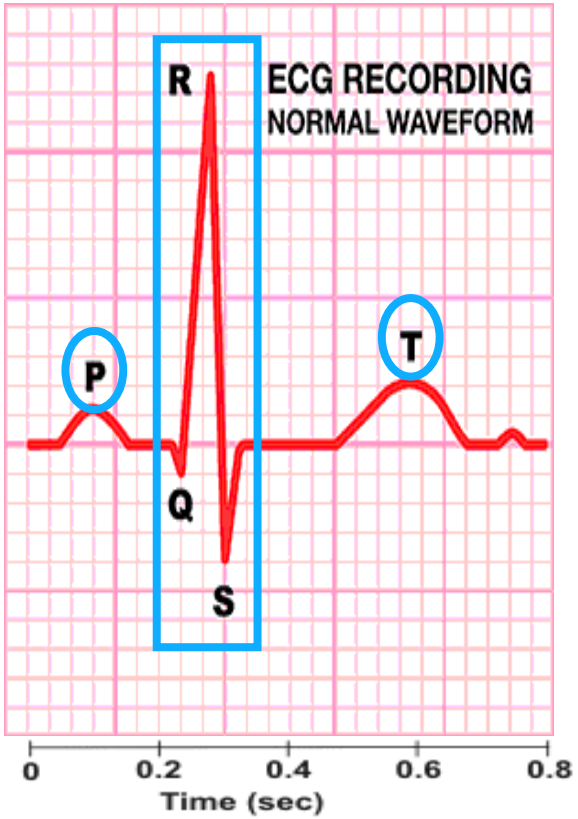
Electrocardiogram (ECG)



Electrocardiogram

- Electrical potentials generated by the current can be recorded
- Placing electrodes of electrocardiograph on the skin at certain locations.
- This recording is electrocardiogram (ECG).

Electrocardiogram (ECG)



An Electrocardiogram

Graphically reflects:

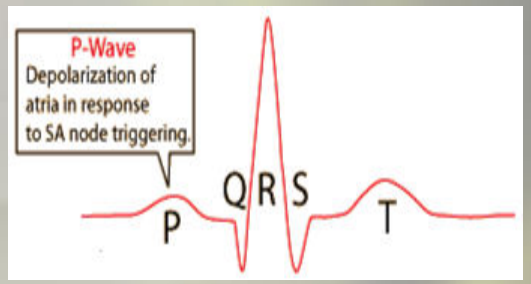
- Depolarization and repolarization as wave deflections
- Represent specific events of cardiac cycle
- Wave deflections are designated as:
  - a P wave
  - a QRS complex
  - a T wave

Electrocardiogram (ECG)

P Wave

- Caused by electrical potentials when atria depolarize
- Occurs just prior to contraction of atria
- Ventricles are in diastole during the expression of P wave.

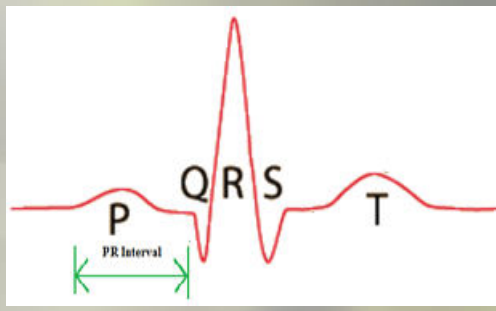
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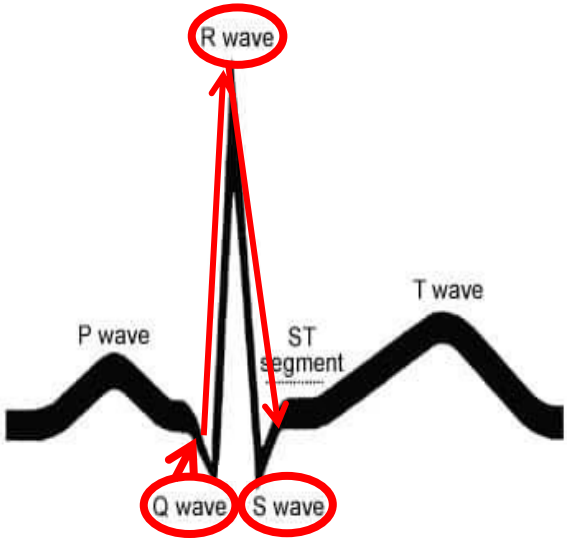
Electrocardiogram (ECG)

The P-R Interval

- Period of time from start of P wave to beginning of the QRS complex.
- Indicates amount of time required for SA depolarization to reach the ventricles



Electrocardiogram (ECG)



QRS Complex

- Comprises of three separate waves

It

- Begins as a short downward deflection (Q-wave)
- Continues as a sharp upward spike (R wave)
- Ends as a downward deflection (S wave)

Electrocardiogram (ECG)

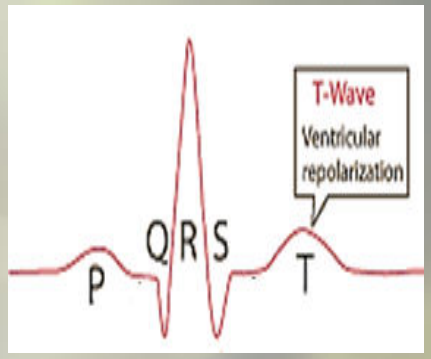
QRS Complex

- Indicates depolarization of ventricles
- During this interval, ventricles are in systole i.e. blood is being ejected from the heart

Electrocardiogram (ECG)

T Wave

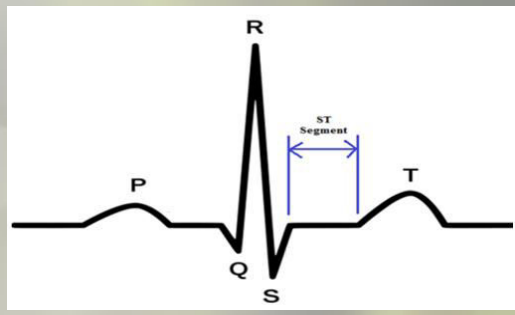
- Known as “repolarization wave”
- Caused by potentials generated when ventricles recover from state of depolarization and repolarize



Electrocardiogram (ECG)

S-T Segment

- Represents time duration b/w completion of depolarization of ventricle and initiation of its repolarization
- This segment should be flat in a normal ECG



Electrocardiogram (ECG)

Uses of Electrocardiogram

- Any heart disease that disturbs electrical activity produces characteristic changes in one or more of ECG waves
- Understanding these wave-deflection patterns is clinically important

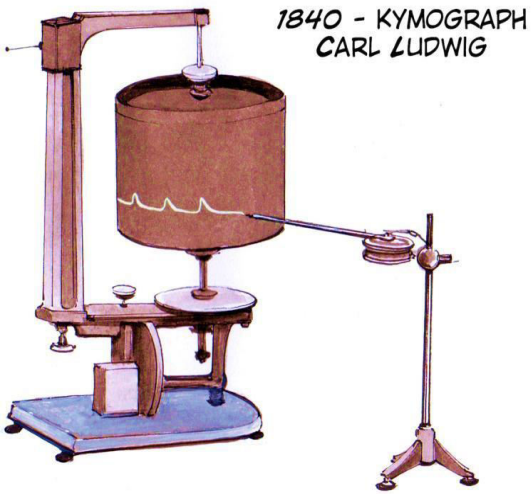
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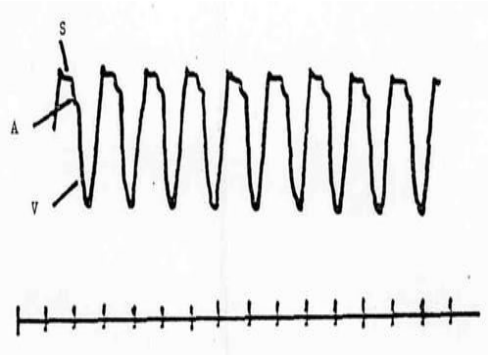
Cardiovascular Physiology

Kymography

LECTURE NO 128

Kymography





Kymograph

- “A device that graphically records changes in the mechanical activities of animal tissues”
- The term comes from Latin word meaning "wave writer“—referring to graphical records produced by the instrument

Kymography

Uses of Kymograph

Used in physiological experiments related to:

- Study of skeletal muscle contractions (twitch and tetanus)
- Cardiac muscle activities (cardiac cycle)
- Measurement of blood pressure
- Rate of respiration

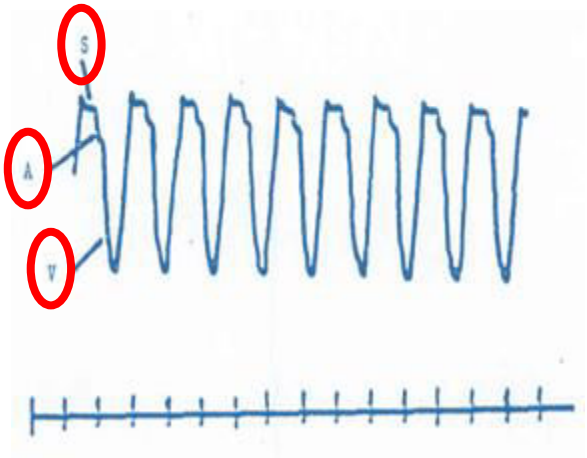
Kymography



Kymograph Apparatus

- Consists of a revolving drum to which a writing stylus is attached
- Stylus records changes over time on a paper wrapped around the drum as the drum revolves

Kymography



A typical kymograph of frog's heart beat showing sinus (s) auricular (a) and ventricular (v) beats.

Kymography of Frog's Heart

- Generally carried out in physiology labs

Provides understanding of:

- Cardiac cycle of frog
- Gives clues of the site of origin of heart beat and its control
- Coordination of contractions of chambers of heart

Kymography

Kymography of Frog's Heart

Kymography also helps to study the effects of on cardiac output:

- Temperature
- Various ions
- Acetylcholine
- Epinephrine

END

Cardiovascular Physiology

Introduction to Hemodynamics

LECTURE NO 129

Introduction to Hemodynamics

Hemodynamics

"The study of physical laws that explain the relationship between pressure and flow of blood through blood vessels of circulatory system"

Introduction to Hemodynamics

Principles of Hemodynamics

- In animals with a closed circulation, blood flows in a continuous circuit

Introduction to Hemodynamics

Blood Returned = Blood Pumped

- Blood is an incompressible fluid
- Volume of blood returning back to heart (each minute) must be equal to the cardiac output (volume pumped out each minute)

Introduction to Hemodynamics

Velocity of Flow

- The velocity of flow at any point is inversely related to total cross-sectional area of the blood vessel.

$$V = Q/A$$

- V = velocity (cm/s)
- Q = blood flow (ml/s)
- A = cross sectional area (cm^2)

Introduction to Hemodynamics

Velocity of Flow

- Blood flow velocity is highest where cross-sectional area is smallest (arteries)
- Lowest velocities occur where cross-sectional area is largest (capillaries)

Introduction to Hemodynamics

Velocity of Flow

- Highest velocities occur in aorta and pulmonary artery
- Velocity falls in capillaries
- It rises again through veins

Introduction to Hemodynamics

Significance of Low Velocity of Flow

- Slow flow of blood in capillaries is time consuming
- Significant for exchange of substances between blood and tissues

Introduction to Hemodynamics

Vascular Resistance

- Blood vessels offer resistance to flow (vascular resistance)
- VR must be overcome to create flow through circulatory system
- Vasoconstriction and greater viscosity increases VR
- Vasodilation and lower viscosity decreases VR

Introduction to Hemodynamics

Vascular Resistance

VR is related to:

- Vessel radius
- Vessel length
- Blood viscosity

Used in calculations of:

- Blood pressure
- Blood flow
- Cardiac function

Introduction to Hemodynamics

Blood Pressure

- Produced due to pumping action of heart
- As pumping action is pulsatile, so is the blood pressure
- BP in systemic arteries varies during each heartbeat, as:
  - Systolic BP (high)
  - Diastolic BP (low)

END

Cardiovascular Physiology

Laminar and Turbulent Flow

LECTURE NO 130

Laminar and Turbulent Flow

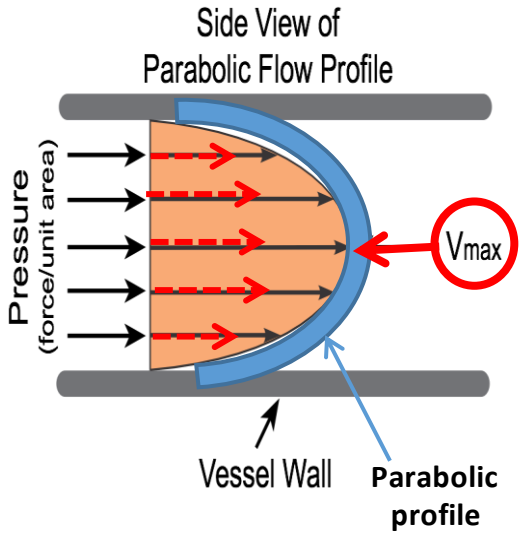
Laminar and Turbulent Flow

- Blood flow is affected by smoothness of inner lining of blood vessels
- Texture of blood vessels results in either turbulent or laminar (smooth) flow

Laminar and Turbulent Flow

Laminar Flow in Smaller Vessels

- Flow is streamlined and continuous
- Characterized by a parabolic velocity profile across the vessel
- Flow occurs in layers at different velocities
- Flow is zero at the wall and maximal at the center
- A pressure difference supplies the force required to slide adjacent layers to past each other



Laminar and Turbulent Flow

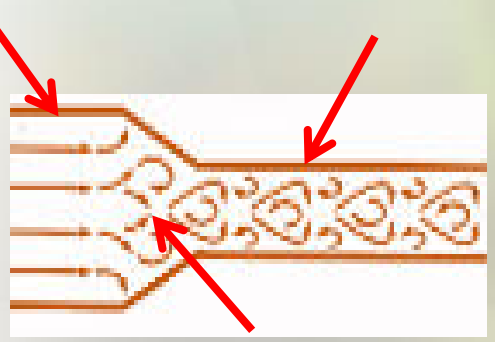
Laminar Flow in Larger Vessels

- Pulsatile
- More complex velocity profile than continuous laminar flow
- Blood is accelerated and slowed with each heartbeat

Laminar and Turbulent Flow

Pulsatile Laminar Flow

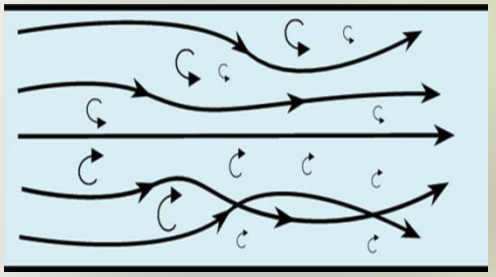
- Reason
- Larger vessel walls are elastic
- Expand and relax with pressure oscillation
- Direction of flow also reverses near the heart as aortic valves shut



Laminar and Turbulent Flow

Turbulent Flow

- Direction of fluid movement is not aligned
- Requires higher energy to move blood through a vessel



Laminar and Turbulent Flow

Turbulent Flow

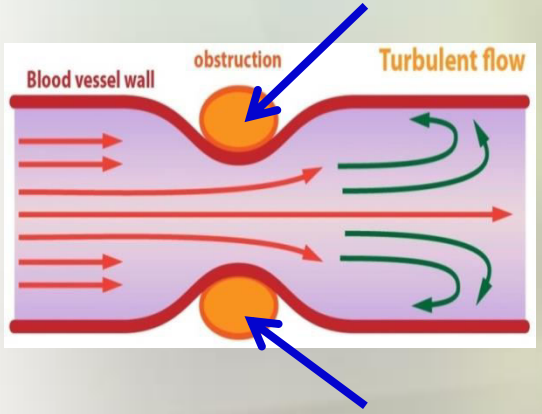
- Highest turbulence in proximal portions of aorta and pulmonary artery
- At the time of ventricular contraction
- During backflow of blood when pulmonary and aortic valves close

Laminar and Turbulent Flow

Turbulent Flow

- Turbulence also happens if smoothness is reduced by any obstruction e.g.

  Buildup of fatty deposits on arterial walls



Laminar and Turbulent Flow

Turbulent Flow

- Uncommon in peripheral, undivided vessels with smooth walls
- Occurs in some situations e.g.

  During very high blood velocities (during strenuous exercise)

END

Cardiovascular Physiology

Relationship Between Pressure and Flow

LECTURE NO 131

Relationship Between Pressure and Flow

Pressure Gradient Determines Flow

- "Difference in blood pressure between two points in a flow path establishes a pressure gradient"
- Pressure gradient determines the direction of flow
- From high to low pressure

Relationship Between Pressure and Flow

Resistance to Flow is Overcome by Pressure

- When heart contracts, pressure in the ventricles increases
- This pressure is used to overcome resistance to flow through the vessels

Relationship Between Pressure and Flow

Role of Kinetic Energy

- Flow of blood depends both on pressure and kinetic energy
- When blood is ejected into the aorta—pressure is converted into kinetic energy
- This energy sets the blood into motion

Relationship Between Pressure and Flow

Role of Kinetic Energy

- Kinetic energy is highest in aorta
- Kinetic energy is negligible in capillaries
- So velocity of flow is highest in aorta and lowest in capillaries

Relationship Between Pressure and Flow

Poiseuille's Law

- Describes relationship between pressure and flow in a rigid tube

Poiz wills

Relationship Between Pressure and Flow

Poiseuille's Law

$$Q = \frac{(P_1 - P_2)\pi r^4}{8L\eta}$$

- It states that the flow rate of a fluid, Q, is directly proportional to:
  - Pressure difference ($P_1 - P_2$) along the length of tube
  - Fourth power of the radius of the tube (r^4)
- and inversely proportional to:
  - Tube length (L)
  - Fluid viscosity (η)
- As flow rate Q is proportional to r^4, very small changes in vessel diameter have a profound effect on flow rate

Relationship Between Pressure and Flow

Poiseuille's Equation and Blood Flow

- Poiseuille's equation applies to steady flows in straight rigid tubes
- Blood vessels are not rigid tubes
- Also, blood pressure and flow are pulsatile

Relationship Between Pressure and Flow

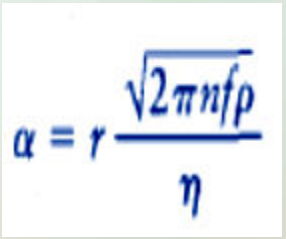
Poiseuille's Equation and Blood Flow

- Doesn't accurately describe pressure-flow relationship in blood vessels
- It is used in modified form for blood flow in vessels

Relationship Between Pressure and Flow

Modified Poiseuille's Equation

- Includes calculating and adding a non-dimensional constant α

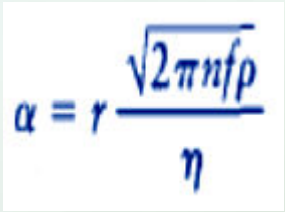


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- α indicates deviation from Poiseuille's law in blood vessels

Relationship Between Pressure and Flow

Poiseuille's Equation For Blood Flow



In this equation:

- ρ = density of blood
- η = viscosity of blood
- f = frequency of oscillation
- n = order of harmonic component
- r = radius of vessel

rho

END

Cardiovascular Physiology

Vascular Resistance to Flow

LECTURE NO 132

Vascular Resistance to Flow

Vascular Resistance

"The resistance that must be overcome to push blood through the circulatory system and create flow"

Vascular Resistance to Flow

Vascular Resistance

Types

- Systemic vascular resistance (SVR)
- Pulmonary vascular resistance (PVR)

Uses

SVR Used in calculations of

- Blood pressure
- Blood flow
- Cardiac function

Vascular Resistance to Flow

Factors affecting SVR

- Elasticity of vessel wall
- Diameter of vessel
- Decreasing vessel diameter (vasoconstriction) increases SVR
- Increasing vessel diameter (vasodilation) decreases SVR

Vascular Resistance to Flow

Calculation of Vascular Resistance

- Calculated by a modified form of the Poiseuille's equation:

$$R = 8L\eta / \pi r^4$$

- R = resistance to blood flow
- L = length of the vessel
- η = viscosity of blood
- r = radius of blood vessel

Vascular Resistance to Flow

END

Calculation of Vascular Resistance

$$R = 8L\eta / \pi r^4$$

It is evident that:

- Resistance to flow is inversely proportional to fourth power of radius of the vessel
- So, is minimal in larger vessels, but maximum in narrow capillaries

Midterm Term Syllabus
Lecture 01 to 132

Efforts By

AARISH ALI (+96569002641)

Contact us to join our esteemed group

“All About Biotech”

Note:

Students must follow the course announcement for midterm syllabus & any up-gradation in the course.

REMEMBER IN PRAYERS

KEEP SMILING

SAY NO TO CURRUPTION

HELP TO OTHERS

Regards,

AARISH ALI & TEAM

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